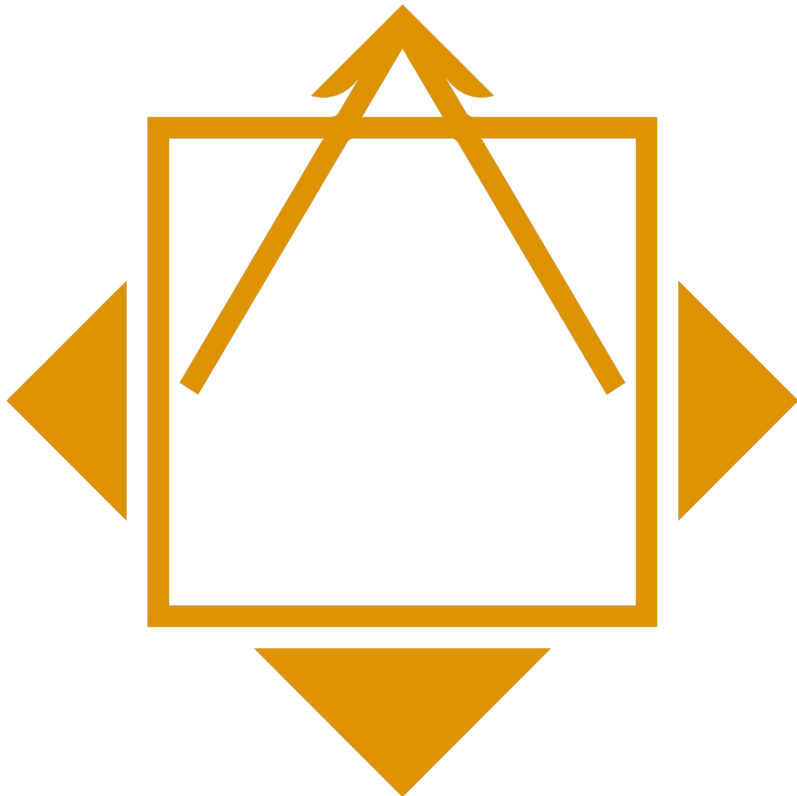




PRUSA WIRE 2026.R1

A Prusa MK3/MK4 to Voron Switchwire Conversion
by Positron 3D

Updated 2026-06-12

CAUTION



Before you begin on your journey, a word of caution.

In the comfort of your own home you are about to assemble a robot. This machine can maim, burn, and electrocute you if you are not careful.

Please do not become the first Prusawire fatality. There is no special Discord flair for that.

Please, read the entire manual before you start assembly.

As you begin building, please check our Discord channels ([#prusawire-questions](#) and [#prusawire-discussions](#)) for any tips and post any questions that may have.

Have fun!

The Prusawire Team and Positron 3D

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INTRODUCTION

CREDITS

Thank you to all the awesome people and companies that have made this possible.

- Positron 3D Team
- Nomads Galaxy, *initial idea*
- Ella Fox, *the architect of Prusawire Beta 1*
- Prusa Research - Prusa i3
- Voron Design - Switchwire
- Jason @ LDO Motors - Part sponsorship
- Levendigs & KB-3D - Part sponsorship
- Sirayatech Filaments
- The many early access and beta testers

We hope you enjoy this build as much as we enjoyed making it.

- The Prusawire Team

@Safe // @erikbuild // @TheNexusAvenger

INTRODUCTION

PART PRINTING GUIDELINES

We recommend the Voron Design *Part Printing Guidelines* for all structural parts.

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

INFILL TYPE

Lines, Grid, Gyroid, Honeycomb, Triangle or Cubic

MATERIAL

ABS or ASA

INFILL PERCENTAGE

Recommended: 40%

LAYER HEIGHT

Recommended: 0.2mm

WALL COUNT

4 walls (0.4mm nozzle)
3 walls (0.5mm nozzle)

EXTRUSION WIDTH

Recommended: Forced 0.4mm

SOLID TOP/BOTTOM LAYERS

Recommended: 5

SUBSTITUTIONS

You may, if desired, substitute something like PETG for non-structural parts (e.g. Screen, aesthetic pieces), but we **strongly recommend** ABS or ASA for all structural parts.

Glass Fiber or Carbon Fiber materials can be used, but note that these materials are stiffer (ASA-GF, ASA-CF) and can sometimes be more brittle (like PC-CF).

INTRODUCTION

FILE NAMING

By this time you should have already downloaded our STL files from the [Prusawire GitHub](#). You might have noticed that we have used a unique naming convention for the files. This is how to use them.



PRIMARY COLOR

Example:
front_grill_left.stl

These files will have nothing at the start of the filename

ACCENT COLOR

Example:
[a]_front_grill_logo_plates.stl

We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

QUANTITY REQUIRED

Example:
[a]_rubber_foot_block_x4.stl

If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP



<https://discord.com/invite/positron>



INTRODUCTION

REPORTING ISSUES

Should you find an issue in the documentation or have a suggestion for an improvement, please consider opening an issue on GitHub (<https://github.com/Positron3D/Prusawire/issues>).

When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

This manual is designed to be a simple reference manual. Building a Prusawire can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.

3D CAD Web Viewer:

<https://cad.positron3d.com>

Prusawire Documentation Site:

<https://prusawire.positron3d.com>



BILL OF MATERIALS (BOM)

The latest BOM can be found on the Prusawire GitHub repository:

<https://github.com/Positron3D/Prusawire/tree/main/BOM>



Hardware kits from partner vendors are available at the top of the BOM listing for convenience.

ASSUMPTIONS

Prusawire is a flexible build. Some accommodations have been designed in for different diameter and length Z rods, different control boards (Prusa Einsy RAMbo, SKR Mini E3), different power supplies (Prusa Black, Prusa Silver, Meanwell), and more. Not all steps presented may be relevant to your build.

It is not possible to document all variations, but notes have been added where appropriate to call out options you may have and resources that may be useful.

This guide describes in detail a “typical” Prusawire build that is re-using a MK3/MK4 Frame, Einsy RAMbo control board, Prusa MK3-style Control Panel, SuperPINDA mount, and LDO NiteHawk-SB + Stealthburner.

HARDWARE

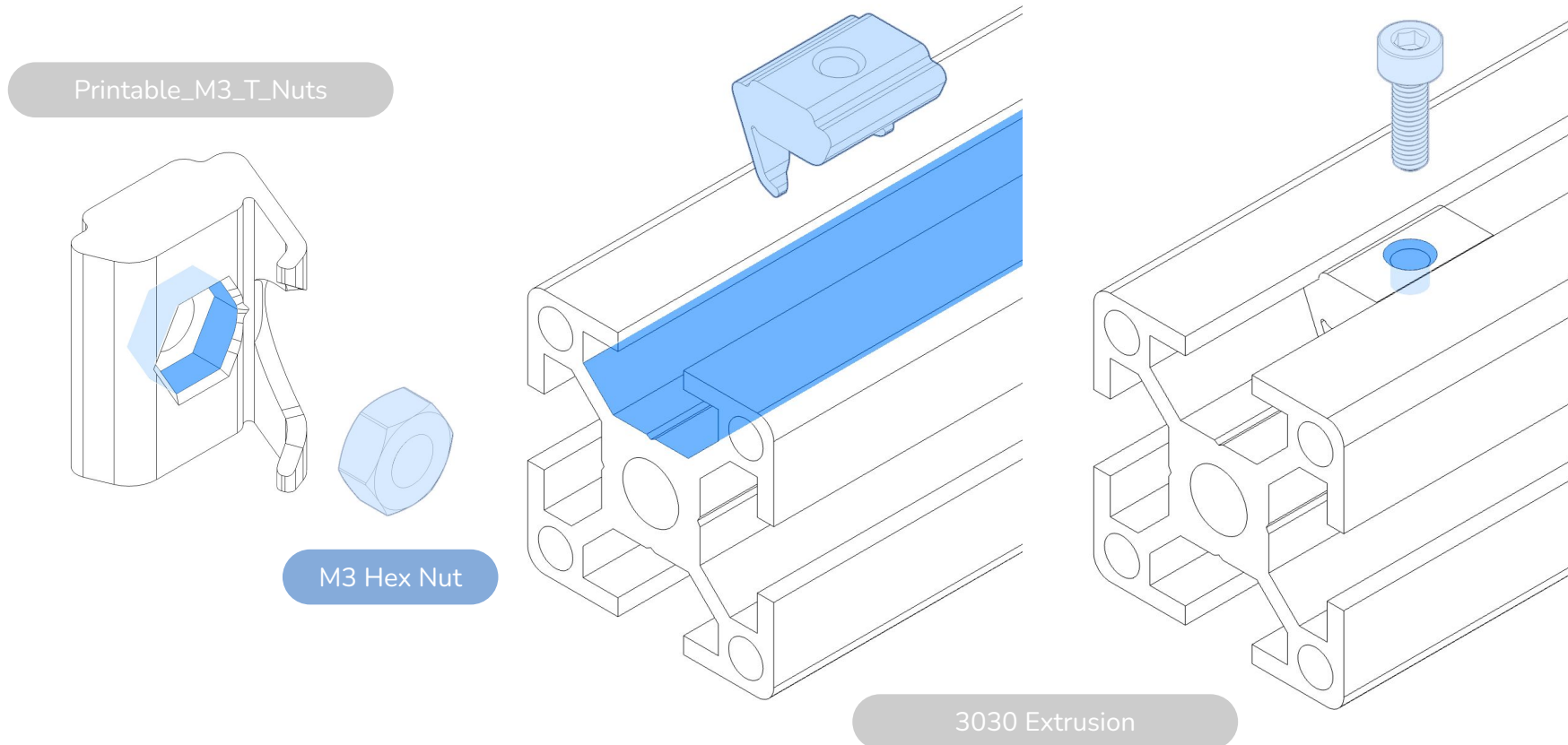
M3 T-Nuts for 3030 Extrusion

PRINTED VERSUS METAL T-NUTS

The Prusawire Team recommends a printed M3 Roll-in T-Nut for 3030 Extrusion as actual hardware is expensive and unnecessary in this application.

Special thanks to Kapplion for the [Printable M3 T-Nut for 3030 Extrusion](#).

The spring “tabs” on the bottom of the T-Nut are designed to bend slightly so you can roll the T-Nut into the extrusion and stay in place.



FRAME



FRAME

FRAME ASSEMBLY

You likely already have a bare, assembled Prusa i3 Frame if you're attempting this build. However, for reference, you can refer to the Prusa MK4, Prusa MK3, and/or Bear Frame assembly guides for frame assembly information:

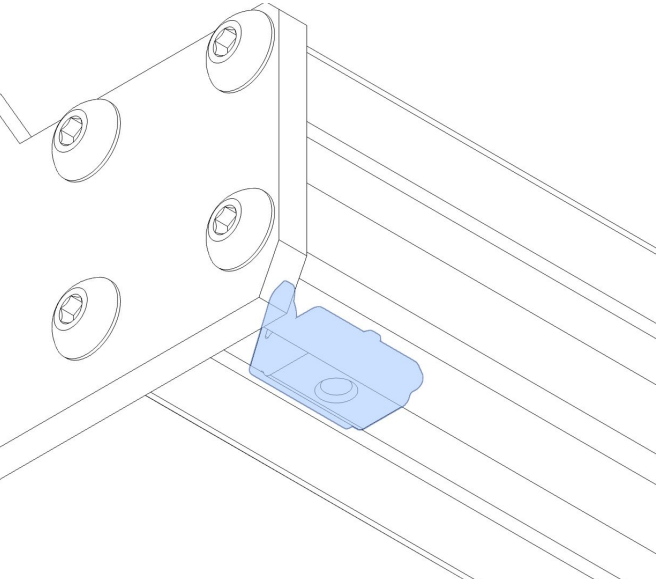
Prusa MK4 Frame: https://help.prusa3d.com/guide/2-frame-assembly_707740#708424
Reference Chapter 2, Steps 1 through 11.

Prusa MK3 Frame: https://help.prusa3d.com/guide/2a-y-axis-assembly_406654
Reference Chapter 2, Steps 1 through 10.

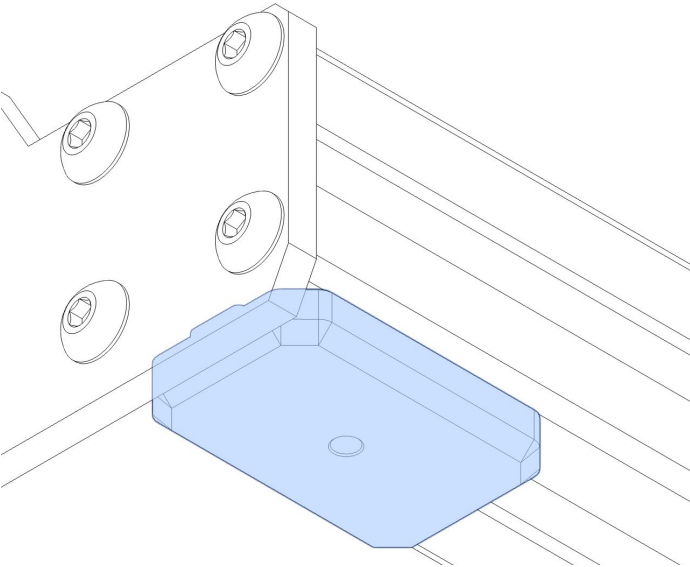
Make sure the frame is square and securely fastened together.



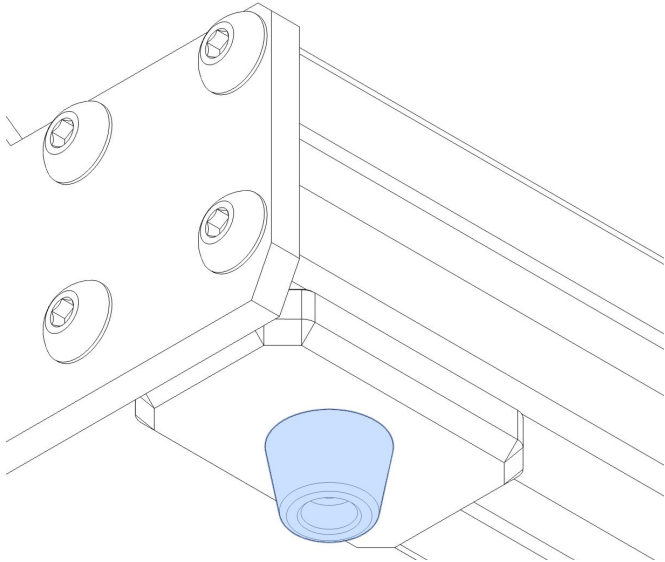
RUBBER FEET



Printable_M3_T_Nuts

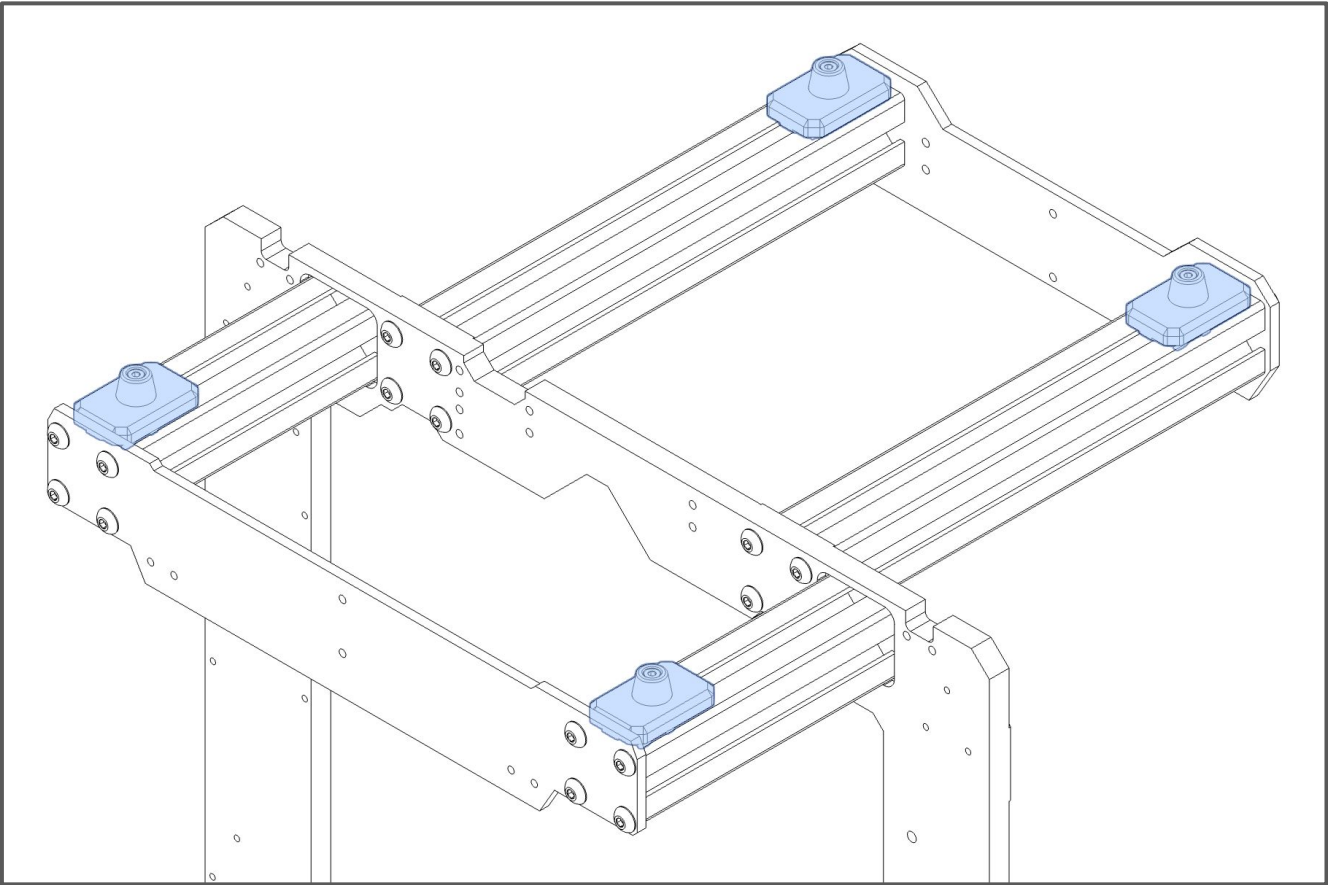
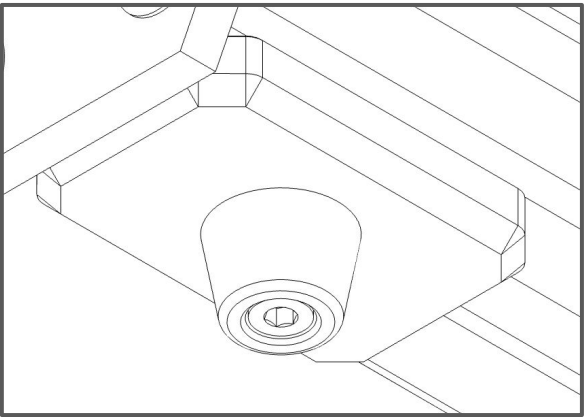
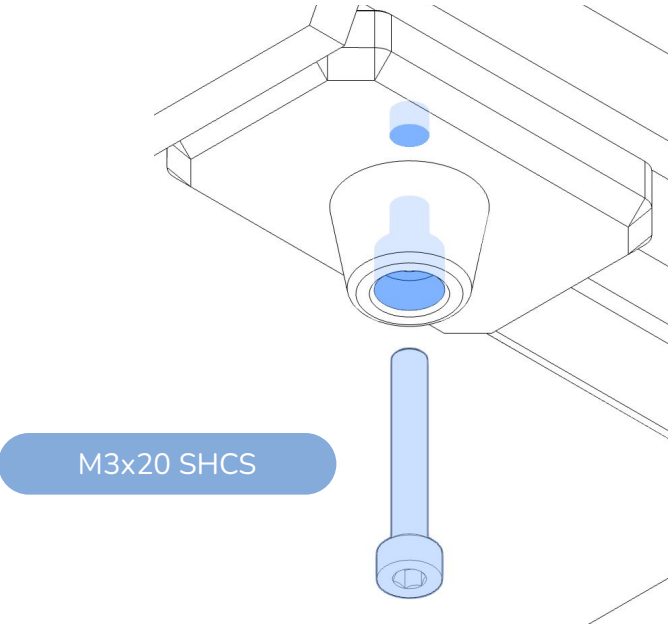


[a]_rubber_foot_block_4x



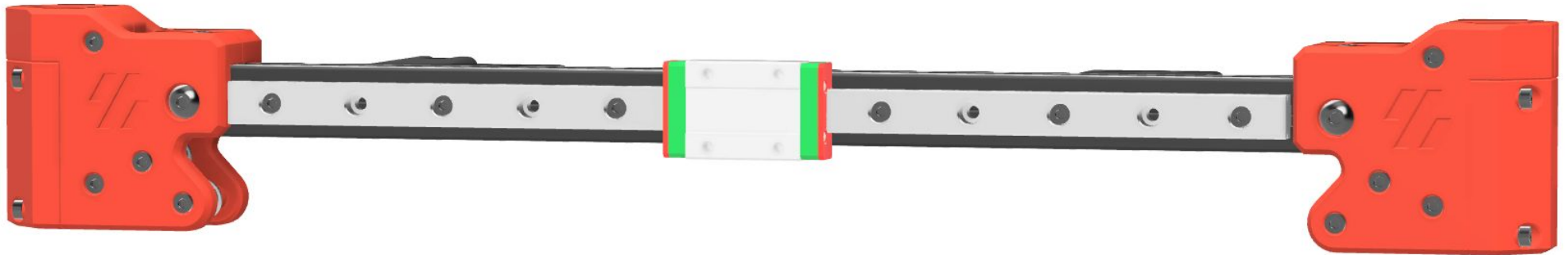
Rubber Foot

RUBBER FEET



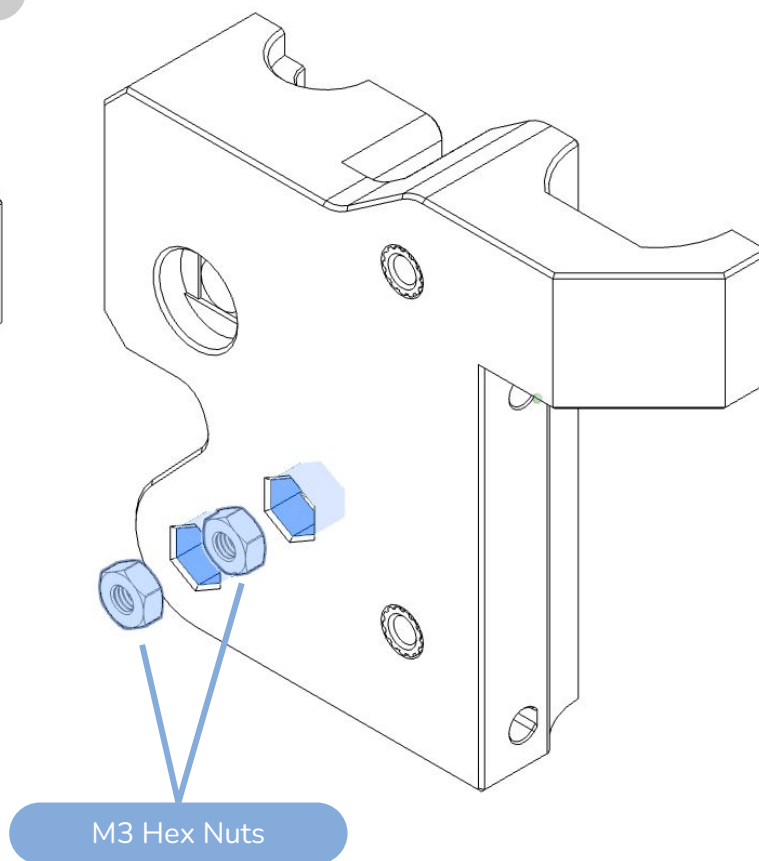
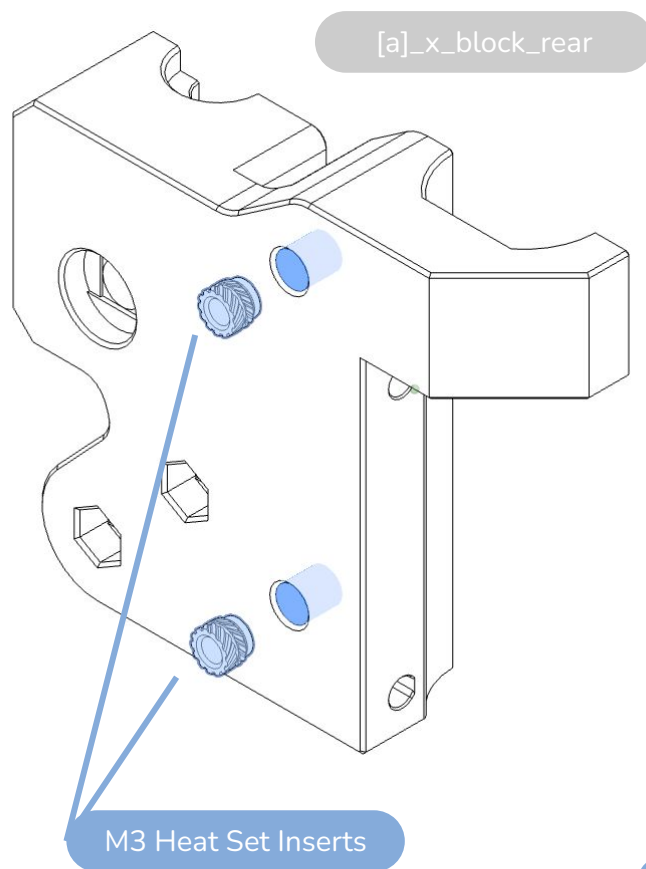
X GANTRY

Overview



X GANTRY

XZ Block - Left

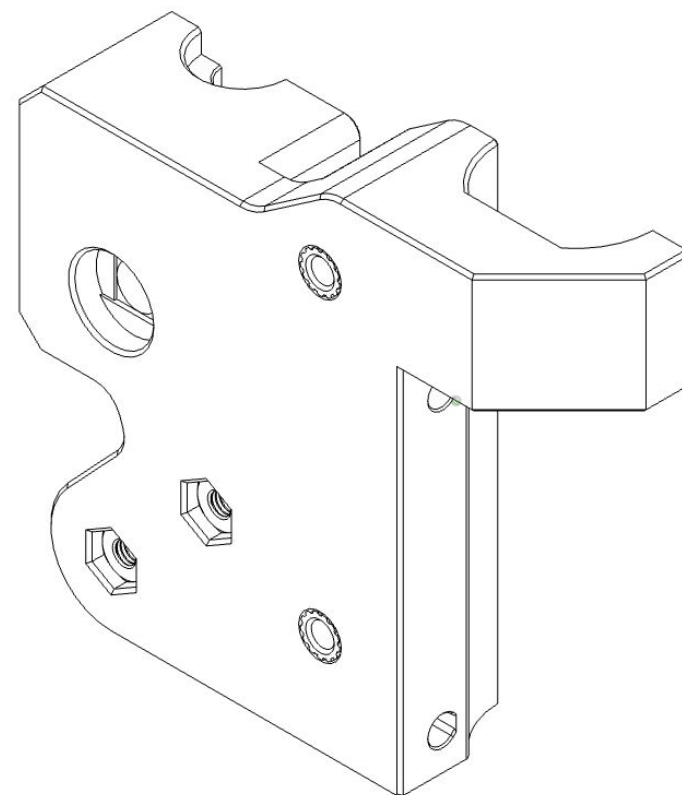


XZ Blocks

Four variants of the XZ Blocks are provided depending on which combination of M3 Square Nut and Belt Gears you are using.

Prusa (slim)-style M3 Square Nuts: M3S	//	Voron V0-style square nuts: V0S
Prusa-spec 16T Gears: 16T	//	Switchwire-spec 20T Gears: 20T

Example: [a]_xz_blocks_M3Sx16T

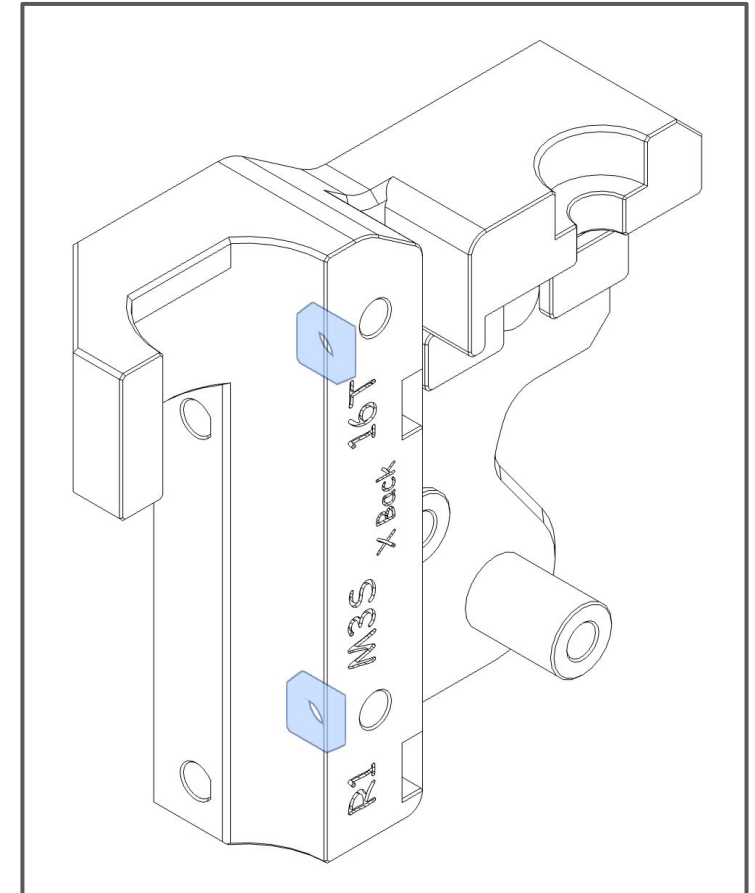
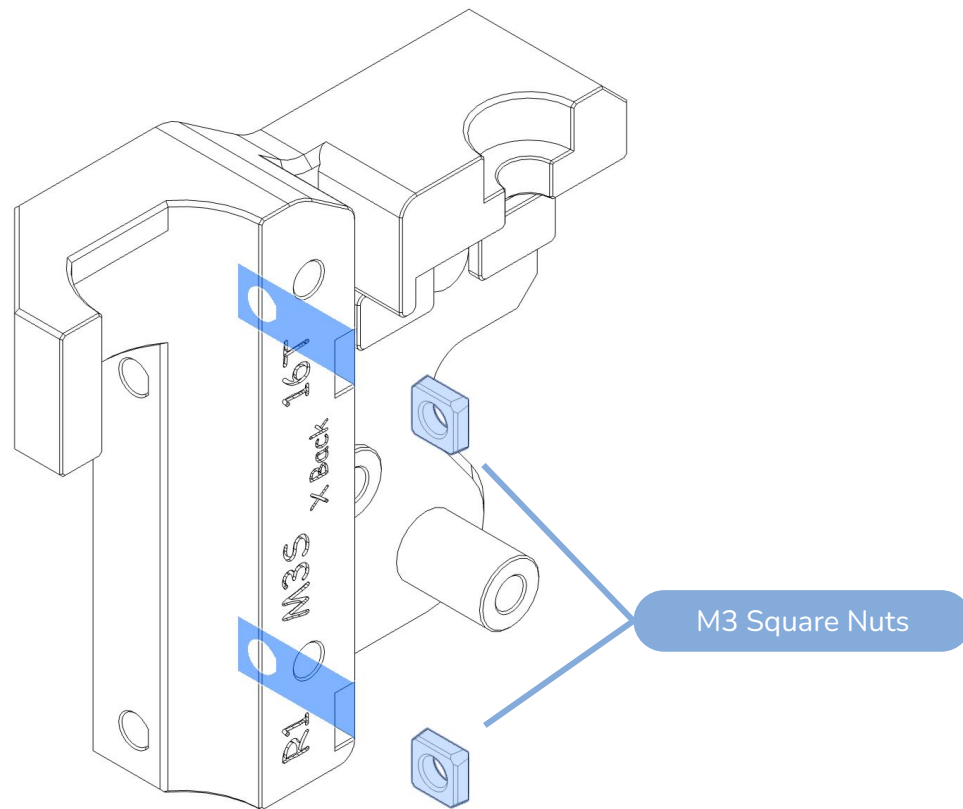


HEAT SET INSERTS

This design relies heavily on heat set inserts. Make sure you have the proper inserts (Voron-spec M3 heat inserts; see the BOM).

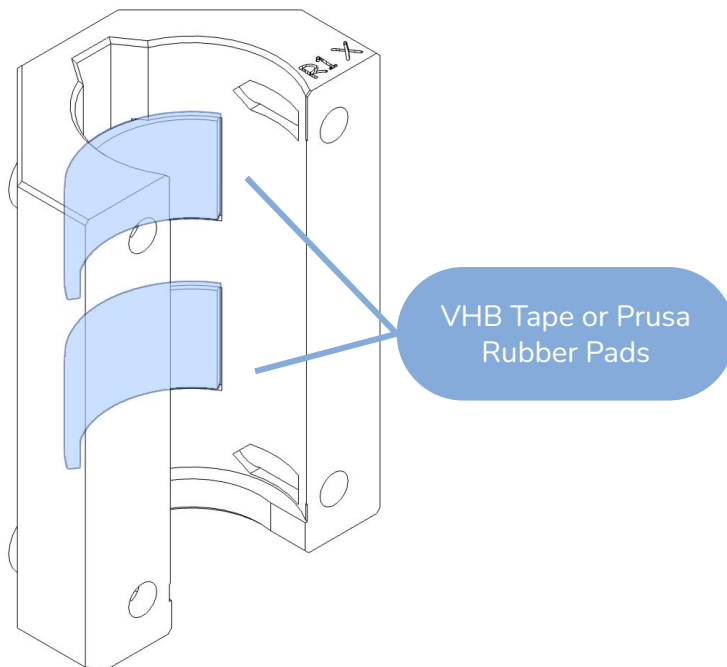
If you've never worked with heat set inserts before we recommend you watch the [linked guide](#).

X GANTRY
XZ Block - Left



X GANTRY

XZ Block - Left



BEARING PADS

You can use the Prusa 20x10x1mm rubber pads if you have them, or just use a similar size cut piece of VHB tape!

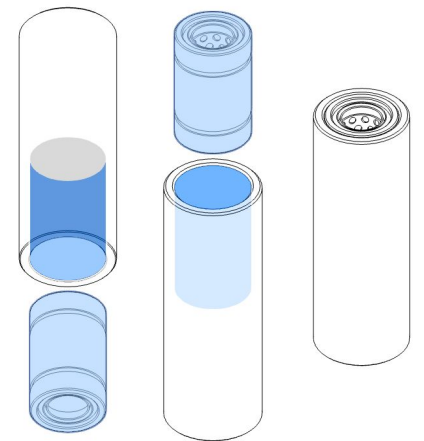
8MM LINEAR RODS AND BEARINGS

To use 8mm Linear Rods and LM8UU Bearings, replace the 1x LM10LUU with 2x LM8UU.

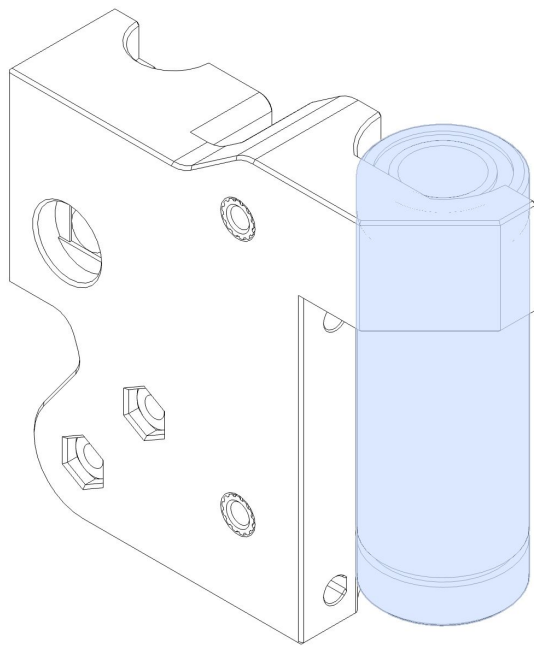
Print **8mm_bearing_adapter_2x_LM8UU** and use around the bearings to adapt the size.

To use 8mm Linear Rods and LM8LUU Bearings, replace the 1x LM10LUU with 2x LM8LUU.

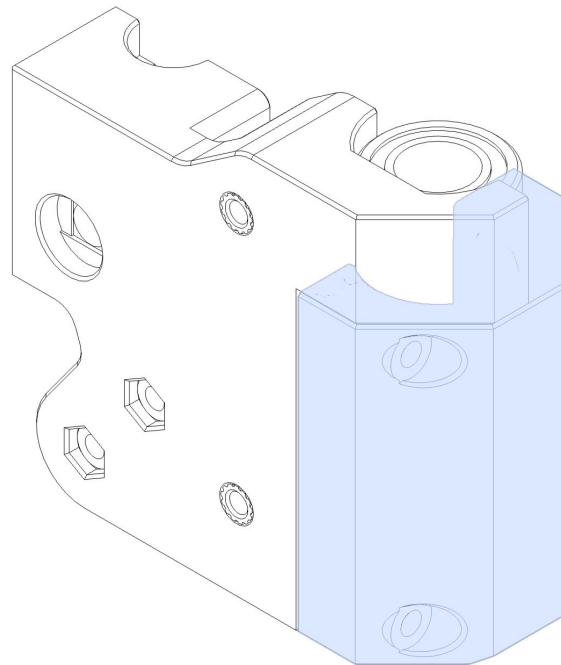
Print **8mm_bearing_adapter_2x_LM8LUU** and use around the bearings to adapt the size.



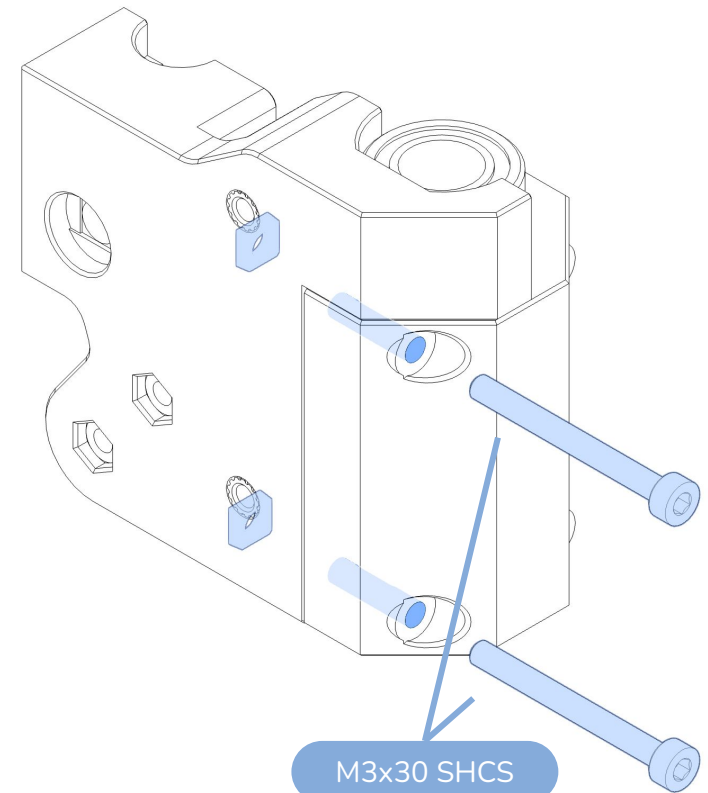
X GANTRY
XZ Block - Left



LM10LUU Bearing

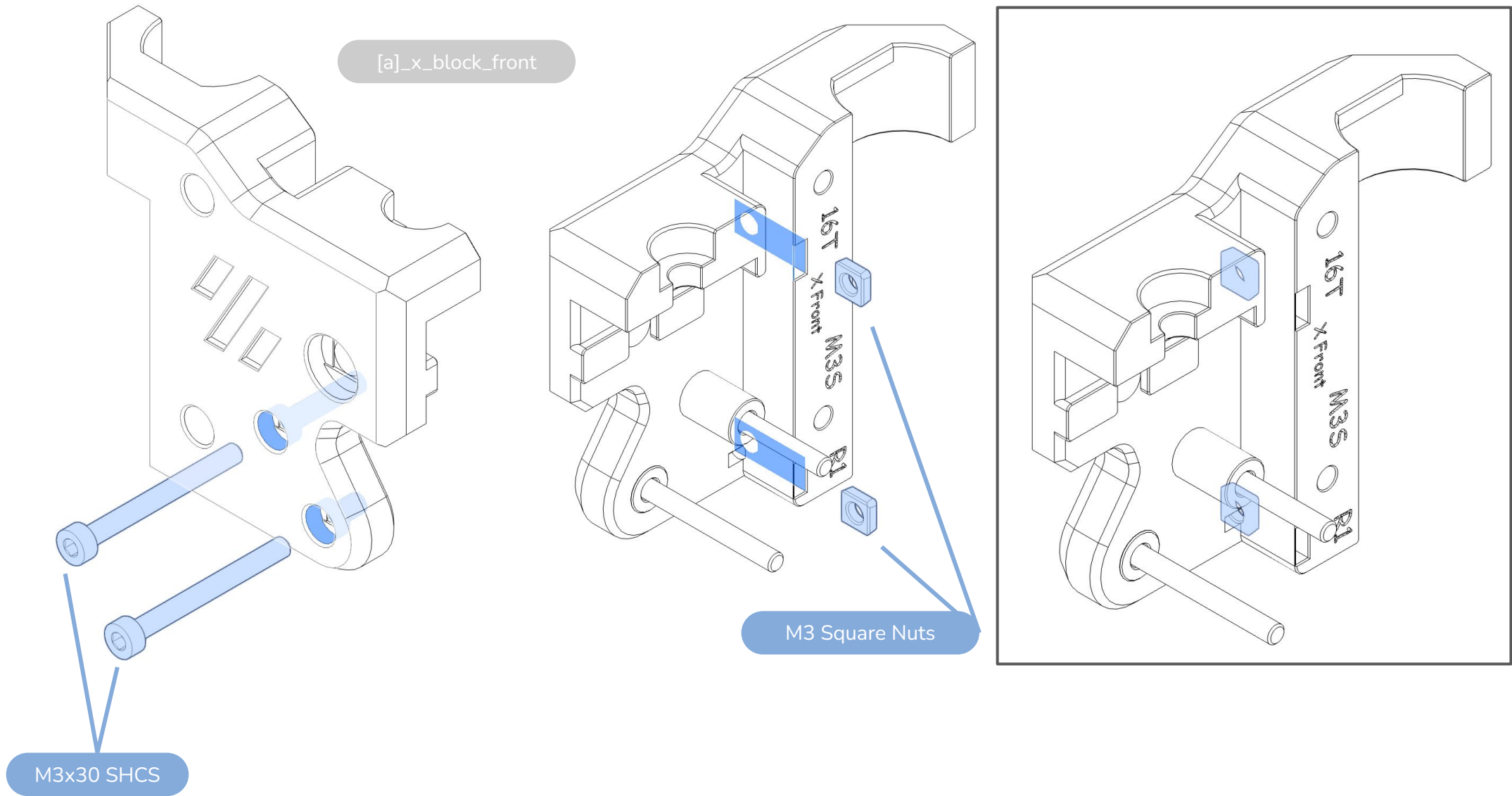


[a]_x_end_bearing_holder

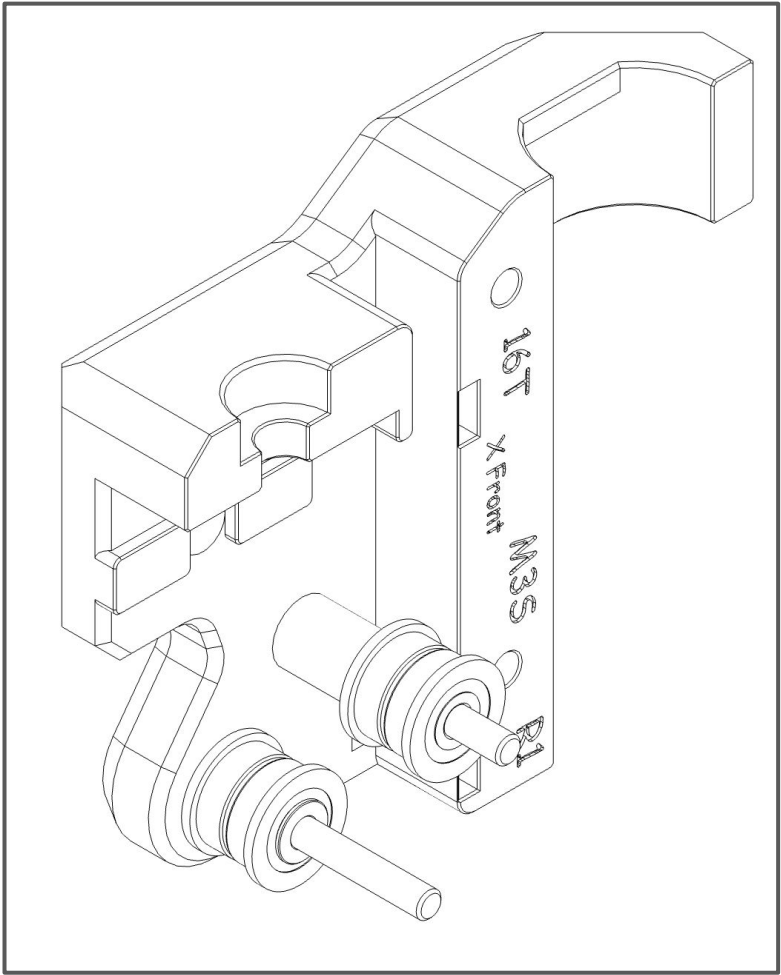
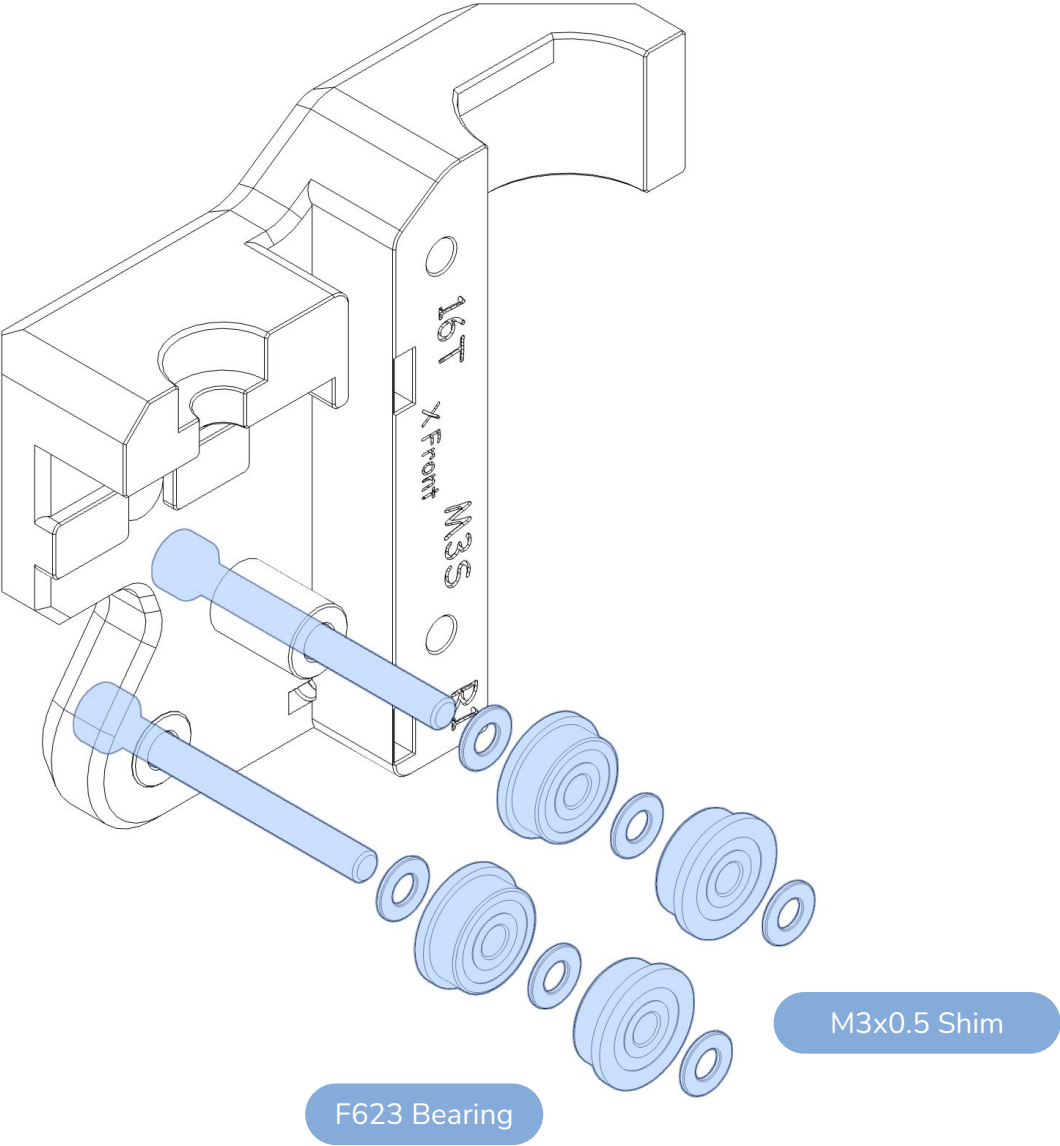


M3x30 SHCS

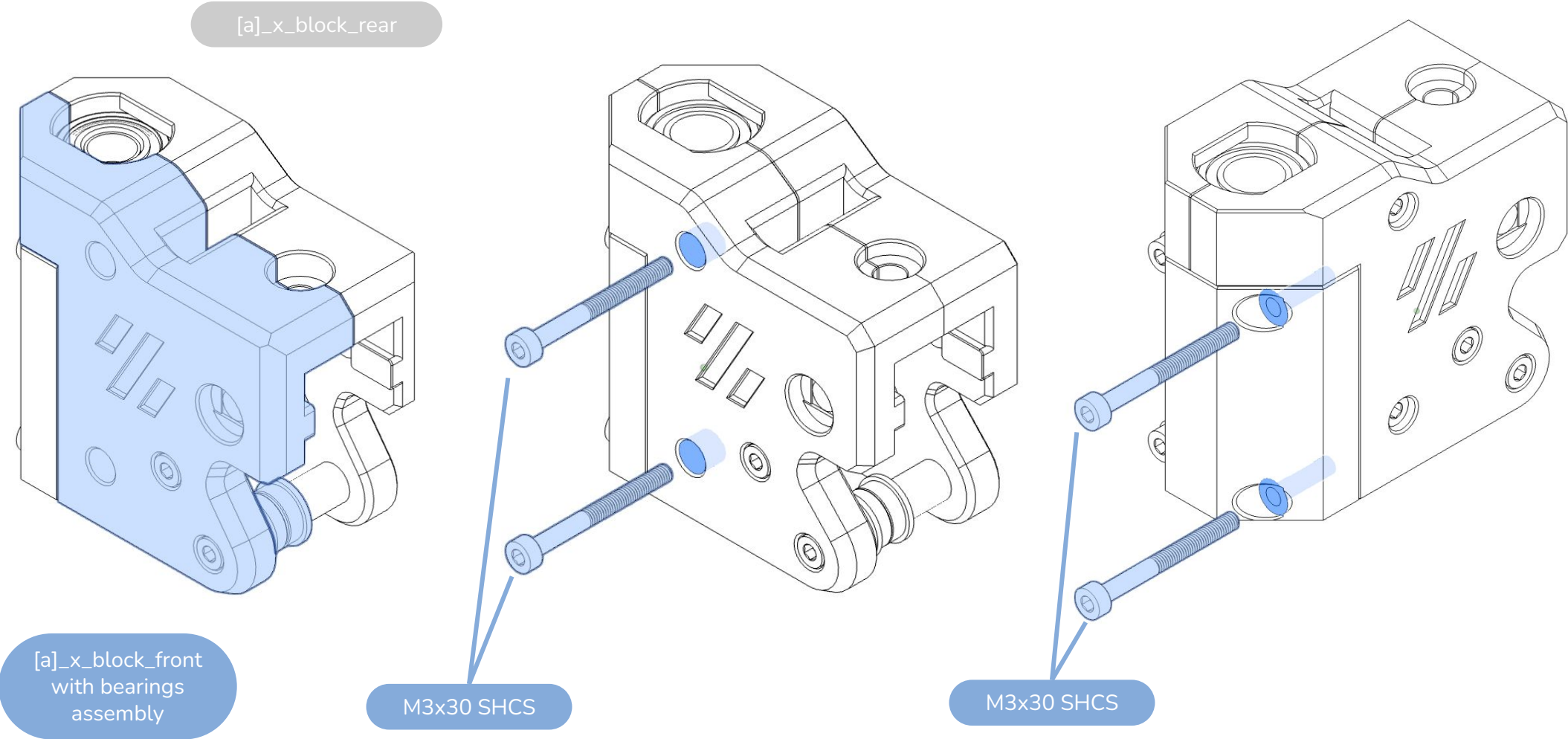
X GANTRY
XZ Block - Left



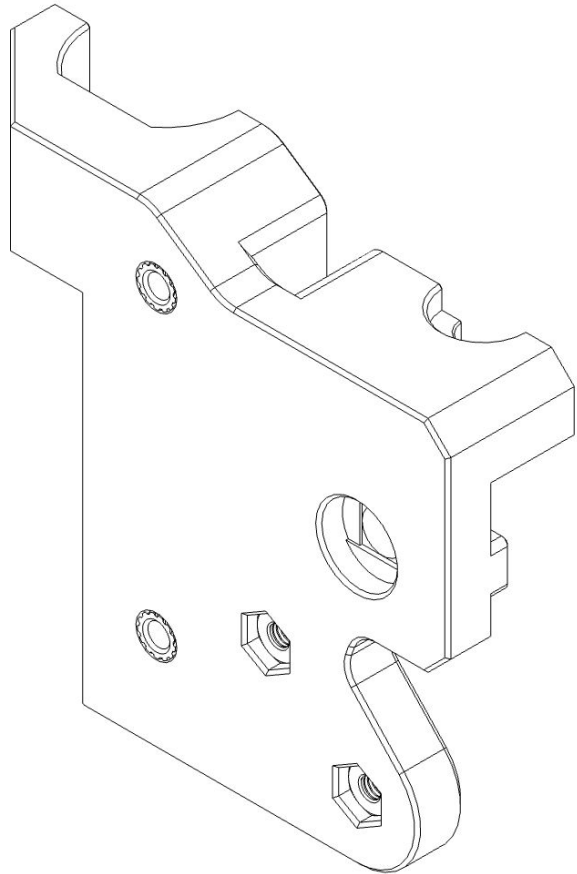
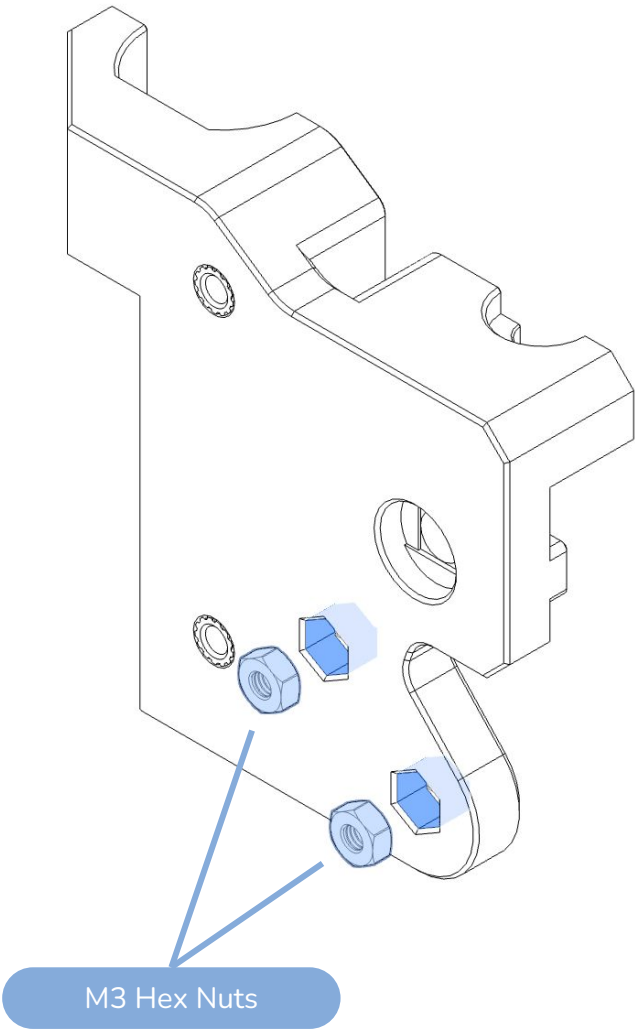
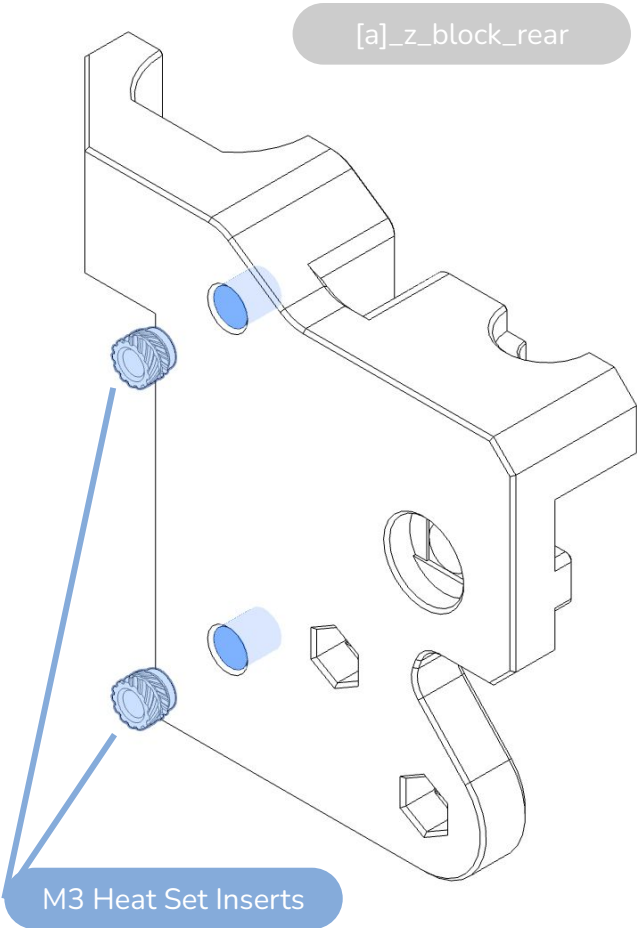
X GANTRY
XZ Block - Left



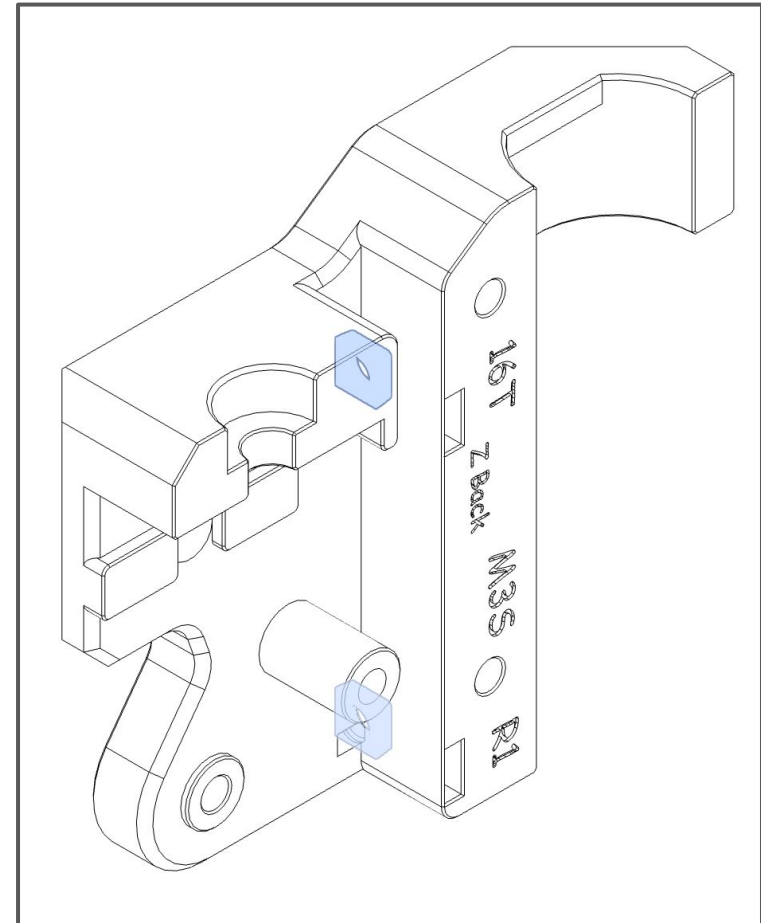
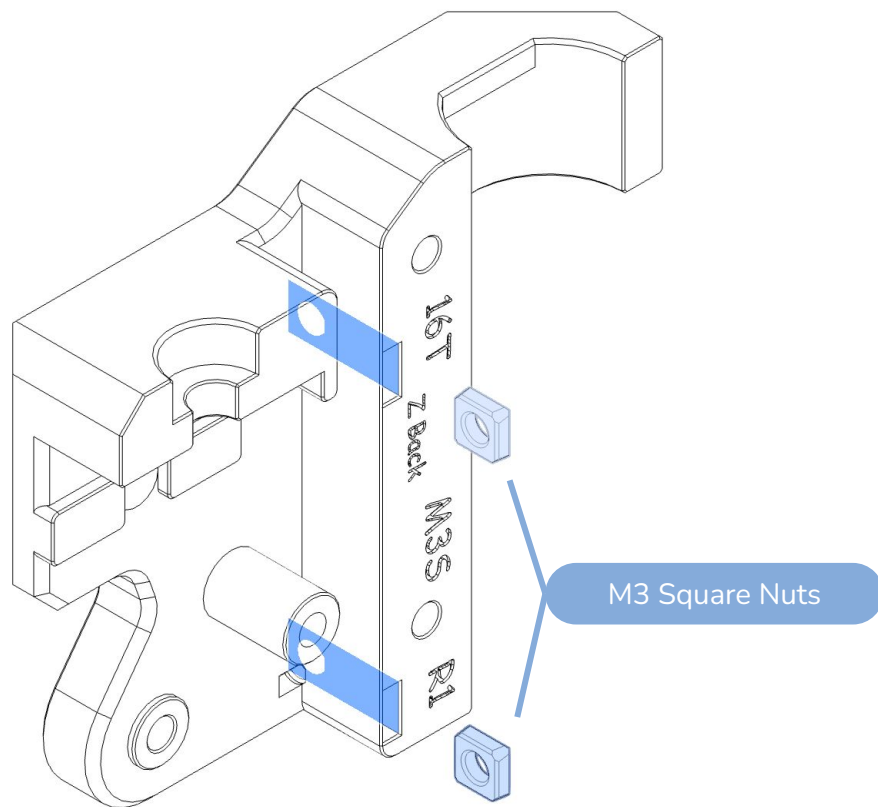
X GANTRY
XZ Block - Left



X GANTRY
XZ Block - Right

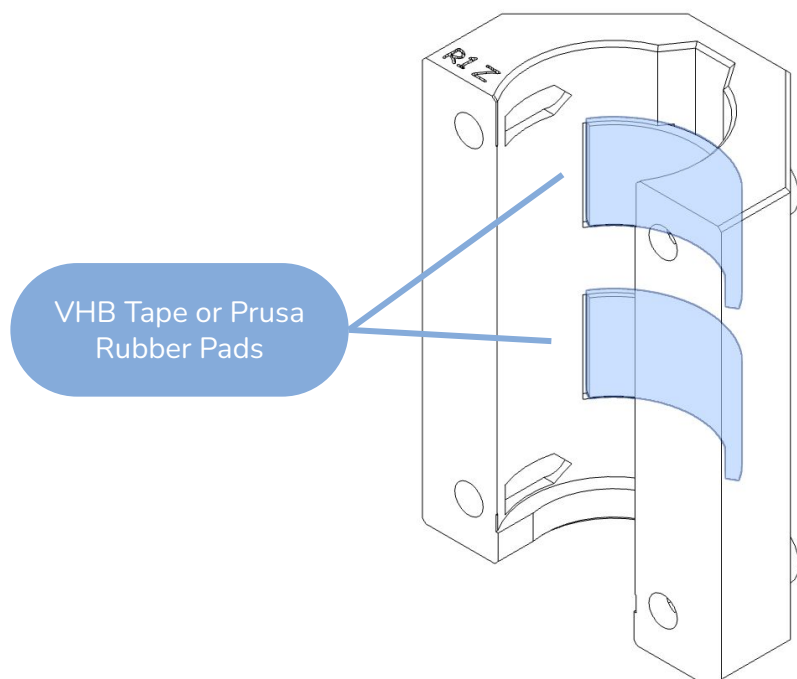


X GANTRY
XZ Block - Right



X GANTRY

XZ Block - Right



BEARING PADS

You can use the Prusa 20x10x1mm rubber pads if you have them, or just use a similar size cut piece of VHB tape!

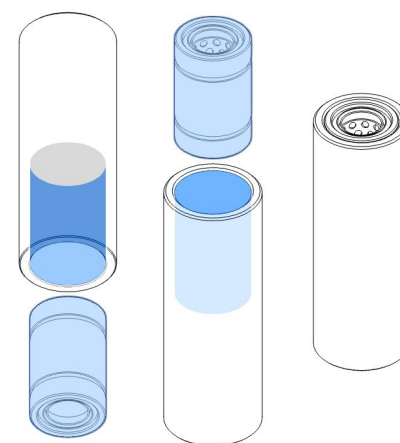
8MM LINEAR RODS AND BEARINGS

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Print **8mm_bearing_adapter_2x_LM8UU** and use around the bearings to adapt the size.

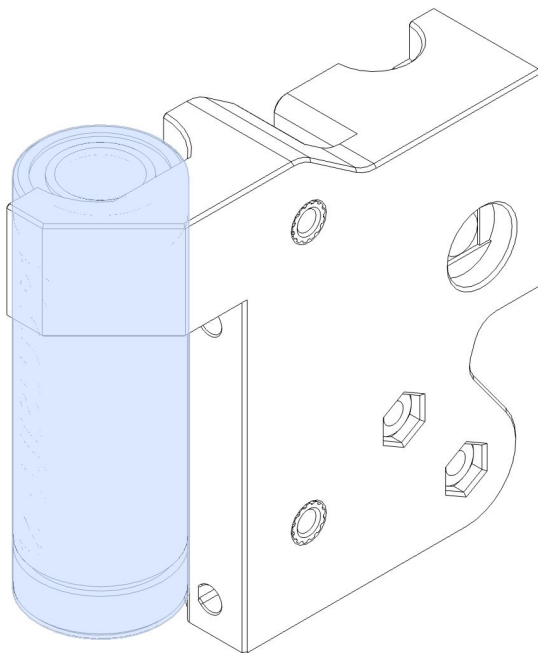
To use 8mm Linear Rods and LM8LUU Bearings, replace the 1x LM10LUU with 2x LM8LUU.

Print **8mm_bearing_adapter_2x_LM8LUU** and use around the bearings to adapt the size.

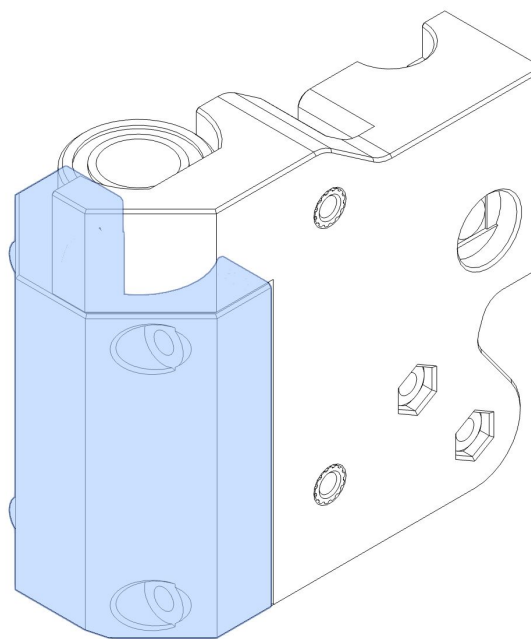


X GANTRY
XZ Block - Right

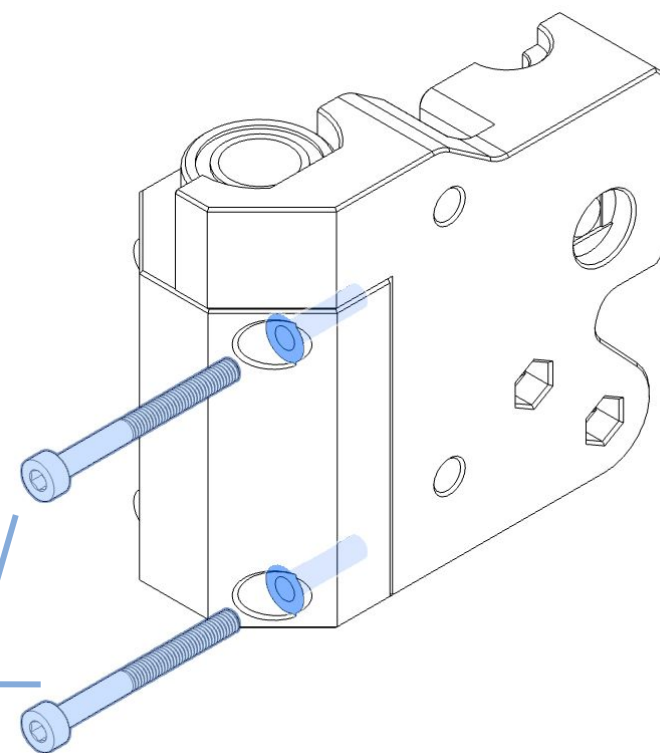
LM10LUU Bearing



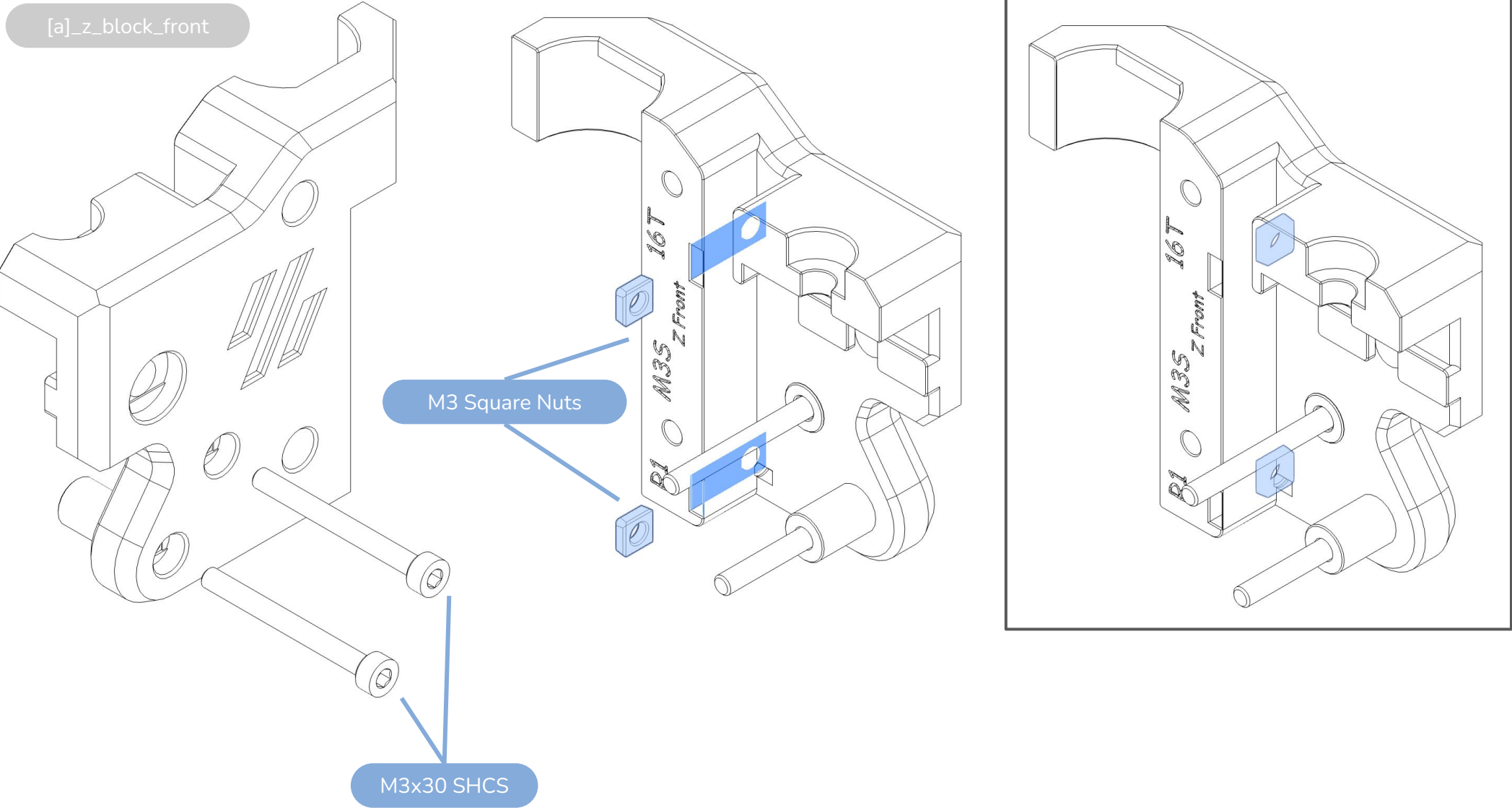
[a]_z_end_bearing_holder



M3x30 SHCS



X GANTRY
XZ Block - Right

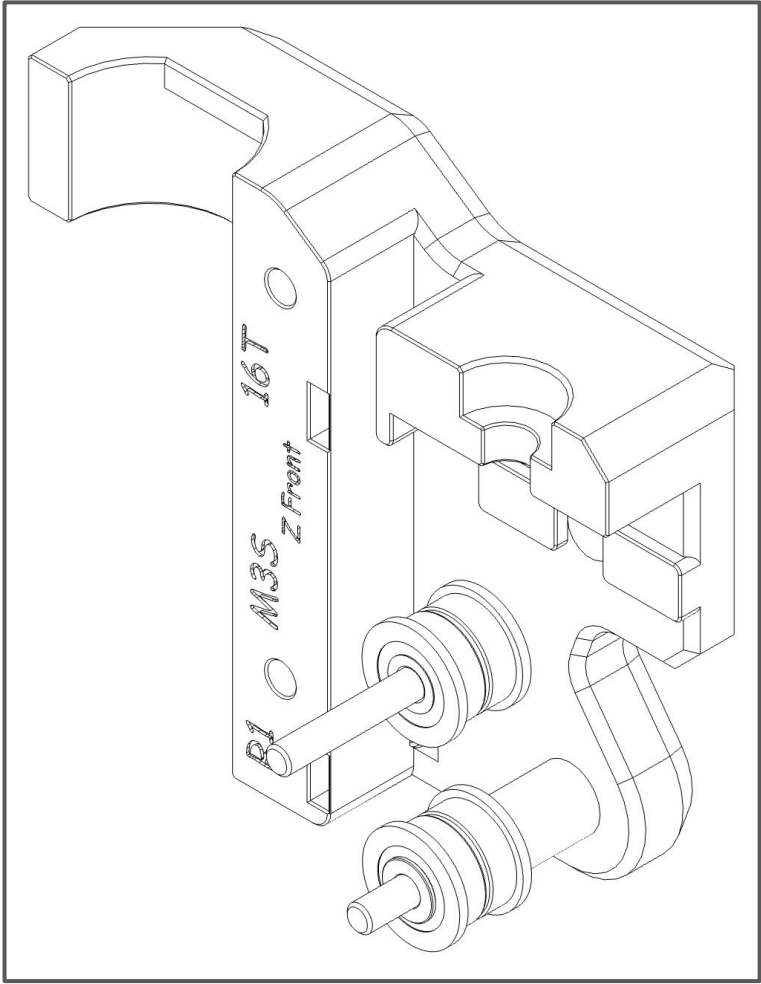
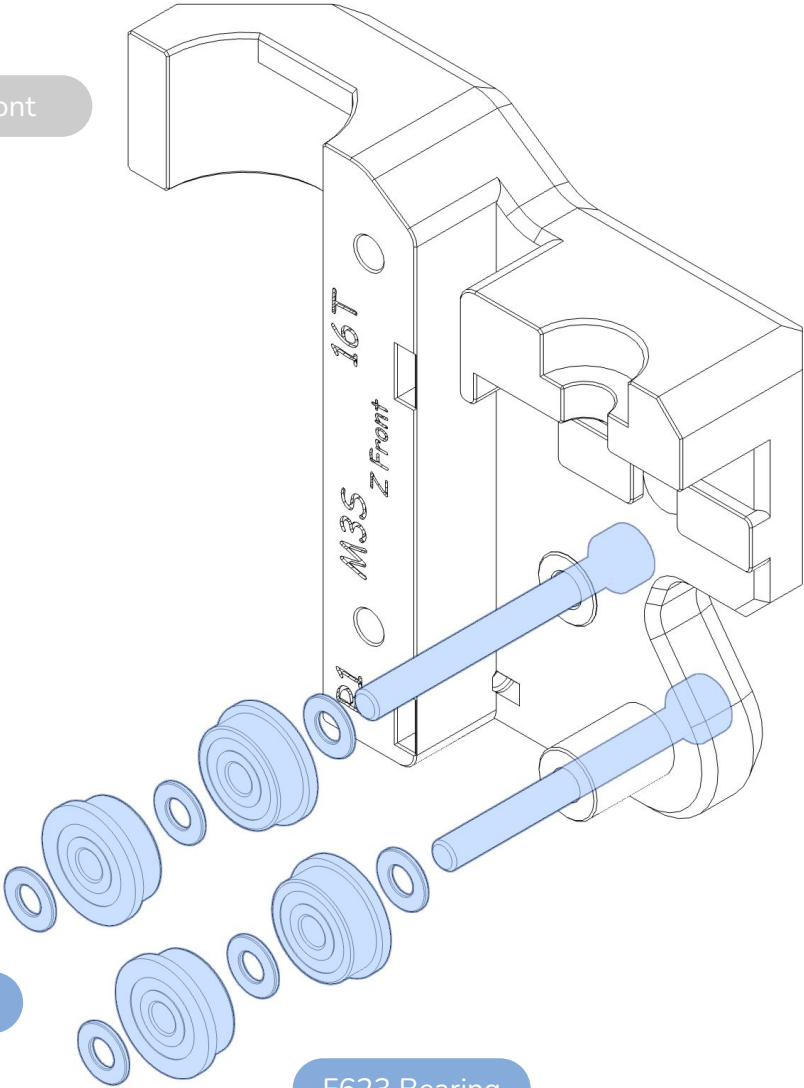


X GANTRY
XZ Block - Right

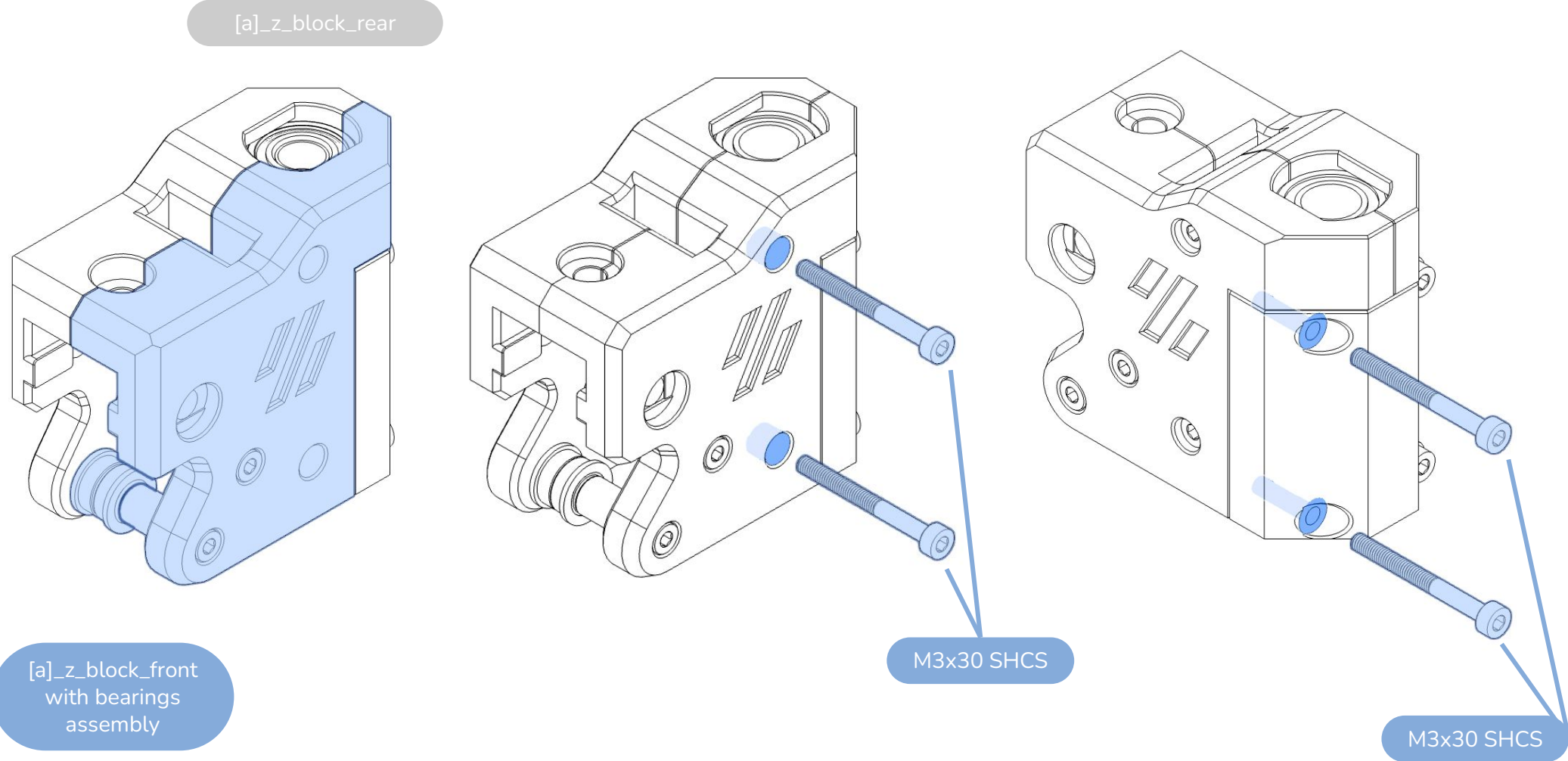
[a]_z_block_front

M3x0.5 Shim

F623 Bearing



X GANTRY
XZ Block - Right



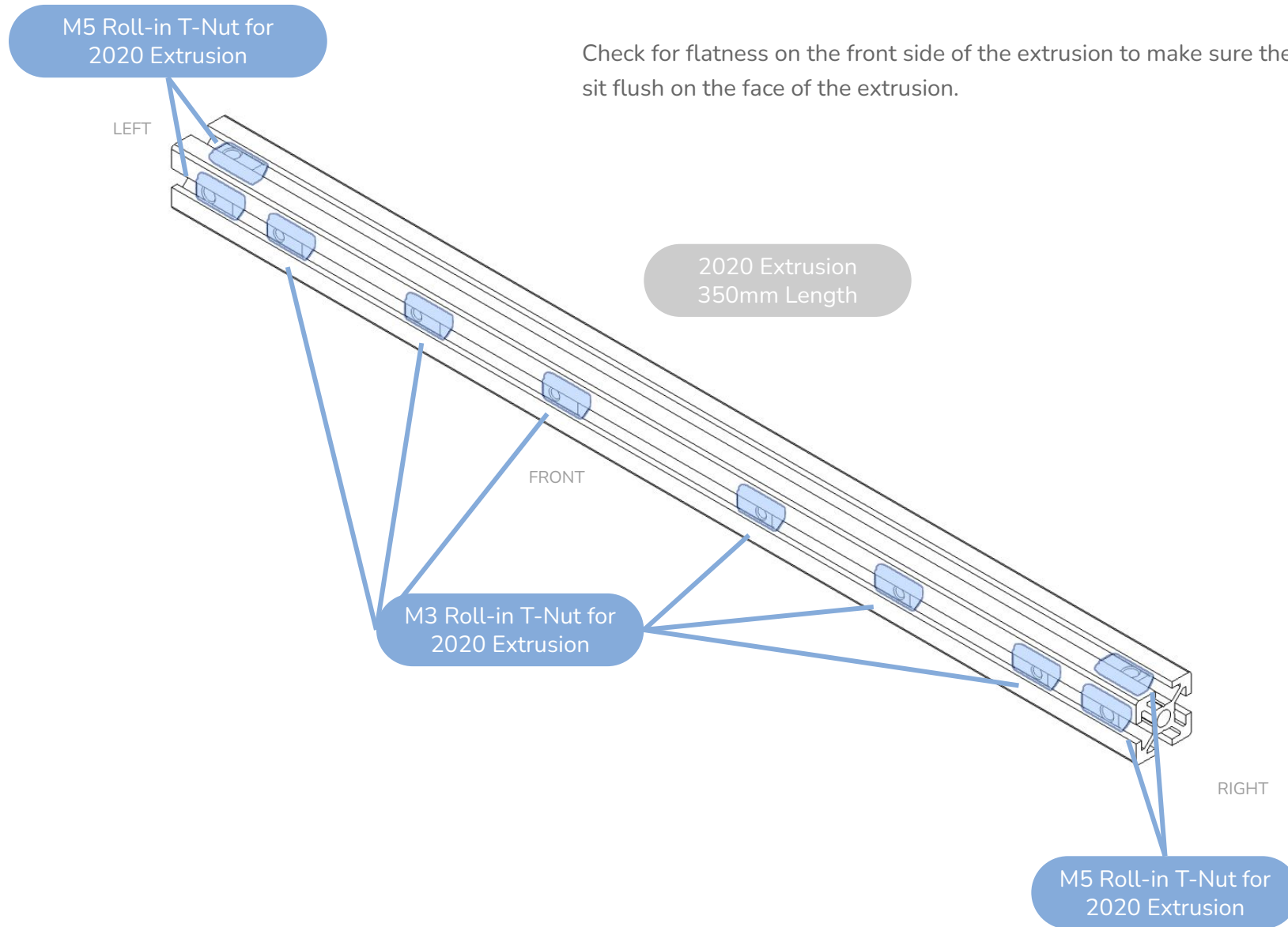
X GANTRY

Extrusion Preparation - Front

EXTRUDE THIS

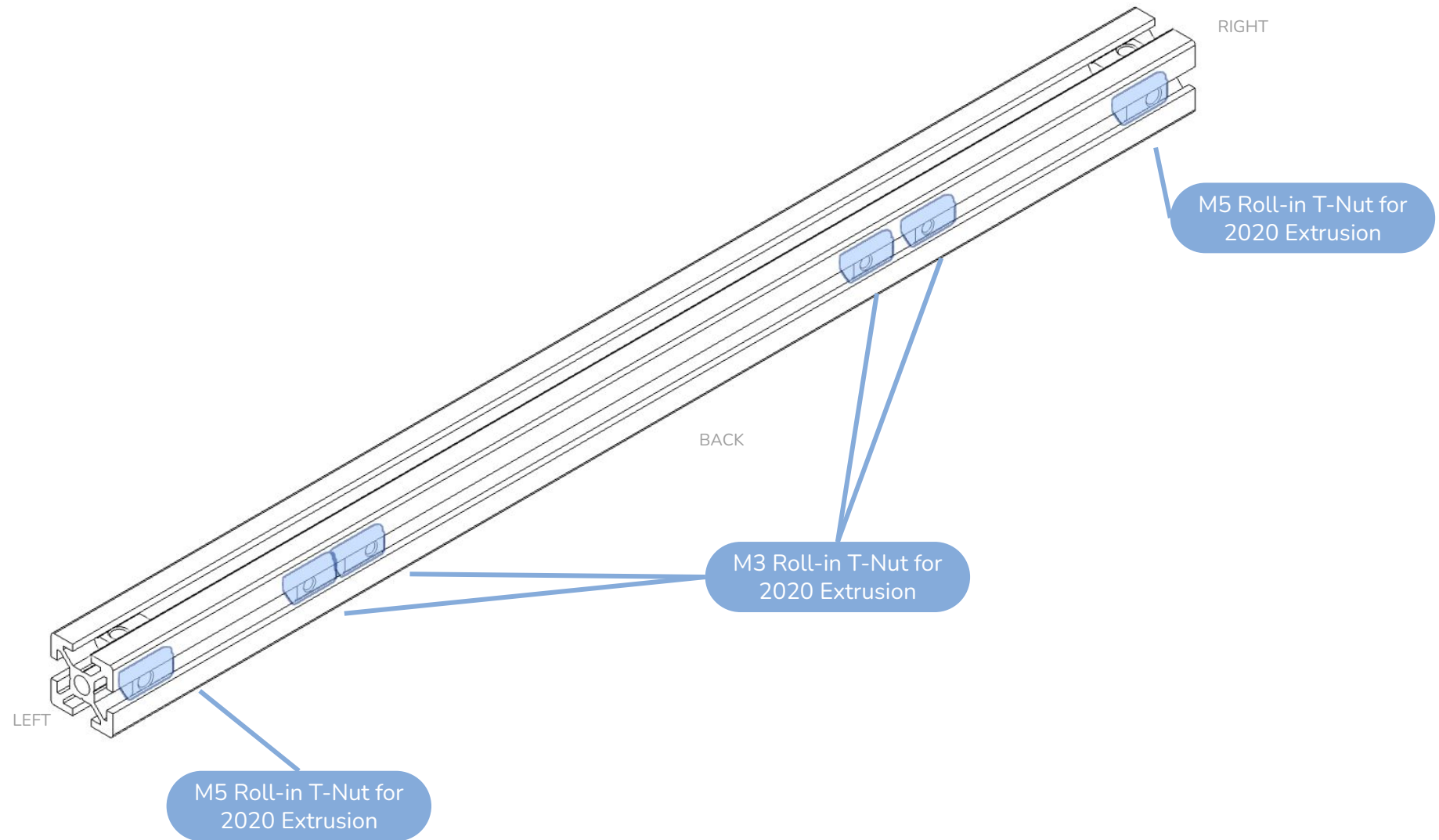
The 2020 Extrusion should be no more than 352mm maximum length! Some tolerance for length is built into the printed parts.

Check for flatness on the front side of the extrusion to make sure the linear rail will sit flush on the face of the extrusion.



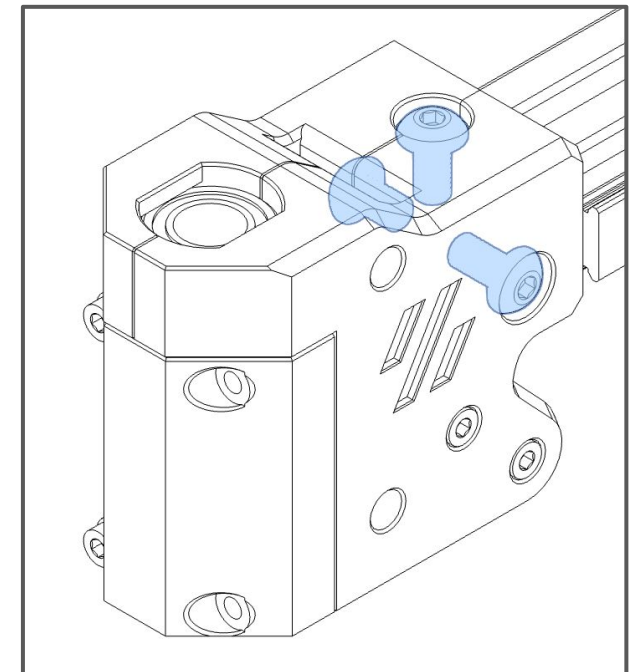
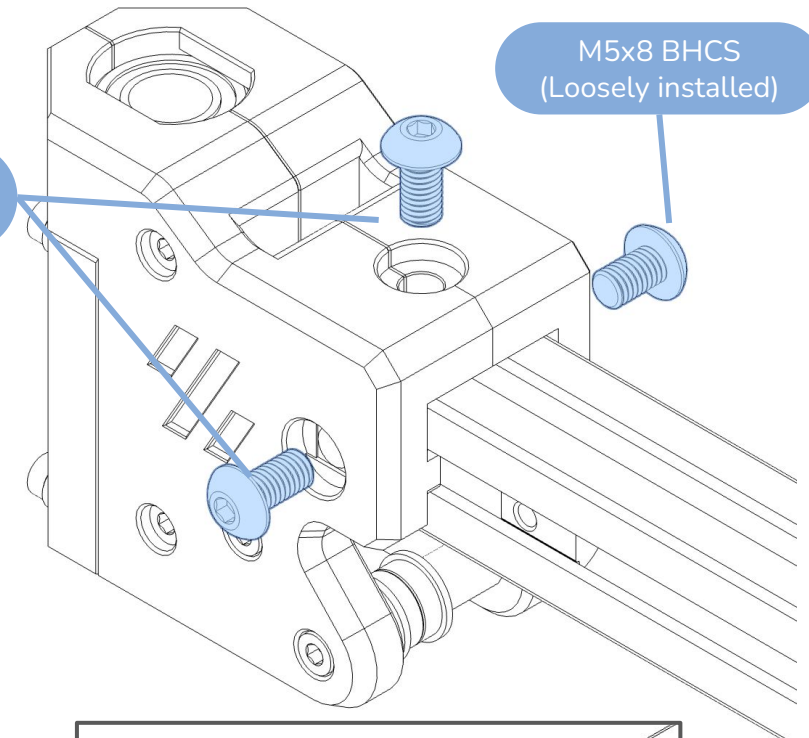
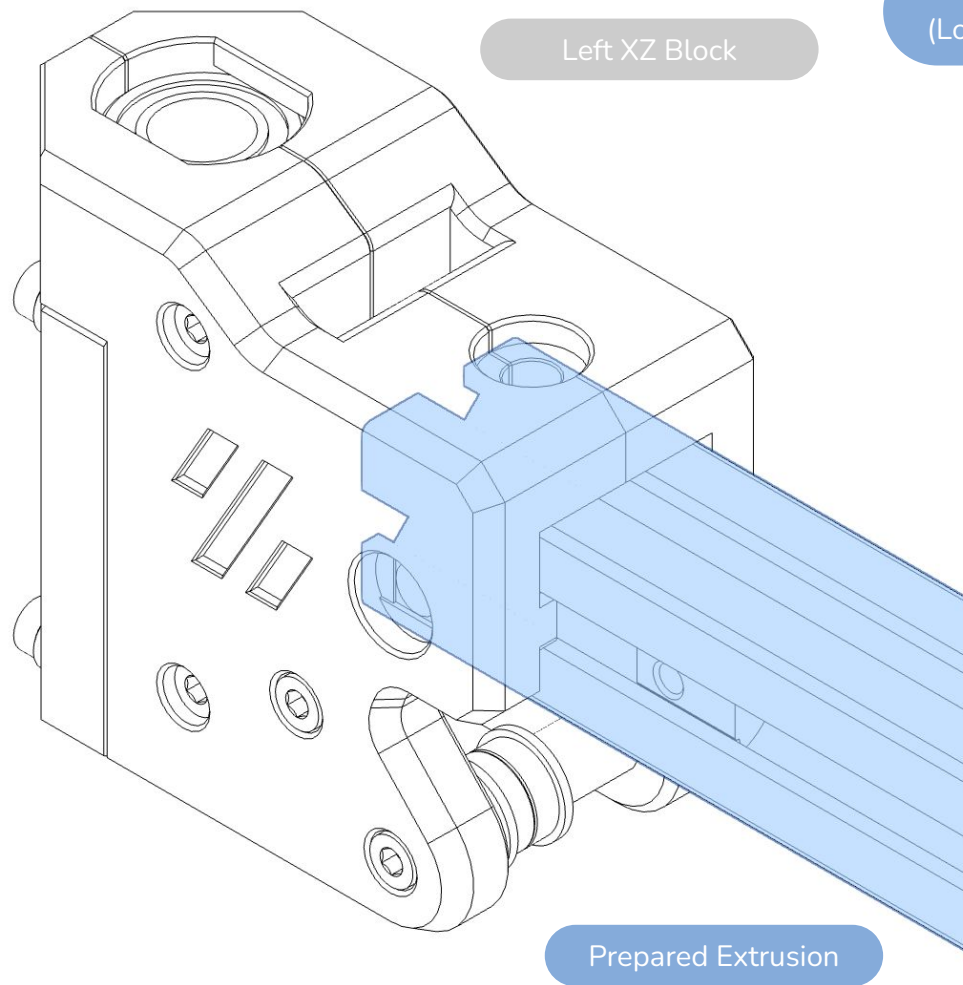
X GANTRY

Extrusion Preparation - Back



X GANTRY

Extrusion + Left XZ Block

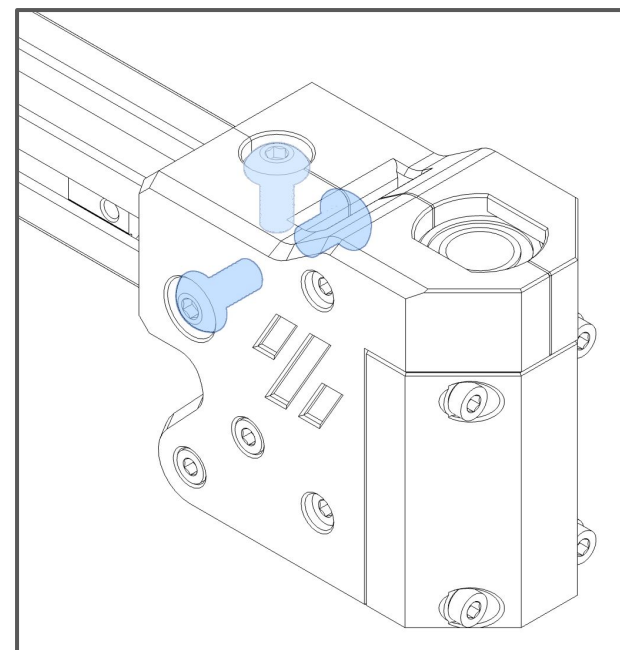
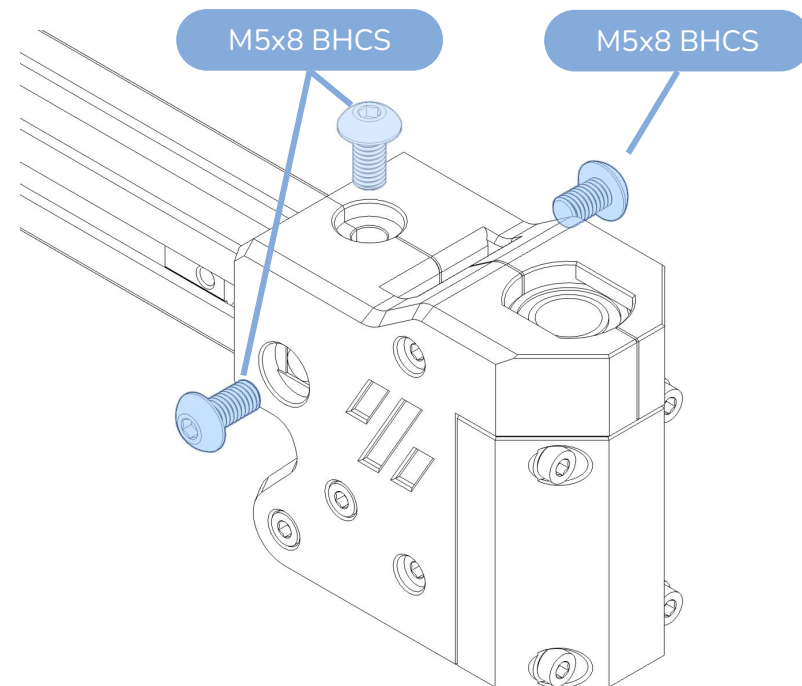
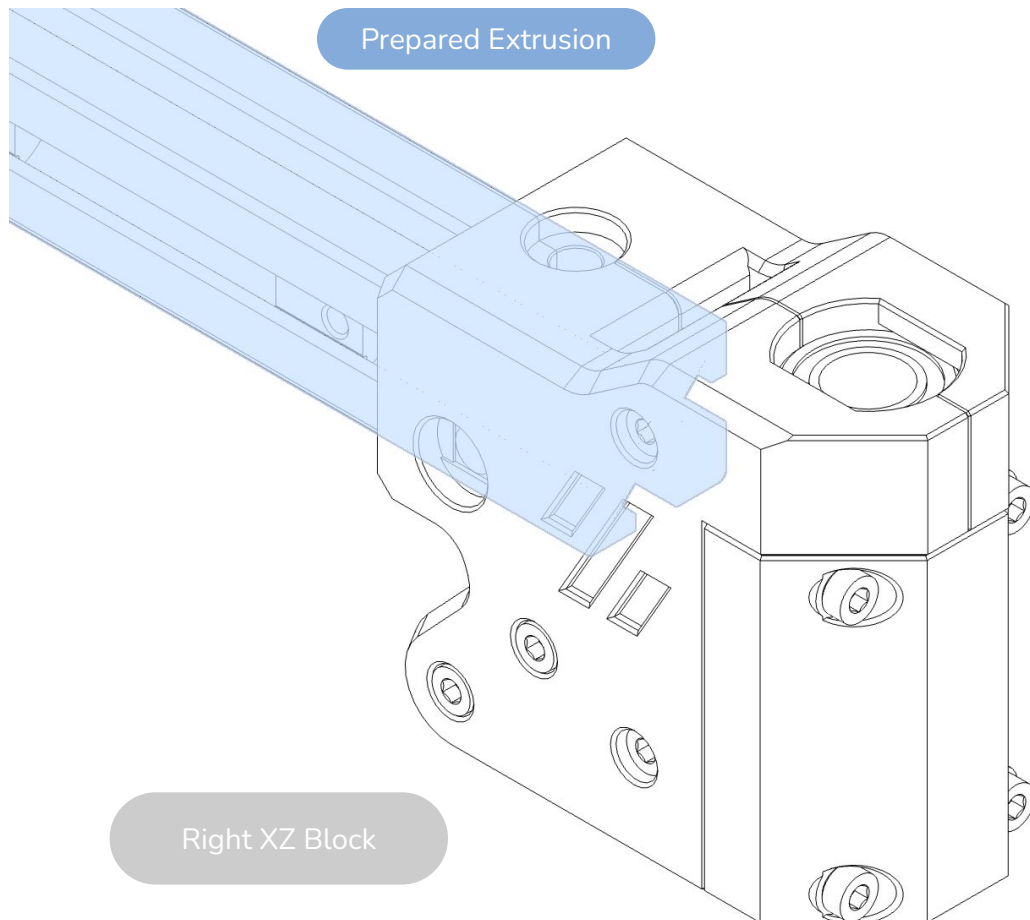


TOLERANCE TIP

The X and Z Block Parts are designed with an extra 1 mm of clearance at each end. That is, it can accommodate the 2020 Extrusion to have a max length of 352mm!

X GANTRY

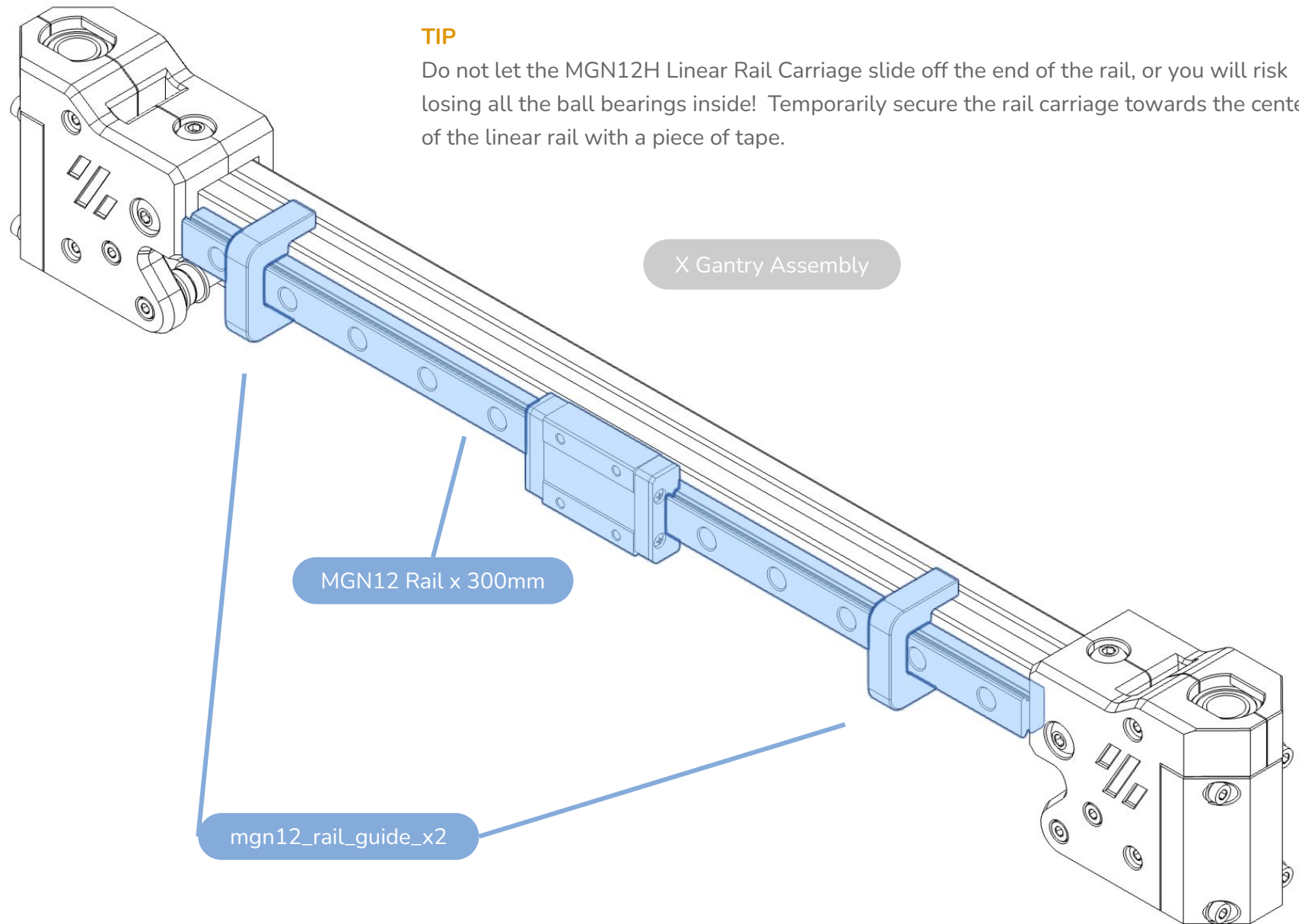
Extrusion + Right XZ Block



X GANTRY Linear Rail

TIP

Do not let the MGN12H Linear Rail Carriage slide off the end of the rail, or you will risk losing all the ball bearings inside! Temporarily secure the rail carriage towards the center of the linear rail with a piece of tape.



POSITIONING

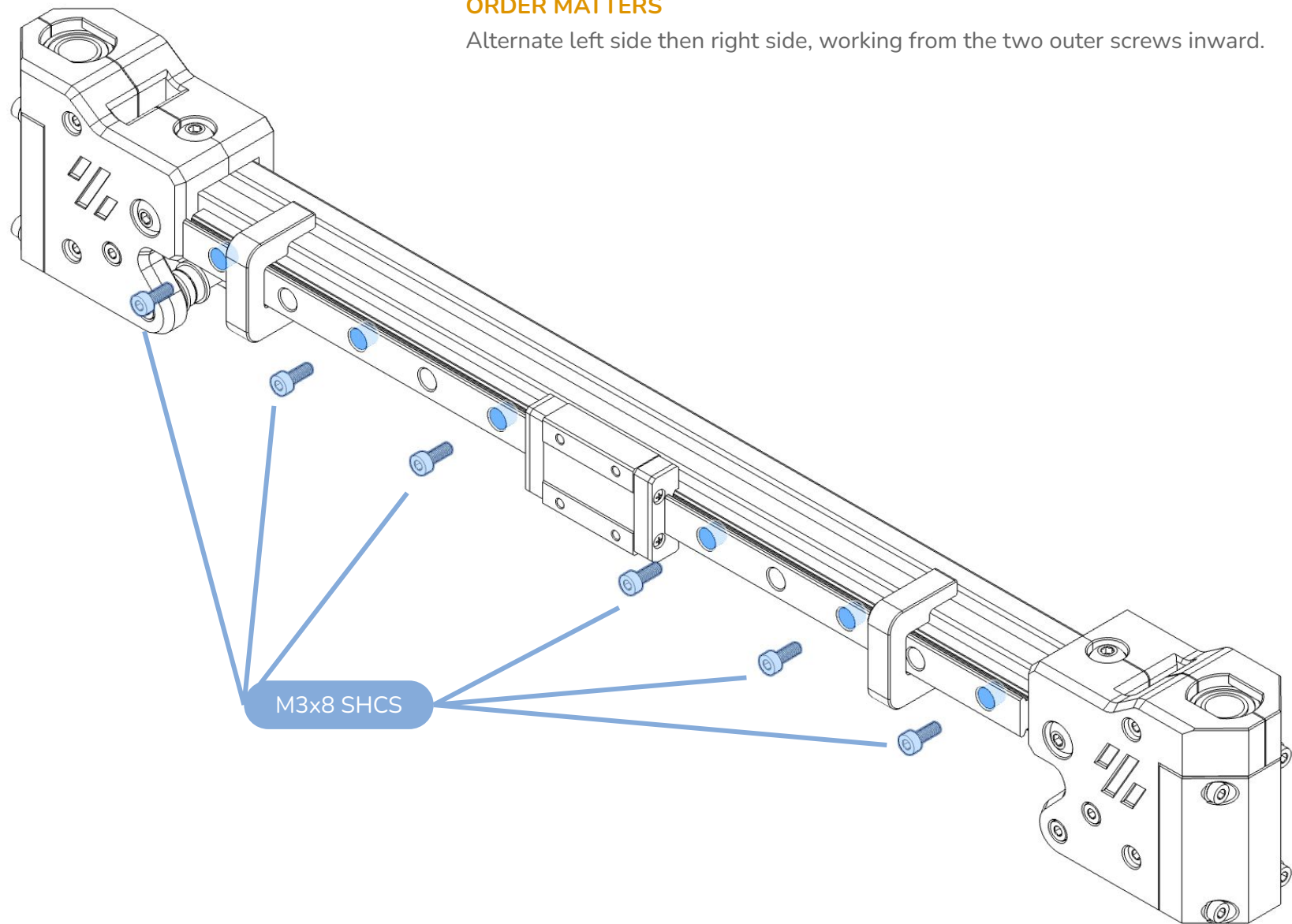
Use the **mgn12_rail_guide_x2** to center the **MGN12 Rail** against the 2020 extrusion profile. The rail should be centered in between the two ends of the extrusion both horizontally (left and right) as well as vertically (up and down).

X GANTRY

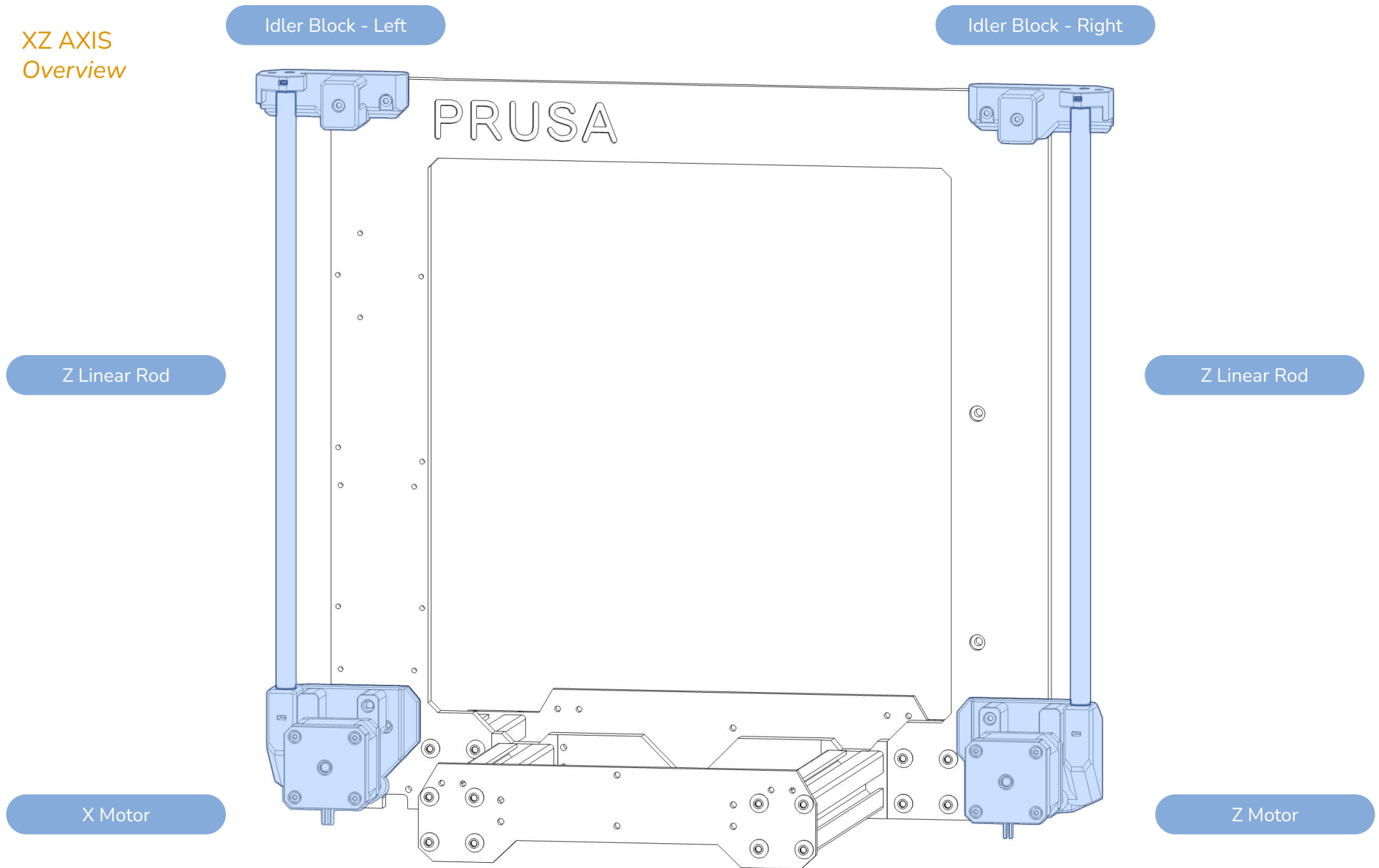
Linear Rail

ORDER MATTERS

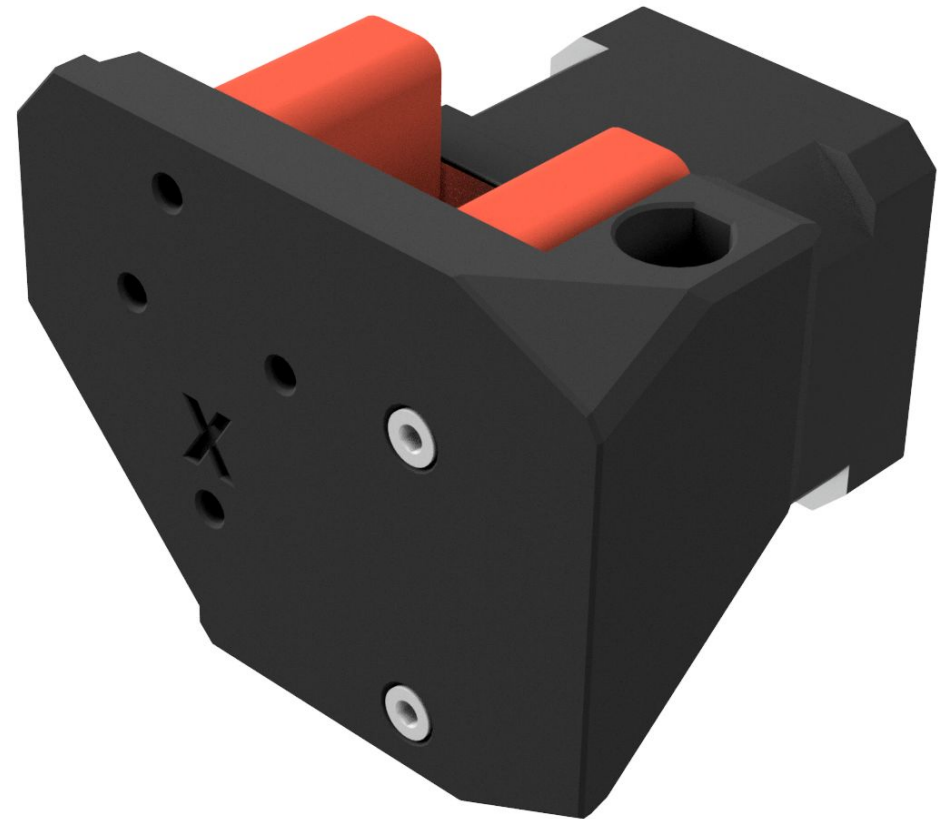
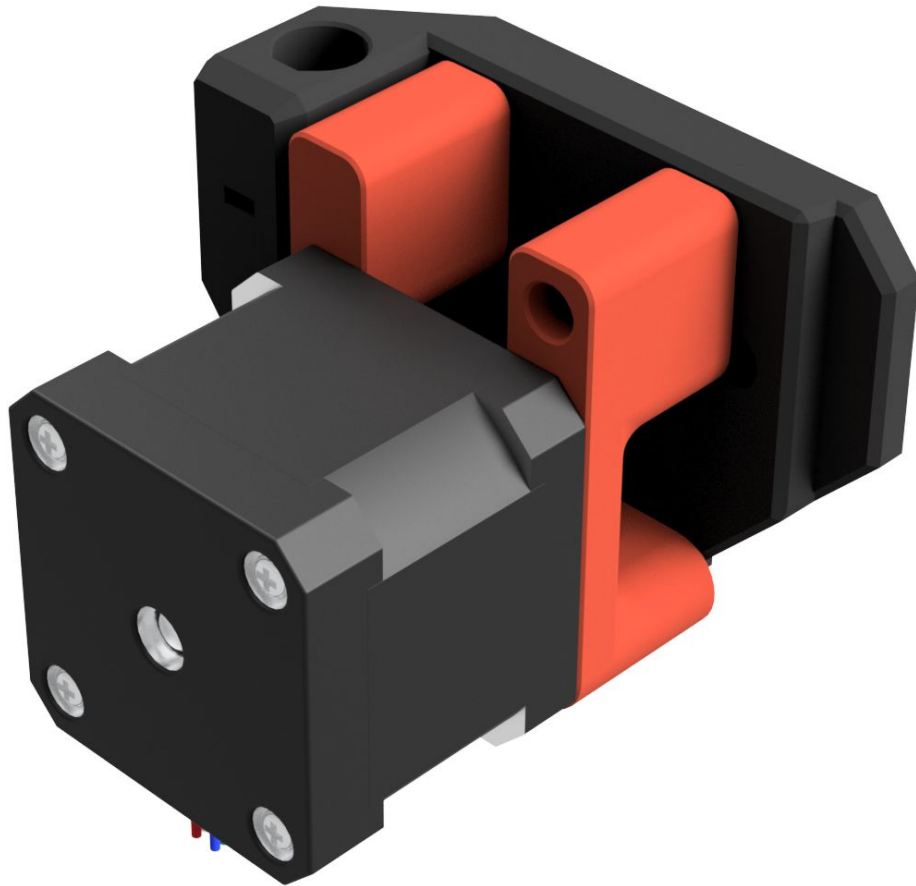
Alternate left side then right side, working from the two outer screws inward.



XZ AXIS Overview

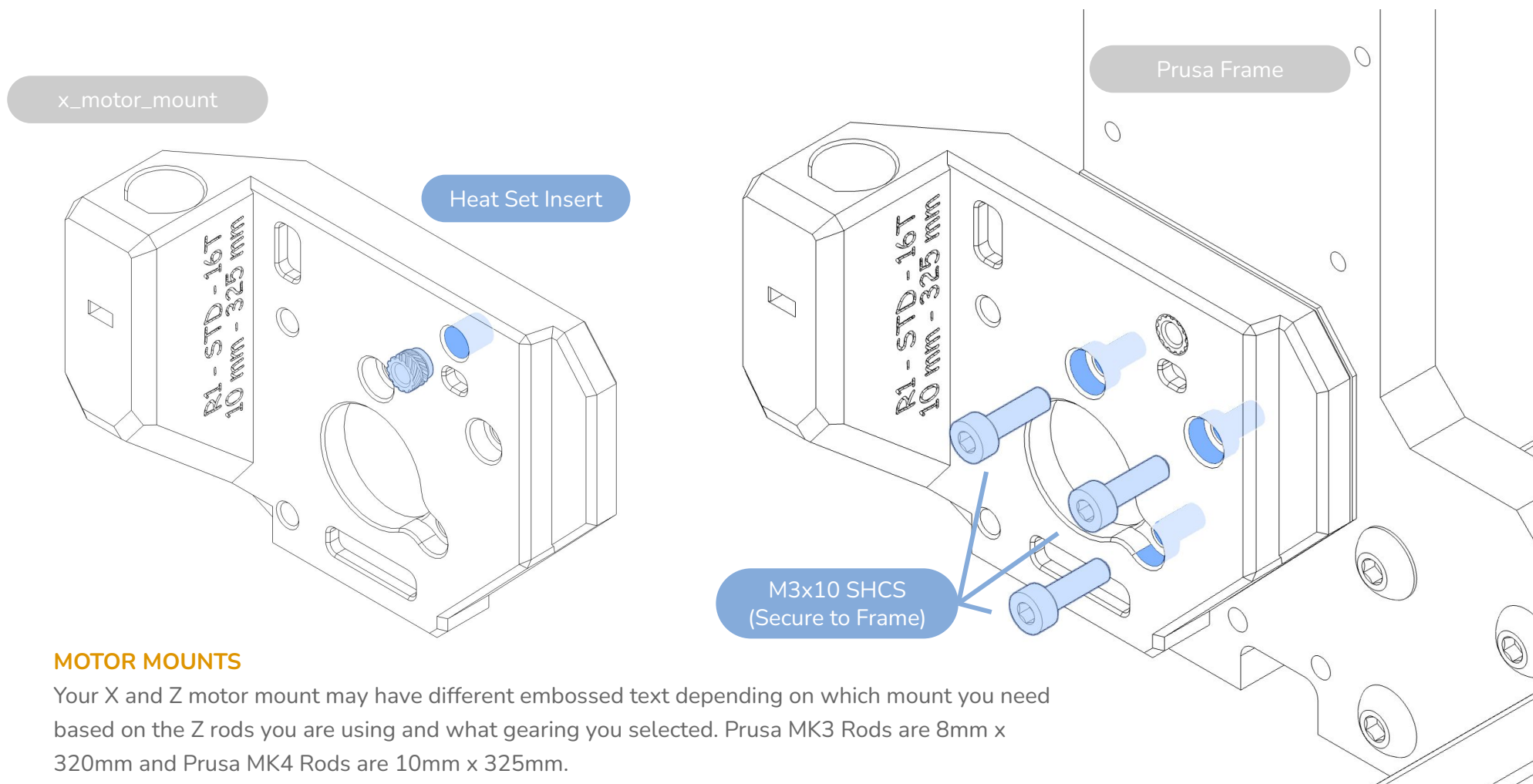


XZ AXIS
X Motor Assembly



XZ AXIS

X Motor Assembly



MOTOR MOUNTS

Your X and Z motor mount may have different embossed text depending on which mount you need based on the Z rods you are using and what gearing you selected. Prusa MK3 Rods are 8mm x 320mm and Prusa MK4 Rods are 10mm x 325mm.

We support a few other rod length combinations to allow flexibility in sourcing. Jump ahead to the section on Linear Rods for more info.

Example: `motor_mounts_{rod-diameter}_{rod-length}_{gearing}`
`motor_mounts_10mm_325mm_20T`

XZ AXIS

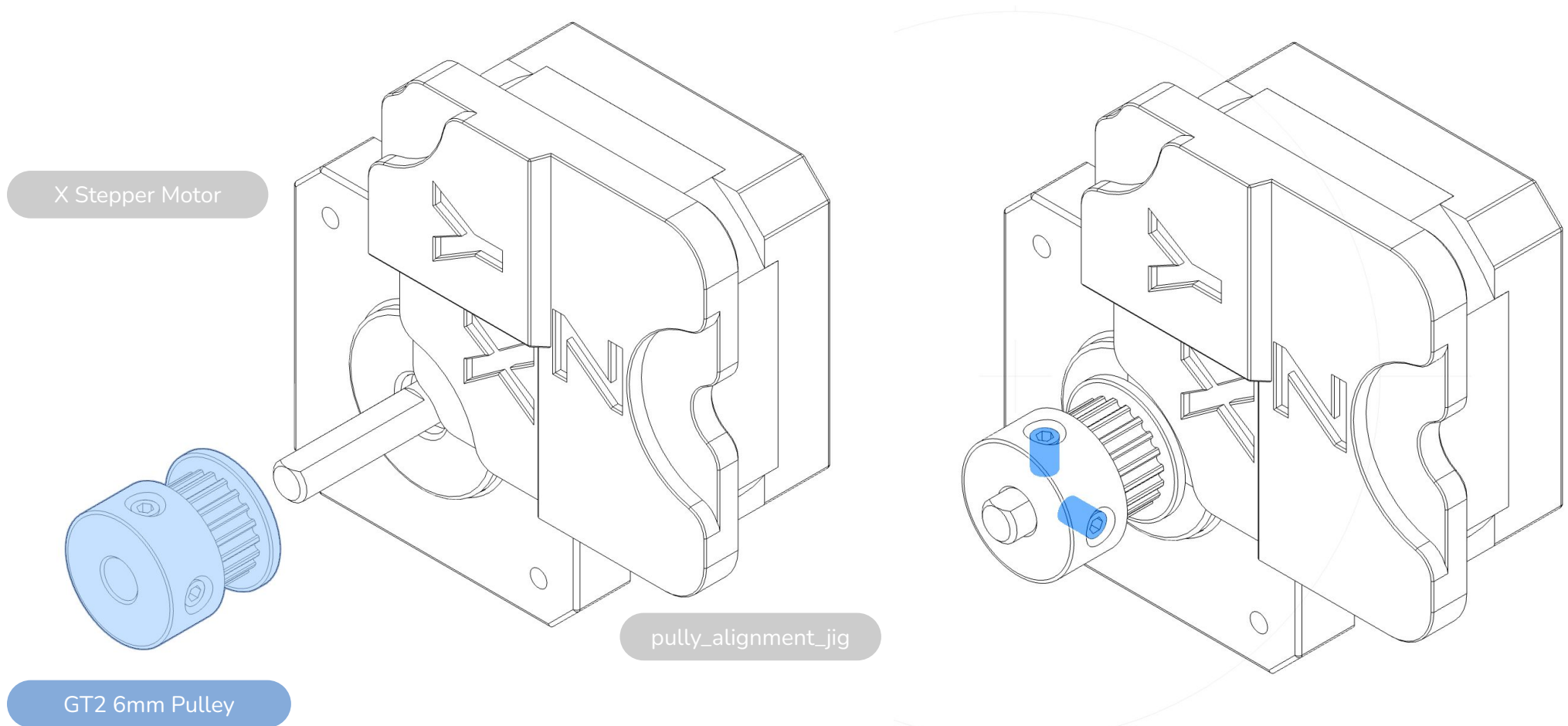
X Motor Assembly

PULLEY ALIGNMENT

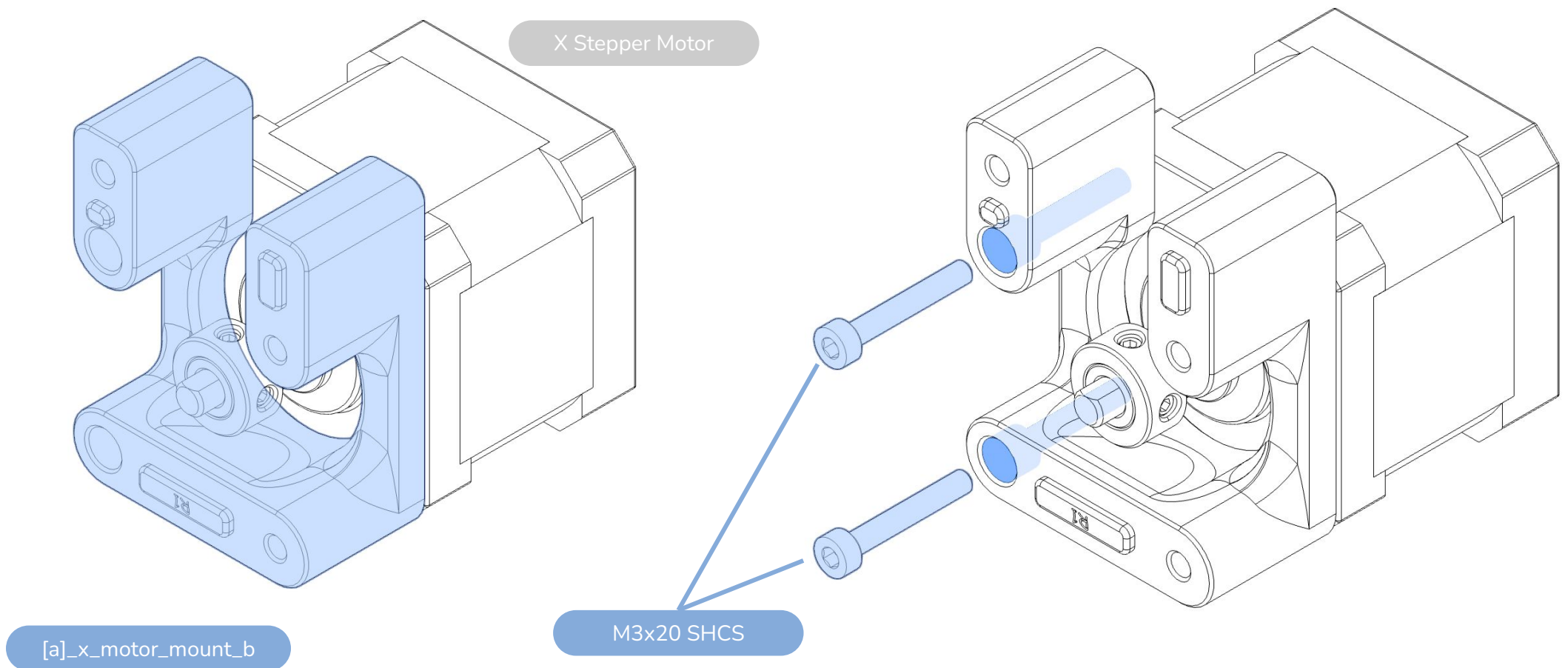
Make sure you use the right alignment jig for your type of pulley! **16T vs 20T.**

Use the printed pulley alignment jig to set the spacing of the GT2 Pulley. Press the small outer curved radius against the stepper motor shaft. Slide the pulley up to the larger relief in the jig. Be sure that one set screw aligns against the flat part of the motor shaft!

Tighten the set screws to secure the pulley. **Note the correct orientation of the pulley!**

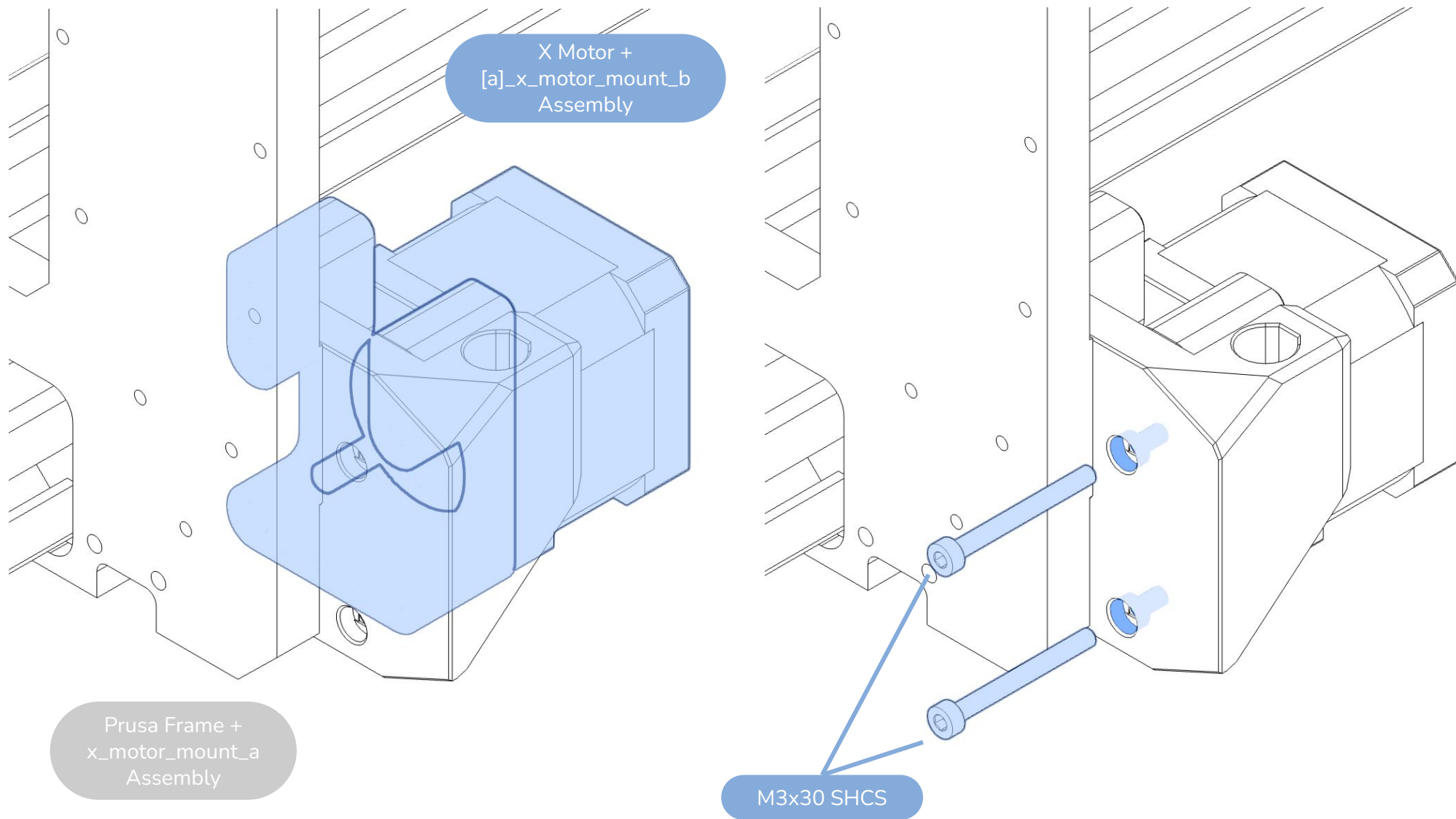


XZ AXIS
X Motor Assembly

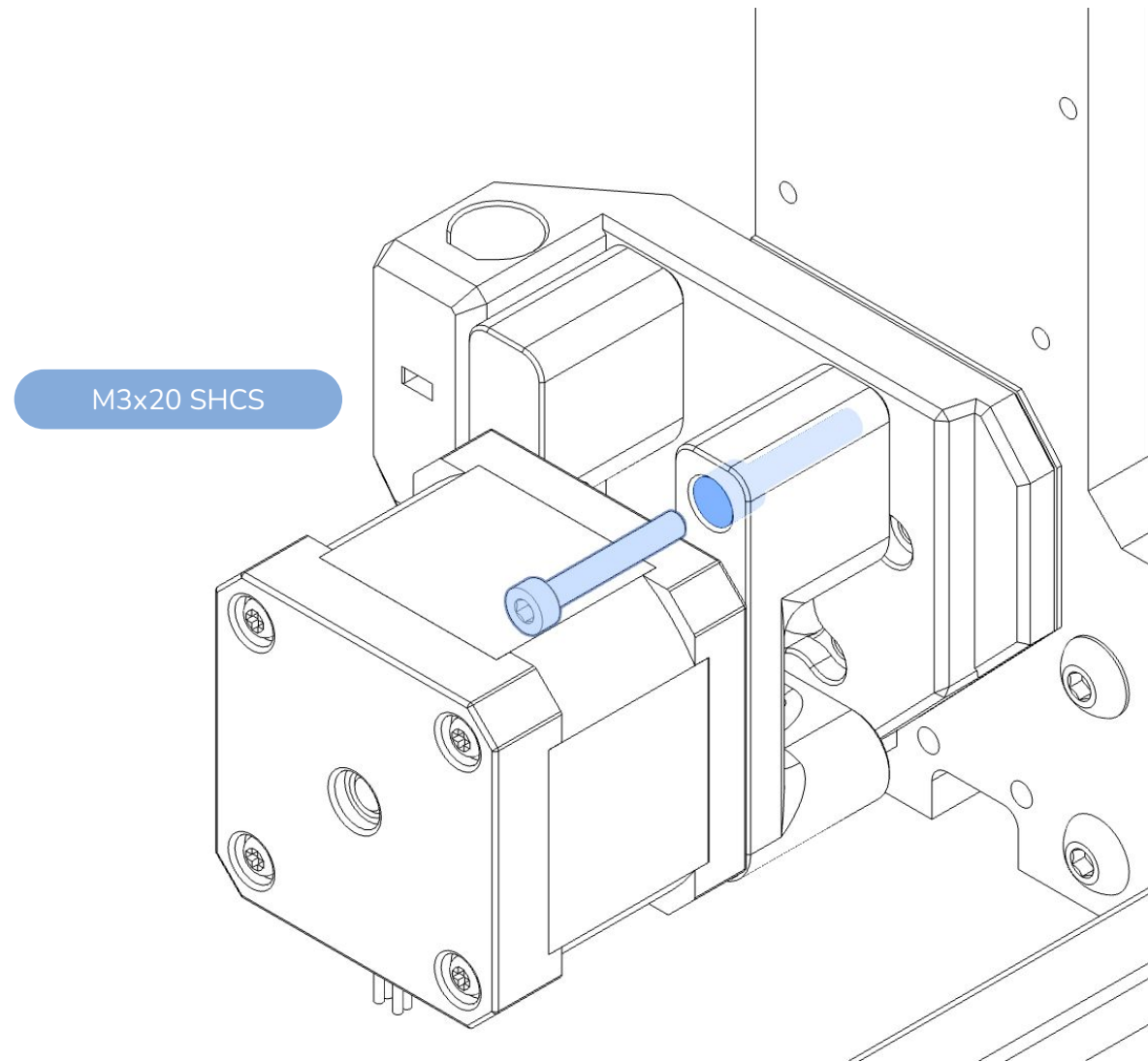


XZ AXIS

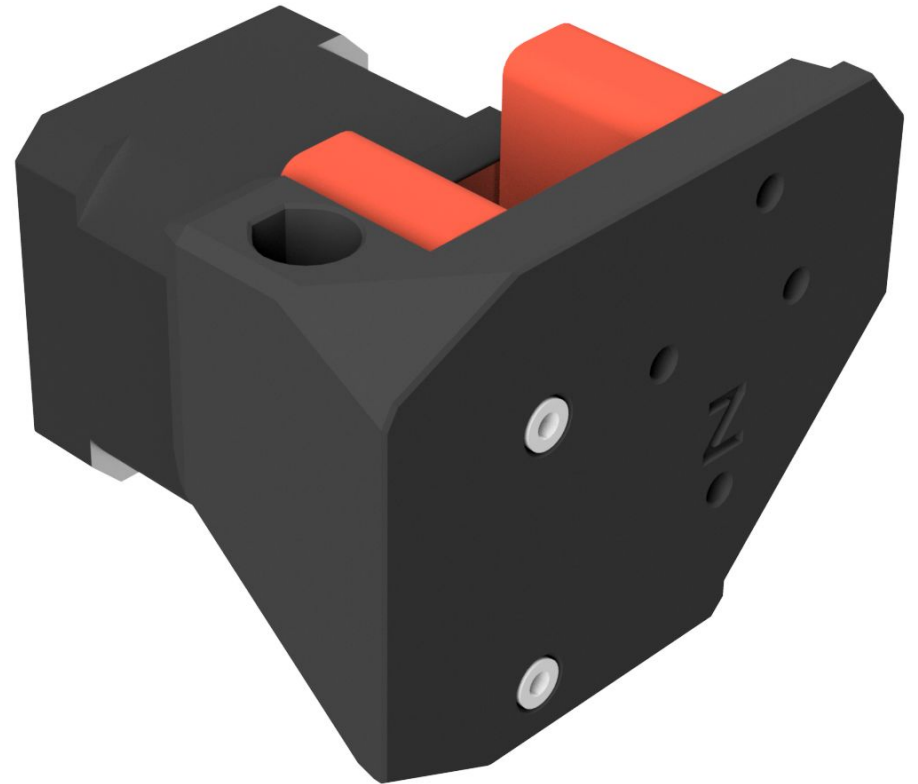
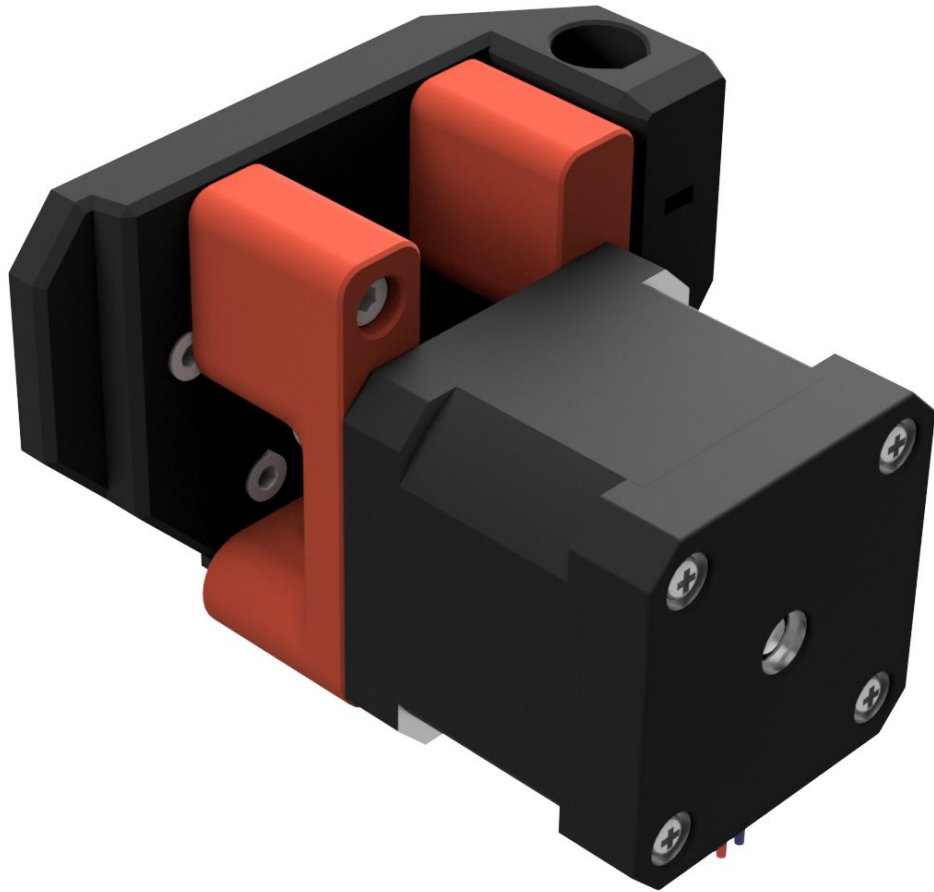
X Motor Assembly



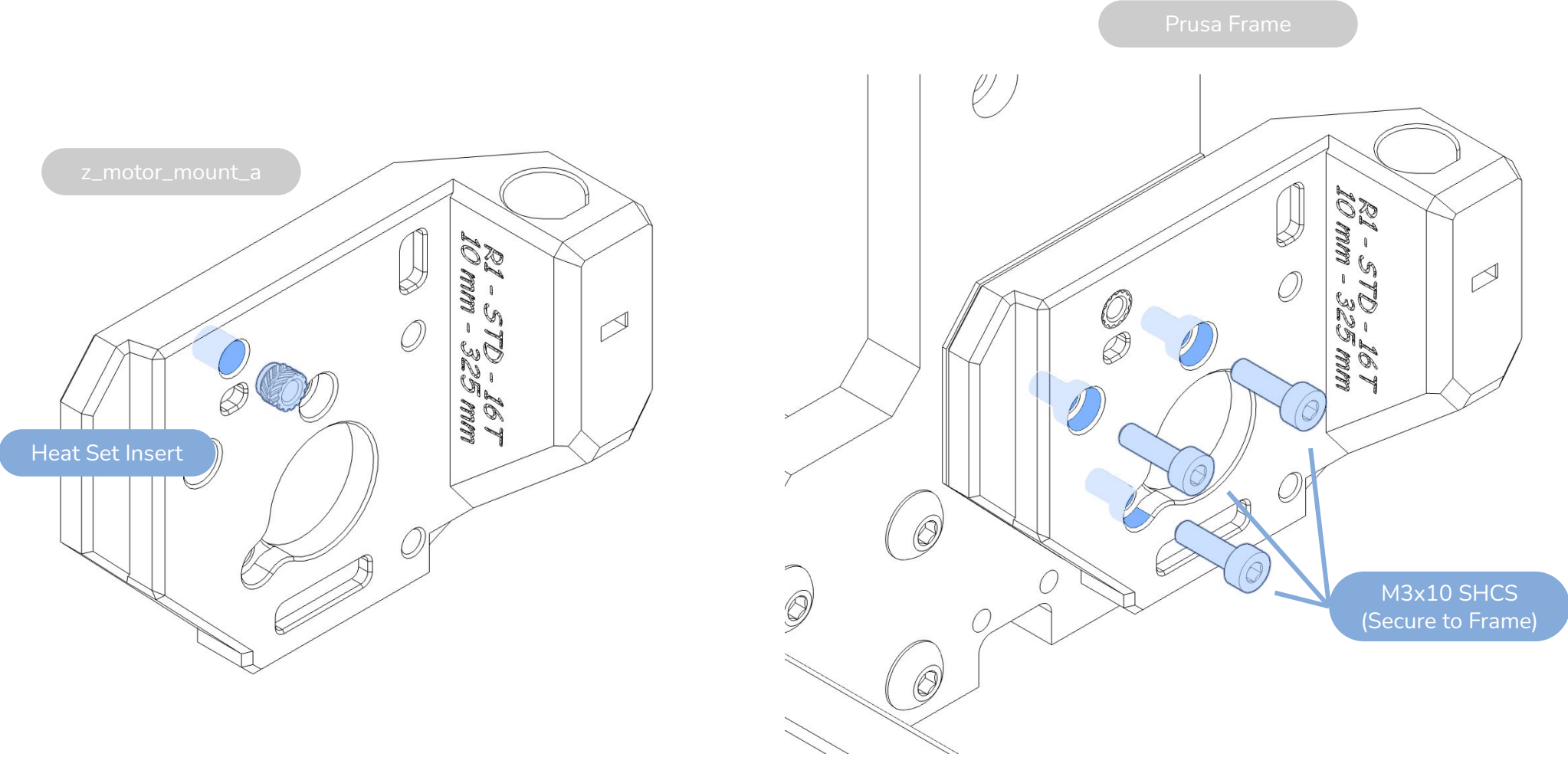
XZ AXIS
X Motor Assembly



XZ AXIS
Z Motor Assembly



XZ AXIS
Z Motor Assembly



XZ AXIS

Z Motor Assembly

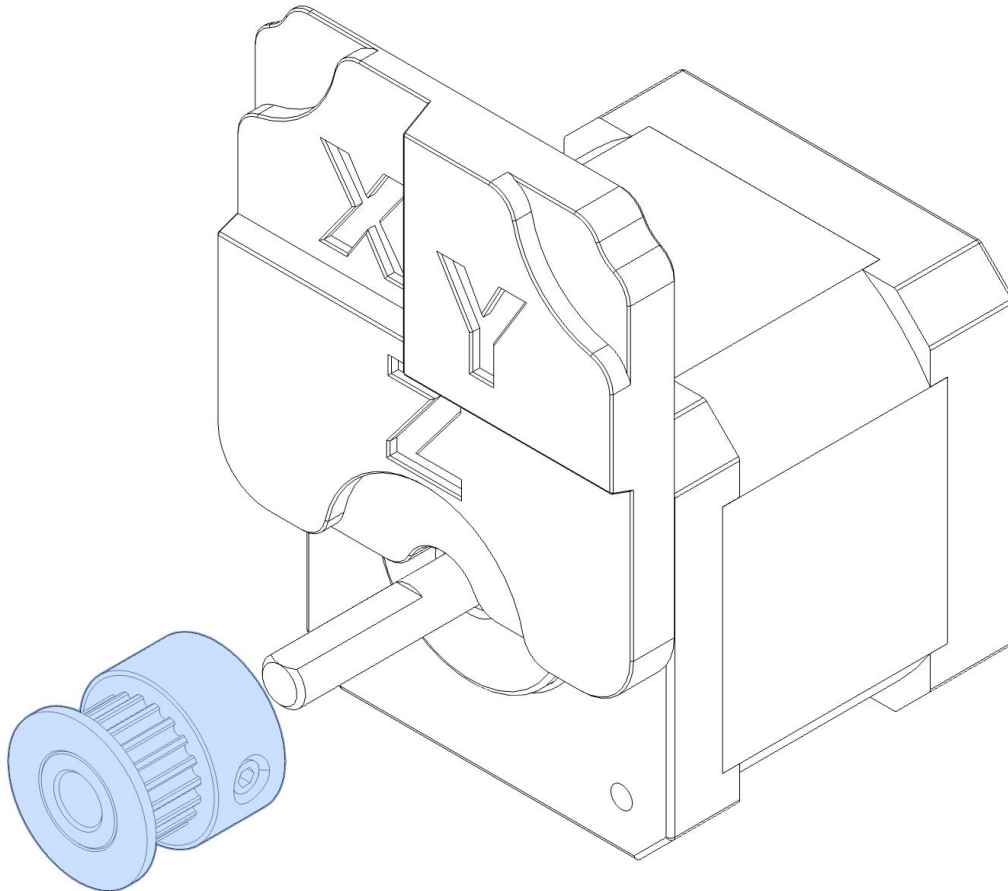
pully_alignment_jig

PULLEY ALIGNMENT

Make sure you use the right alignment jig for your type of pulley! **16T vs 20T.**

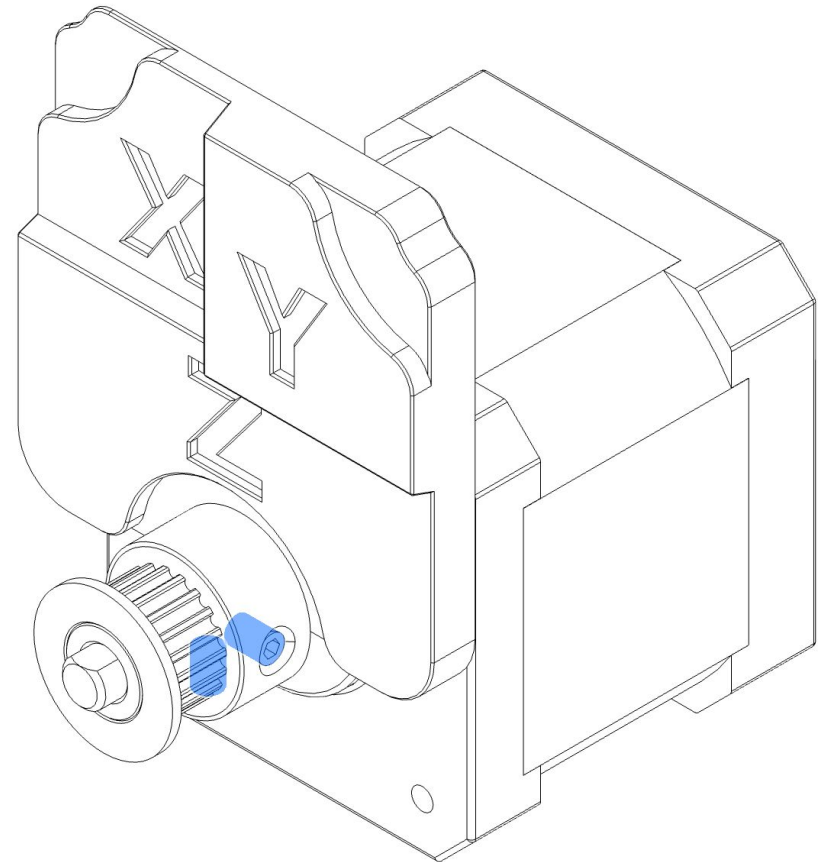
Use the printed pulley alignment jig to set the spacing of the GT2 Pulley. Press the small outer curved radius against the stepper motor shaft. Slide the pulley up to the larger relief in the jig. Be sure that one set screw aligns against the flat part of the motor shaft!

Tighten the set screws to secure the pulley. **Note the correct orientation of the pulley!**

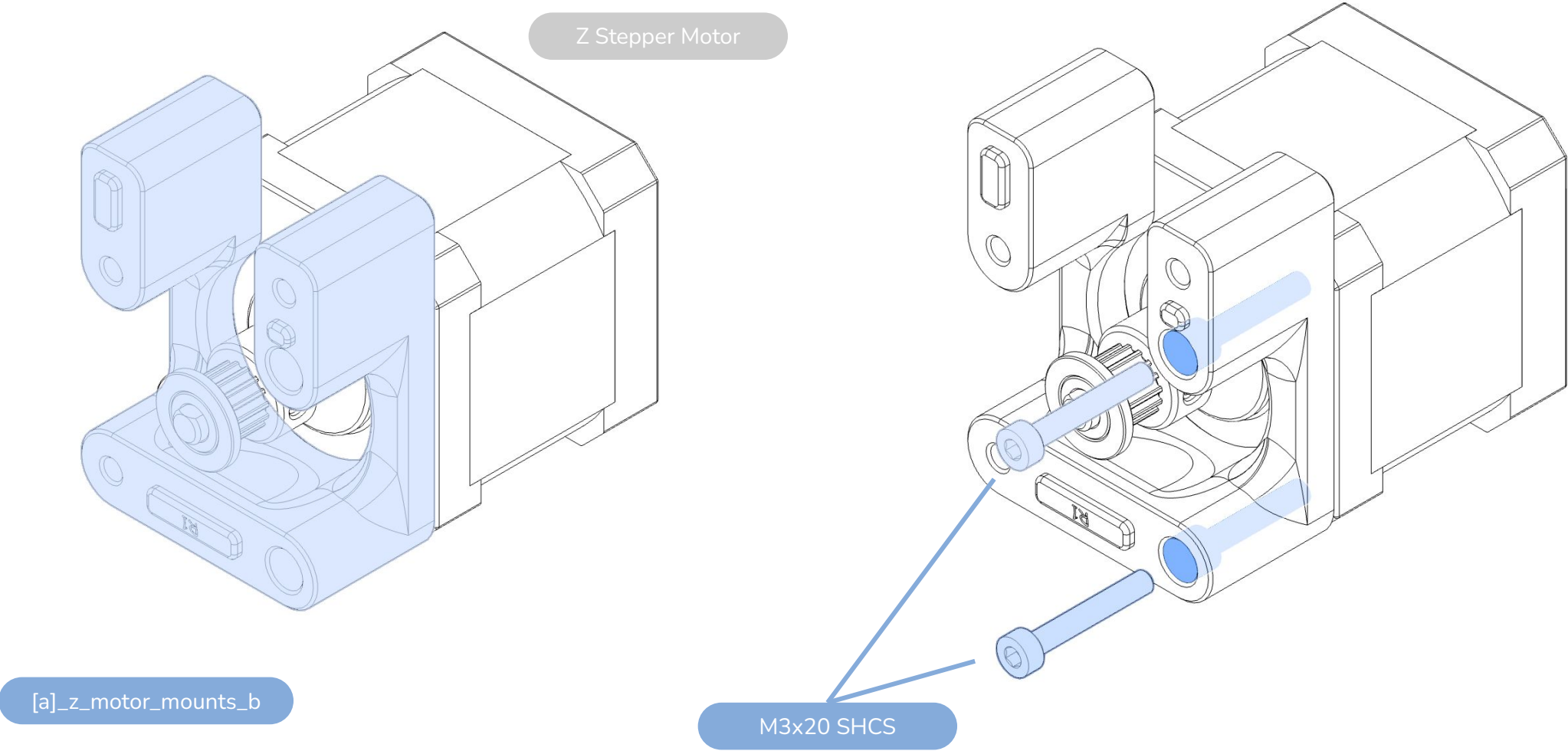


GT2 6mm Pulley

Z Stepper Motor

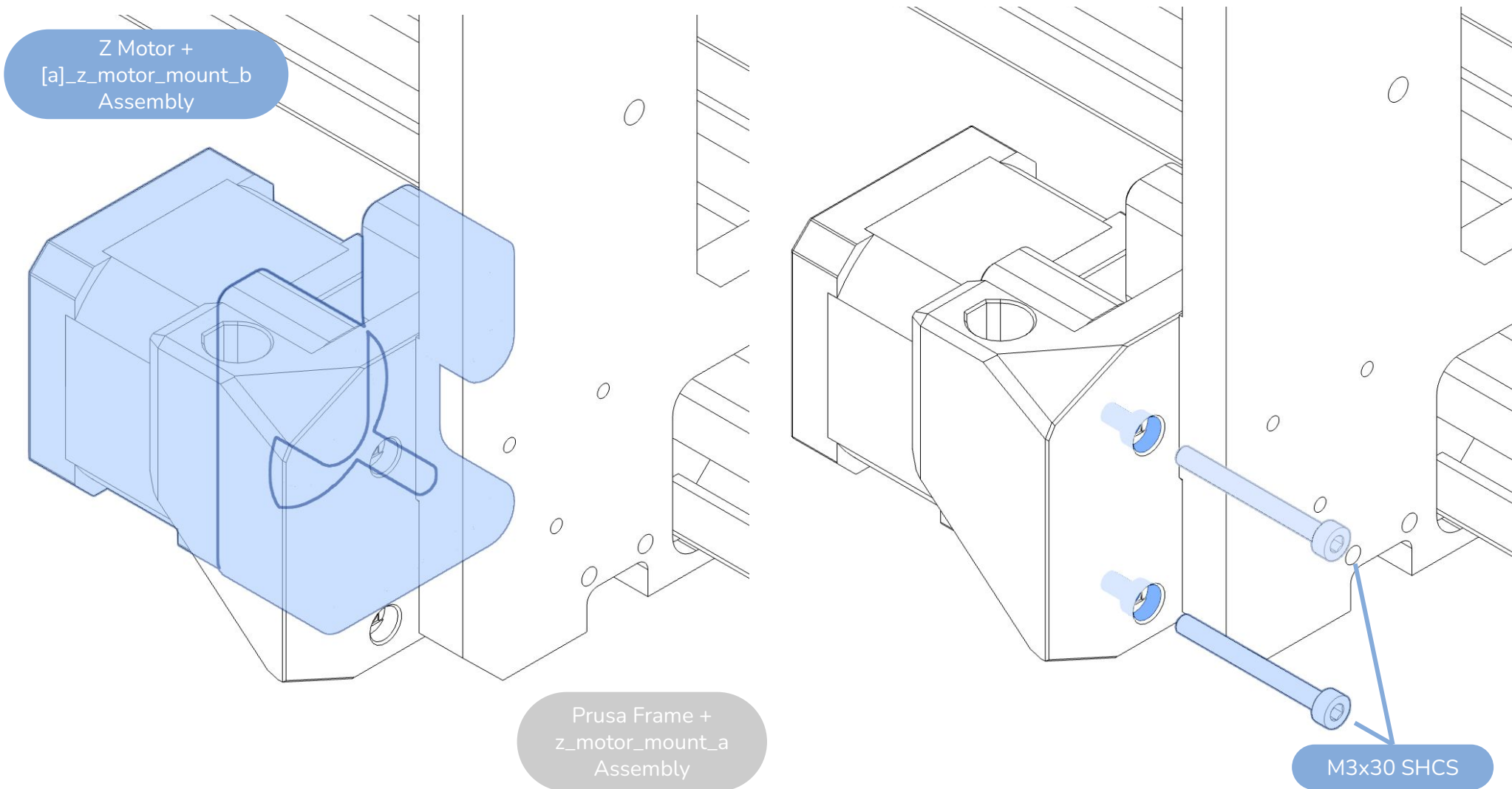


XZ AXIS
Z Motor Assembly

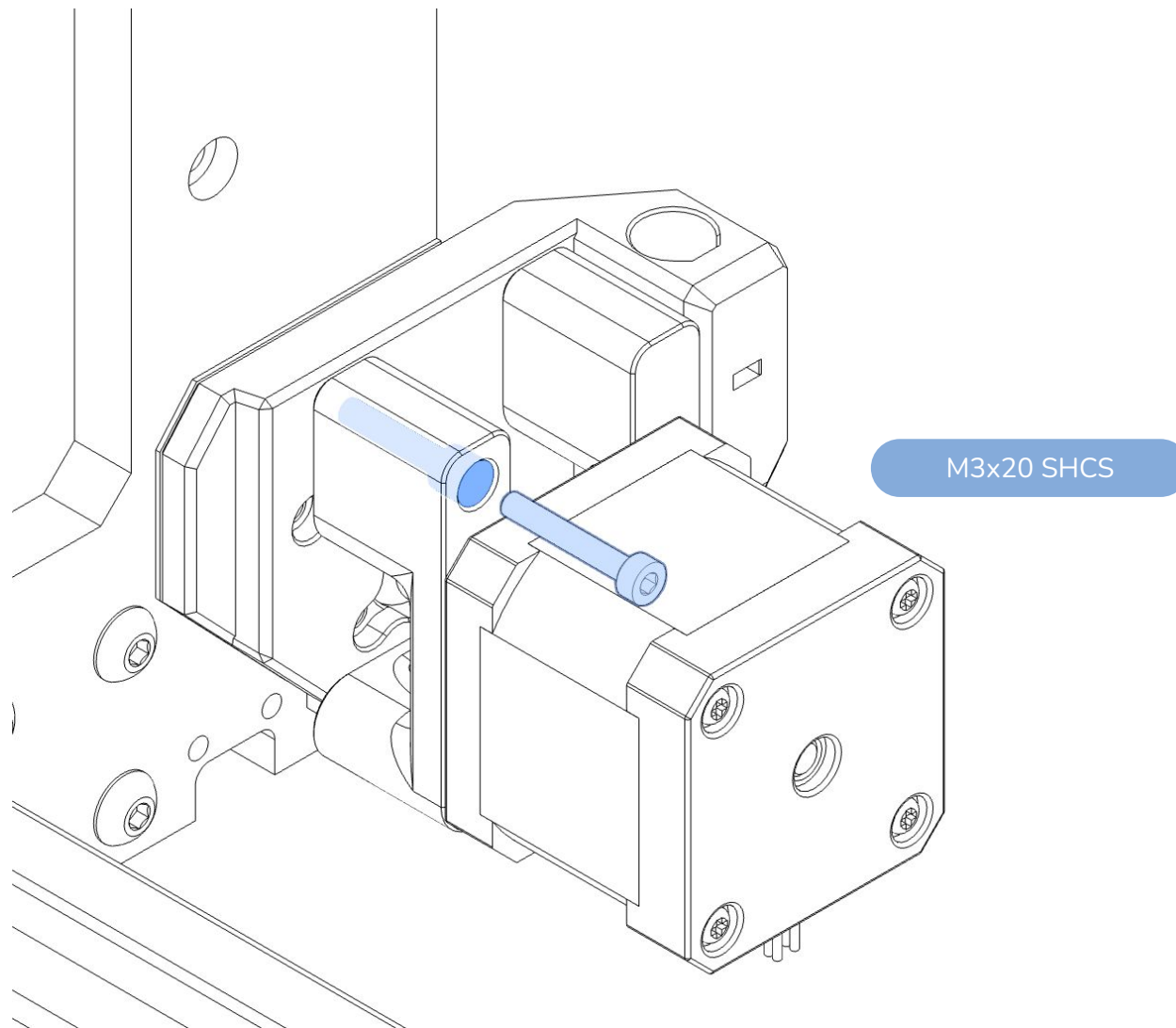


XZ AXIS

Z Motor Assembly

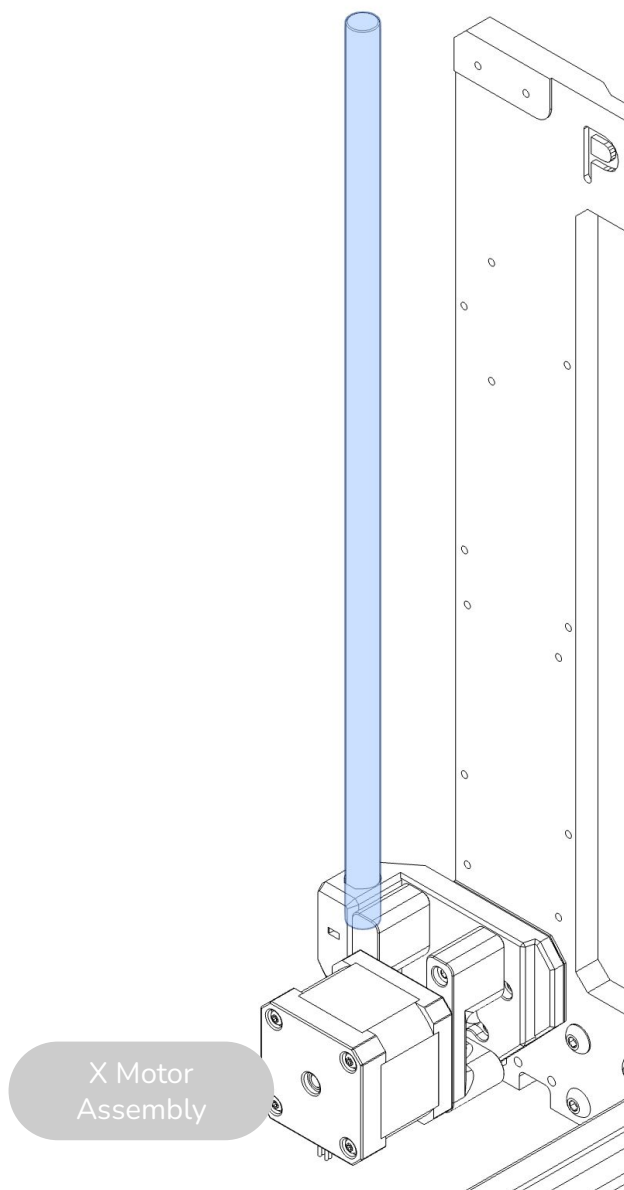


XZ AXIS
Z Motor Assembly



XZ AXIS

Linear Rods



Z LINEAR RODS

NOTE: The Prusawire design is designed to accommodate a range of linear rod diameters and lengths that you may already have. 10mm diameter is preferred for stability, but 8mm (MK3 spec) is extremely usable!

Currently:

10mm x 320mm
10mm x 325mm (Prusa MK4 Stock Z Rods)
10mm x 330mm
10mm x 341mm

8mm x 320mm (Prusa MK3 Stock Z Rods)
8mm x 325mm
8mm x 330mm

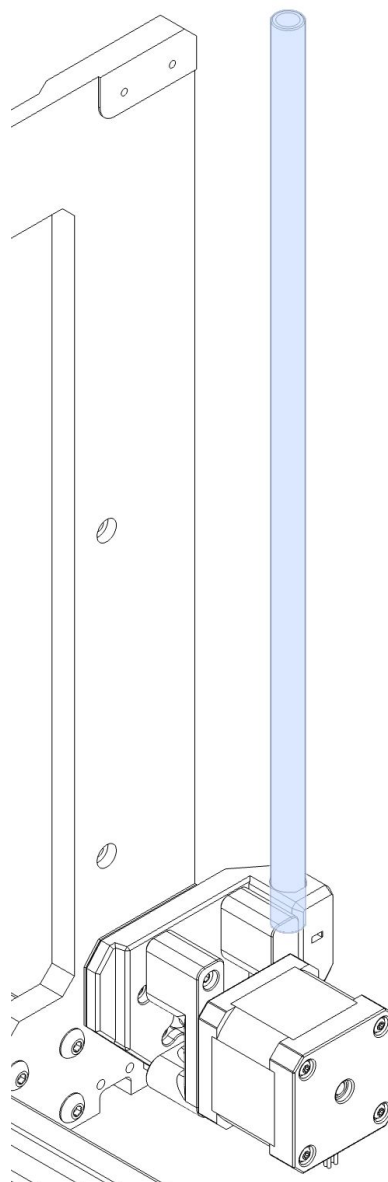
STLs have been provided for the different rod sizes. Choose the version of the follow parts that accommodate your desired Z Linear Rod size:

[a]_x_idler_block_{diameter}

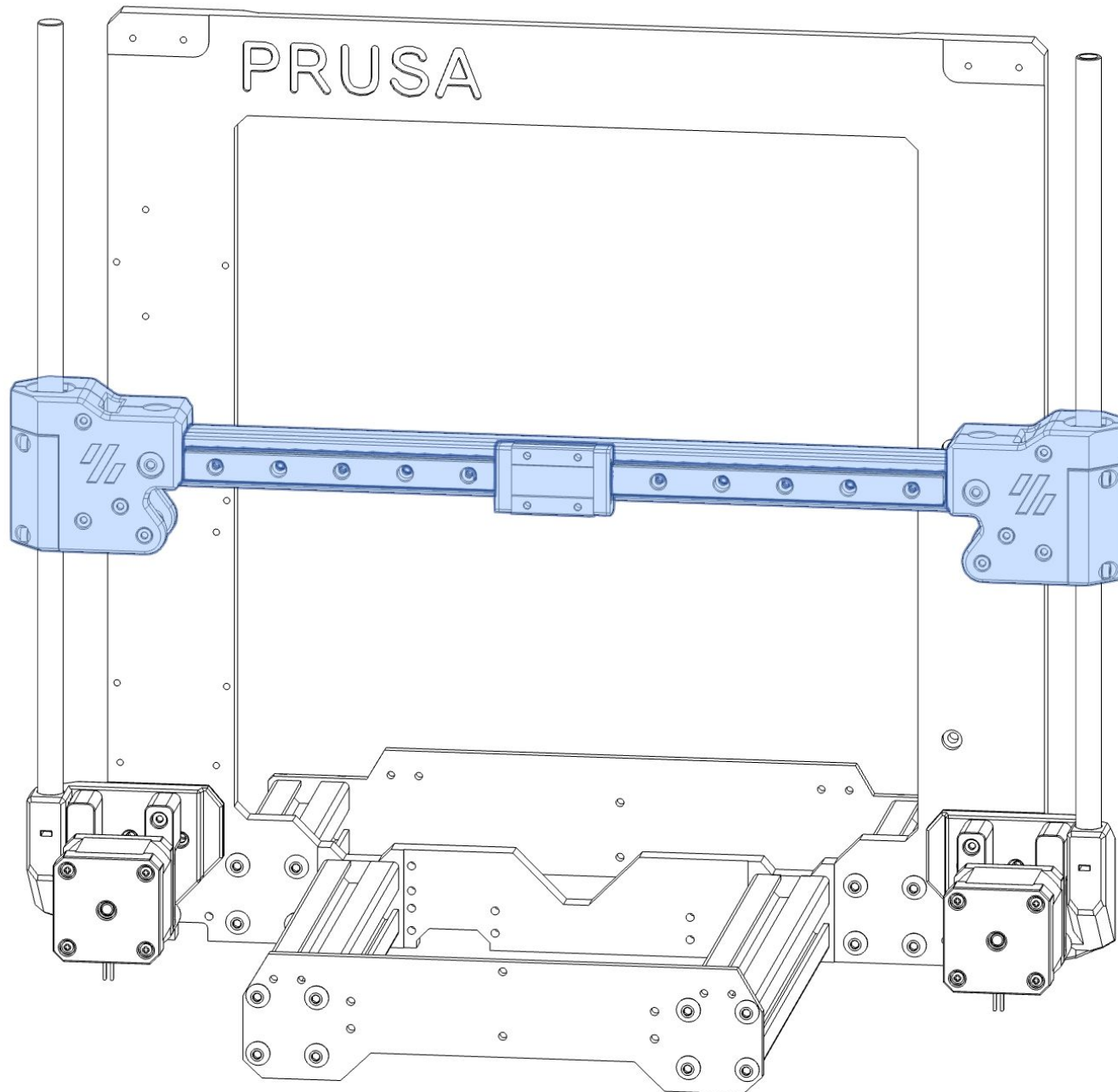
[a]_z_idler_block_{diameter}

x_motor_mount_{diameter}_{length}_{gearing}

z_motor_mount_{diameter}_{length}_{gearing}



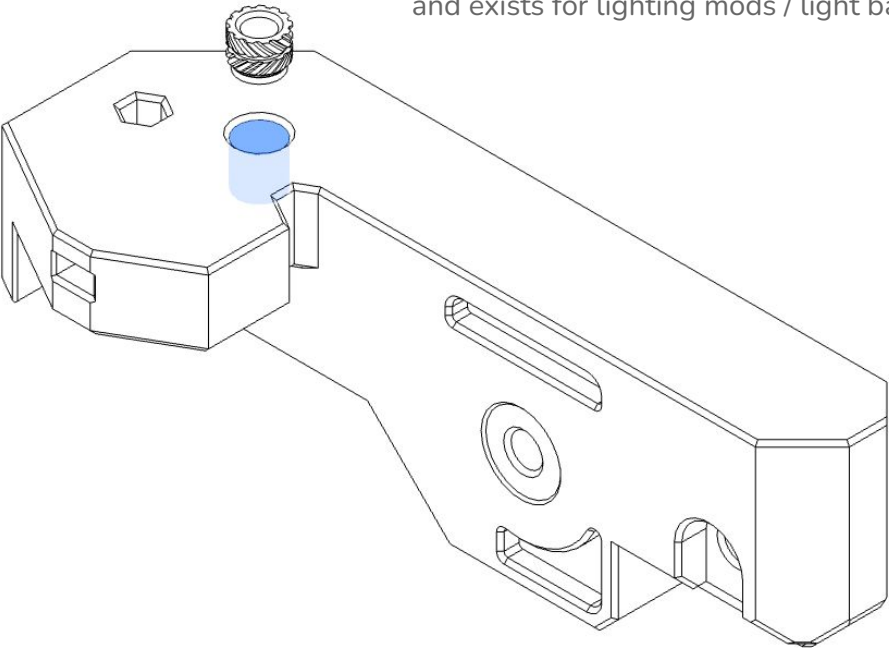
XZ AXIS
X Gantry ON



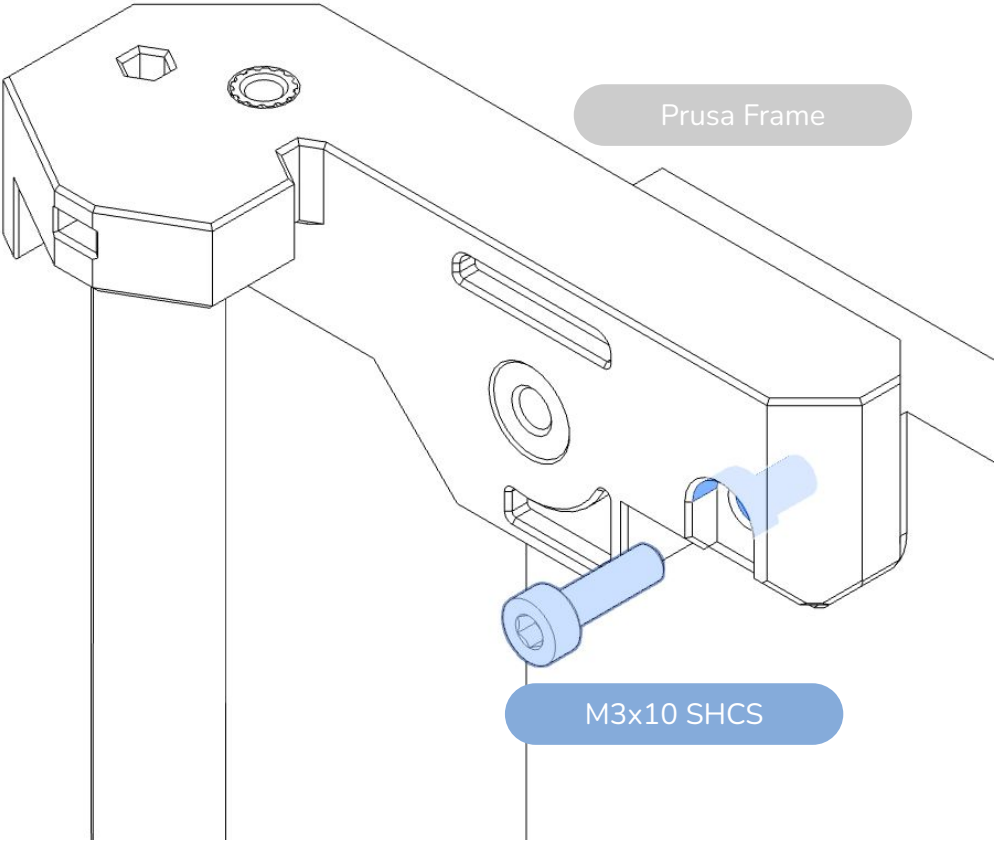
XZ AXIS
Idler Block - Left

M3 Heat Set Insert

LIGHT BAR MOUNT
The Heat Set Insert at the top is **optional** and exists for lighting mods / light bars.



[a]_x_idler_block

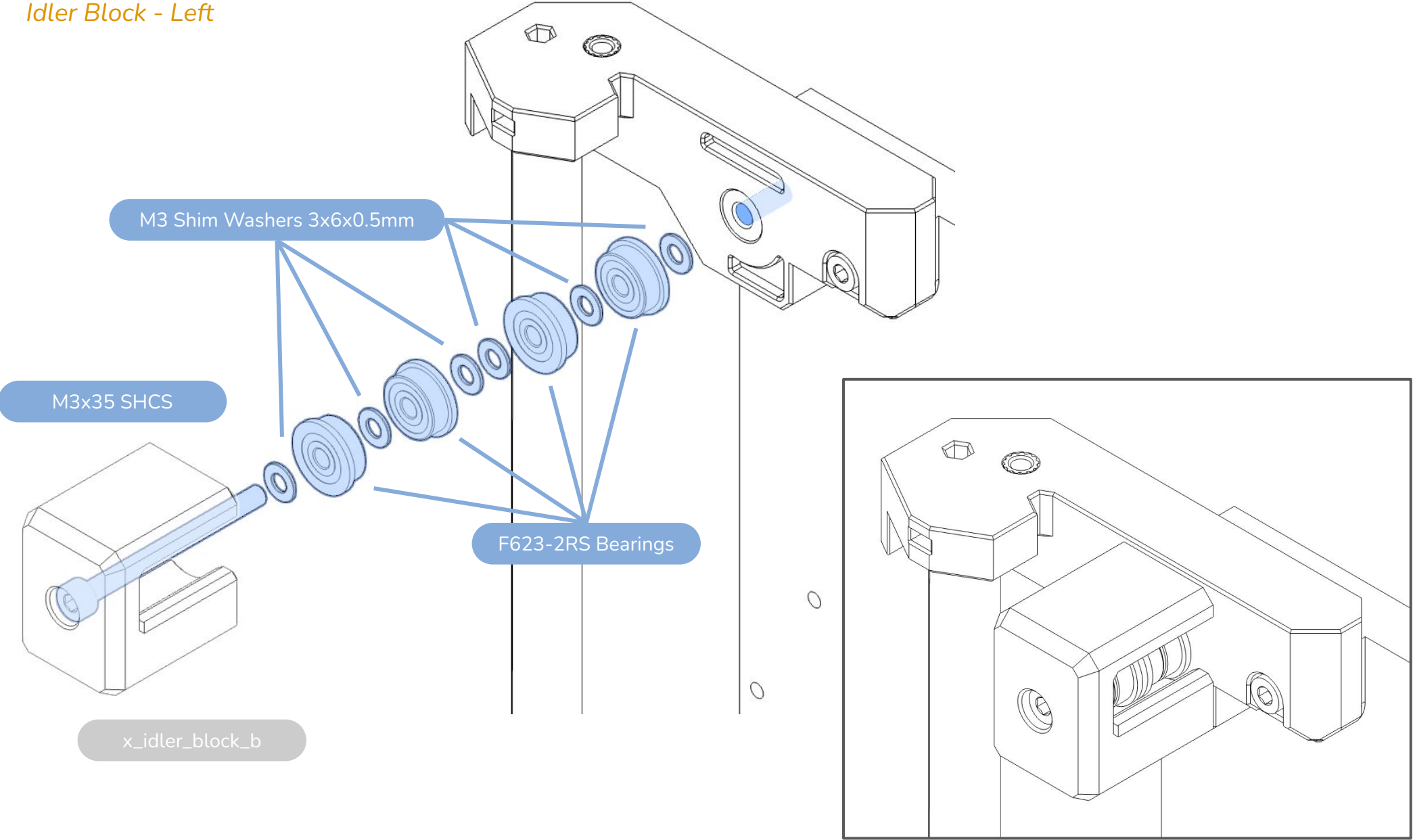


Left Z Rod

M3x10 SHCS

Prusa Frame

XZ AXIS
Idler Block - Left



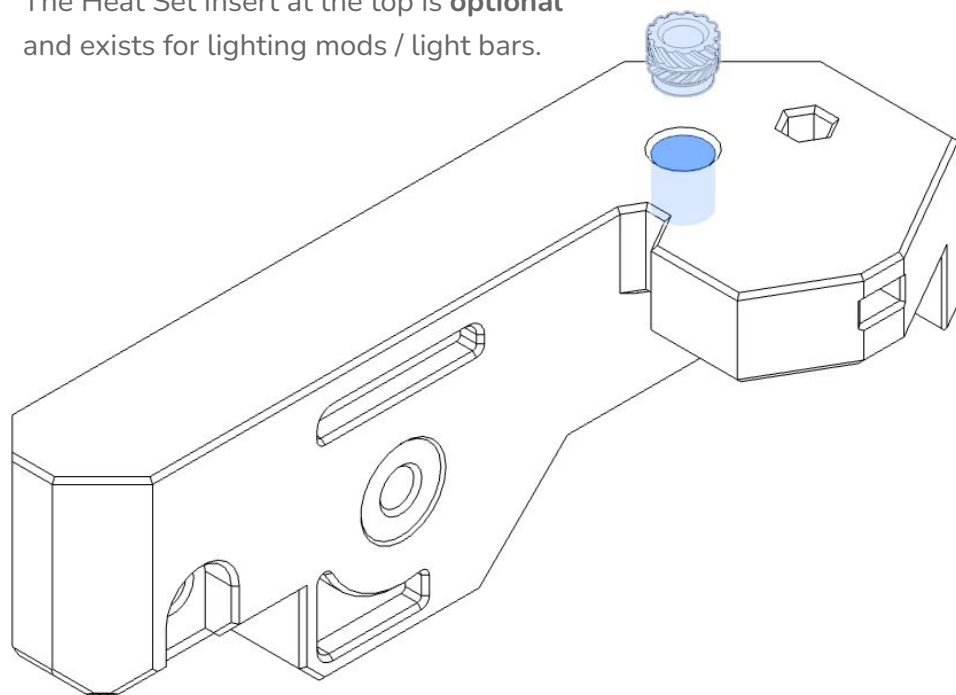
XZ AXIS

Idler Block - Right

LIGHTING MOUNT

The Heat Set Insert at the top is **optional** and exists for lighting mods / light bars.

M3 Heat Set Insert

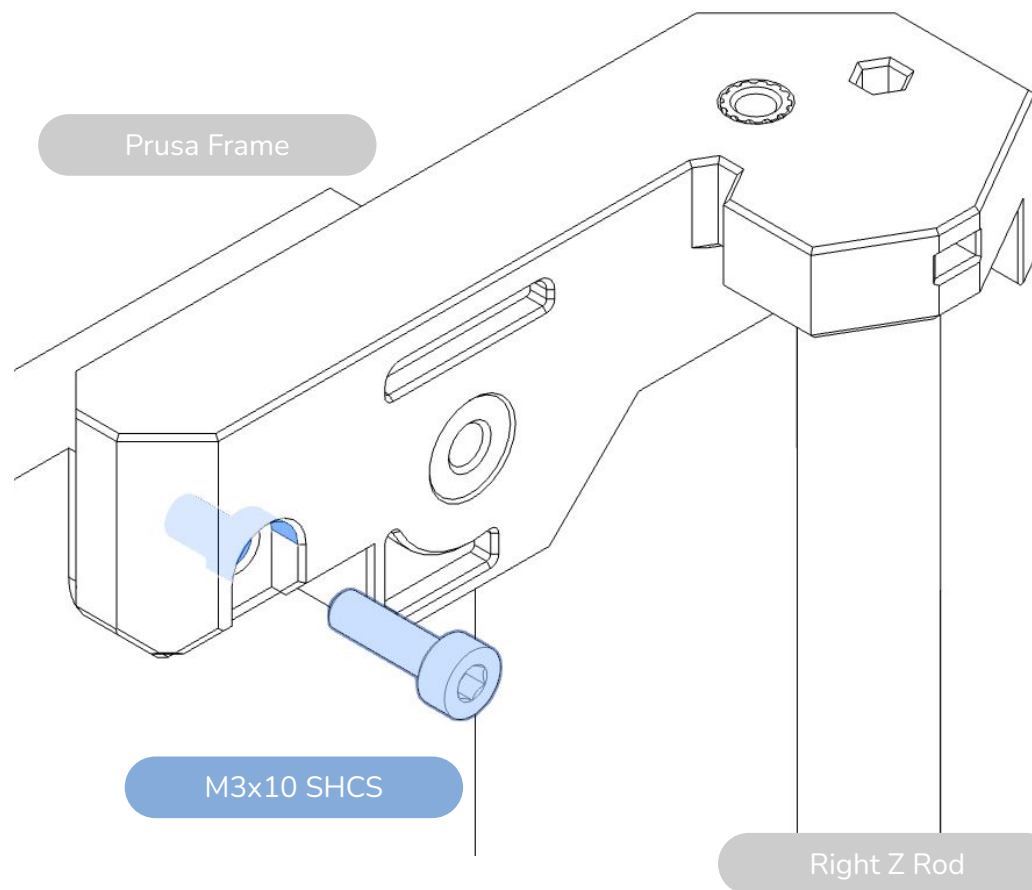


[a]_z_idler_block

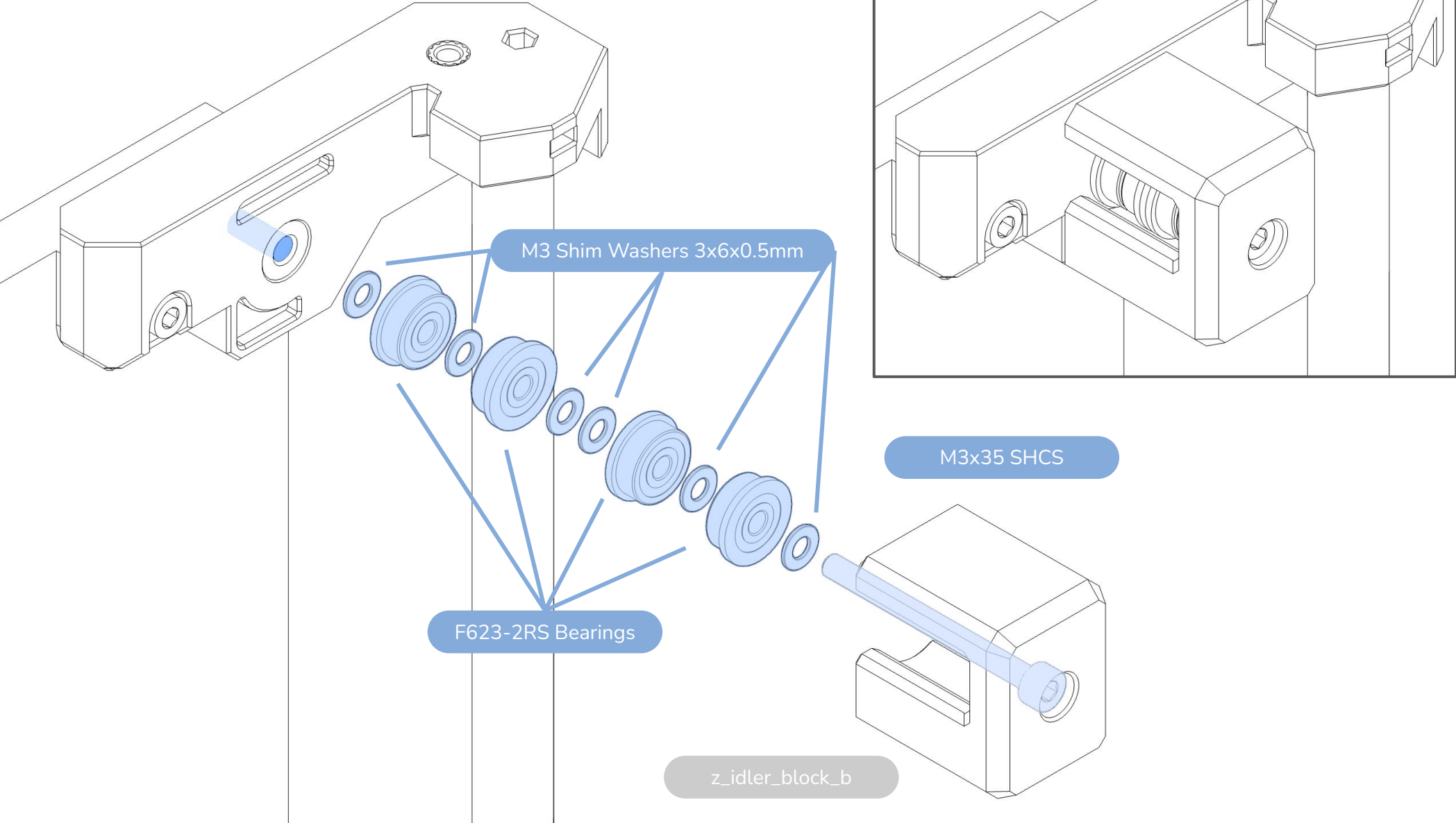
Prusa Frame

M3x10 SHCS

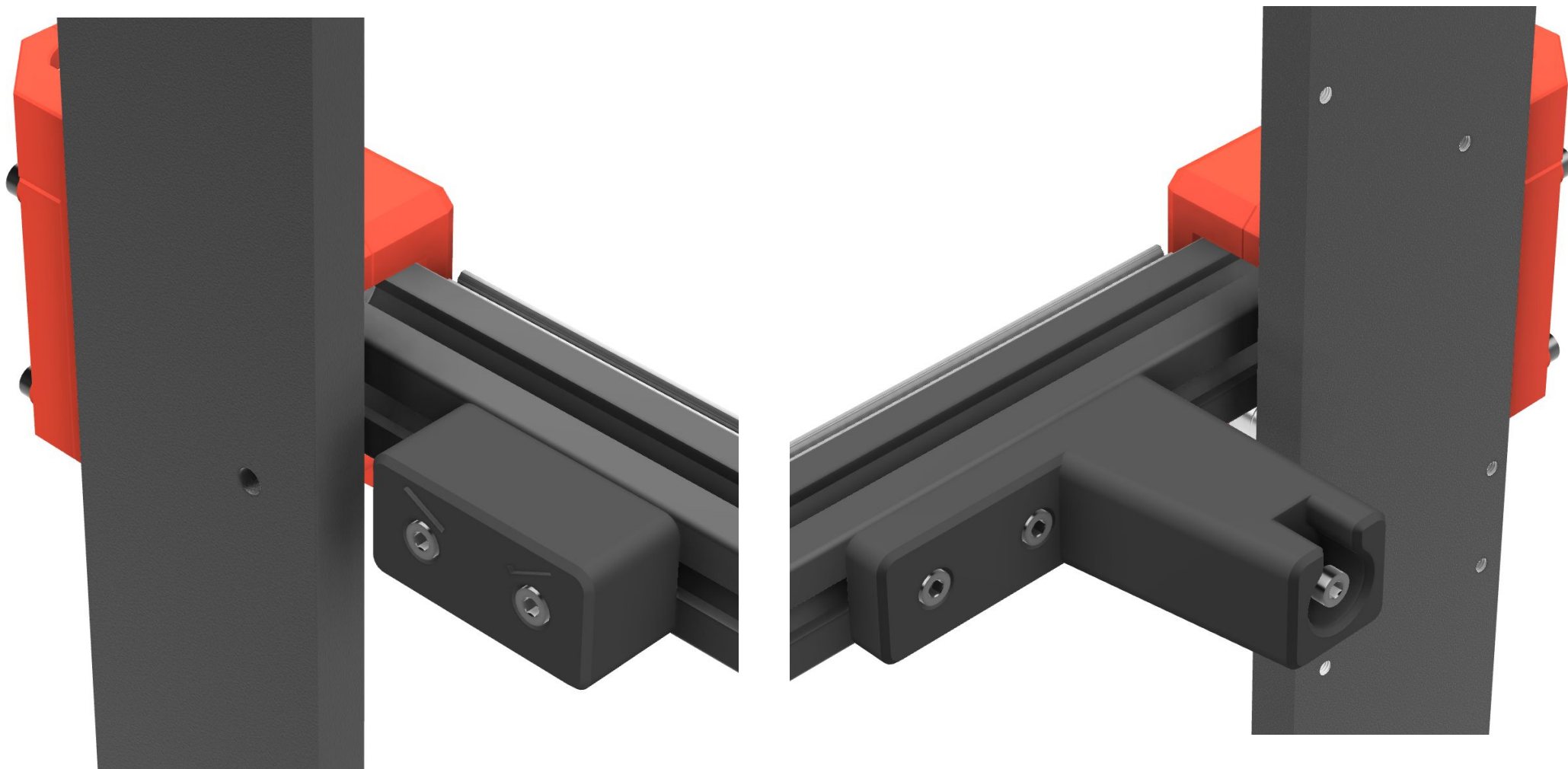
Right Z Rod



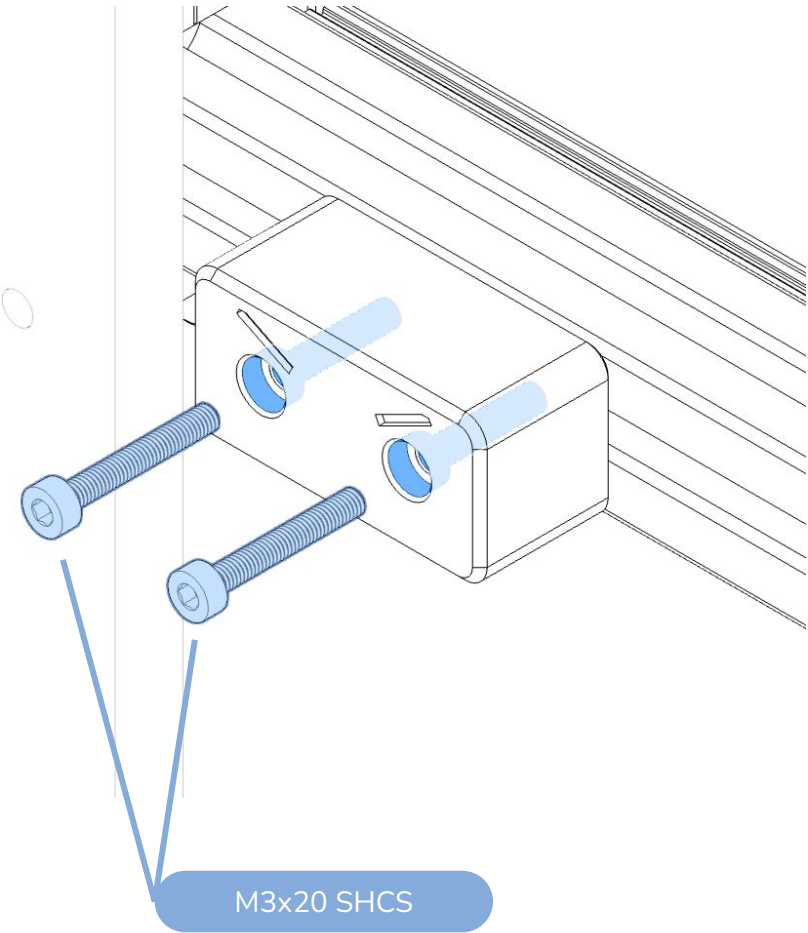
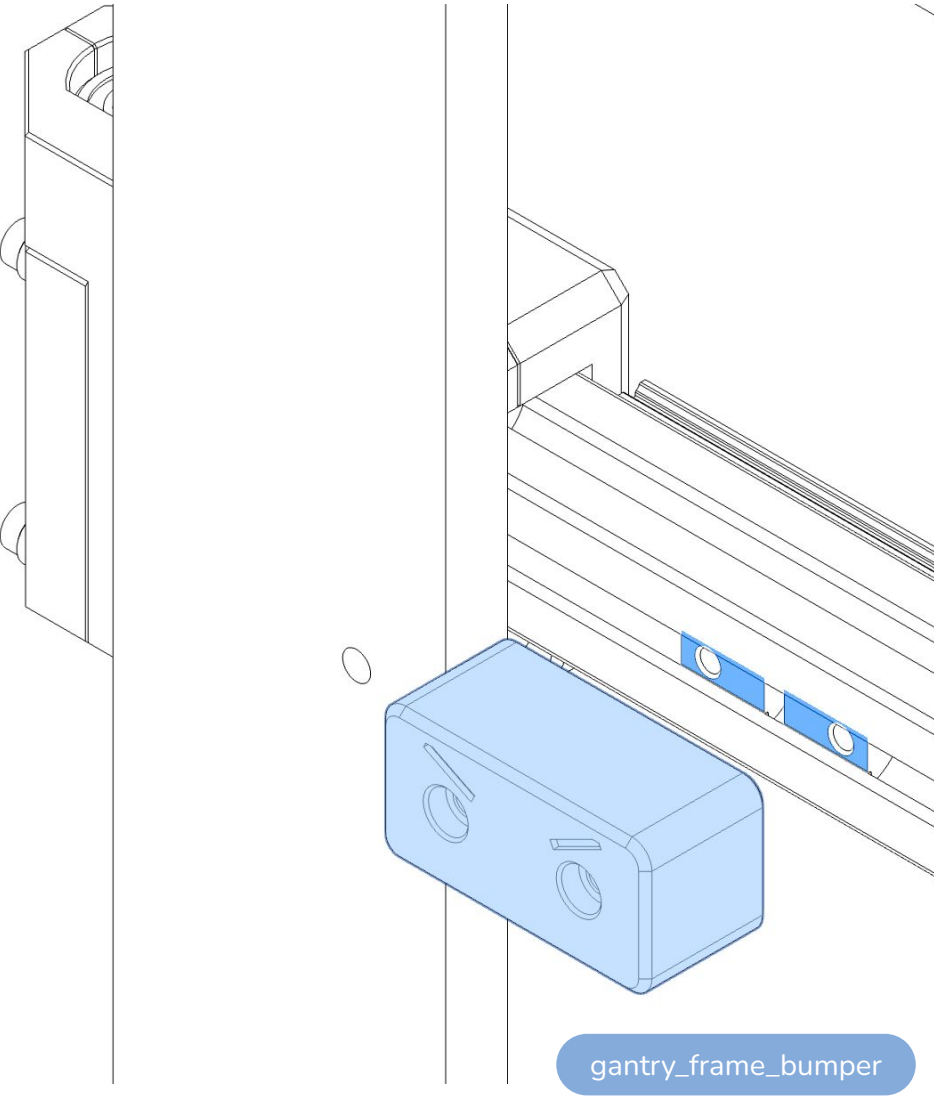
XZ AXIS
Idler Block - Right



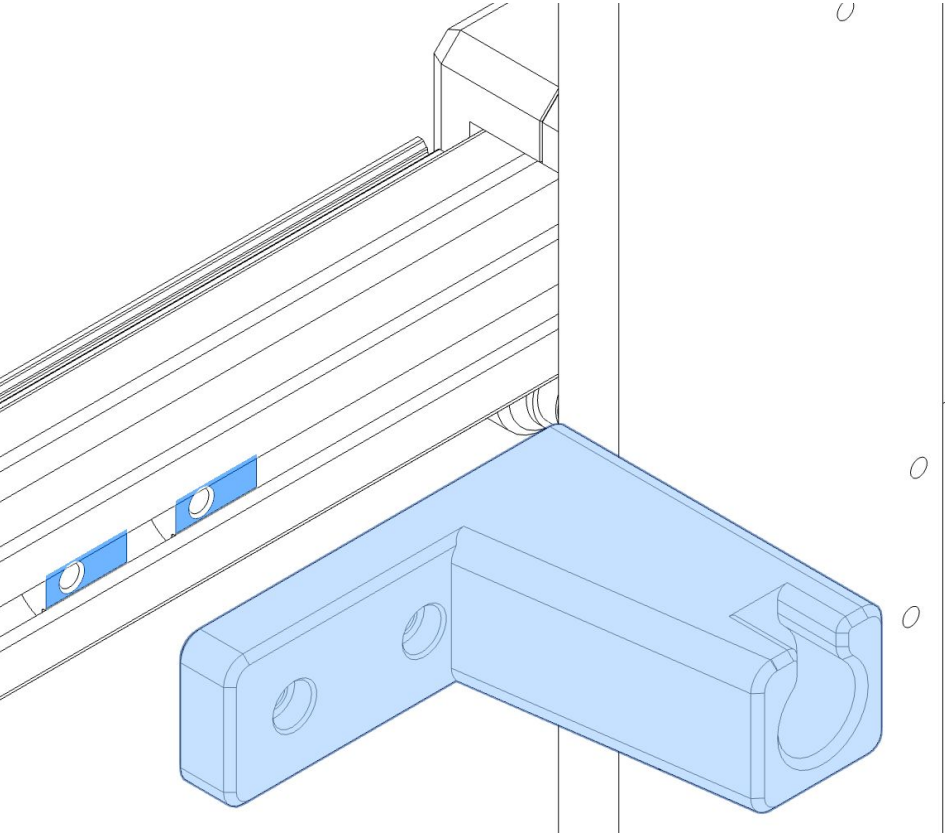
XZ AXIS
Gantry Bumper &
Door Puller Anchor



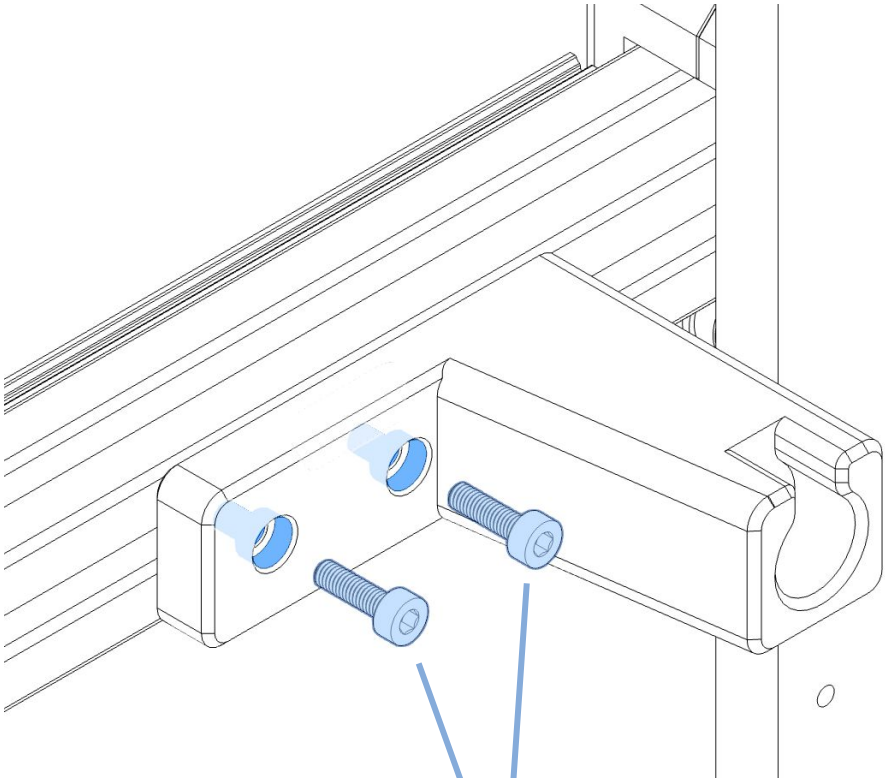
XZ AXIS
Gantry Bumper



XZ AXIS
Door Puller Anchor

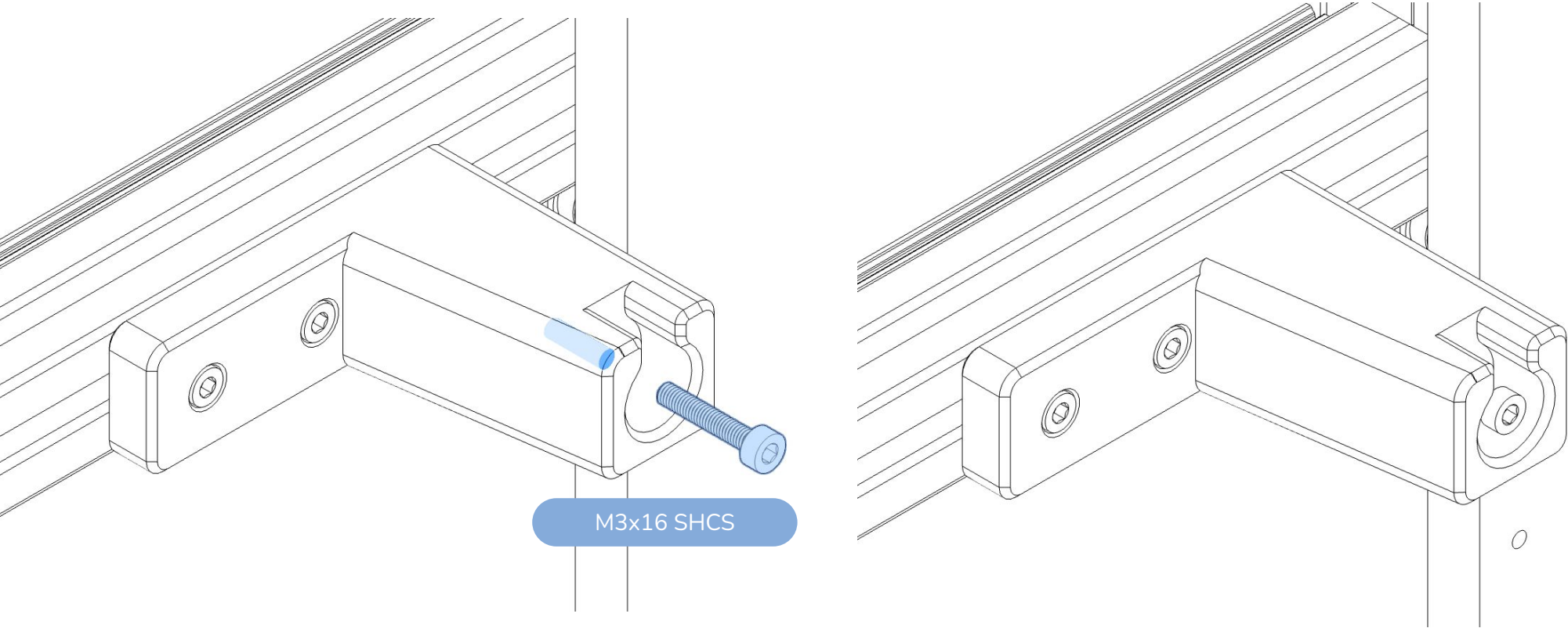


door_puller_anchor



M3x10 SHCS

XZ AXIS
Door Puller Anchor



GANTRY DERACKING

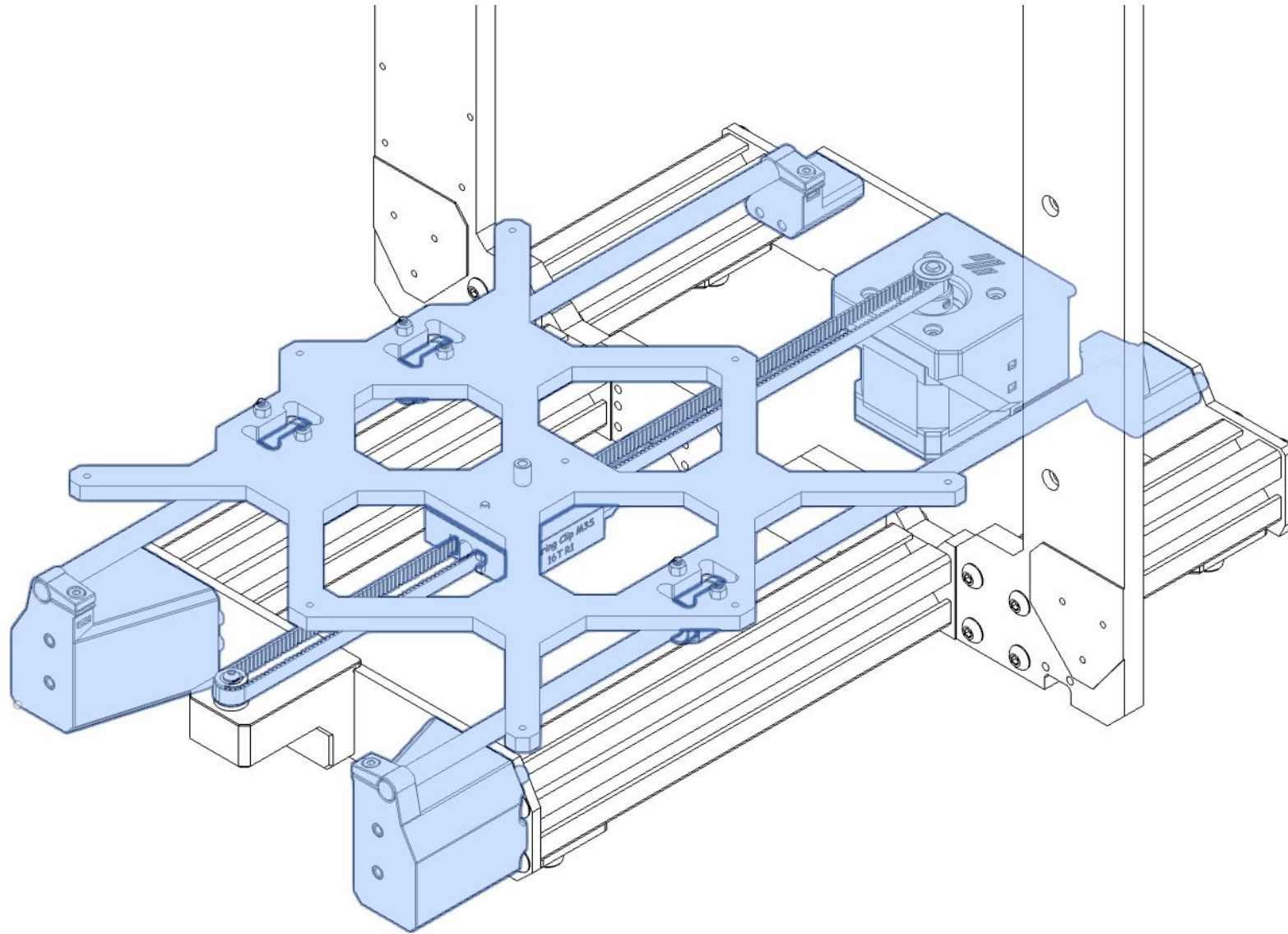
The gantry de-racking procedure is critical to ensure smooth motion in X and Z. The general idea is to loosen certain screws on the gantry, bump the bottom of the gantry against the X and Z motor joints to align, and tighten the screws.

The complete procedure can be found on the Prusawire Documentation Site:

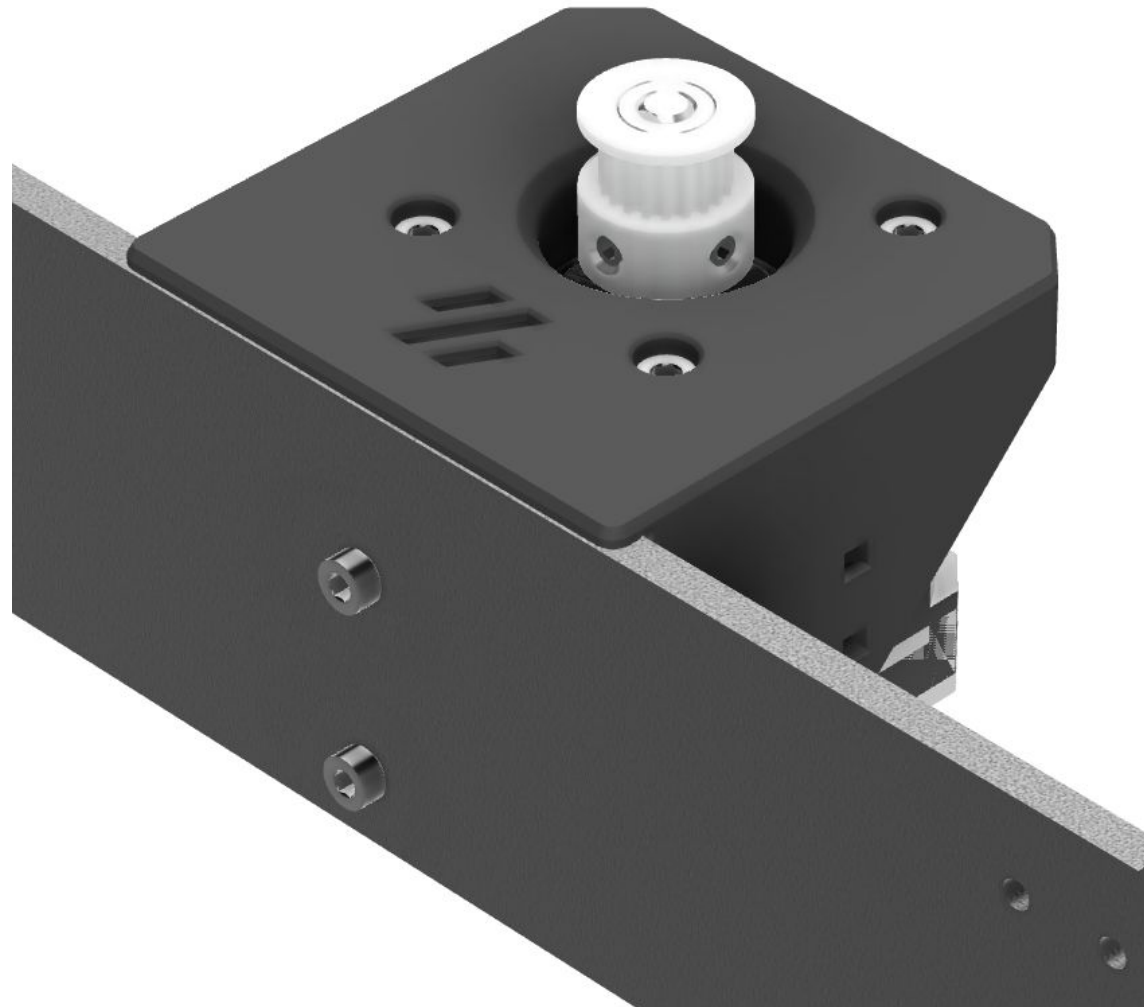
<https://positron3d.github.io/Prusawire/hardware/gantry-deracking.html>



Y AXIS Overview



Y AXIS
Motor Mount

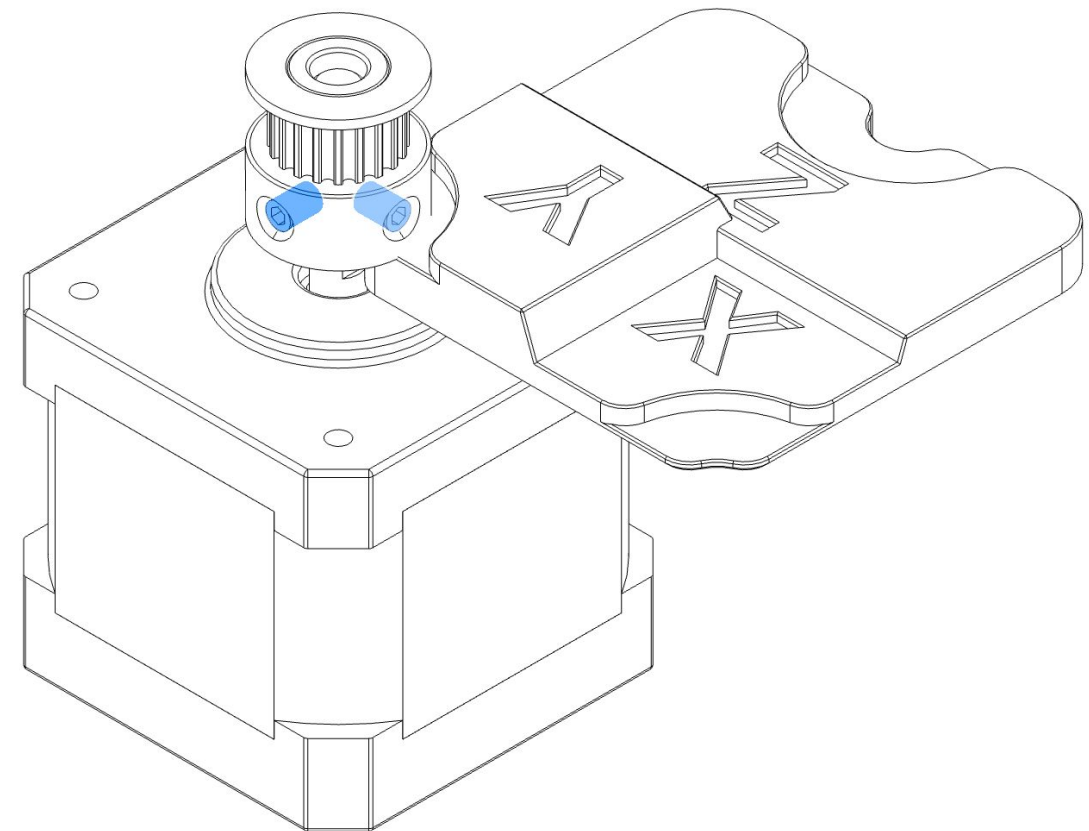
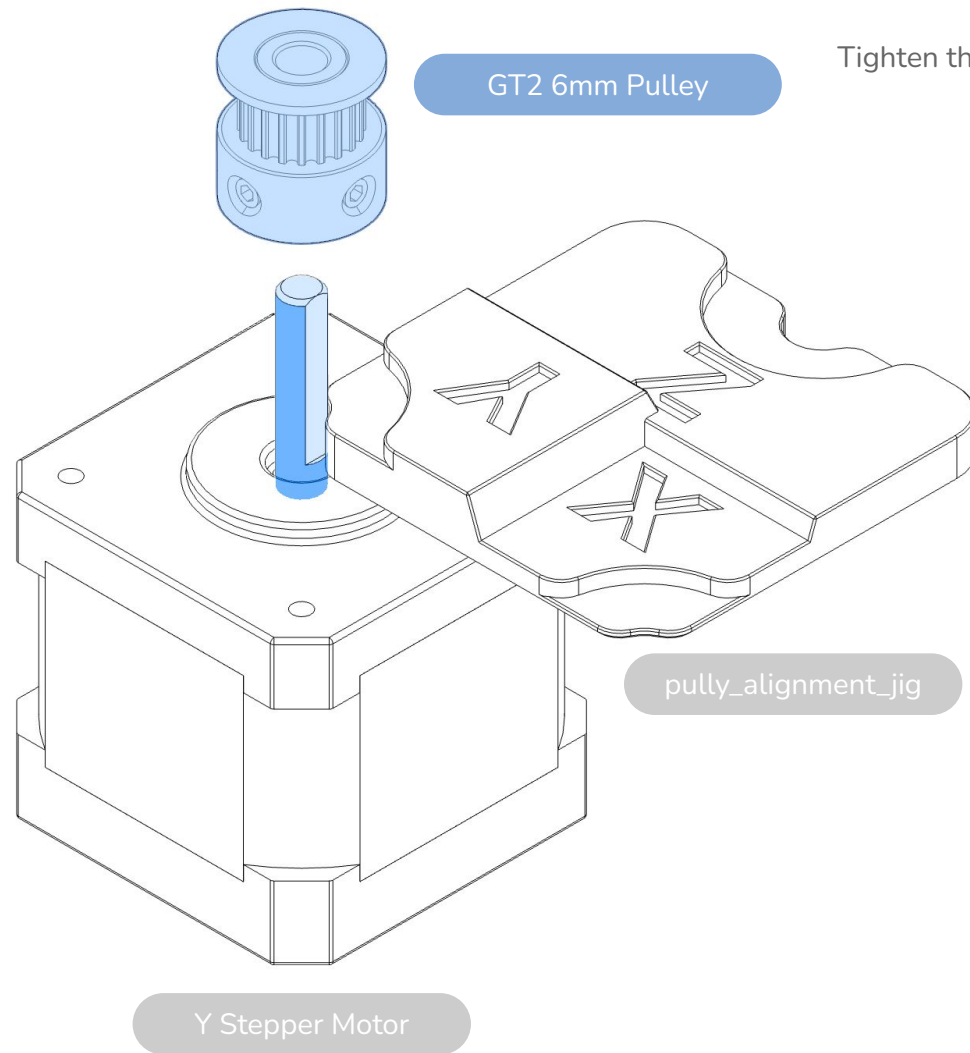


Y AXIS Motor Mount

PULLEY ALIGNMENT

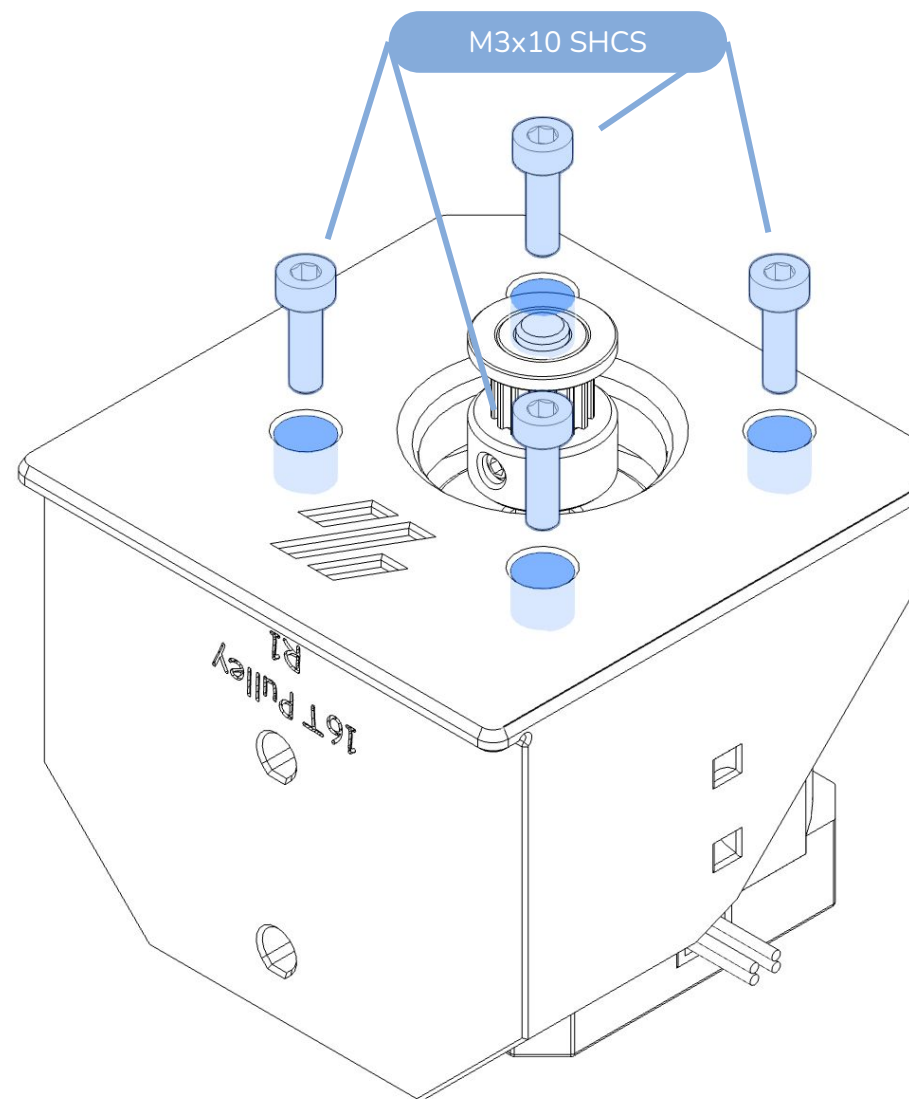
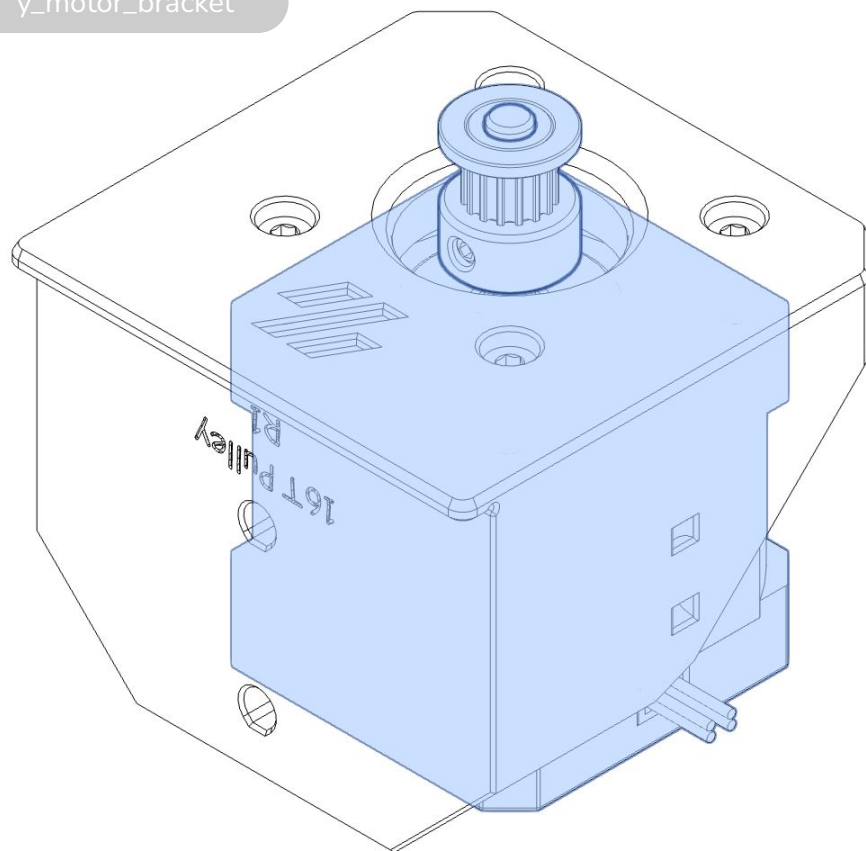
Make sure you use the correct jig! Either 16T (Prusa-spec) or 20T (Switchwire-spec)!

Use the printed pulley alignment jig to set the spacing of the GT2 Pulley. Press the small outer curved radius against the stepper motor shaft. Slide the pulley up to the larger relief in the jig. Be sure that one set screw aligns against the flat part of the motor shaft!



Y AXIS Motor Mount

y_motor_bracket

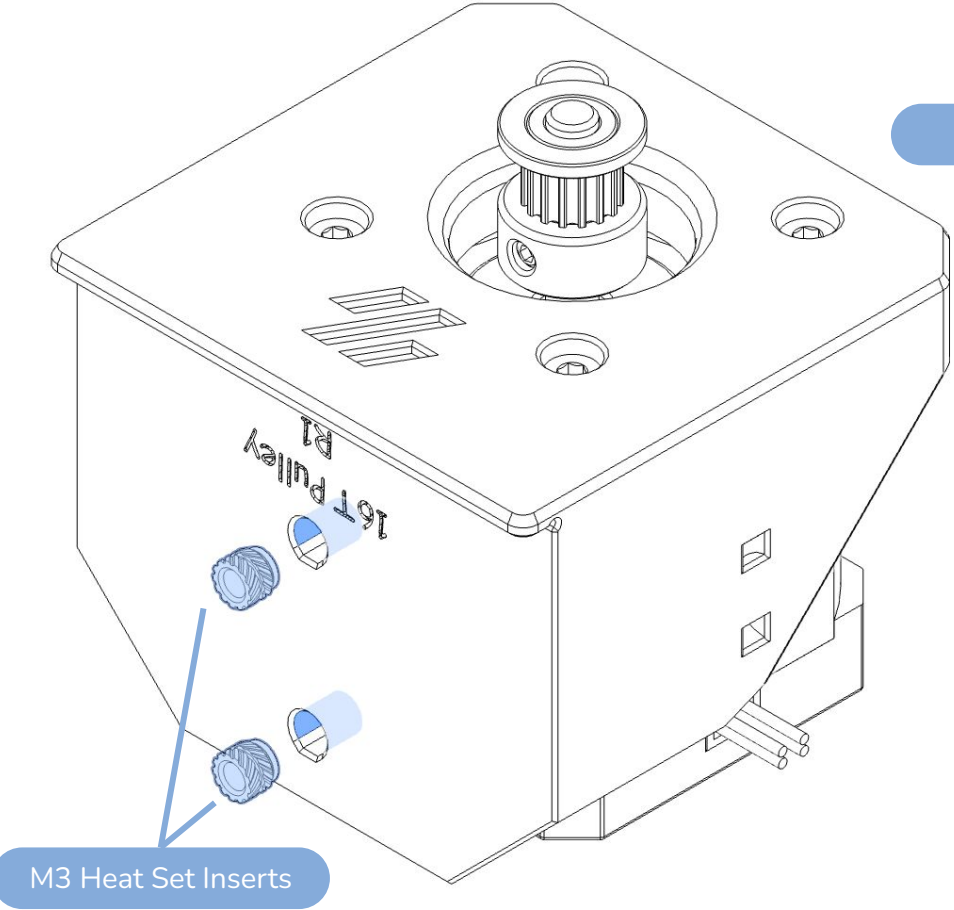


MOTOR MOUNT

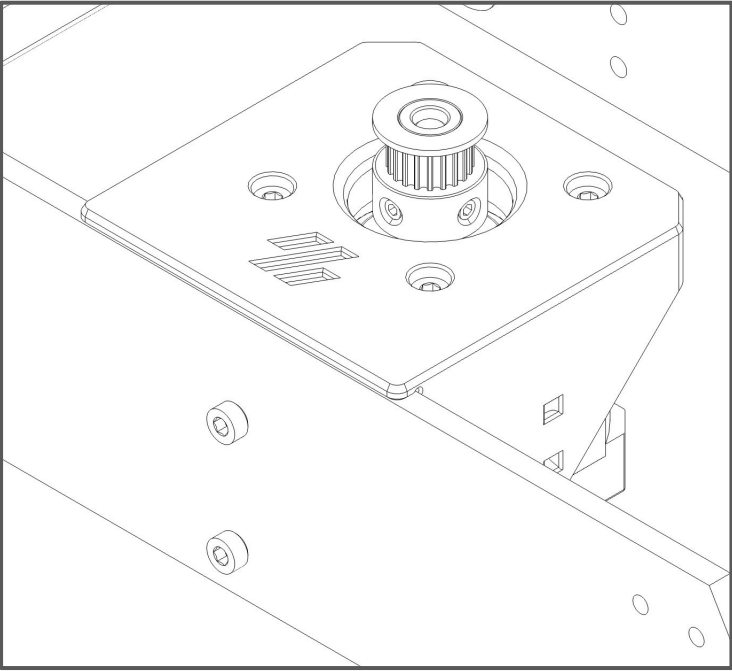
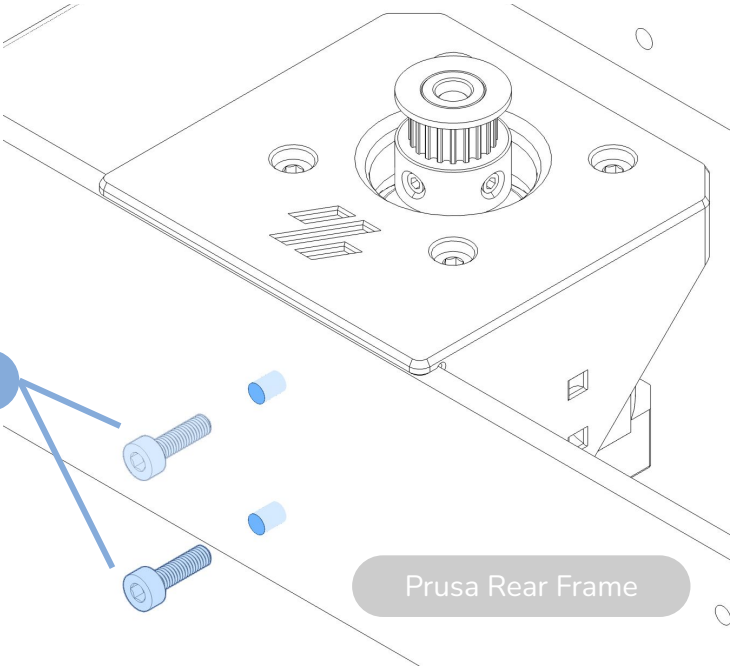
Your X and Z motor mount may have different embossed text depending on which mount you need based on what gearing you selected.

Example: y_motor_bracket_16T_pulley
 y_motor_bracket_20T_pulley

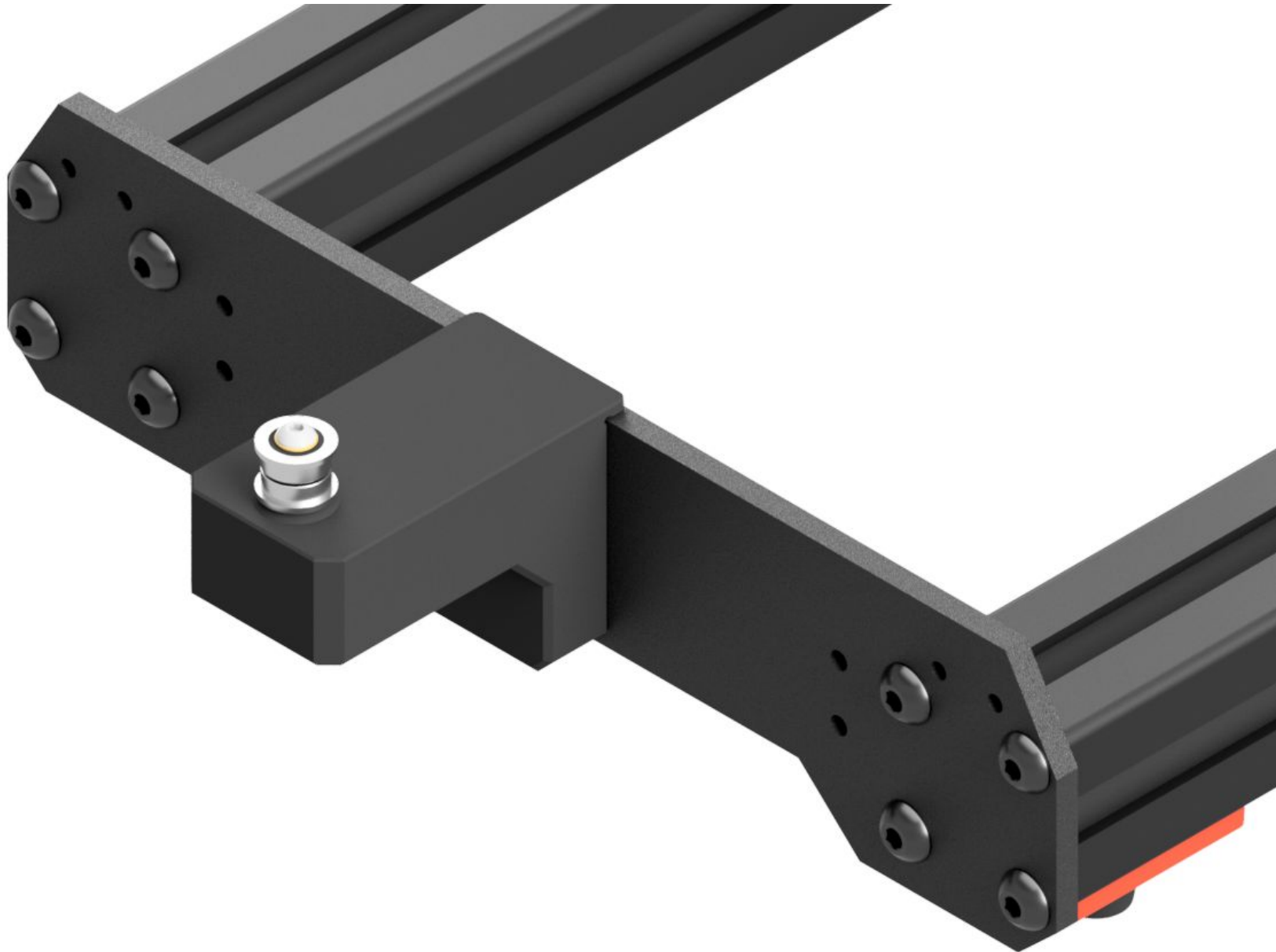
Y AXIS
Motor Mount



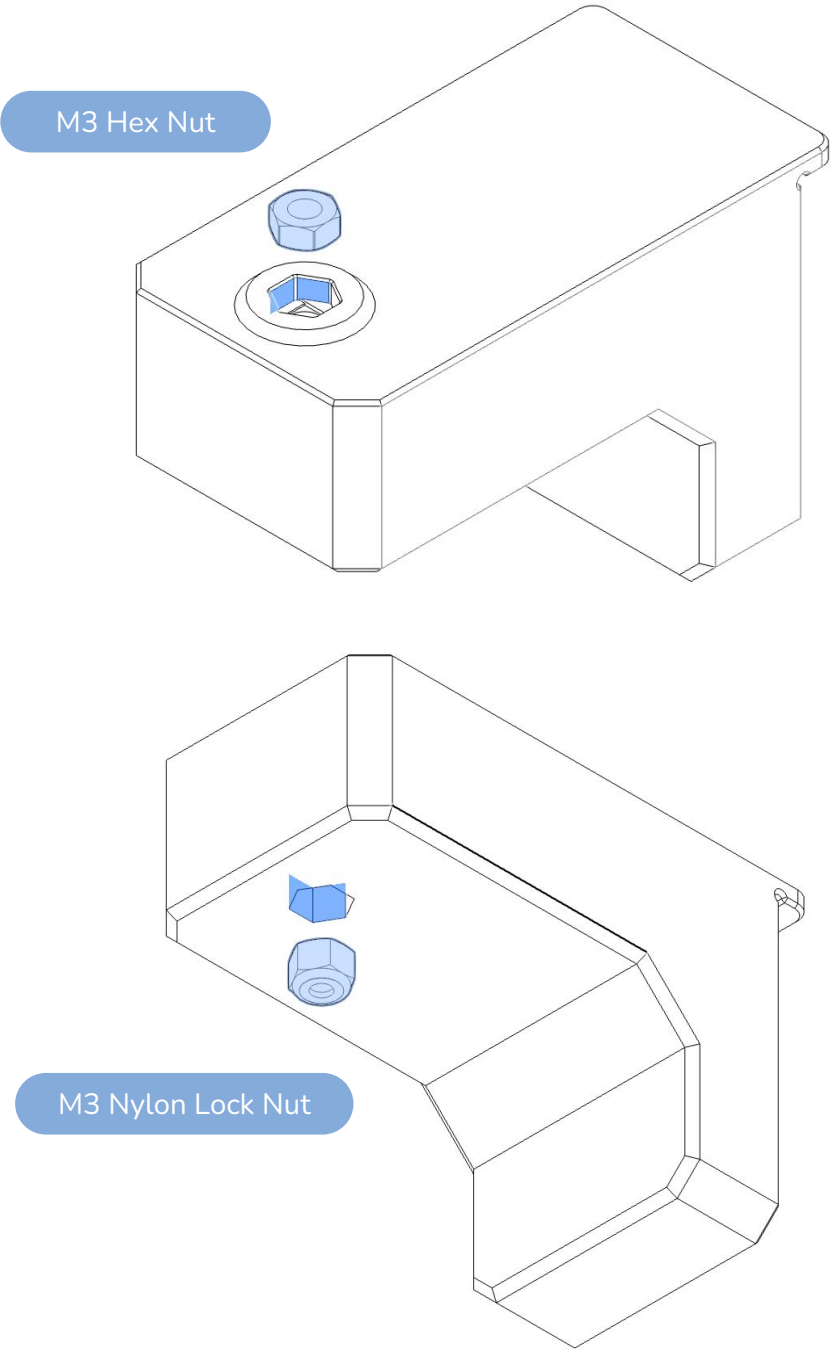
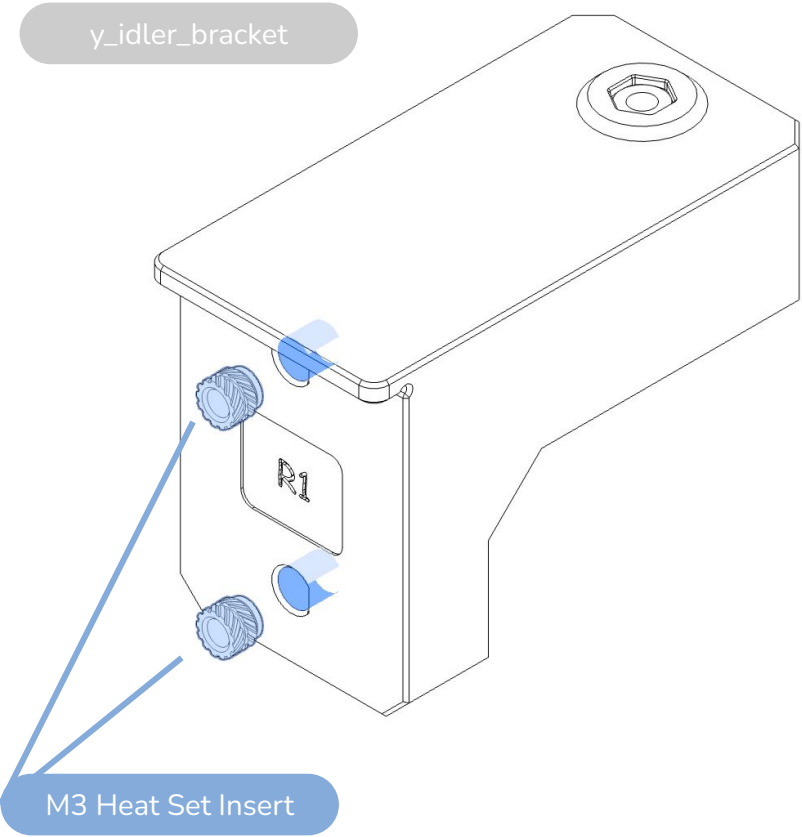
M3x10 SHCS



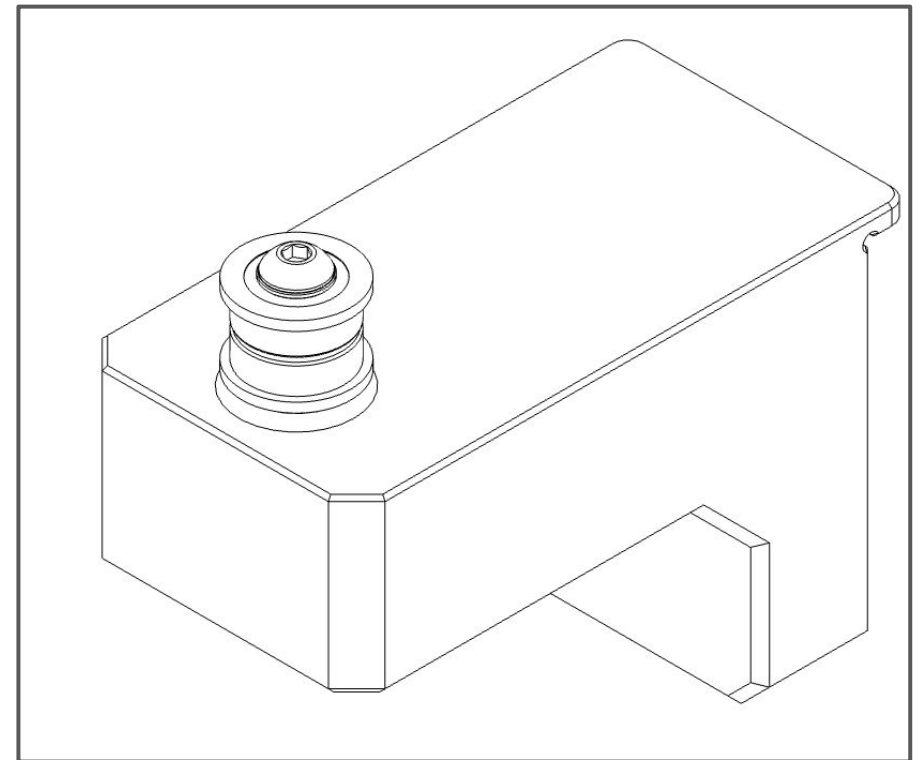
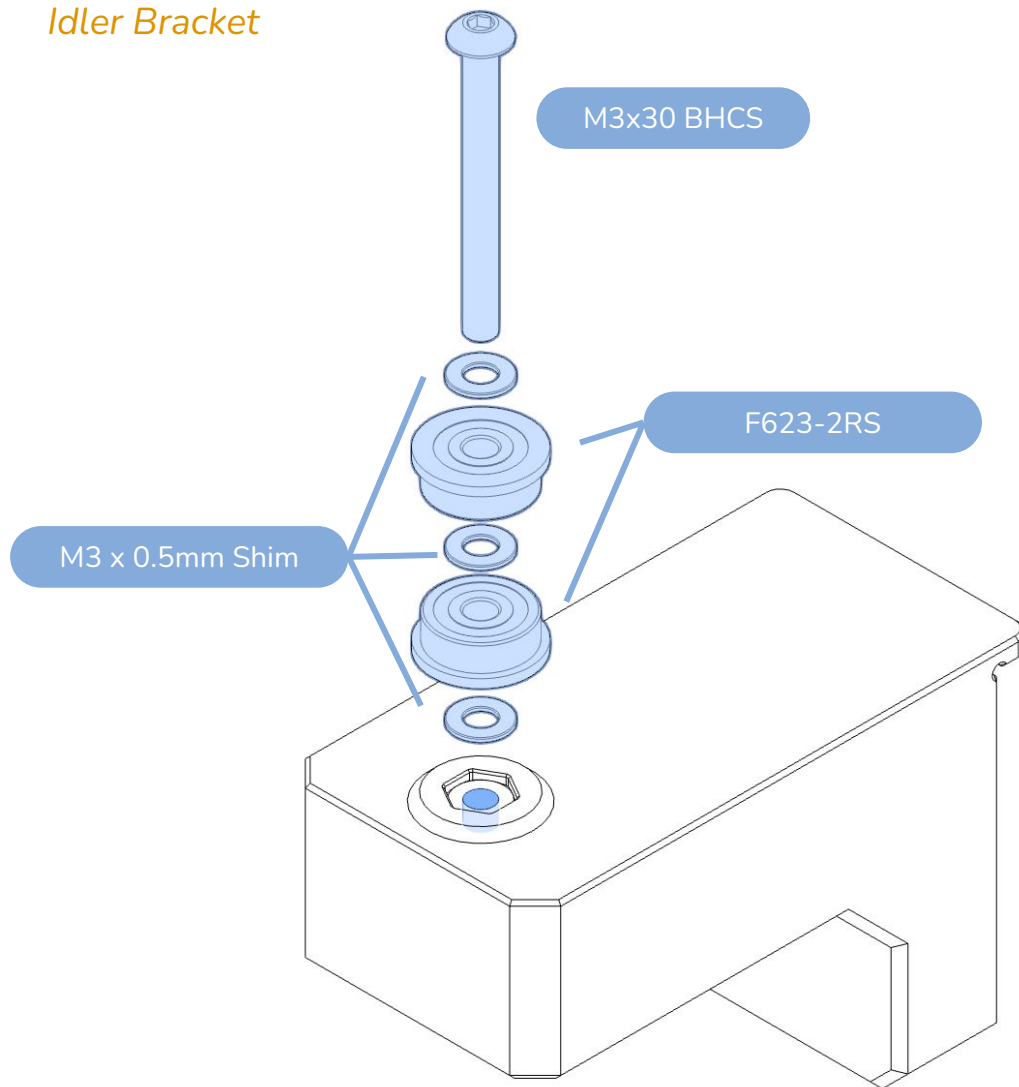
Y AXIS
Idler Bracket



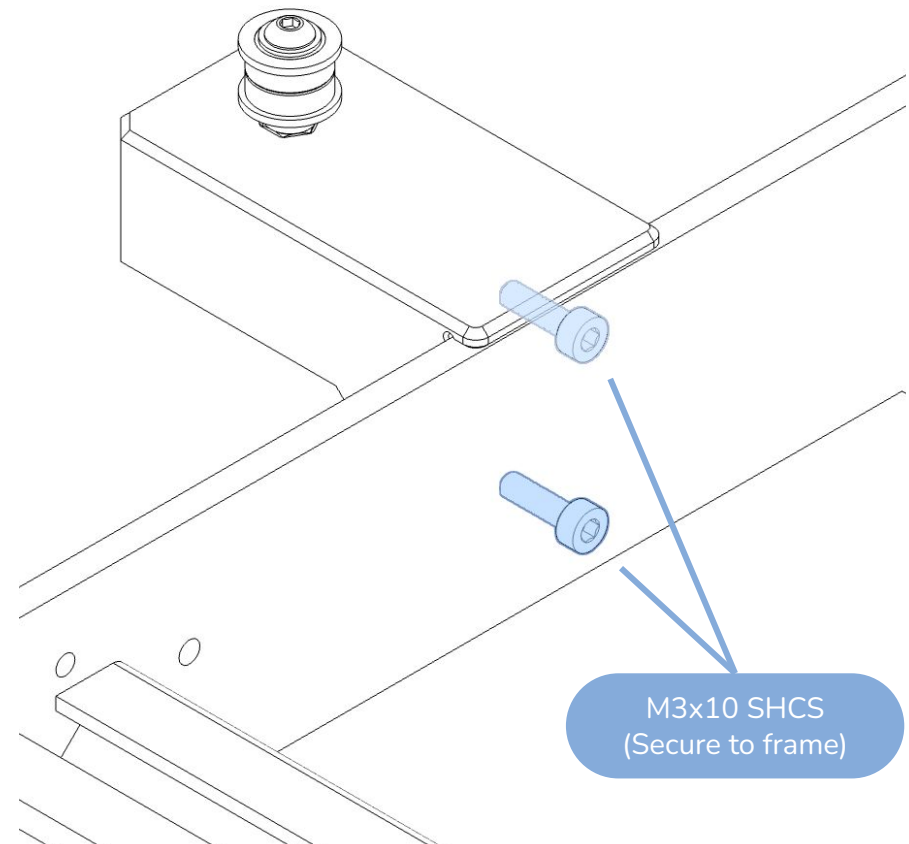
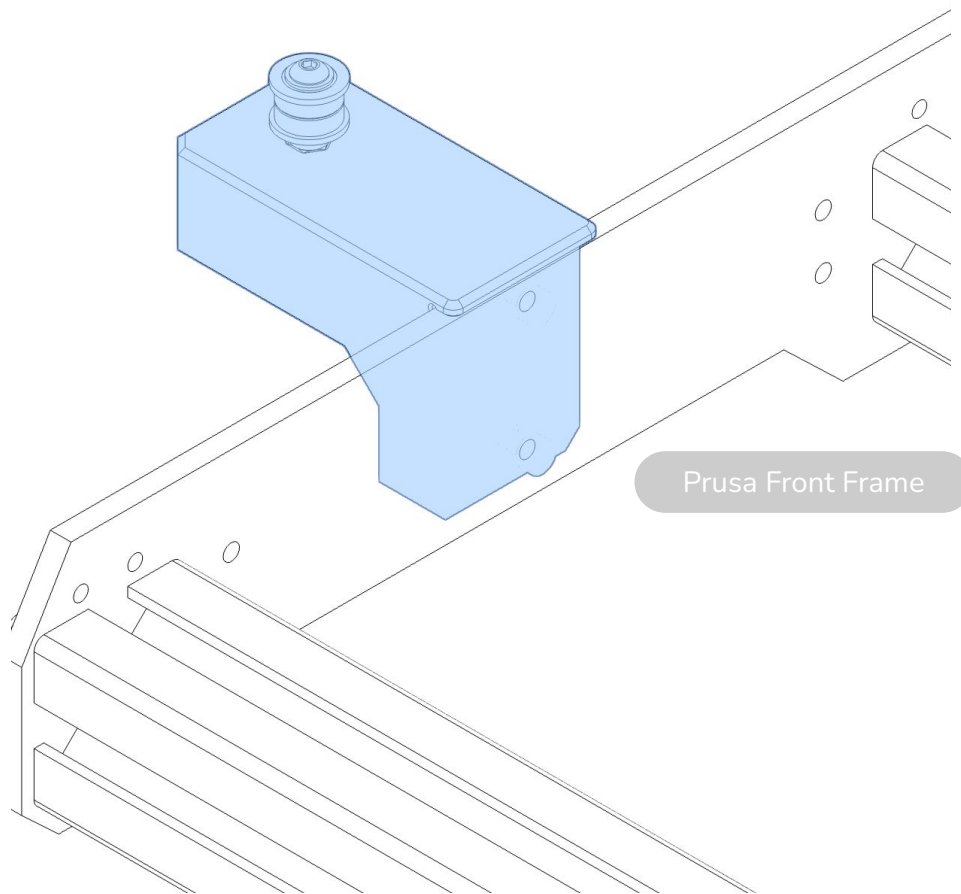
Y AXIS
Idler Bracket



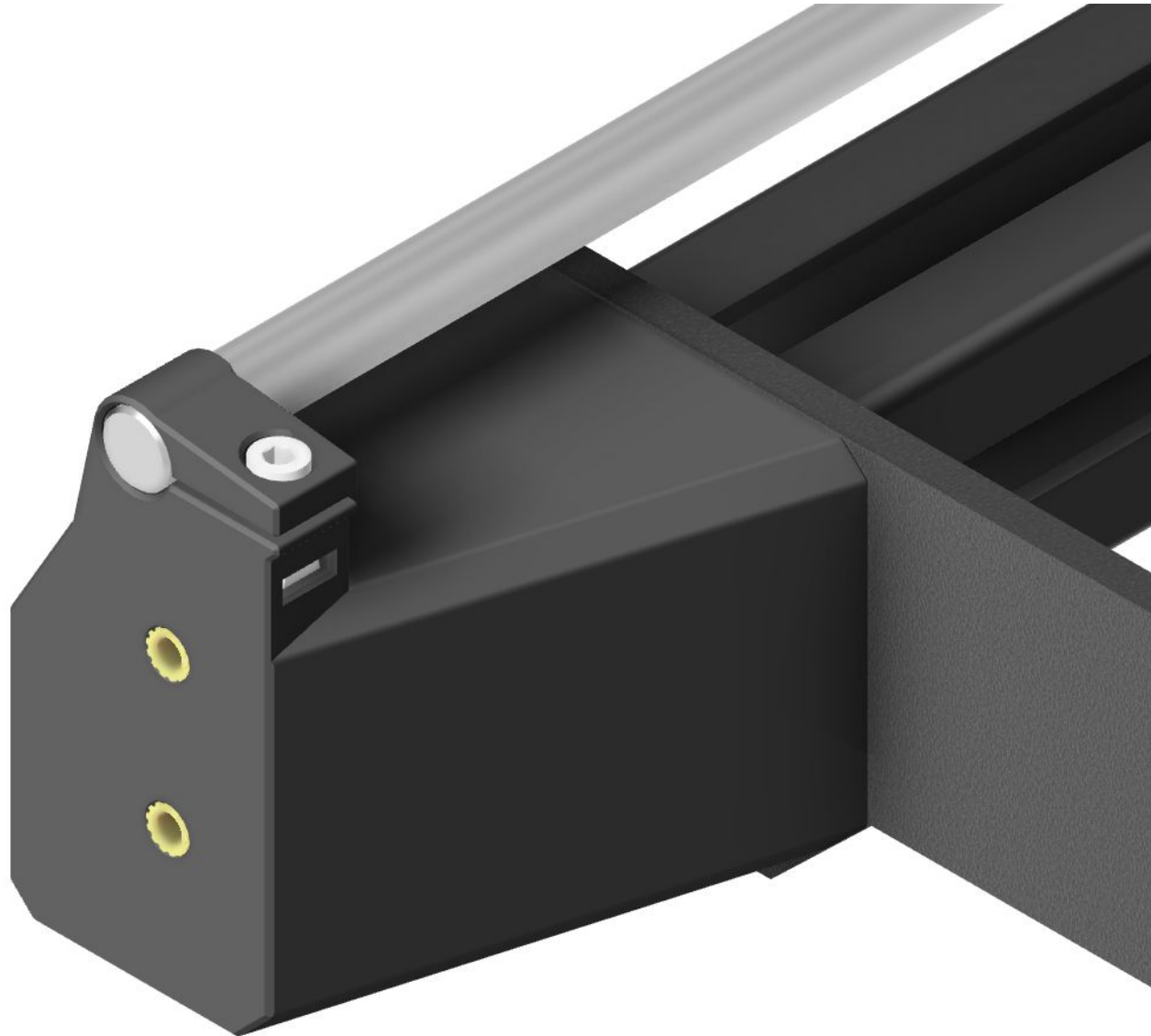
Y AXIS
Idler Bracket



Y AXIS Idler Bracket



Y AXIS
Y Rod Holder - Front Left



Y AXIS

Y Rod Holder - Front Left

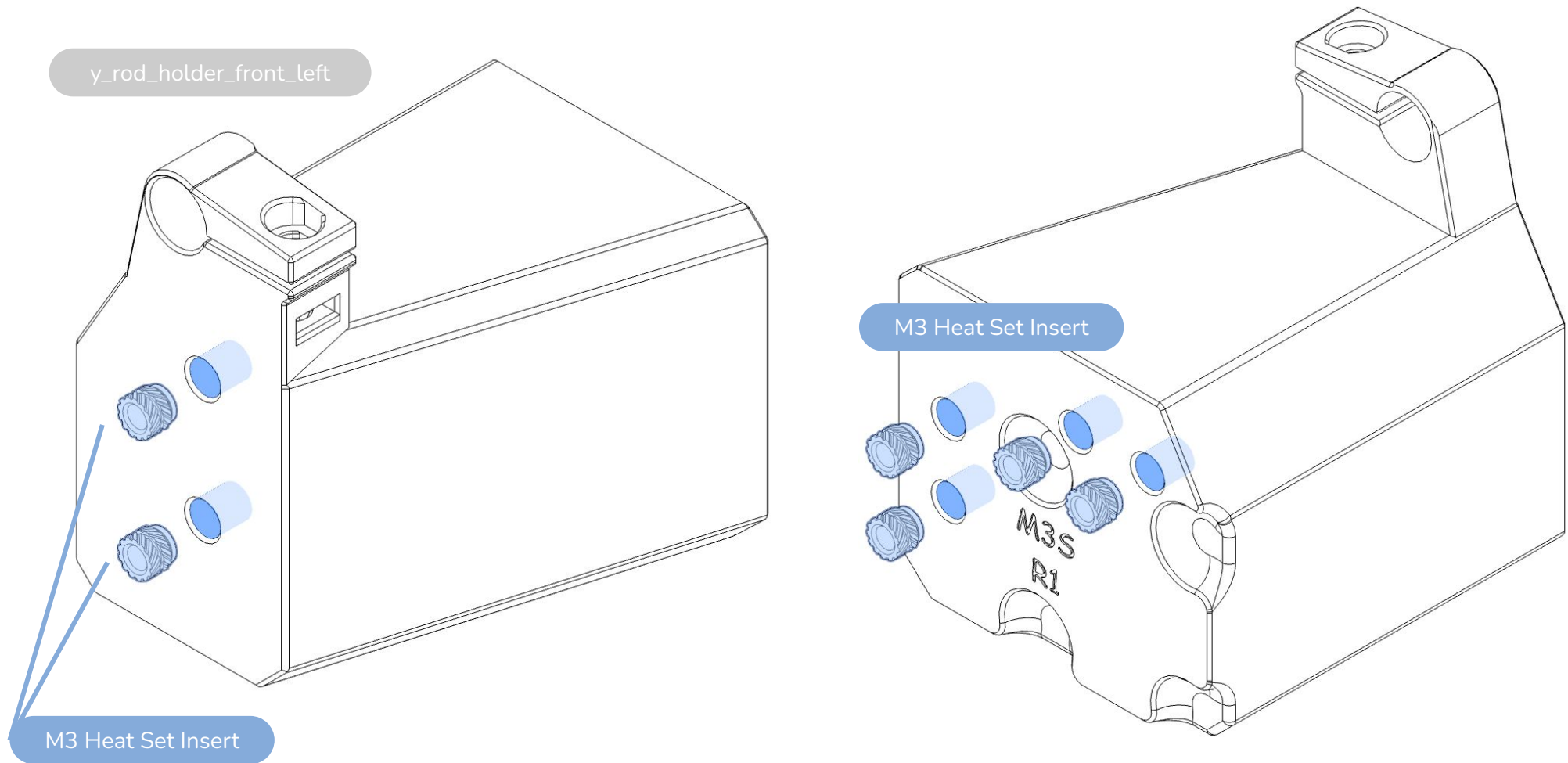
Y Rod Holders

Two variants of the front and rear Y Rod Holders are provided depending on which type of M3 Square Nut you are using.

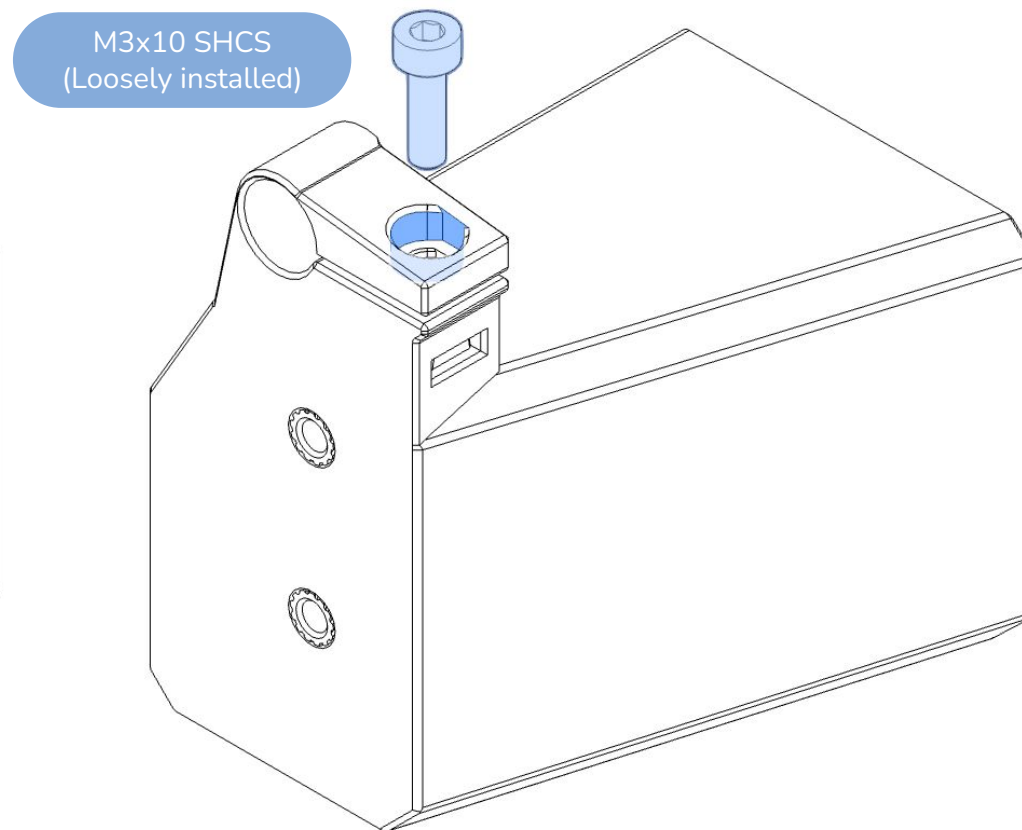
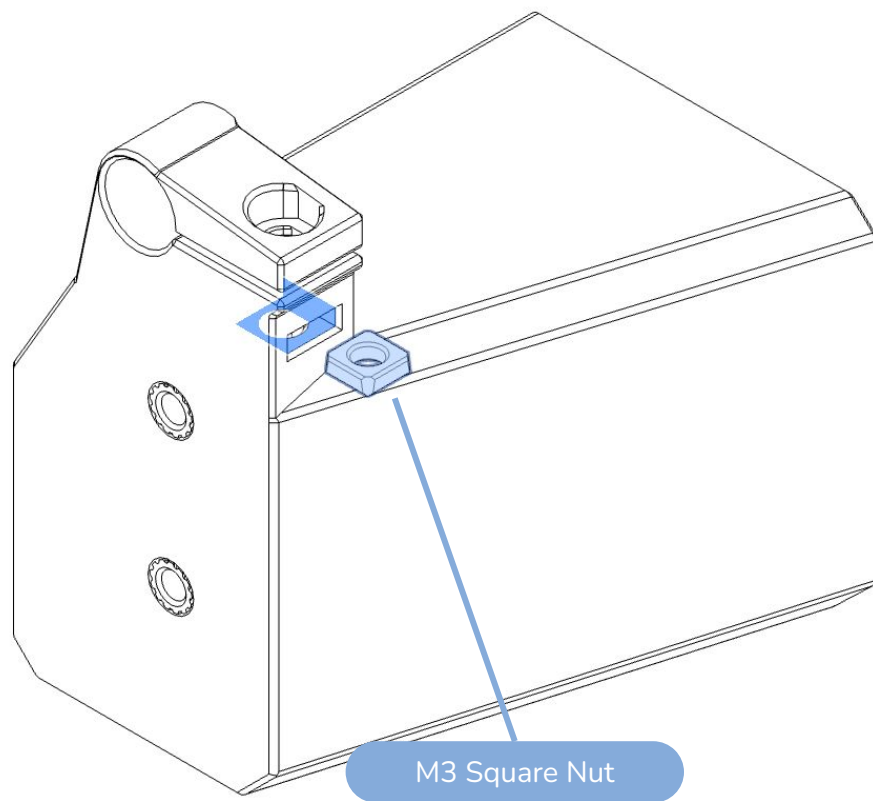
Prusa (slim)-style M3 Square Nuts: **M3S** // Voron V0-style square nuts: **V0S**

Example: `y_rod_holder_front_M3S.stl`

`y_rod_holder_rear_V0S.stl`

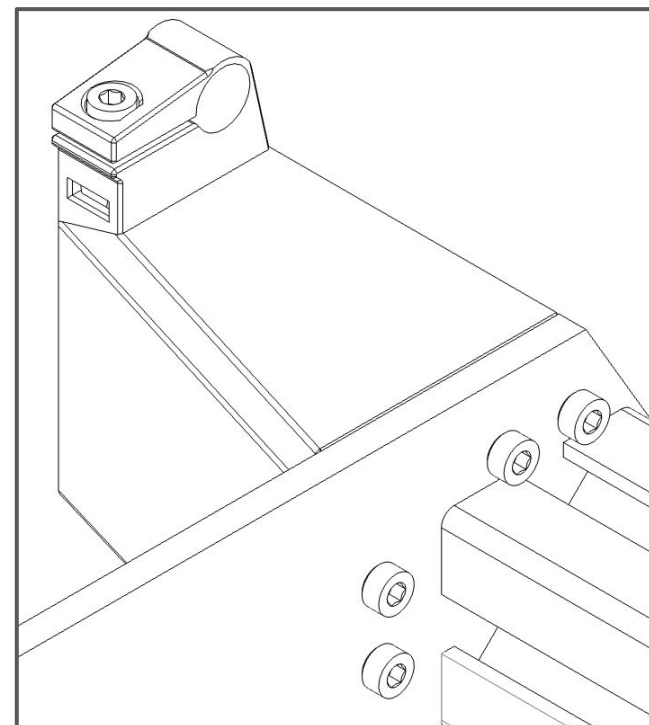
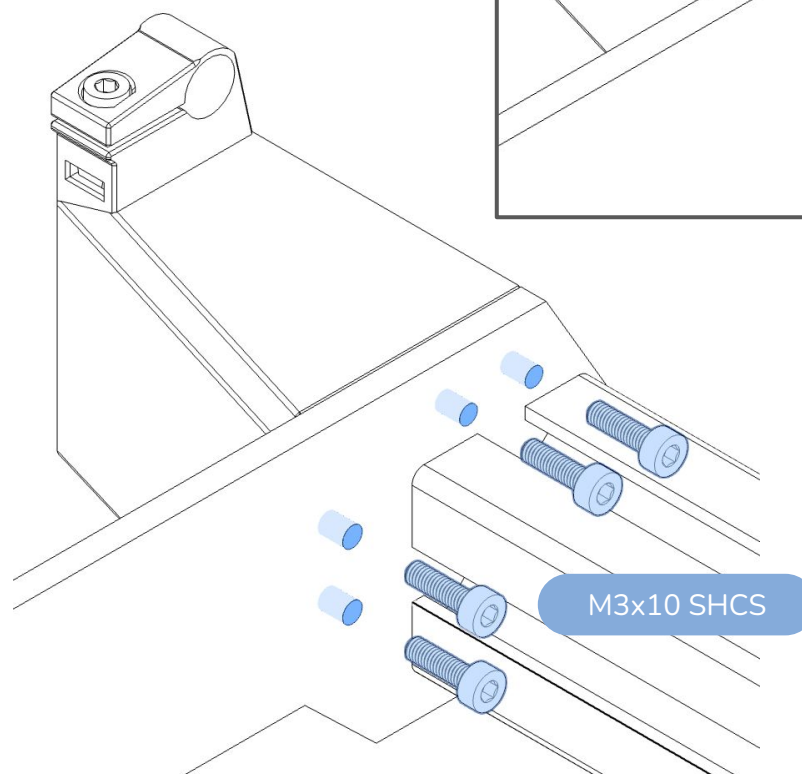
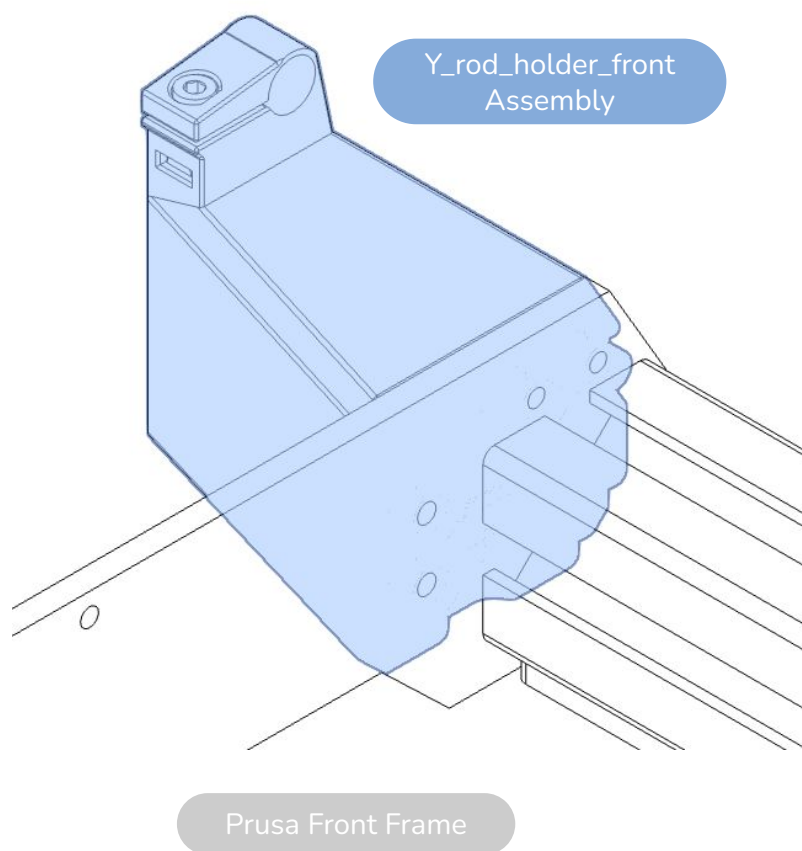


Y AXIS
Y Rod Holder - Front Left



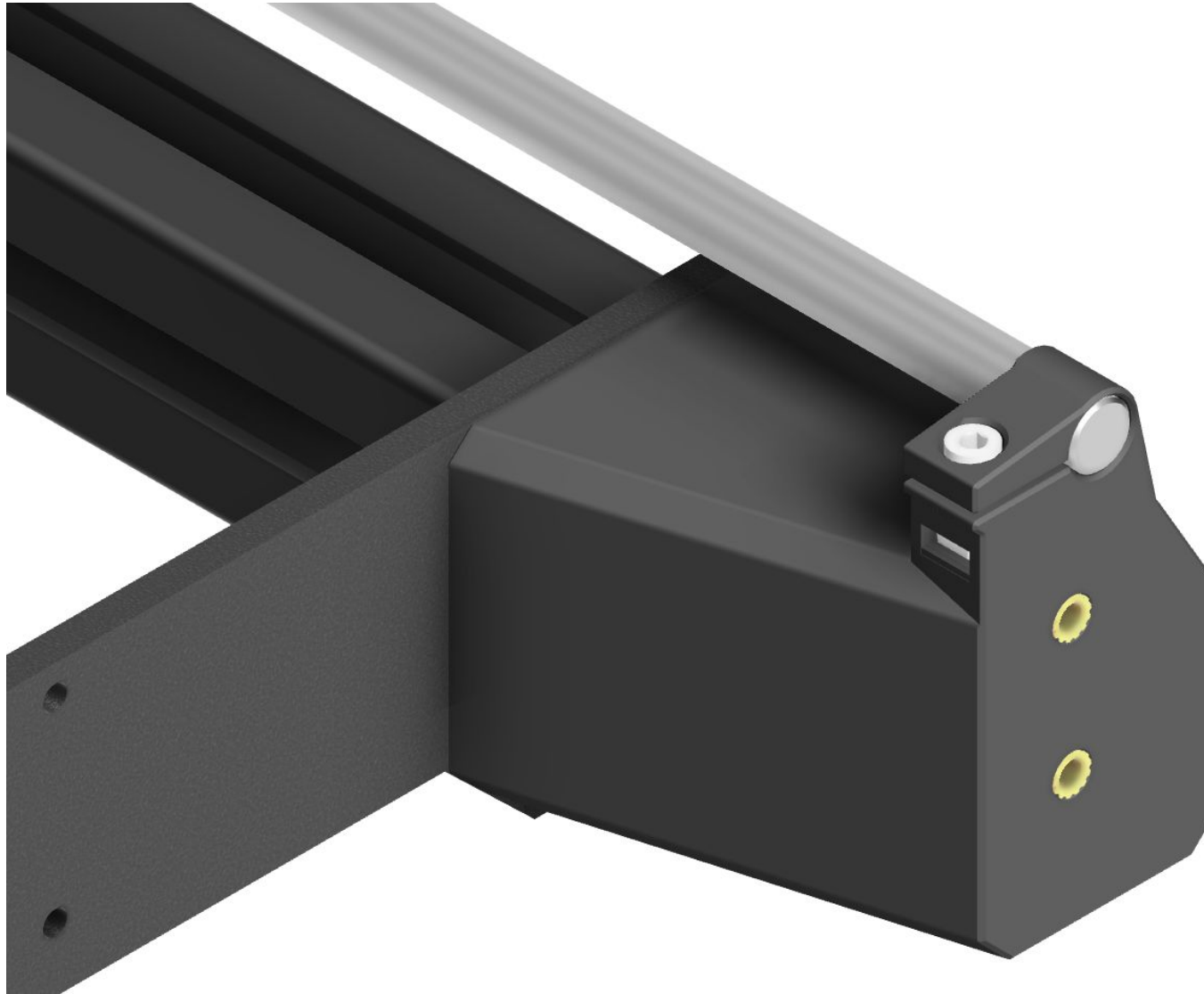
Y AXIS

Y Rod Holder - Front Left



Y AXIS

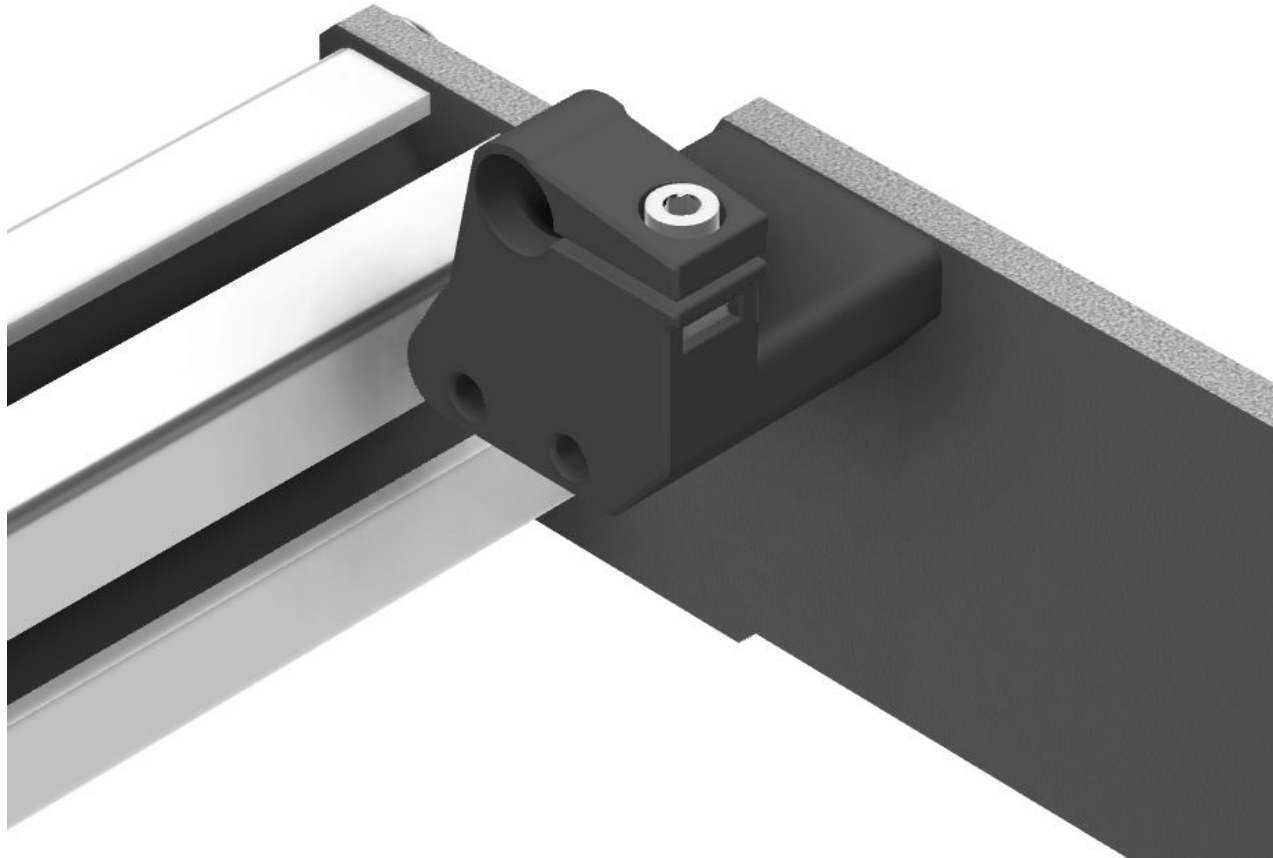
Y Rod Holder - Front Right



Y ROD HOLDER - FRONT RIGHT

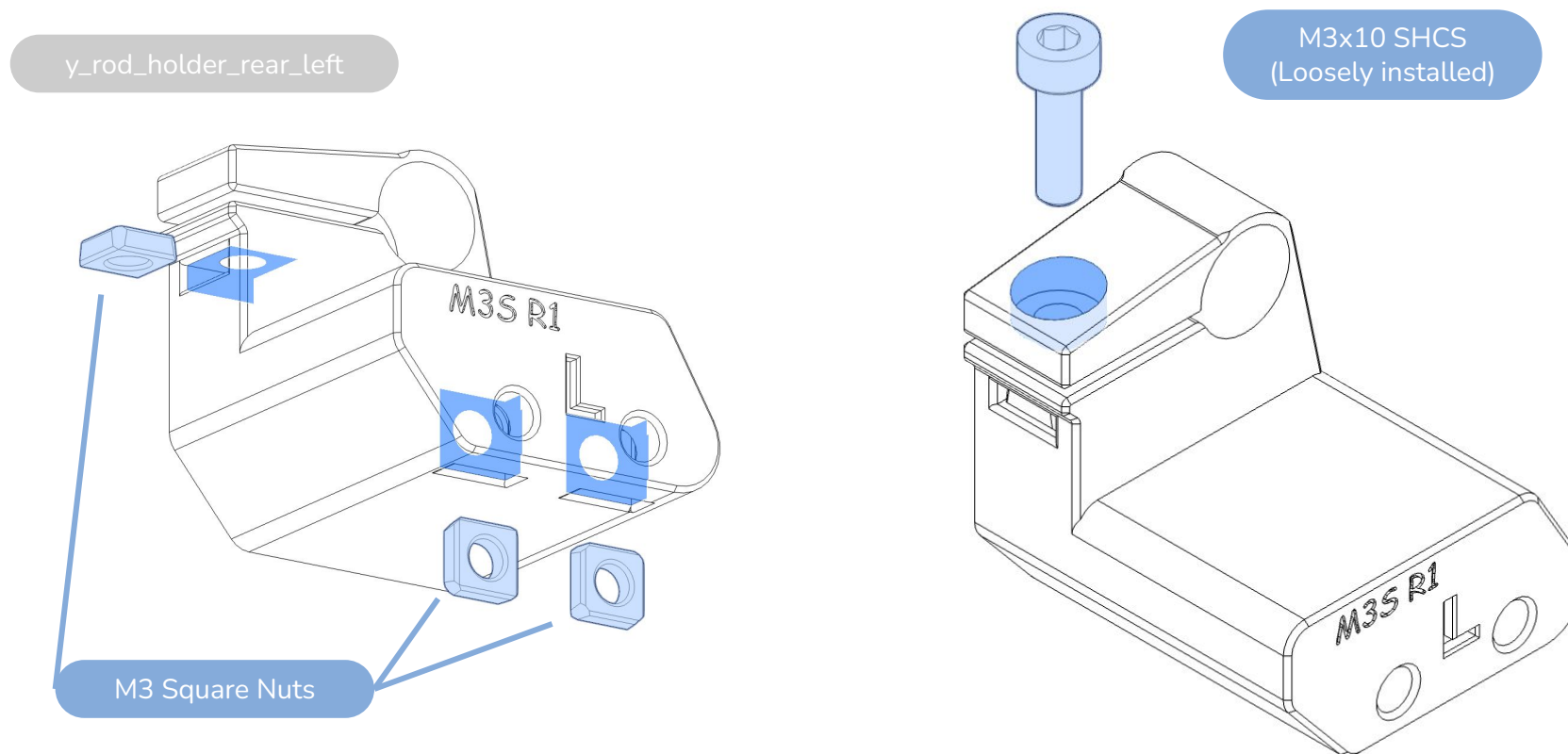
Repeat the steps just shown for **Y Rod Holder - Front Right**. The parts are mirrored, but otherwise identical.

Y AXIS
Y Rod Holder - Back Left



Y AXIS

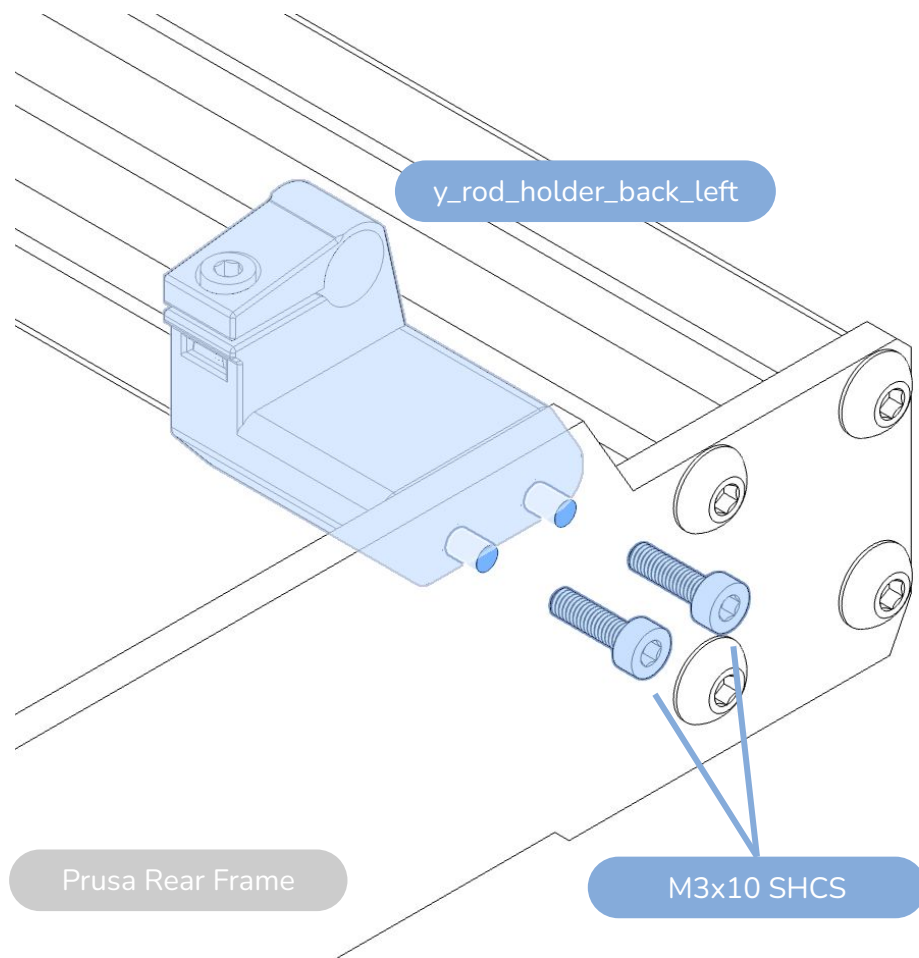
Y Rod Holder - Back Left



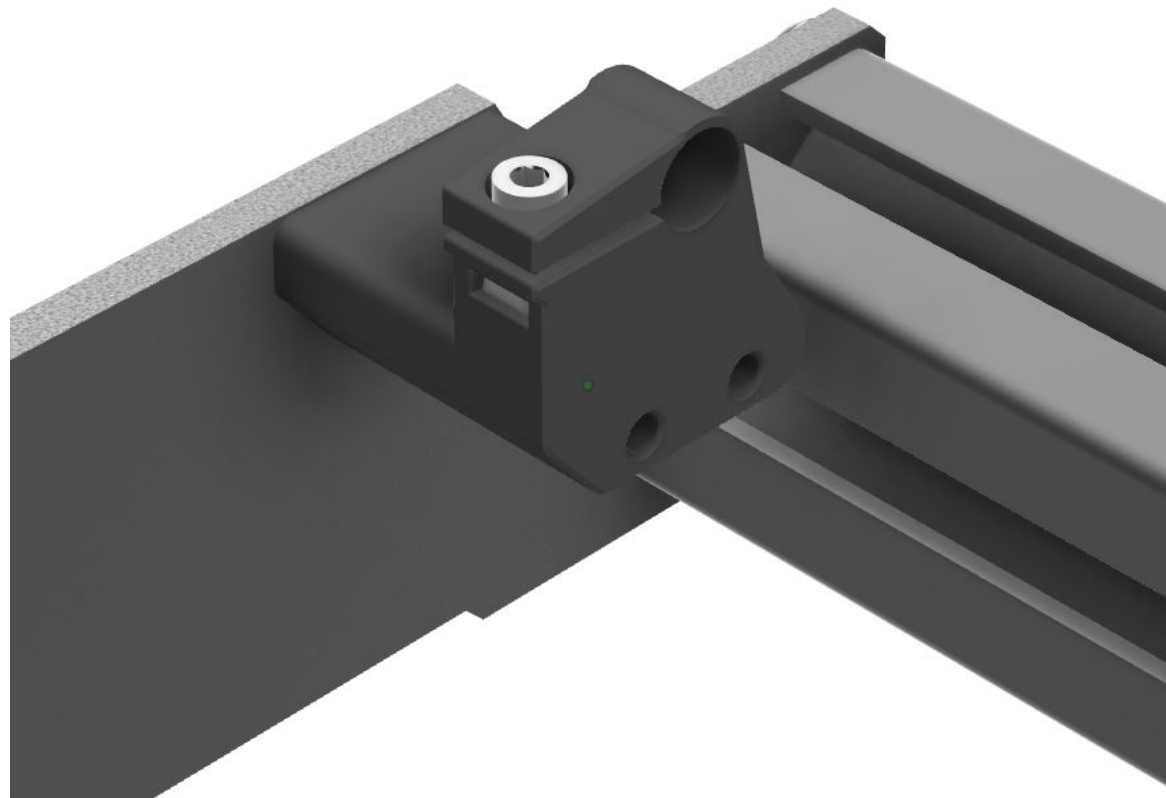
TIP

The two square nuts inserted through the bottom of the **y_rod_holder_rear** may be prone to falling out depending on part tolerances. A small piece of tape can be used to temporarily retain them until the next step!

Y AXIS
Y Rod Holder - Back Left



Y AXIS
Y Rod Holder - Back Right



Y ROD HOLDER - BACK RIGHT

Repeat the steps just shown for **Y Rod Holder - Back Right**. The parts are mirrored, but otherwise identical.

Y CARRIAGE ASSEMBLY

You likely already have an assembled Prusa MK3 or MK4 Y-Carriage.

The Prusawire Team recommends upgrading to the newer bearing clips (introduced with the MK3S+/MK4-style) in lieu of the original MK3 U-Bolts. Both are supported, however.

However, for reference, you can refer to either of the Prusa heatbed assembly instructions for further guidance:

Prusa MK3 Y-Carriage Assembly (with U-Bolts):

https://help.prusa3d.com/guide/2-y-axis-assembly_25488#26528

Reference Chapter 2, Steps 24 through 27.



Prusa MK3S+/MK4 Y-Carriage Assembly (with Bearing Clips):

https://help.prusa3d.com/guide/7-y-carriage-heatbed-assembly_707785#718723

Reference Chapter 7, Steps 12 through 11.

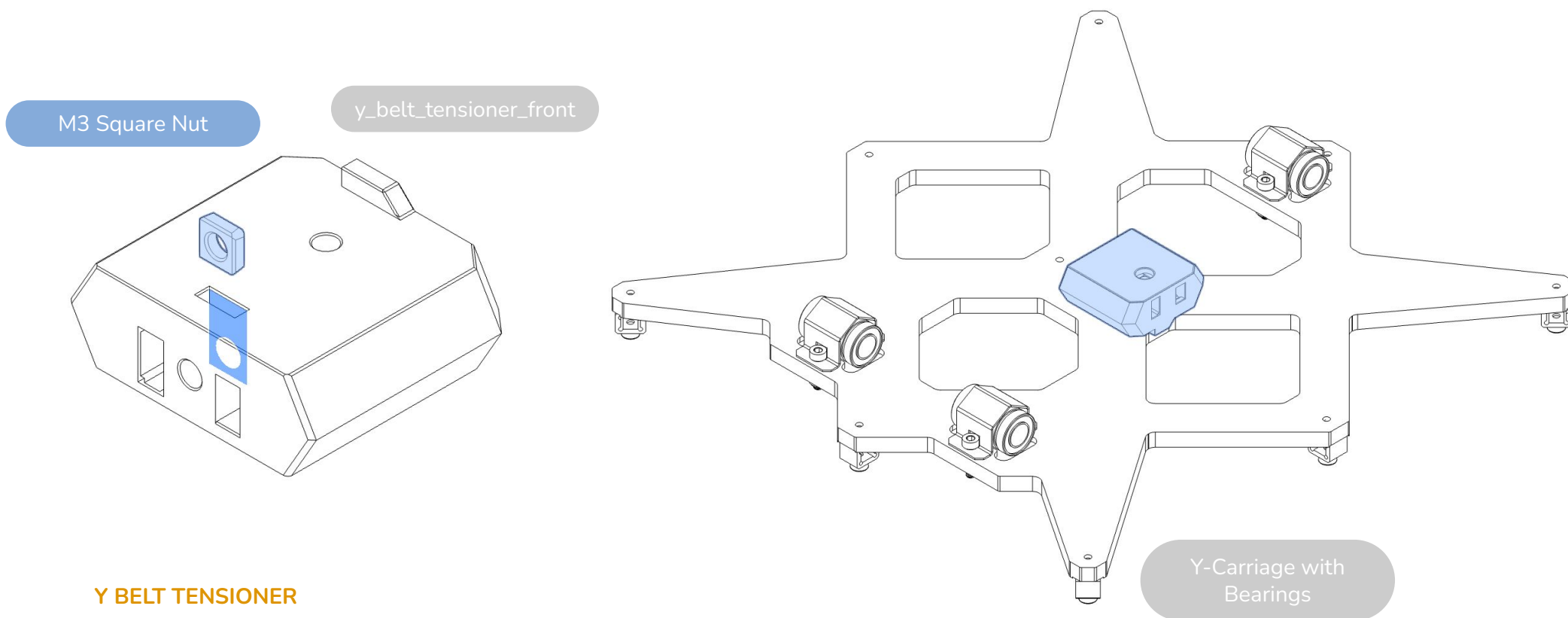


Y AXIS
Y Tensioner



Y AXIS

Y Tensioner



Y BELT TENSIONER

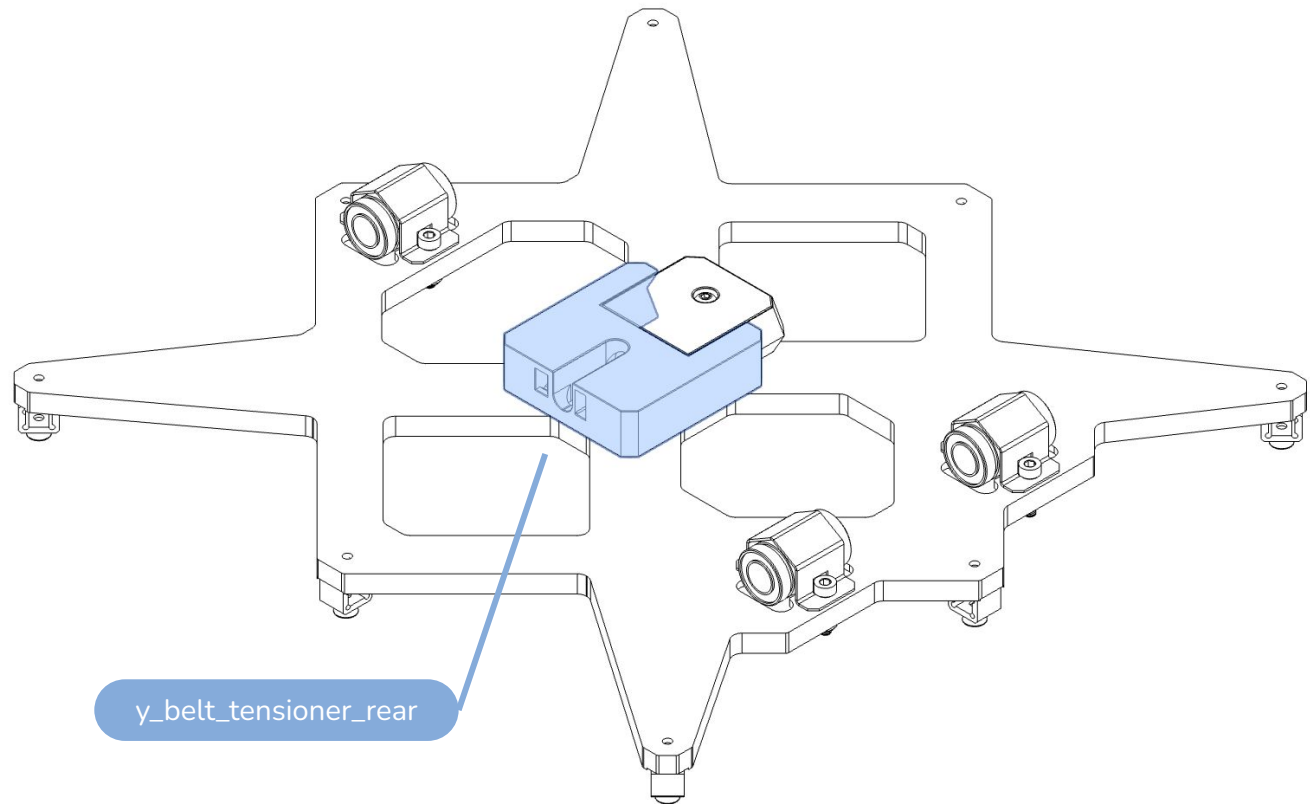
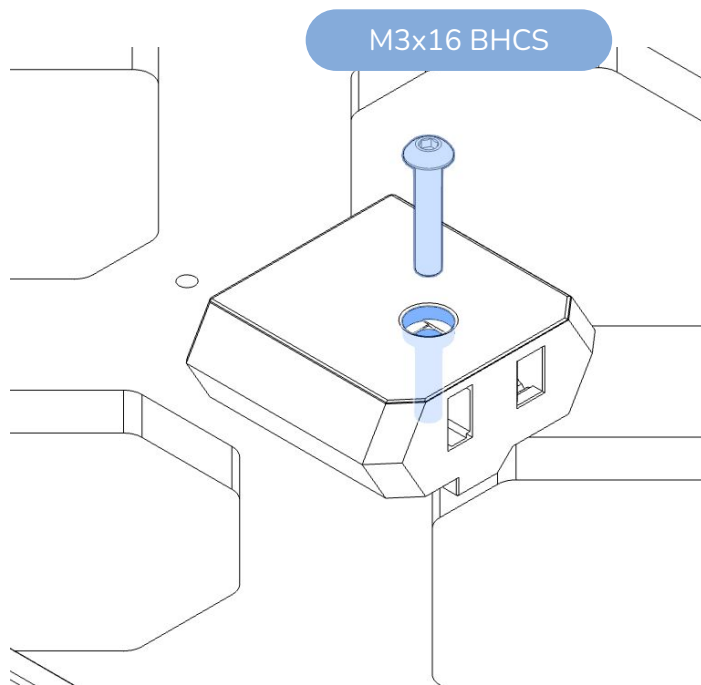
Multiple variants of the **y_belt_tensioner** are provided depending on which type of M3 Square Nut, Bearing Clip, and Motor Pulley you are using.

MK4-style Bearing Clips: Bearing_Clips	//	MK3-style U-Bolts: U_Bolts
Prusa (slim)-style M3 Square Nuts: M3S	//	Voron V0-style square nuts: V0S
Prusa-spec 16T Gears: 16T	//	Switchwire-spec 20T Gears: 20T

Example: **y_belt_tensioner_Bearing_Clips_M3S_16T.stl**

Y AXIS

Y Tensioner

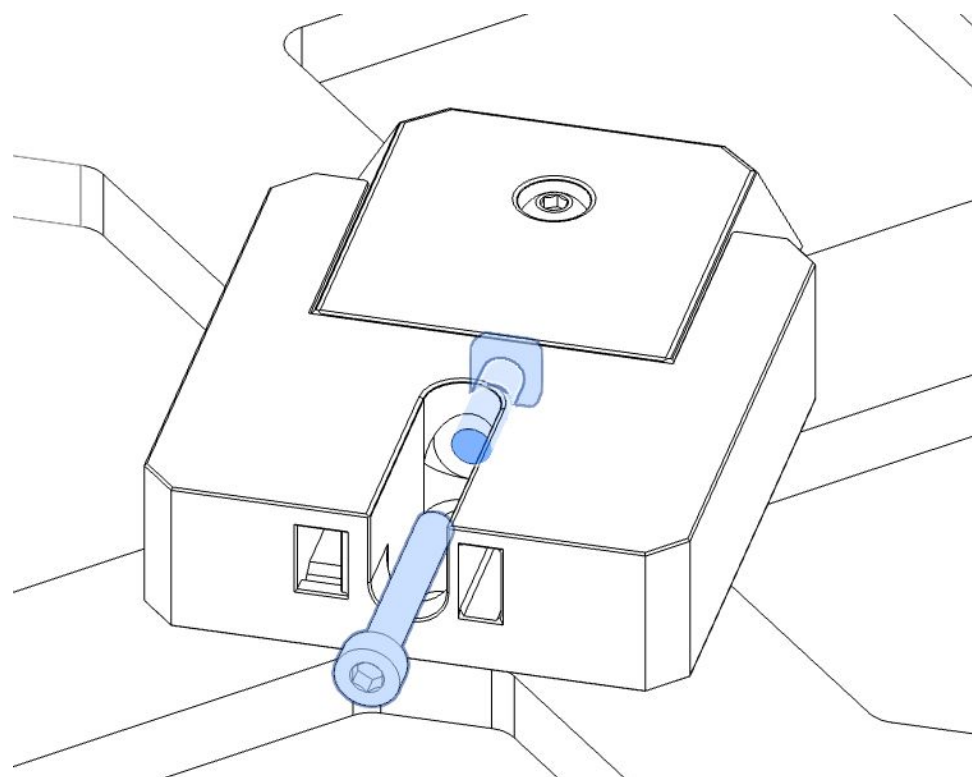


Y AXIS

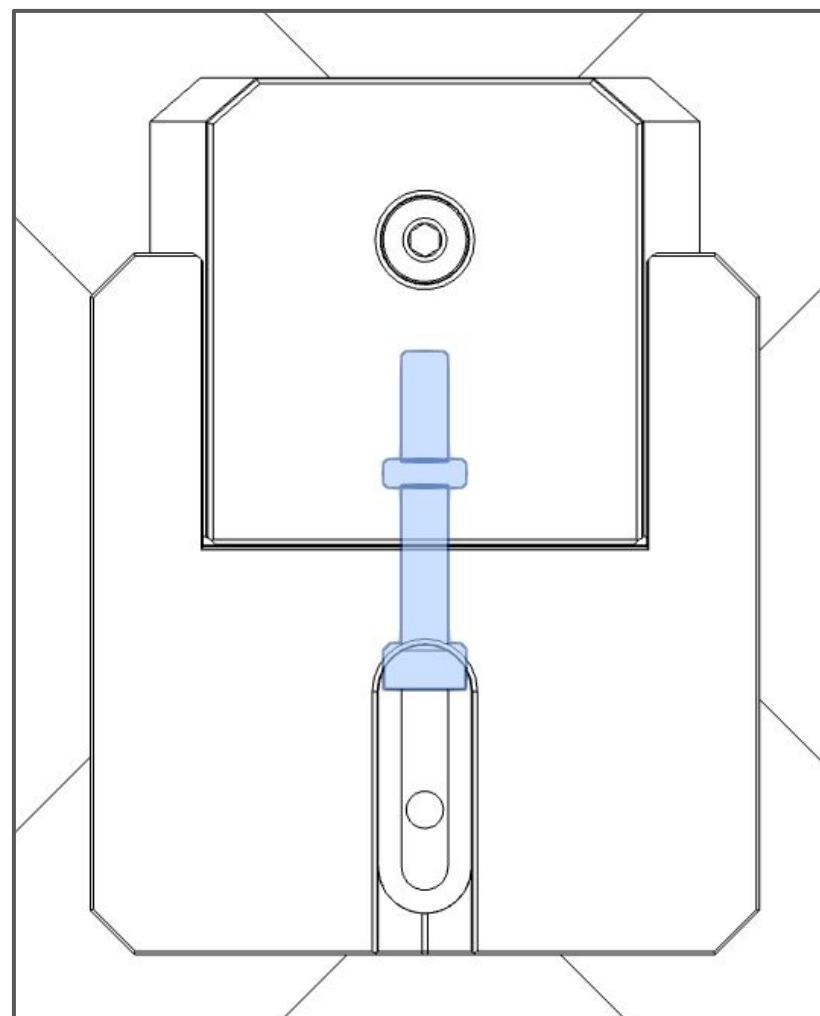
Y Tensioner

TIP

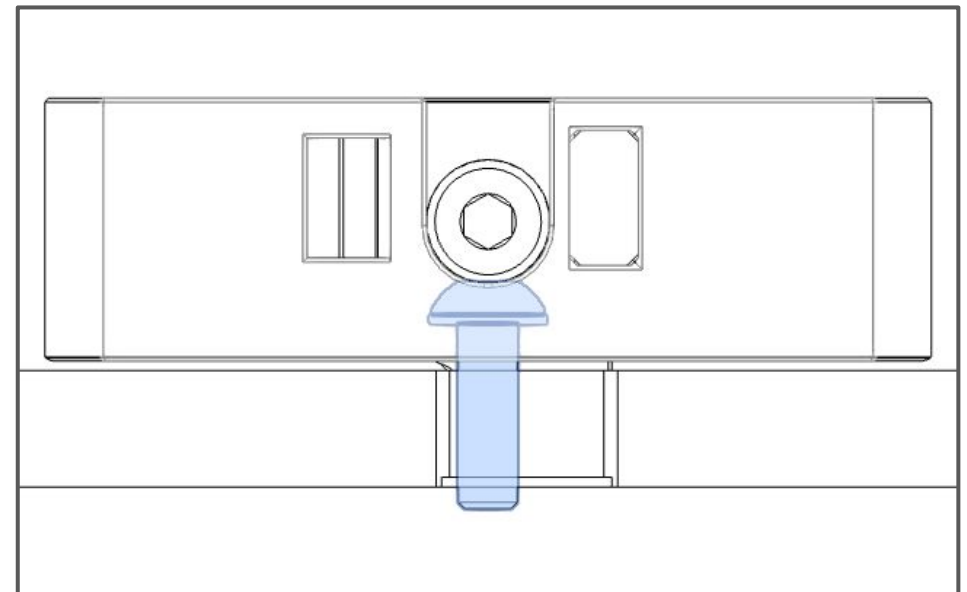
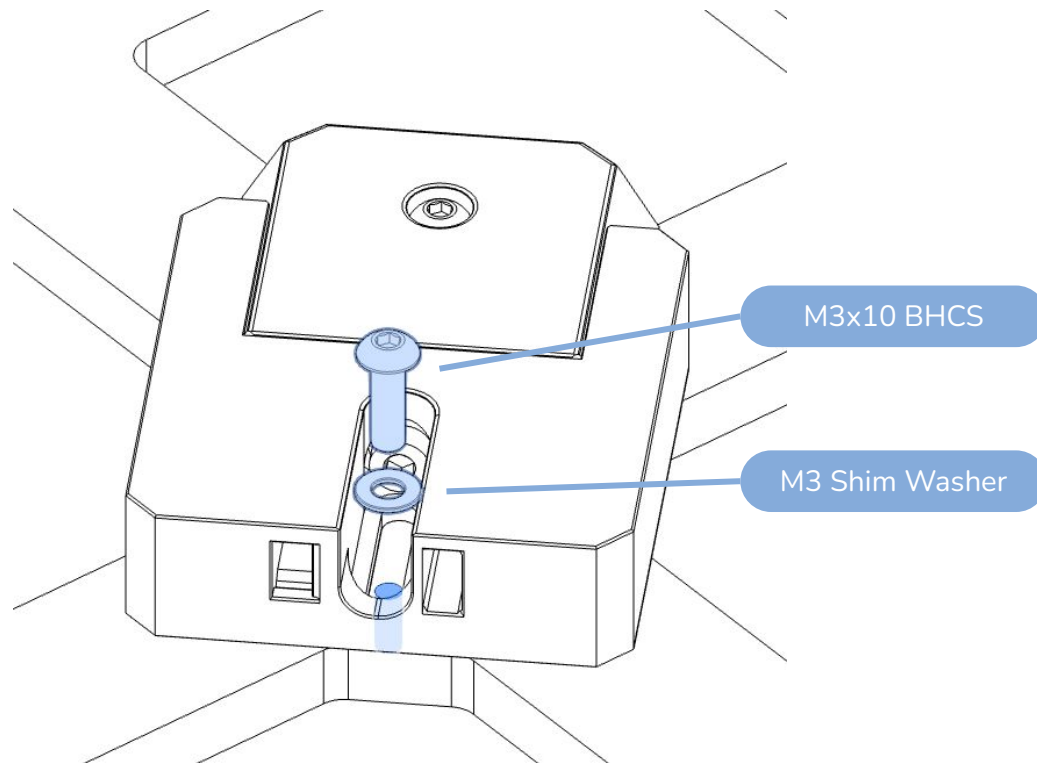
Make sure not to fully tighten the M3x20 SHCS and keep the two tensioner halves somewhere spread apart. This is how you tension the Y-Axis Belt later!



M3x20 SHCS
(Install Loosely)



Y AXIS
Y Tensioner

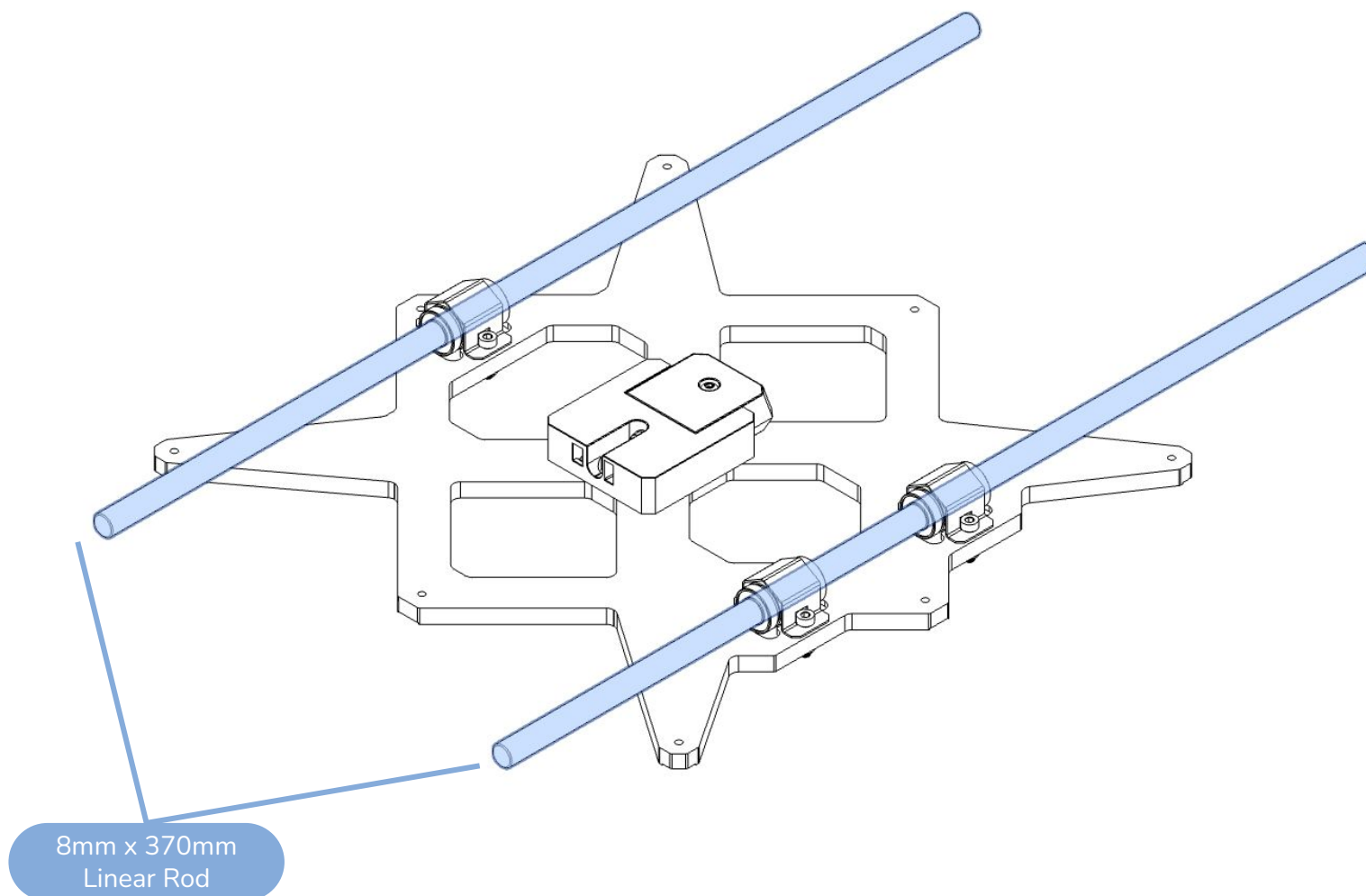


Y AXIS

Y Linear Rods

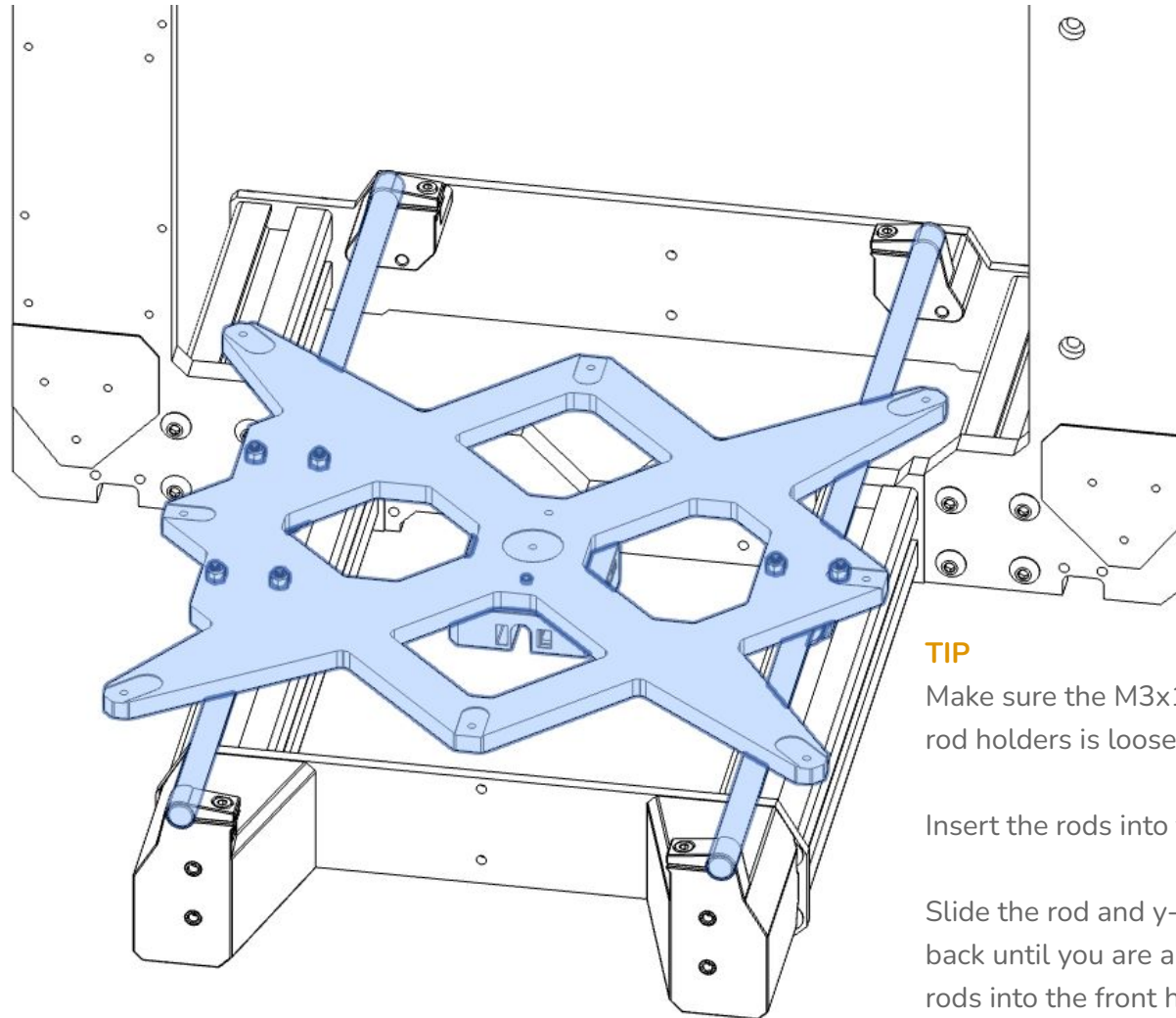
TIP

Make sure the linear rods are clean and debris free before carefully inserting into the bearings.



Y AXIS

Y-Carriage and Rod Assembly



TIP

Make sure the M3x10 SHCS for each of the four rod holders is loose before inserting.

Insert the rods into the rear holders first.

Slide the rod and y-carriage assembly toward the back until you are able to insert the front of the rods into the front holders.

Slide the rods forward until flush with the front of the front rod holders.

Tighten the M3x10 SHCS until snug.

Y-BELT INSTALLATION AND TENSIONING

The Y-Belt must be installed and tensioned properly to ensure smooth and accurate motion.

Extra belt length has been specified in the BOM; it is expected to have some leftover belt afterwards.

DO NOT cut your belt until you certain you have done it correctly. It is okay to leave a little extra length hanging out of the tensioner until final assembly and validation.

The complete procedure can be found on the Prusawire Documentation Site:

<https://prusawire.positron3d.com/hardware/y-belt.html>



HEATBED ASSEMBLY AND MOUNTING

You likely already have an assembled Prusa MK52 heatbed (or other clone bed such as LDO Switchwire) if you're attempting this build.

The MK3 style round bed spacers are fine to use, however, the Prusawire Team strongly recommends the upgrade to the MK4 style bed expansion joints. They are both easier to install/maintain as well as a better design.

However, for reference, you can refer to either of the Prusa heatbed assembly instructions for further guidance:

Prusa MK3 Heatbed Assembly:

https://help.prusa3d.com/guide/7-heatbed-psu-assembly_172872#172915

Reference Chapter 7, Steps 1 through 11 for Heatbed Assembly.

Reference Chapter 7, Steps 12 through 16 for Heatbed Mounting to Y-Carriage.



Prusa MK4 Heatbed Assembly:

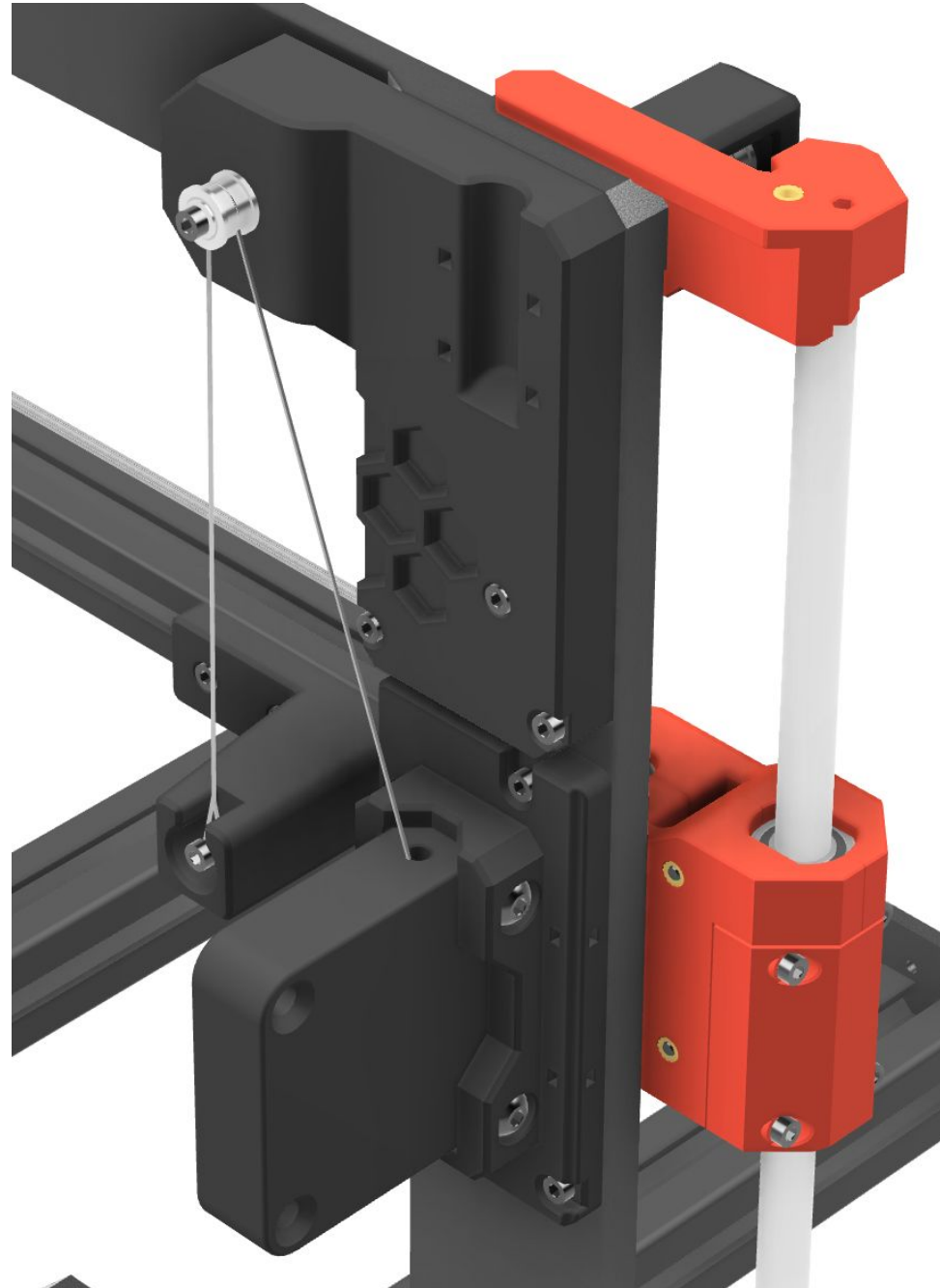
https://help.prusa3d.com/guide/7-y-carriage-heatbed-assembly_707785#717626

Reference Chapter 7, Steps 2 through 11 for Heatbed Assembly.

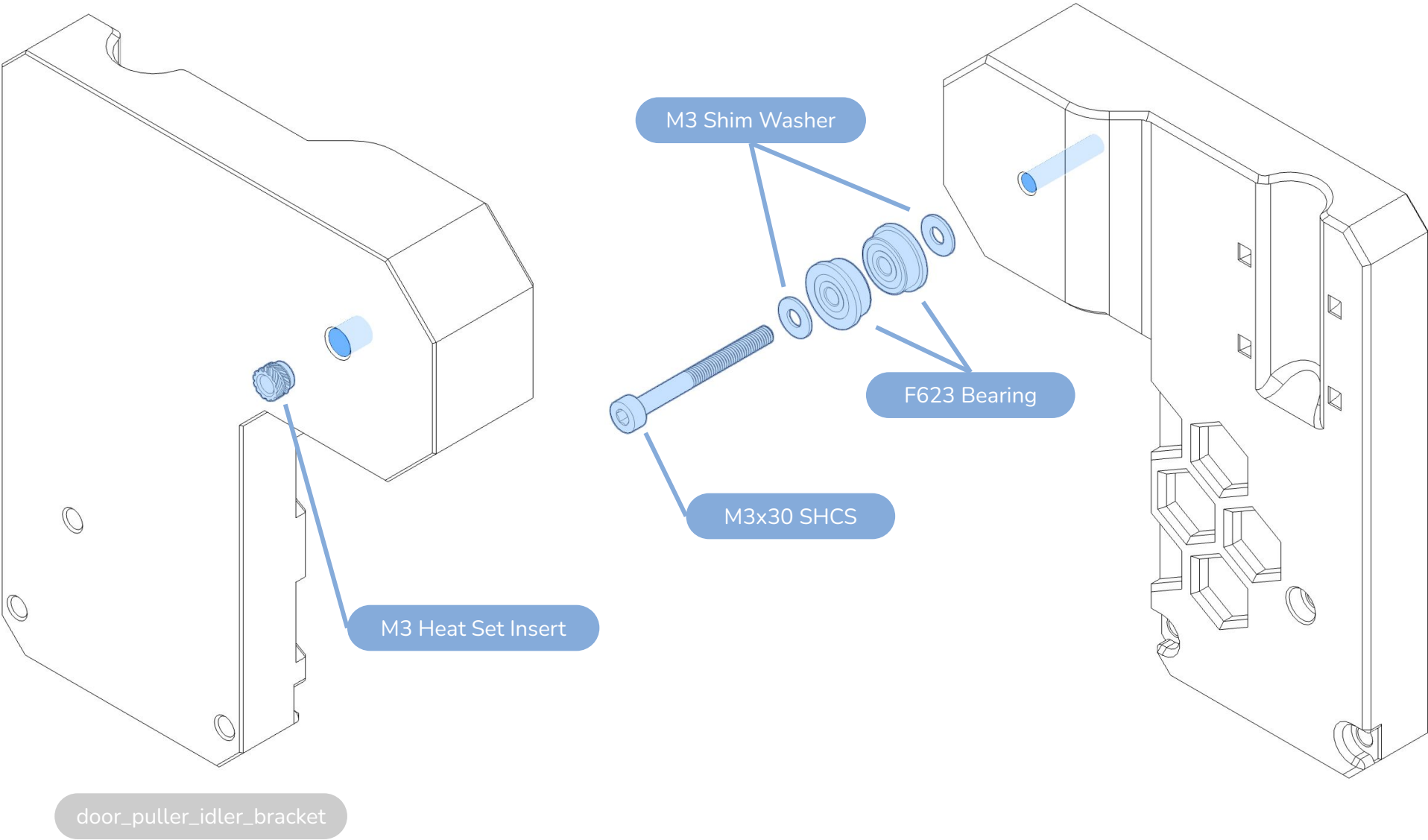
Reference Chapter 7, Steps 33 through 38 for Heatbed Mounting to Y-Carriage.



DOOR PULLER

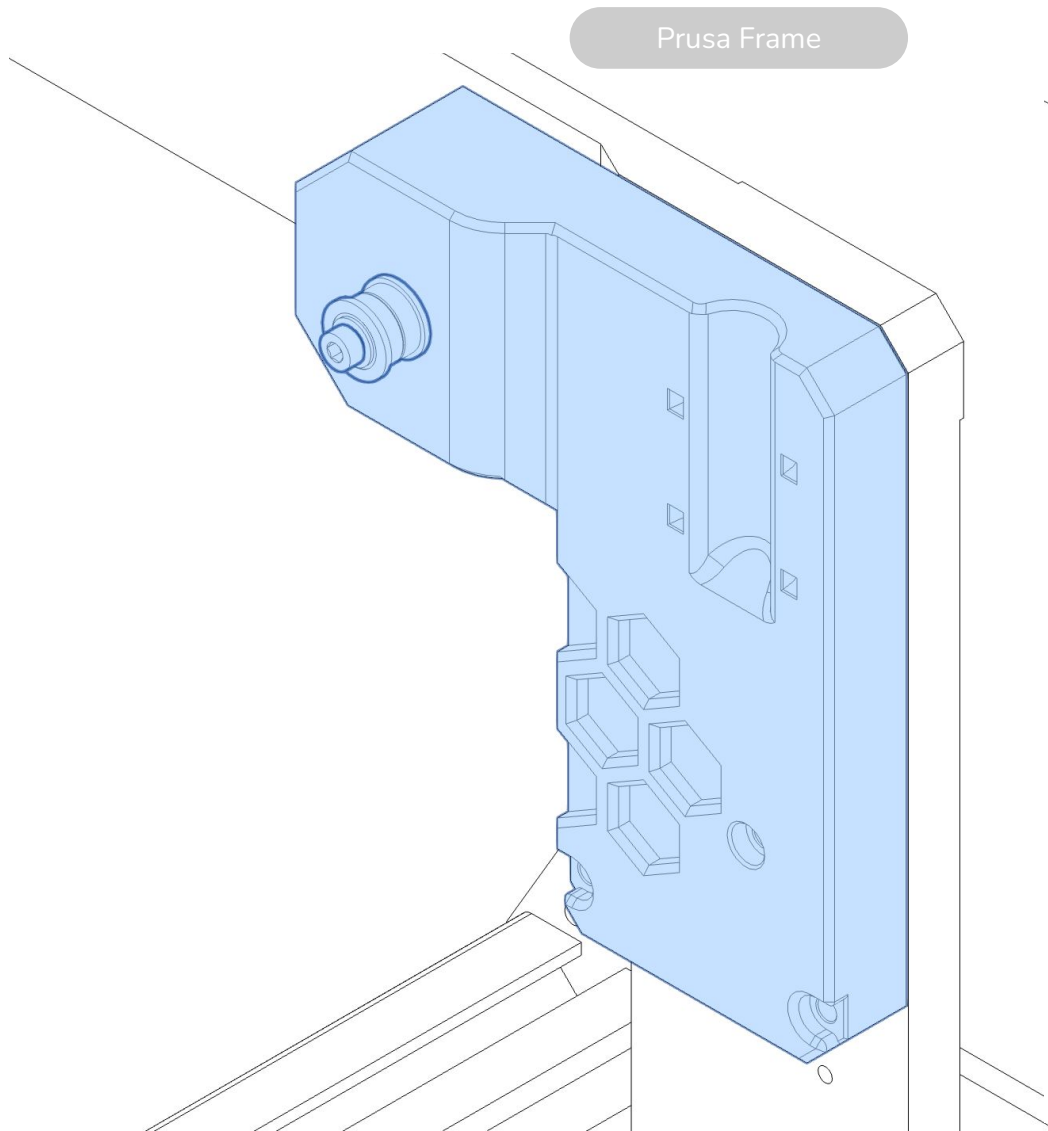


DOOR PULLER
Idler



DOOR PULLER

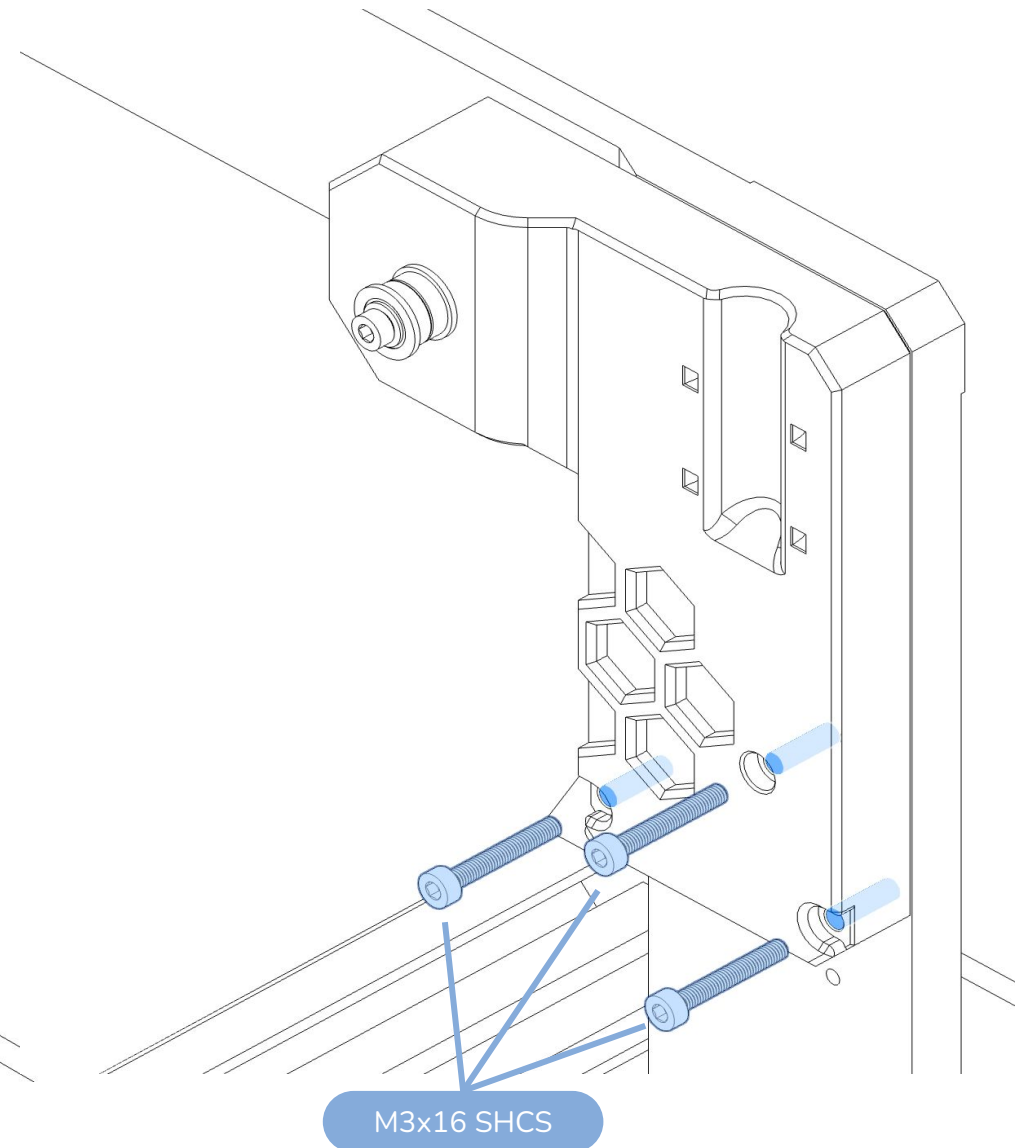
Idler



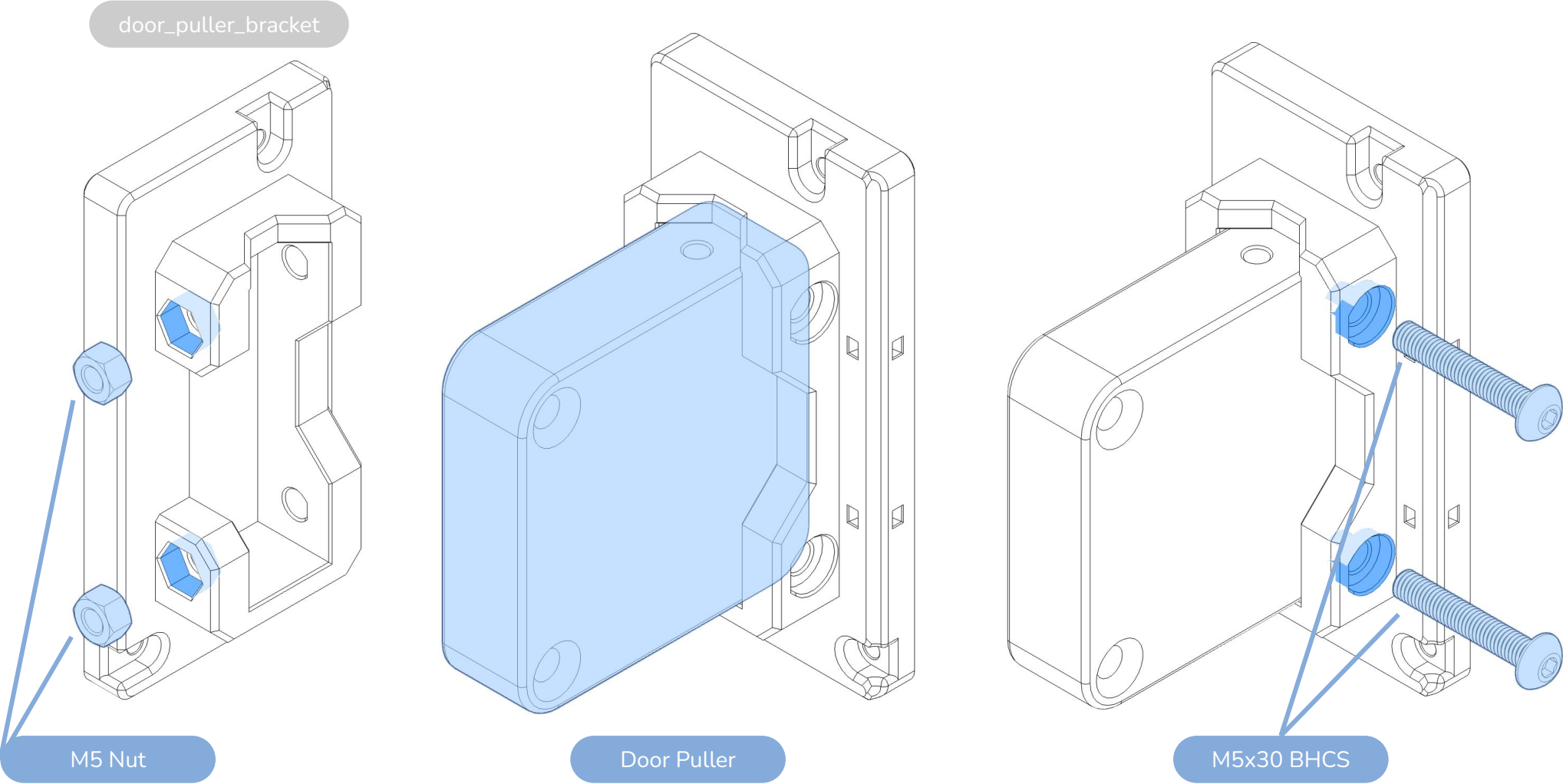
WIRING TIP

The recess in **door_puller_idler_bracket** is designed to allow you to zip-tie your toolhead cabling. This functions as a strain relief and helps route the toolhead cabling down to the control board.

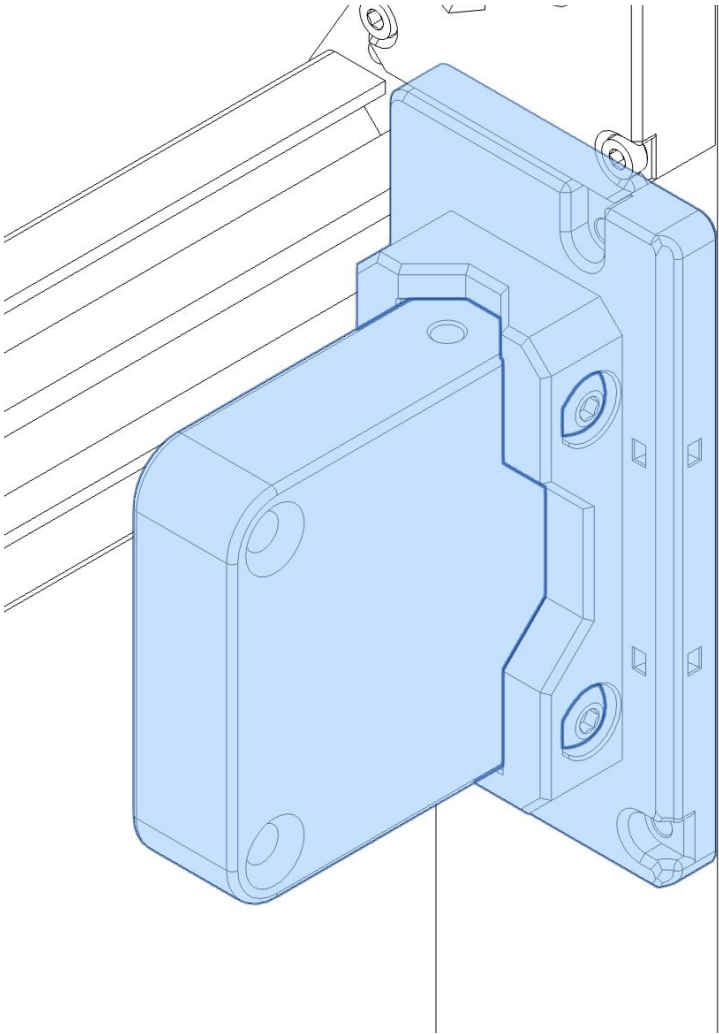
There are zip-tie points on the **door_puller_bracket** as well.



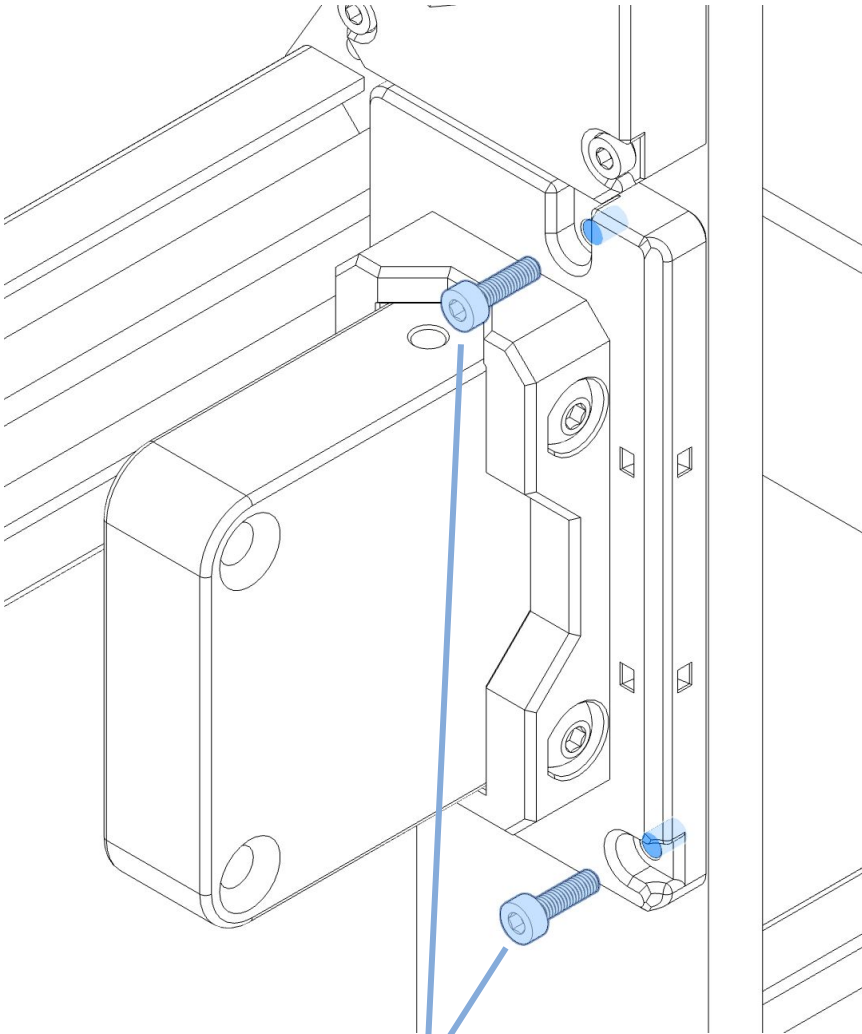
DOOR PULLER
Mount



DOOR PULLER
Mount

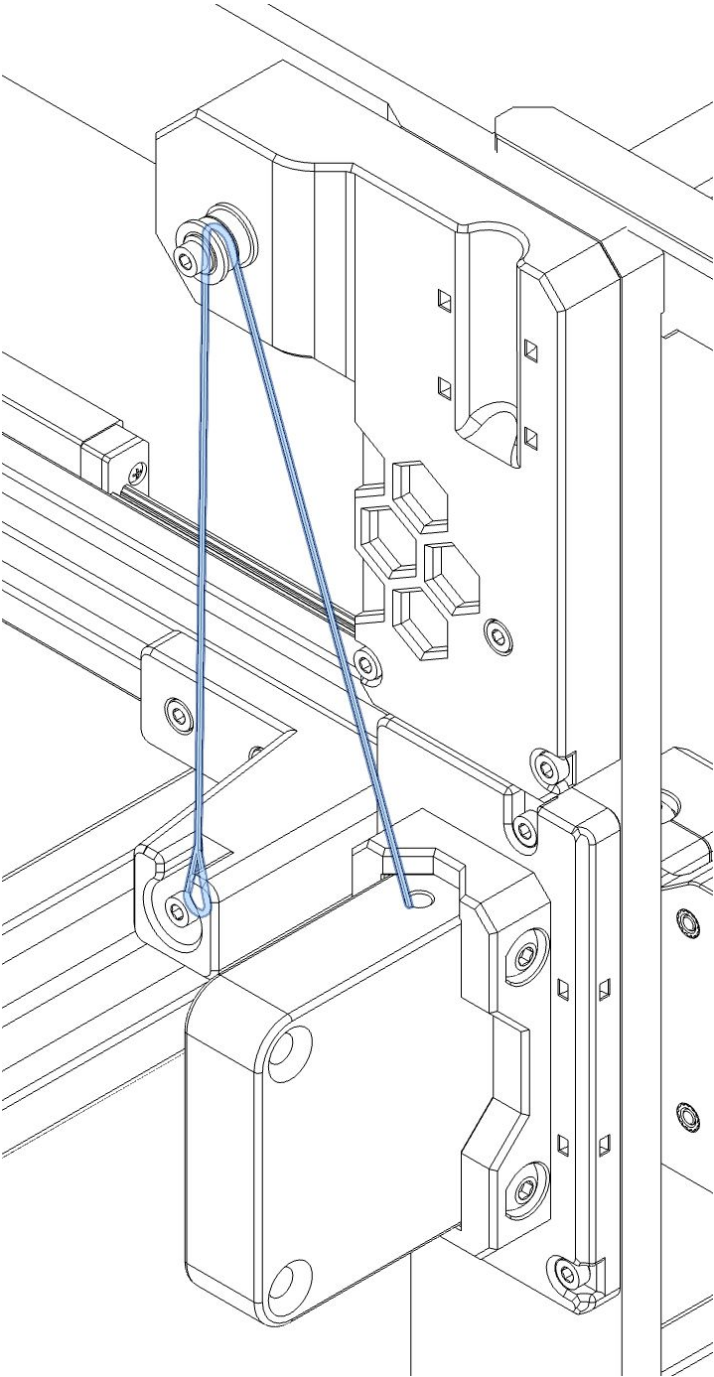
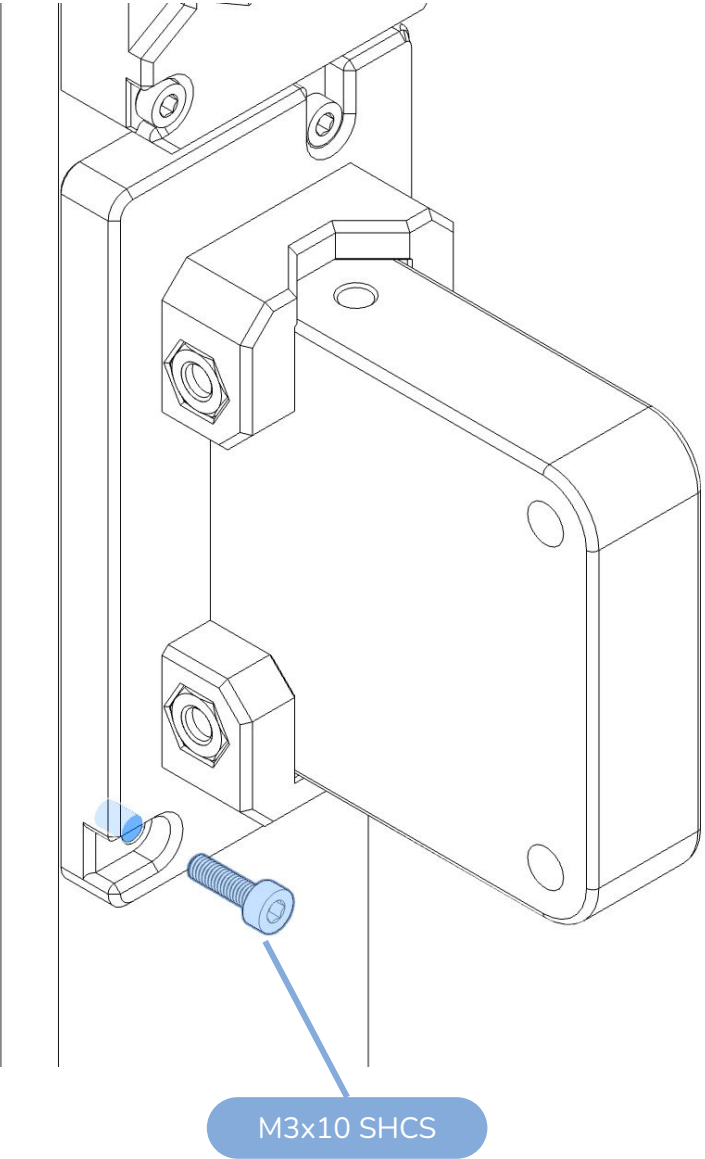


Prusa Frame

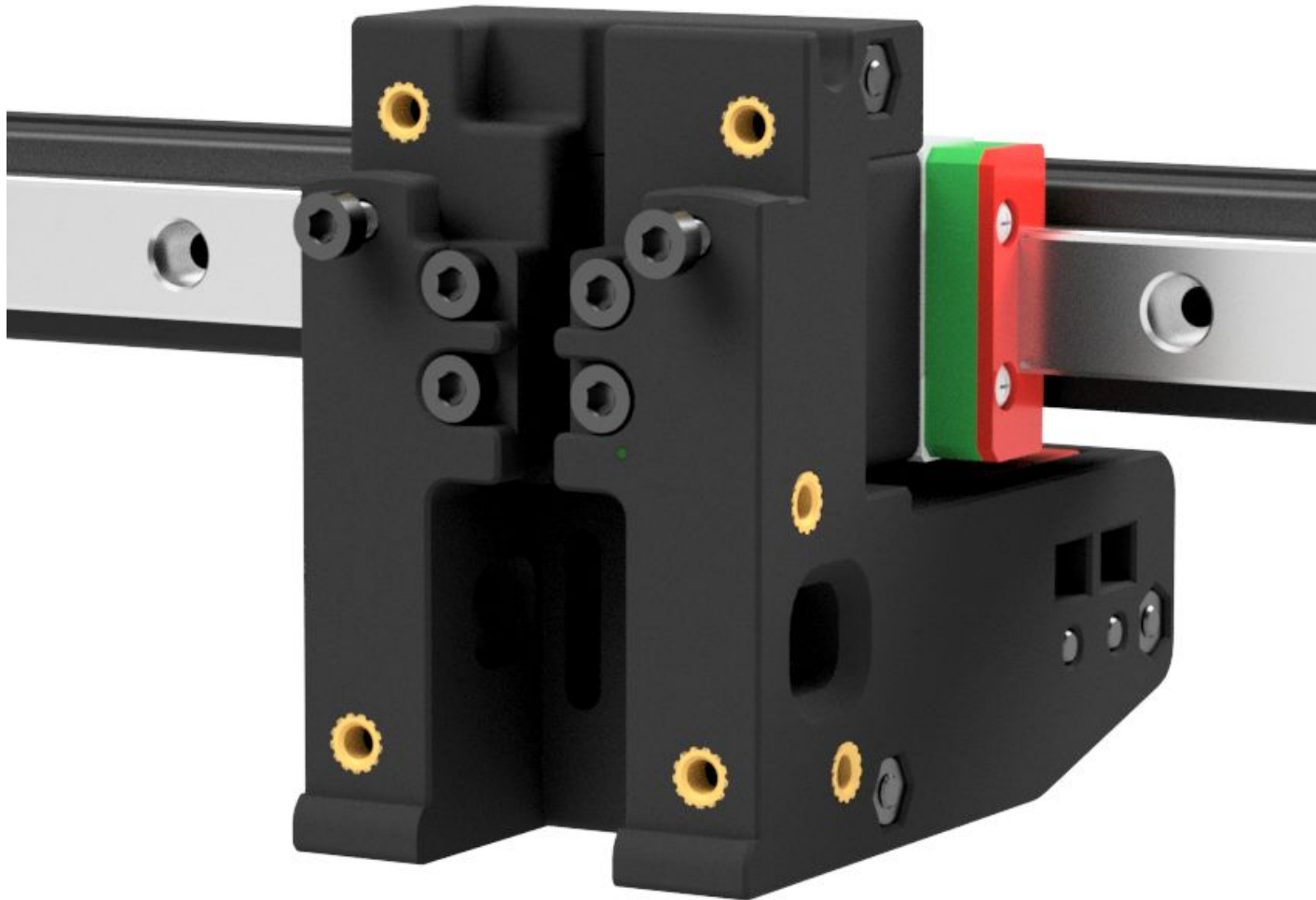


M3x10 SHCS

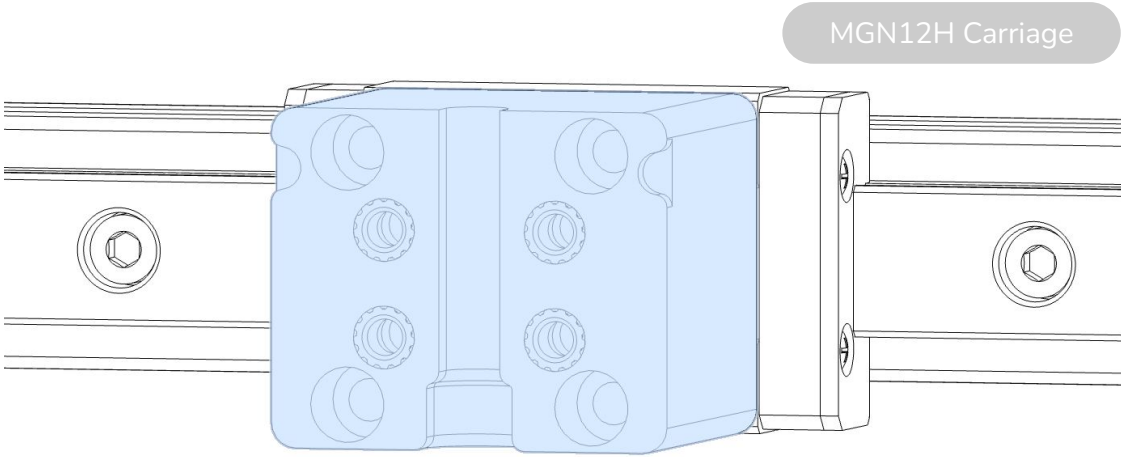
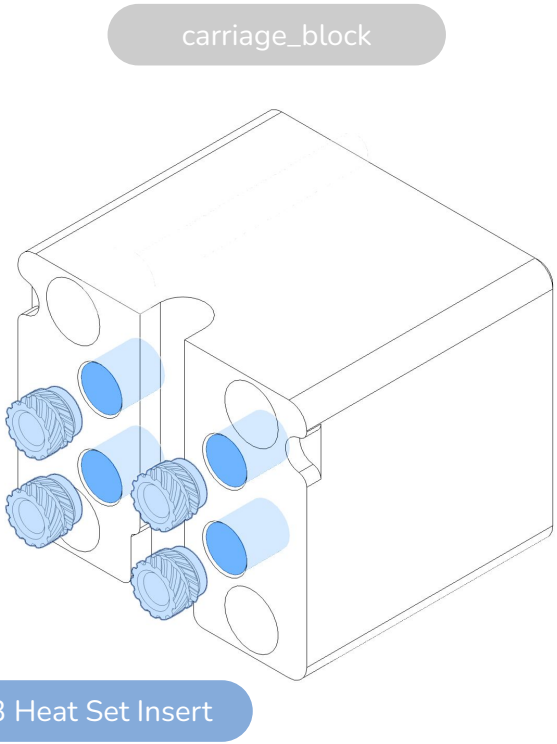
DOOR PULLER
Mount



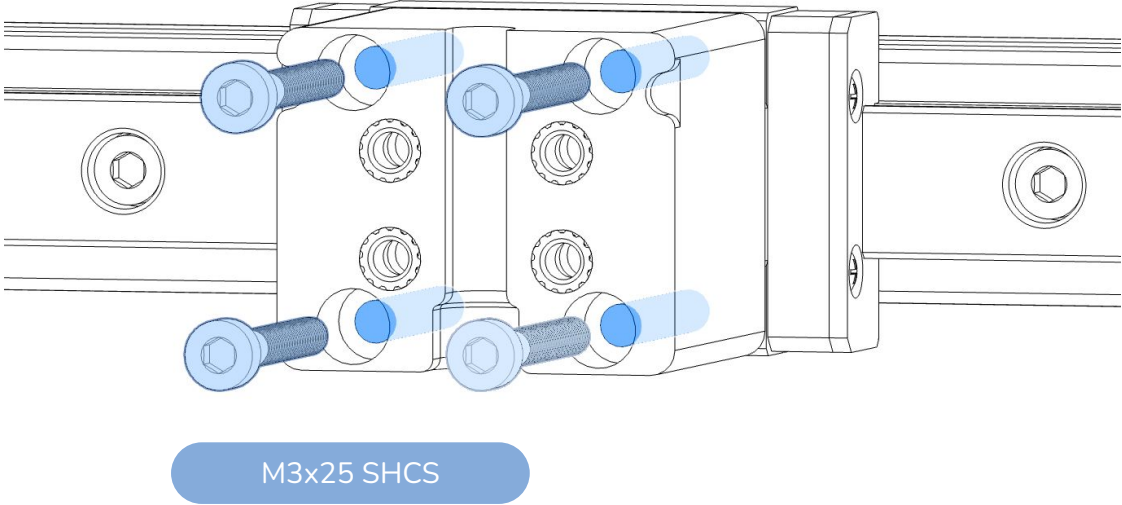
X CARRIAGE



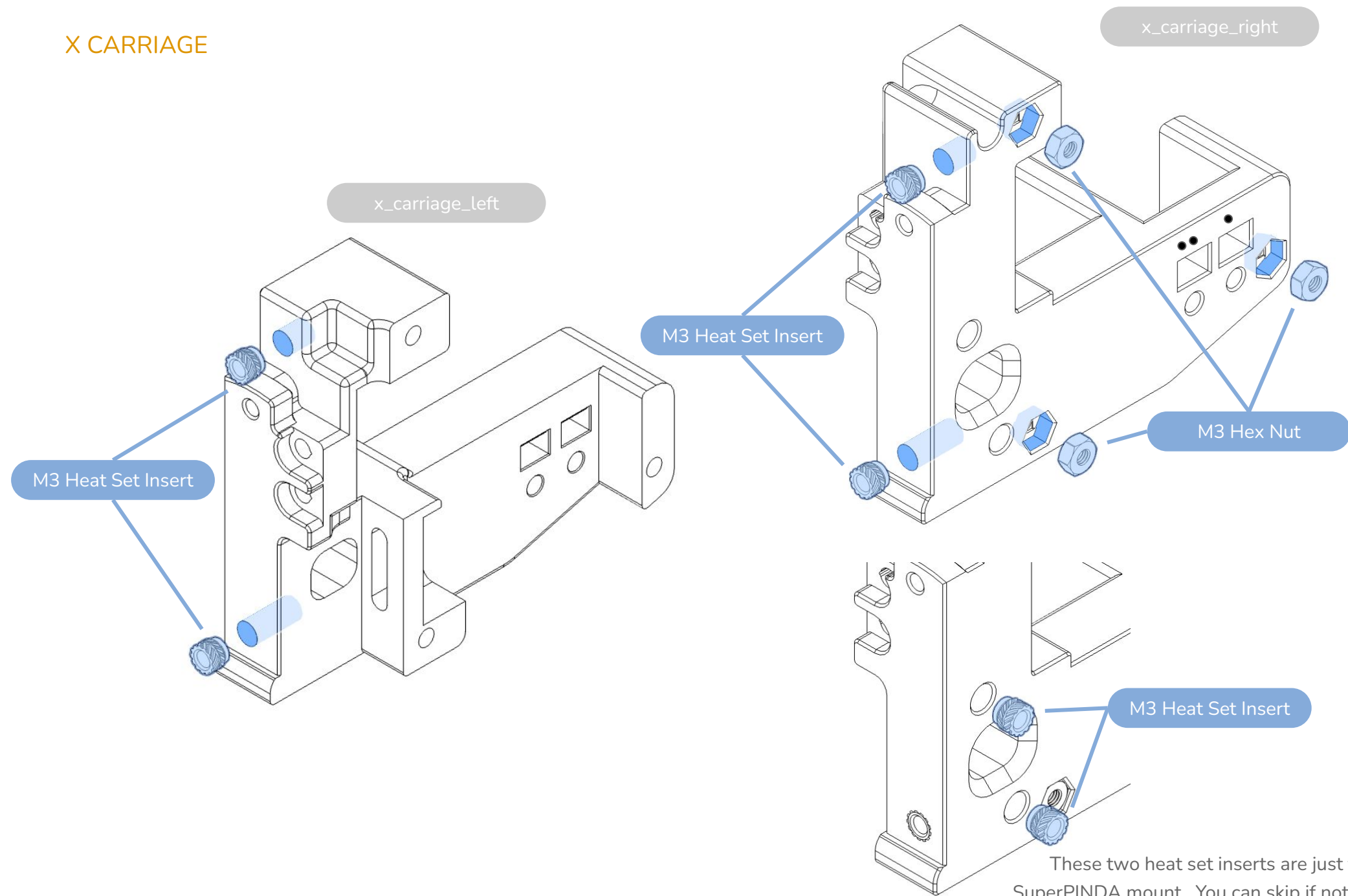
X CARRIAGE



ORIENTATION
Note that the two notches on the left and right sides go UP!



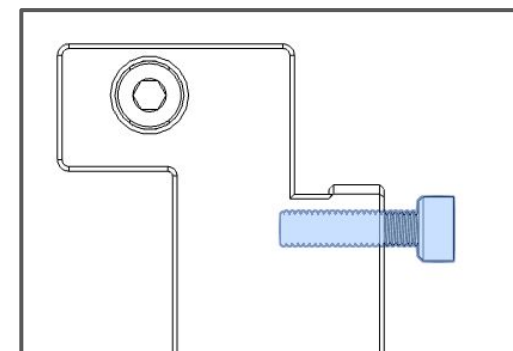
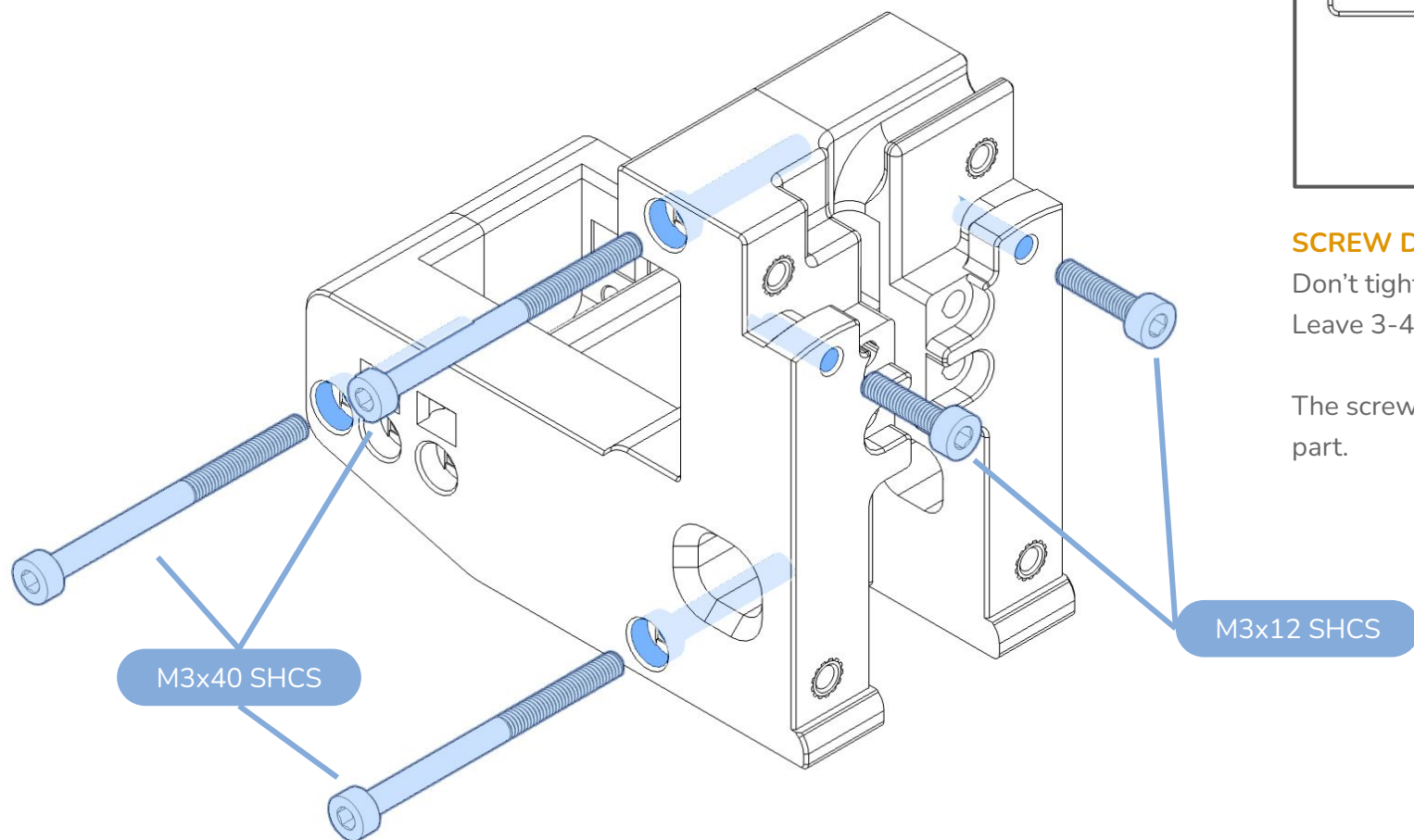
X CARRIAGE



TIP

These two heat set inserts are just for the SuperPINDA mount. You can skip if not using!

X CARRIAGE

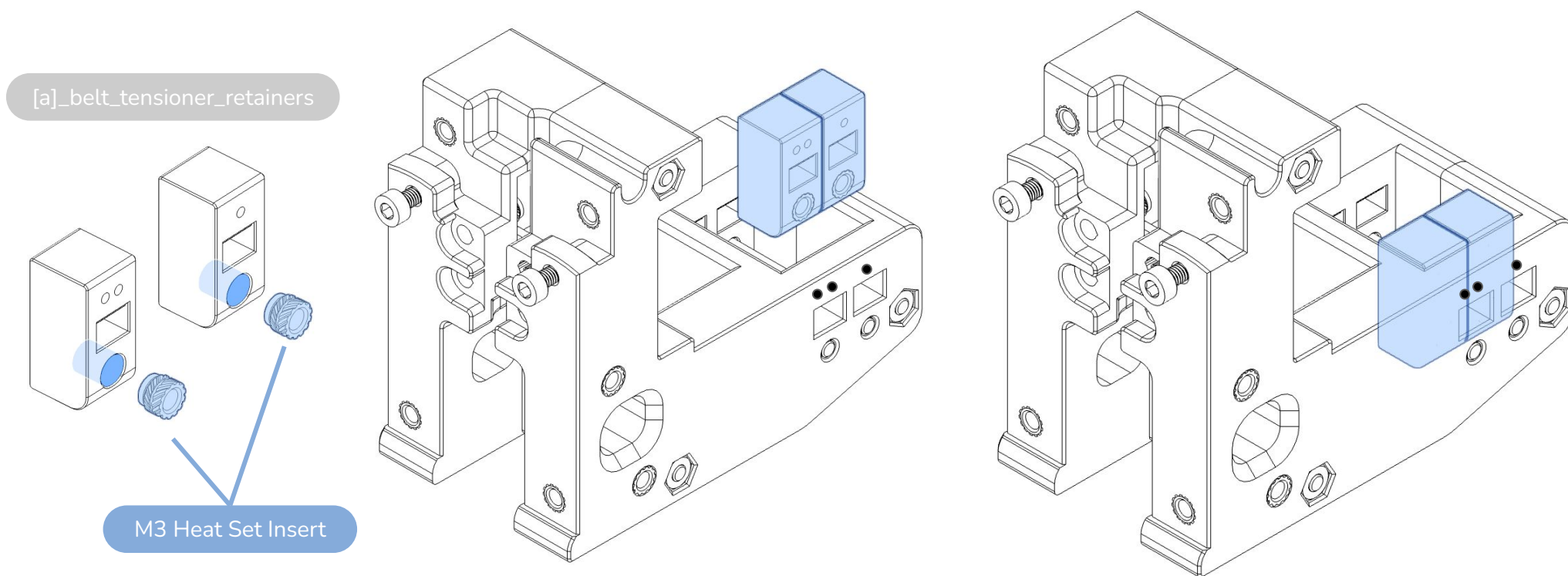


SCREW DEPTH

Don't tighten the M3x12 SHCS all the way.
Leave 3-4mm of exposed thread.

The screws thread directly into the printed part.

X CARRIAGE



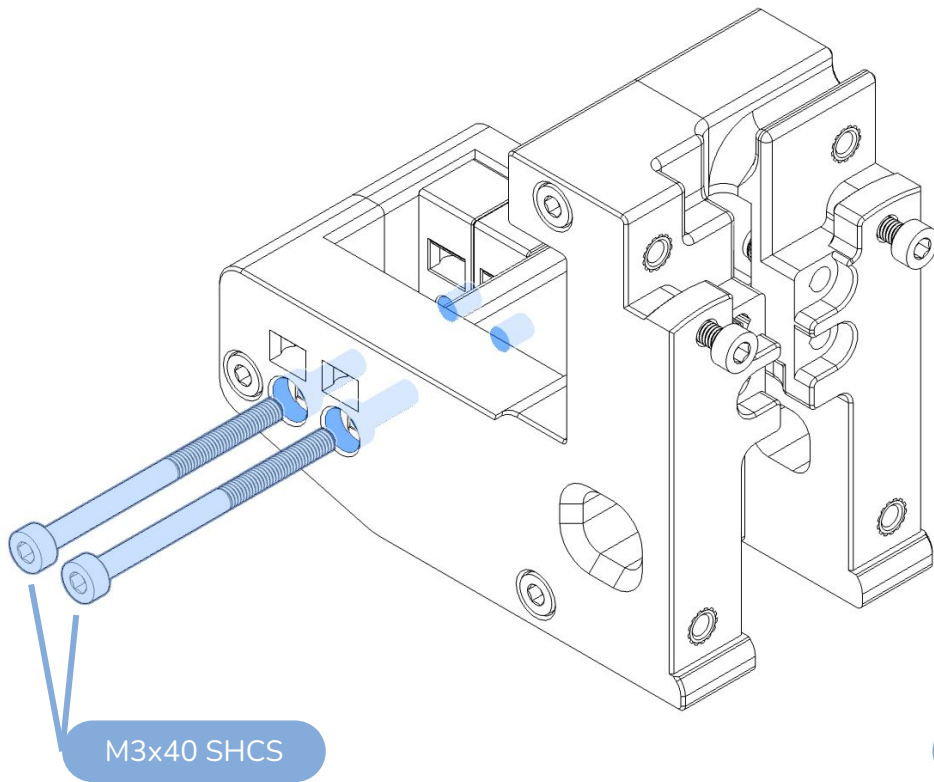
DOTS? DOTS.

x_carriage_right and **[a]_belt_tensioner_retainers** are marked with either one or two dots. Match the position to the dots.

The tensioners are NOT interchangeable!

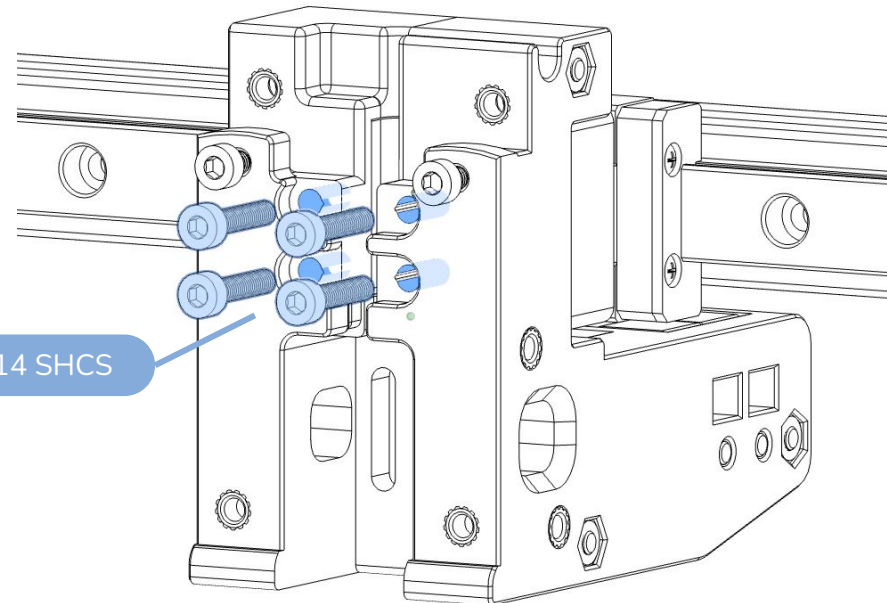
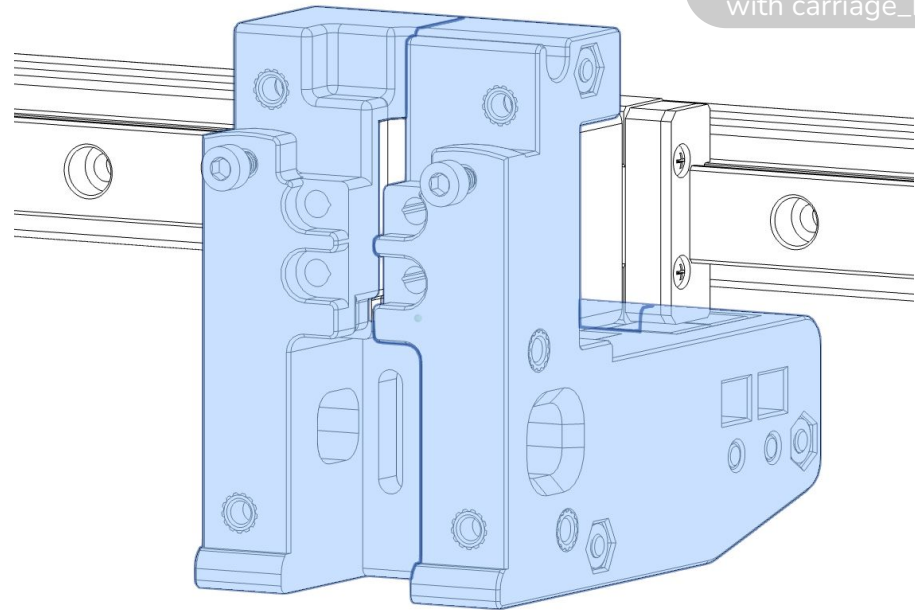
X CARRIAGE

MGN12H Carriage
with carriage_block



SCREW DEPTH

Thread the M3x40 SHCS into the [a]_belt_tensioner parts until just seated. Don't tighten the tensioners just yet, you will do so when installing the XZ Belts.



XZ-BELT INSTALLATION AND TENSIONING

The X-Belt and Z-Belt must be installed and tensioned properly to ensure smooth and accurate gantry motion.

Extra belt length has been specified in the BOM; it is expected to have some leftover belt afterwards.

Match your belt lengths exactly prior to starting this procedure.

DO NOT cut your belt until you certain you have done it correctly. It is okay to leave a little extra length hanging out of the tensioner until final assembly and validation.

The complete procedure can be found on the Prusawire Documentation Site:

<https://prusawire.positron3d.com/hardware/xz-belt.html>



STEALTHBURNER



VORON STEALTHBURNER

The Voron Stealthburner is the officially supported Toolhead for Prusawire.

Other toolheads are possible and are left as an exercise for the user. For example, an experimental AntHead mount is available on the Prusawire GitHub.

Voron Stealthburner Assembly Manual:

https://github.com/VoronDesign/Voron-Stealthburner/blob/main/Manual/Assembly_Manual_SB.pdf



X CARRIAGE

SuperPINDA Mount



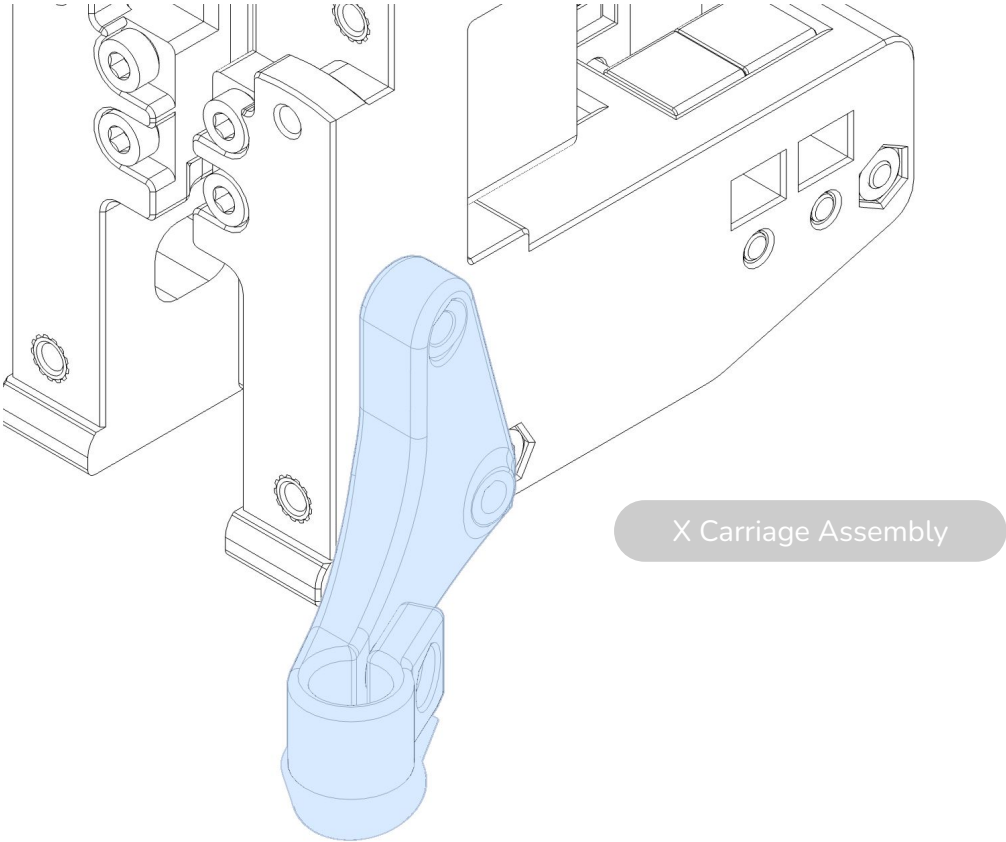
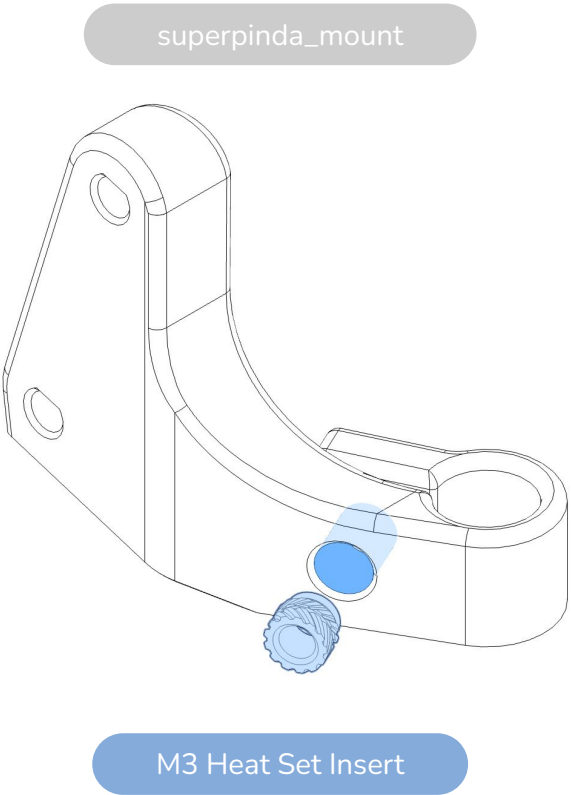
SUPERPINDA

If re-using a Prusa SuperPINDA for bed mesh leveling, utilize the included toolhead mount. Configuration files assume this mounting positioning for bed meshing.

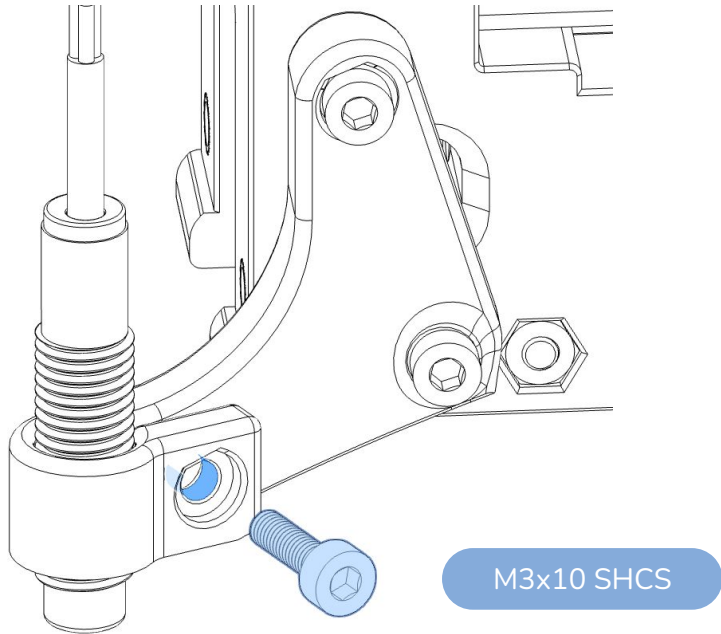
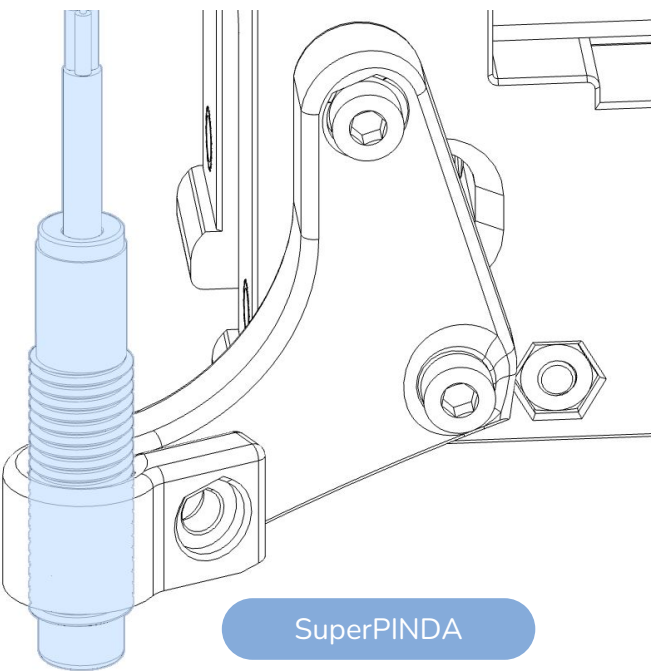
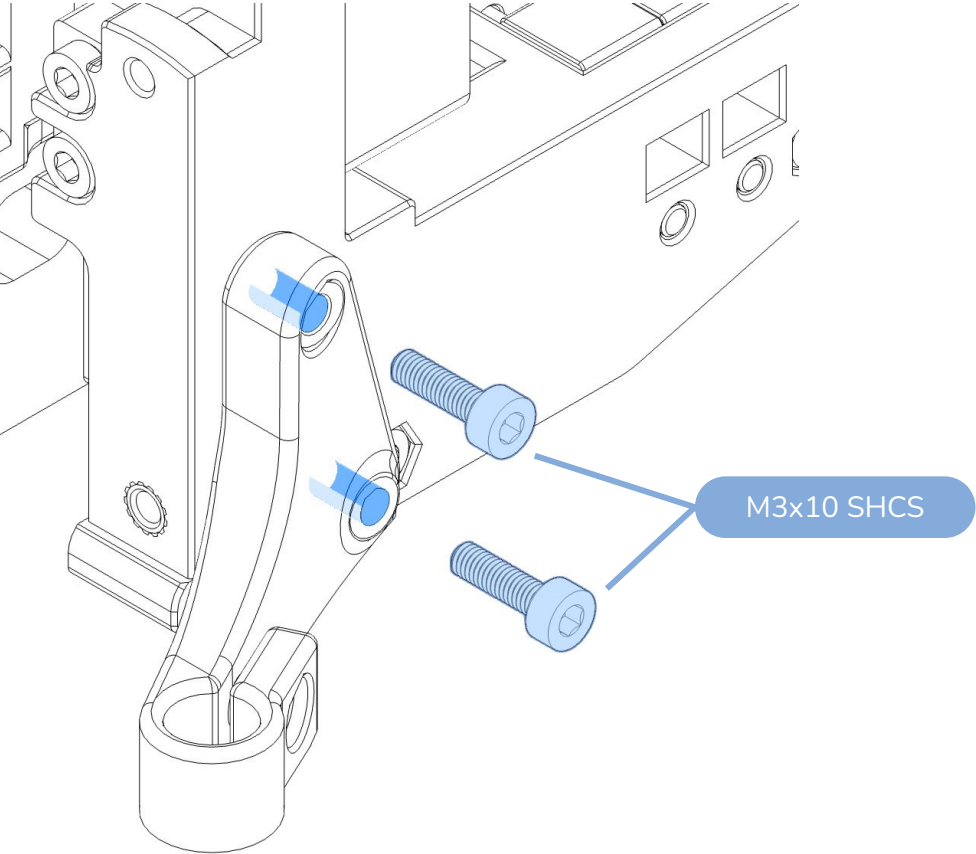
The SuperPINDA should be set so the probe is exactly 1.5mm from the build plate. [A printed spacer](#) can be used to set this more easily.



X CARRIAGE
SuperPINDA Mount



X CARRIAGE
SuperPINDA Mount



FRONT GRILL & SCREEN



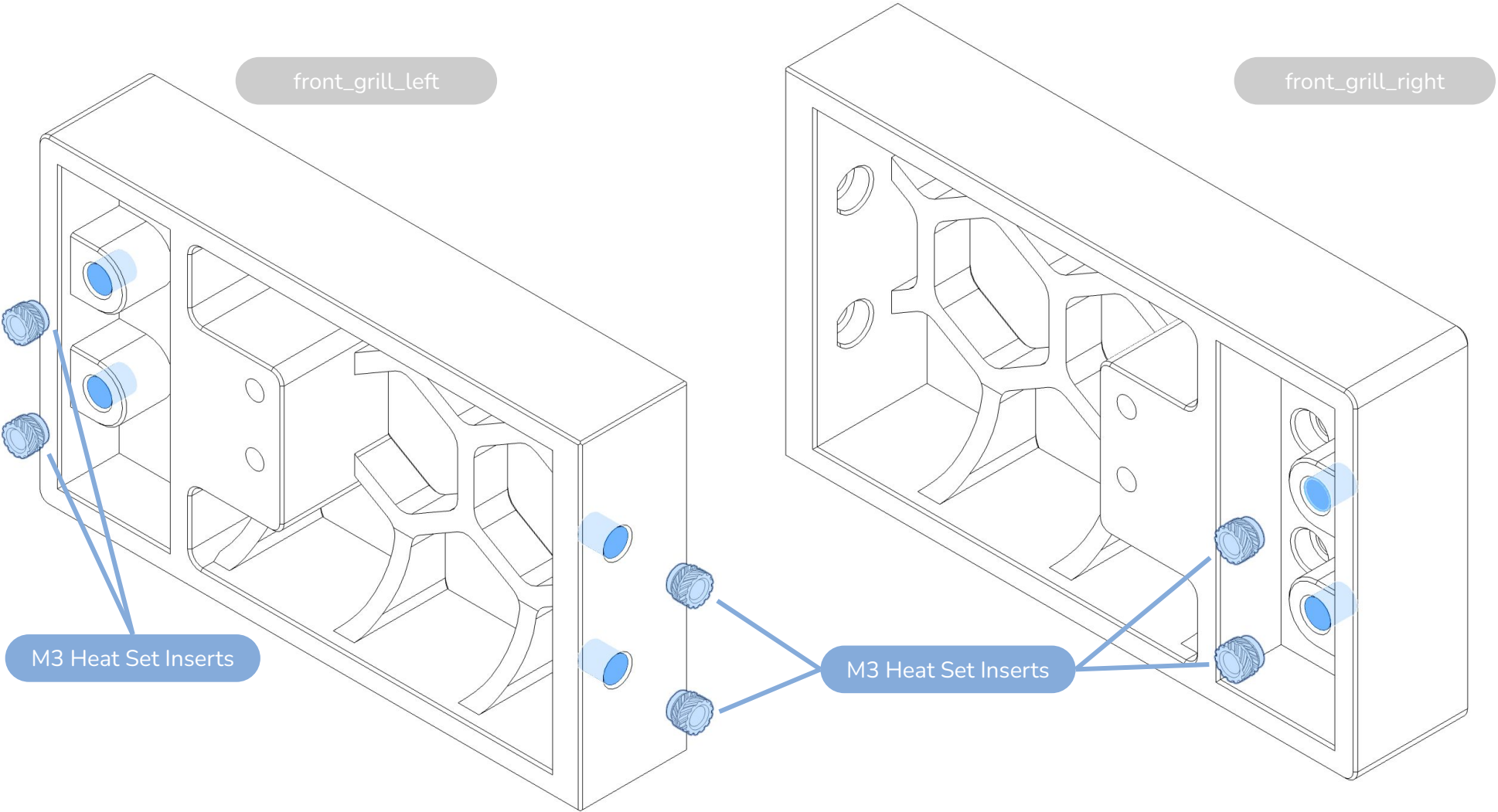
SCREEN OPTIONS

We demonstrate installation and usage of the **Prusa MK2/MK3-era LCD Panel** used with the Einsy RAMbo control board. Other great options exist, including the **BTT HDMI 5** and **BTT PI TFT50 V2.1**.

Check out the [Mods directory on the Prusawire GitHub](#).



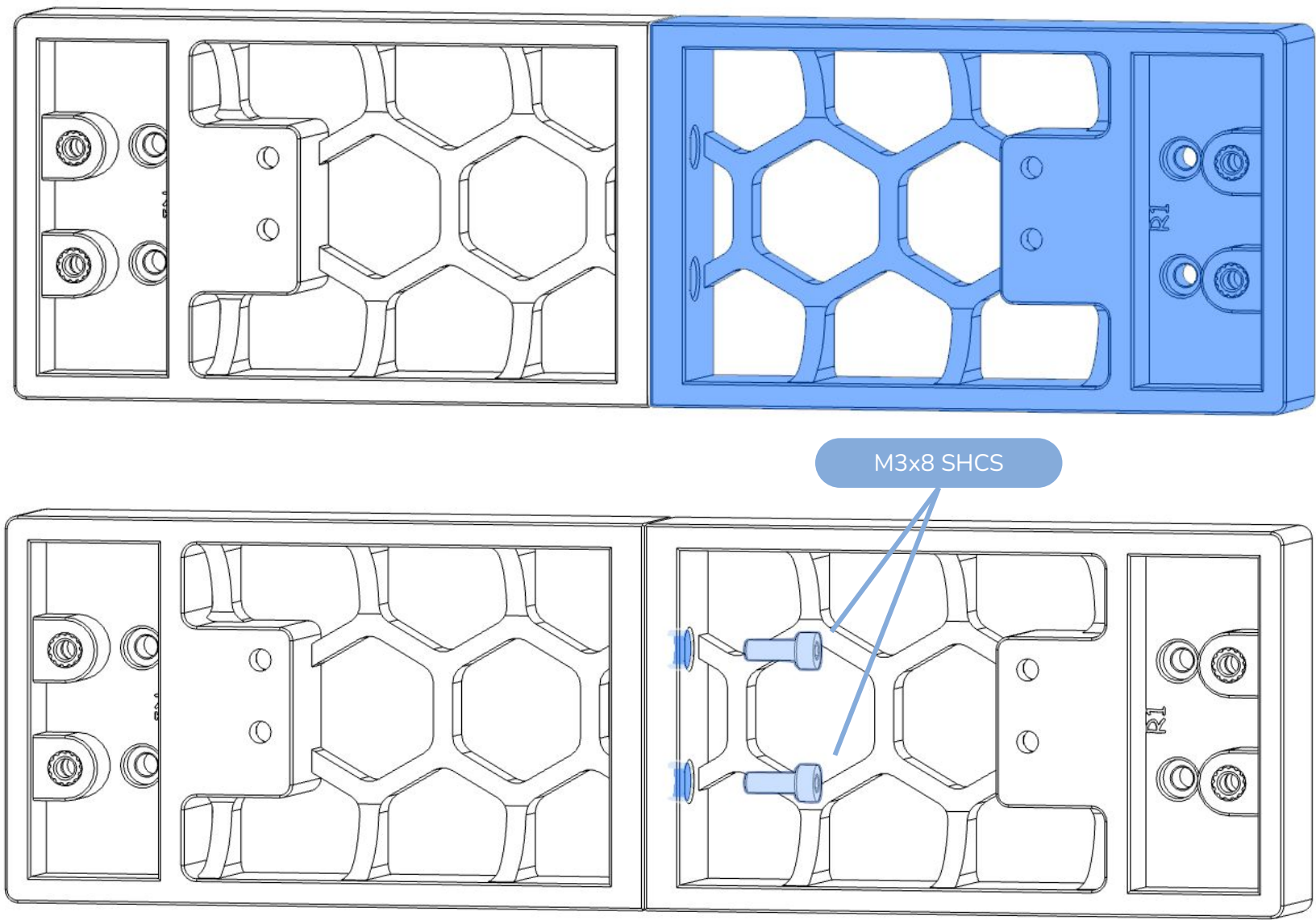
FRONT GRILL



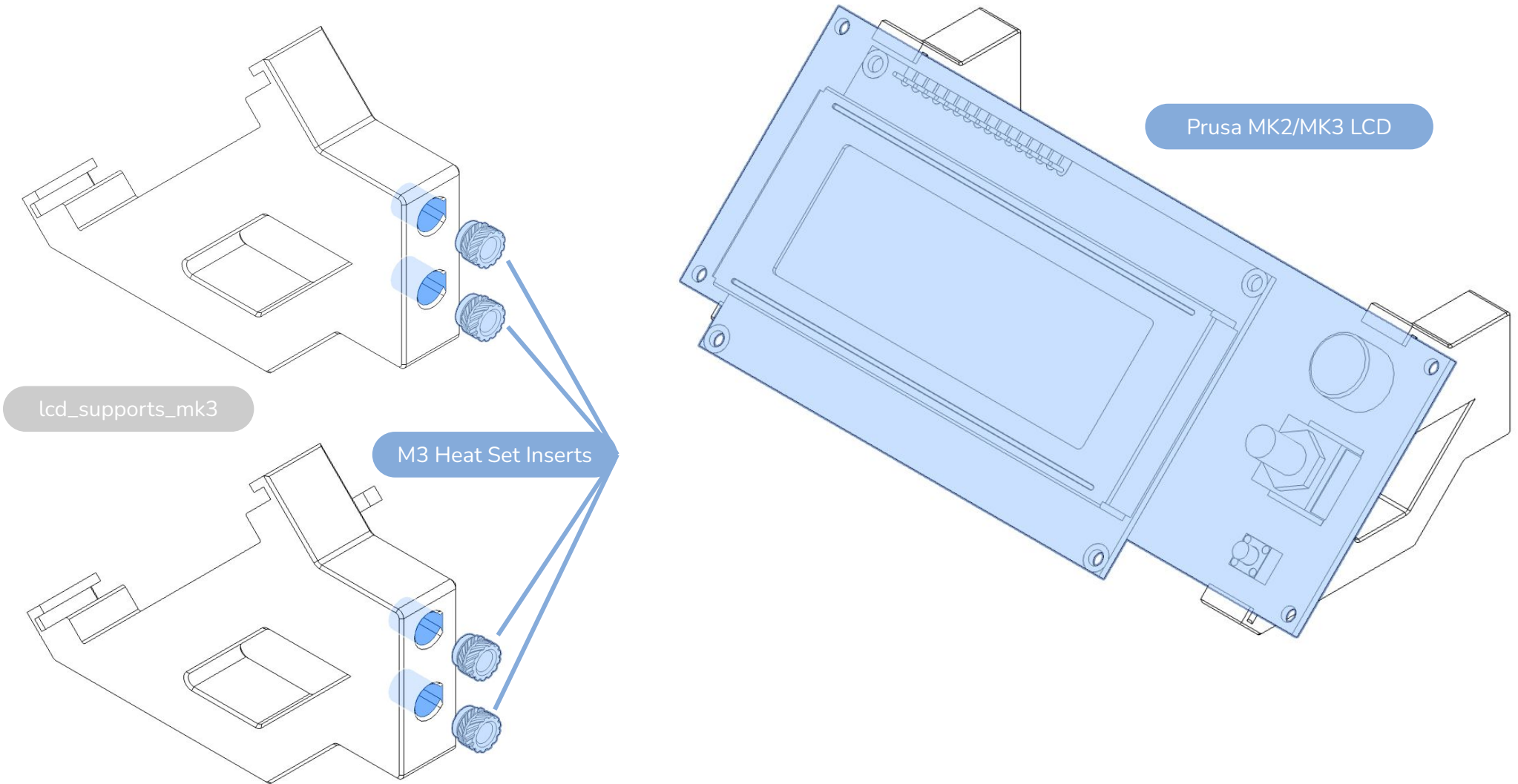
FRONT GRILL

front_grill_left

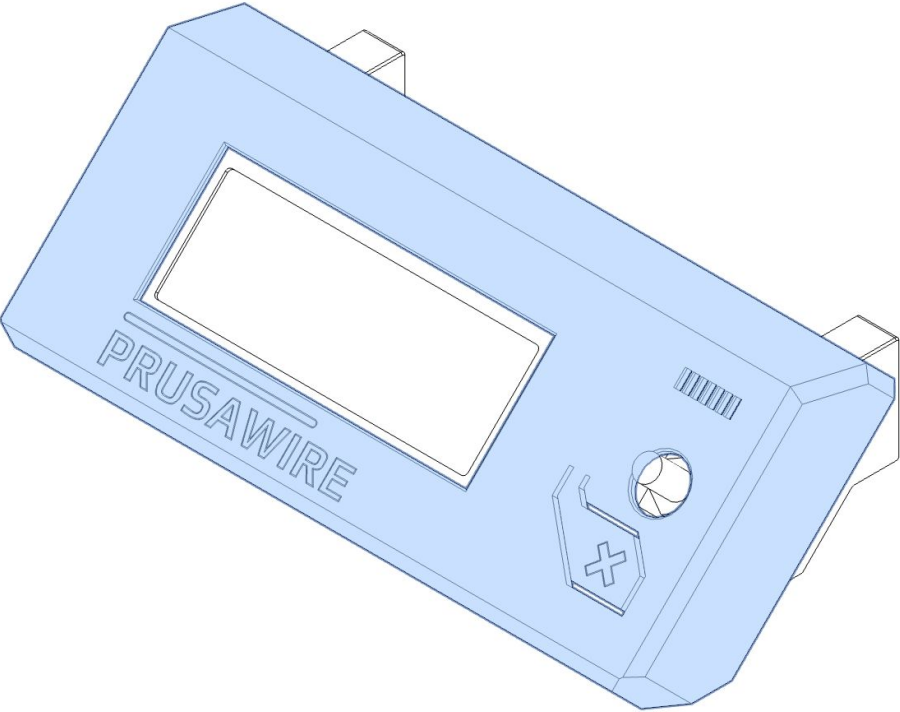
front_grill_right



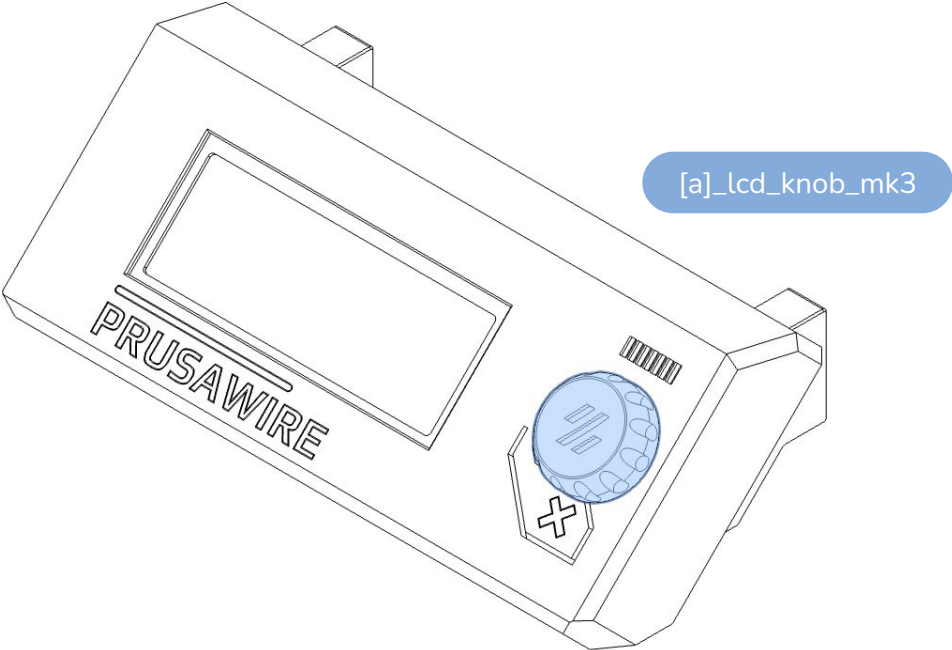
FRONT SCREEN



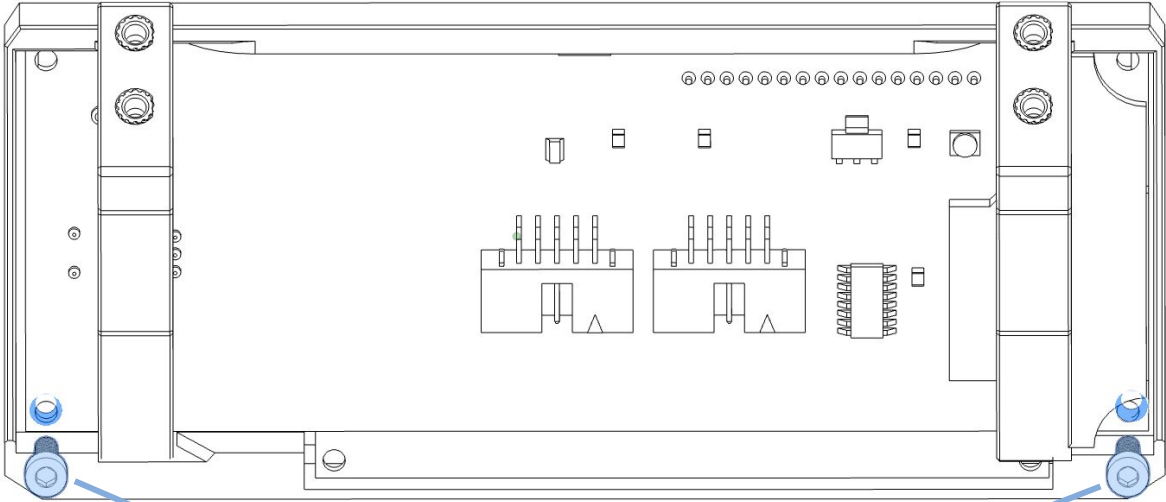
FRONT SCREEN



mmu_lcd_cover_mk3
or
lcd_cover_mk3_single_color



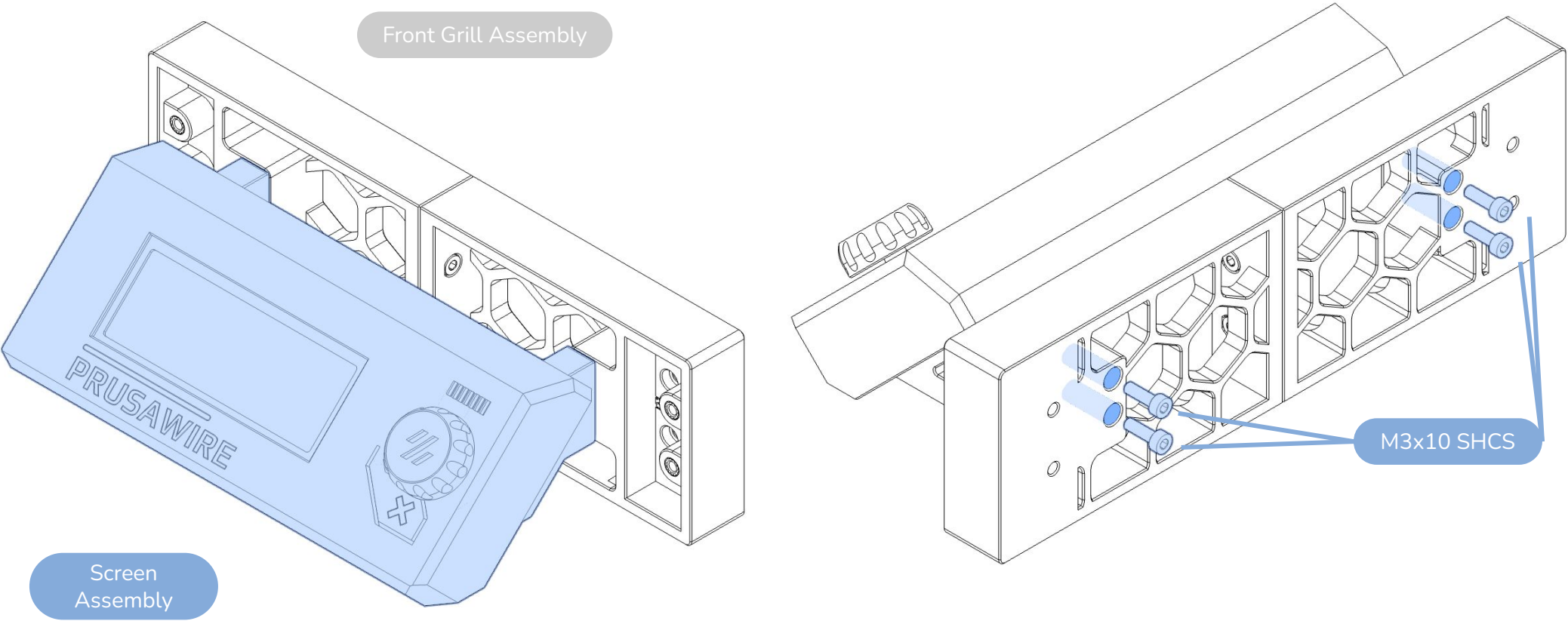
[a]_lcd_knob_mk3



REAR OF LCD

M3x10 SHCS

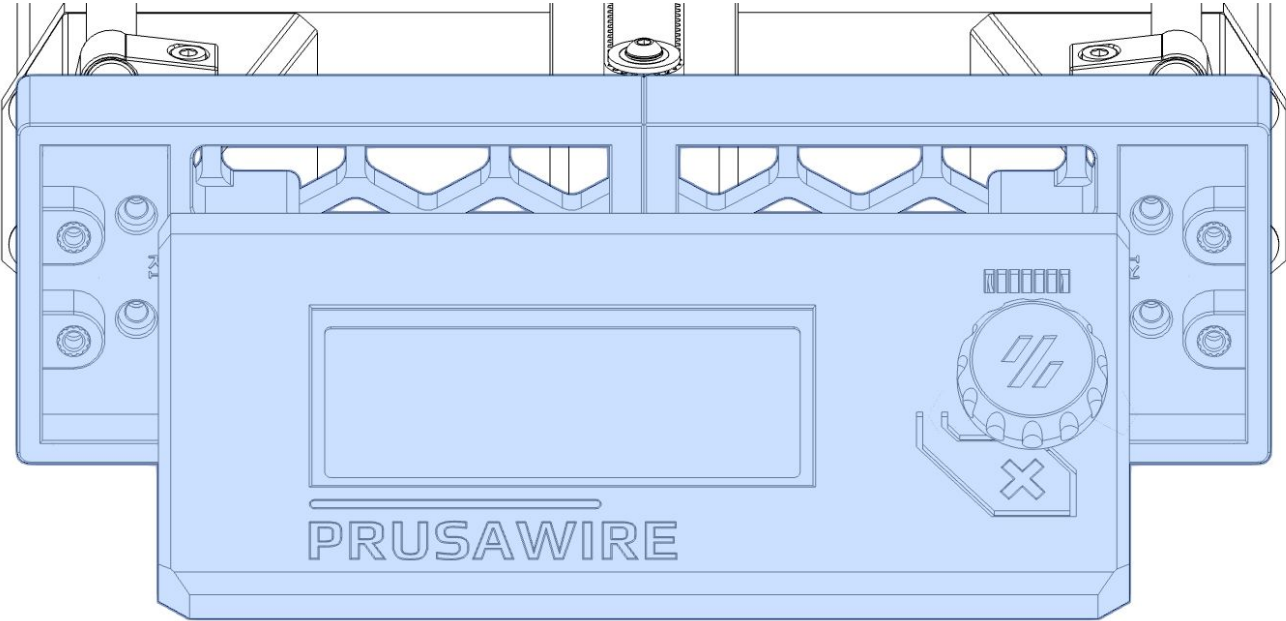
FRONT SCREEN



FRONT GRILL
& SCREEN

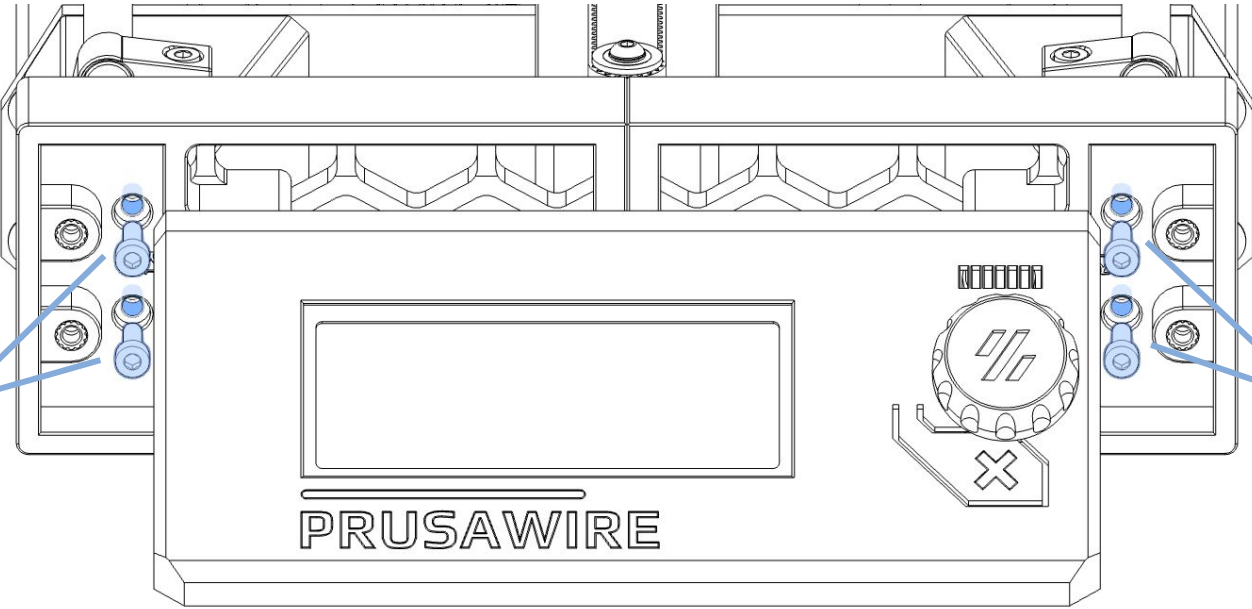
y_rod_holder_front_left

y_rod_holder_front_right

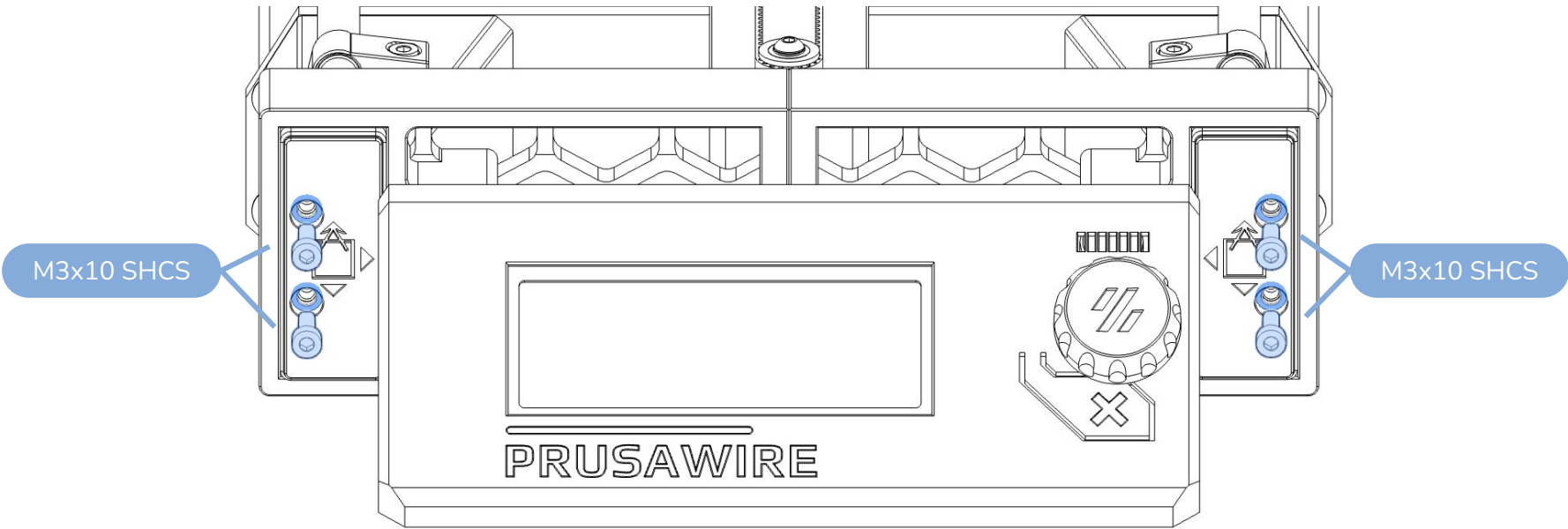
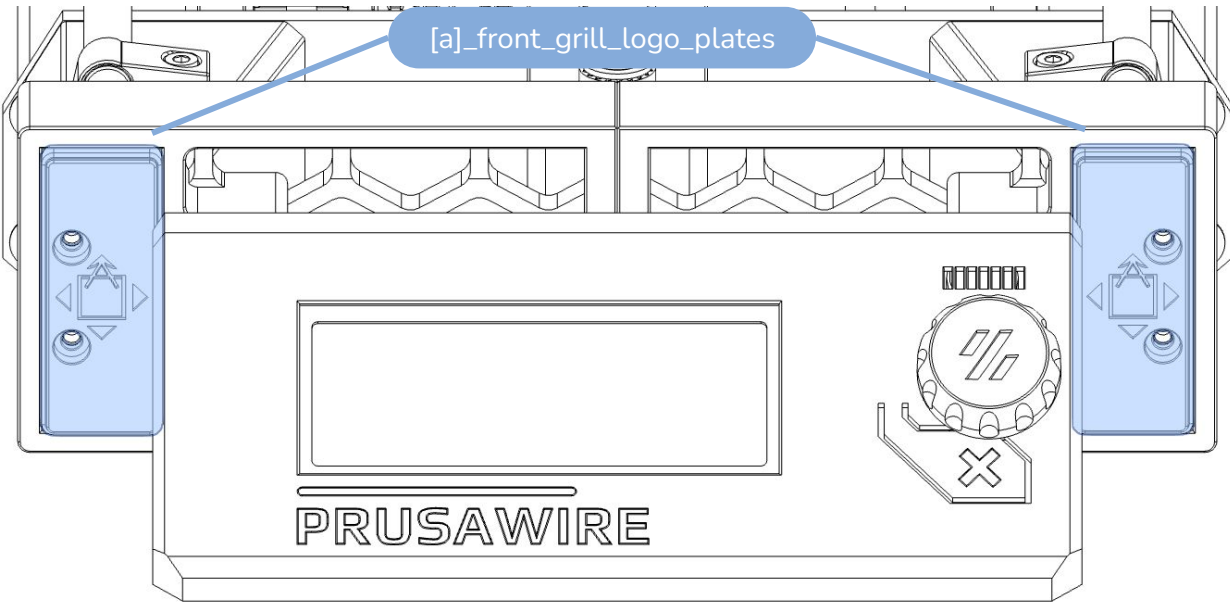


M3x10 SHCS

M3x10 SHCS



FRONT GRILL
& SCREEN



POWER SUPPLY

POWER SUPPLY INSTALLATION

The Power Supply (PSU) is a critical structural member of the Prusa i3 Frame Design. Installation will vary based on your chosen PSU. The BOM assumes hardware for the Prusa Black PSU.

Prusa Black PSU (easiest, most expensive):

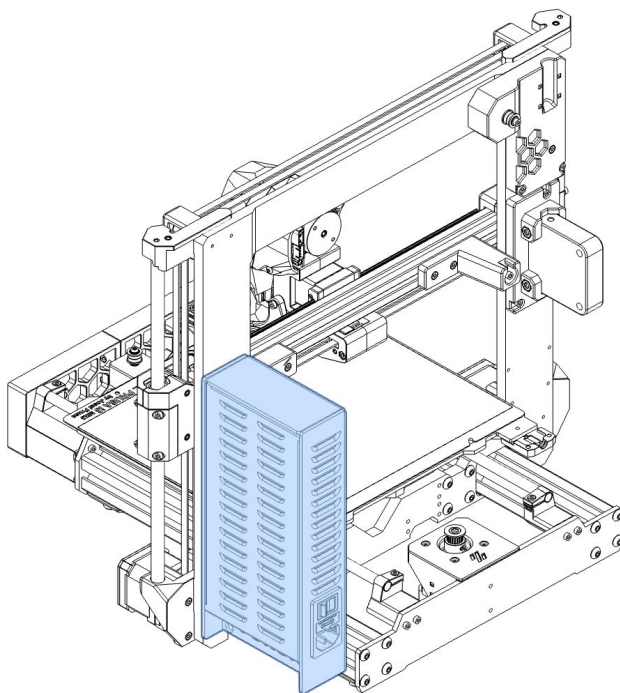
https://help.prusa3d.com/guide/2-frame-assembly_427481#428125 (Chapter 2, Steps 17 through 17)

Prusa Silver PSU (not recommended):

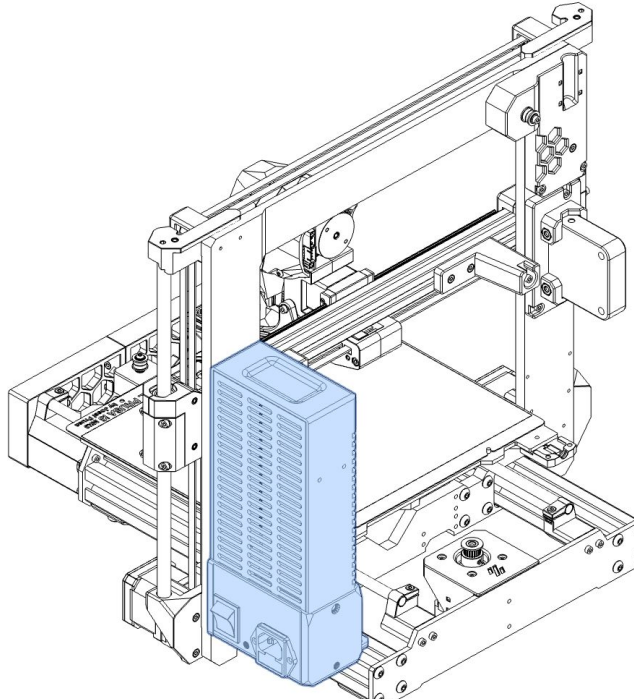
Visit the Mods directory on GitHub.

Meanwell LRS-350 / RSP-320 (more work, a little less expensive):

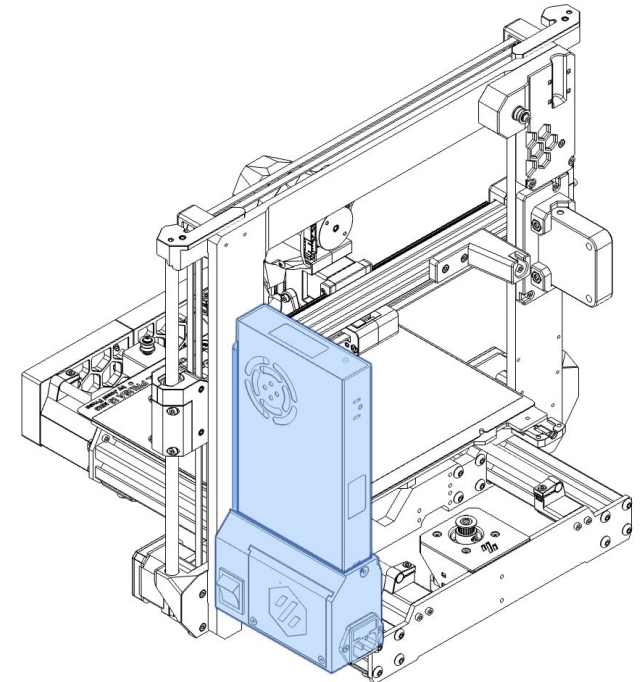
Visit the Mods directory on GitHub.



Prusa Black PSU



Prusa Silver PSU



Meanwell PSU



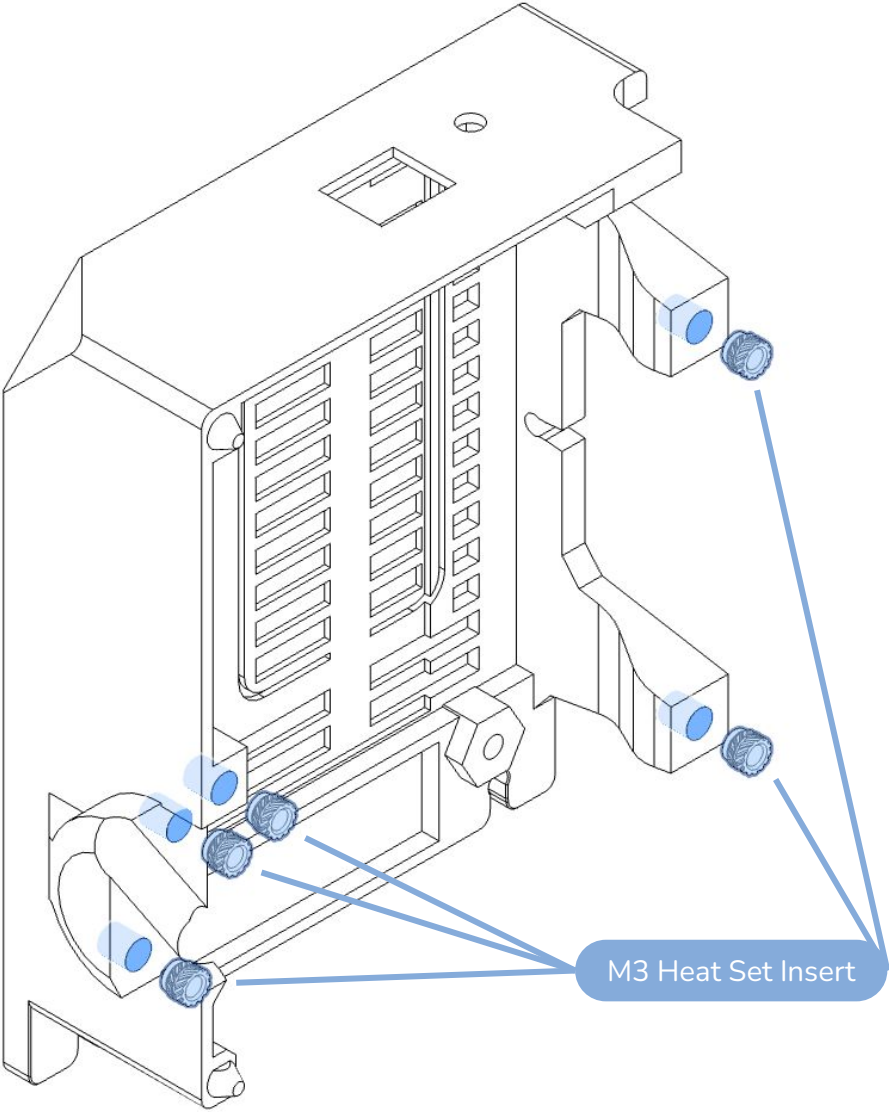
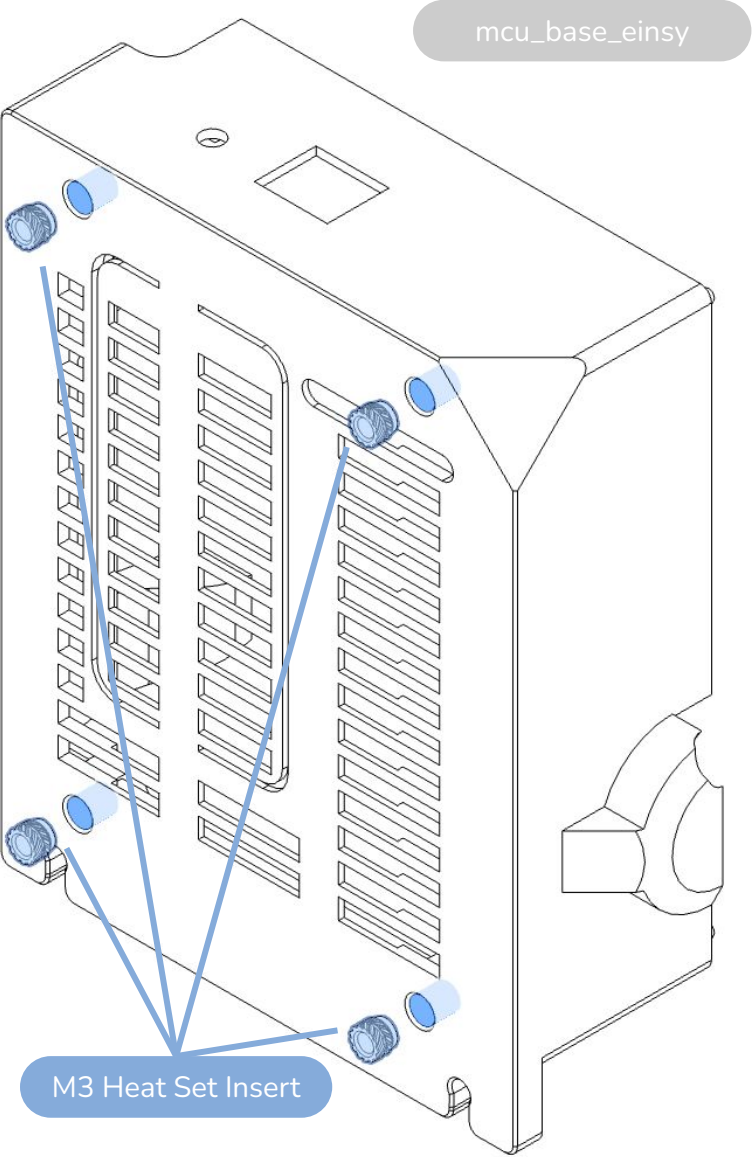
CONTROL BOARD OPTIONS

We demonstrate installation and usage of the **Prusa Einsy RAMbo** and housing. If re-using a MK2/MK3 RAMbo, discard the original housing. The BTT SKR Mini E3 is an excellent (cheaper and more powerful) option as well. Housing assembly for both is similar.

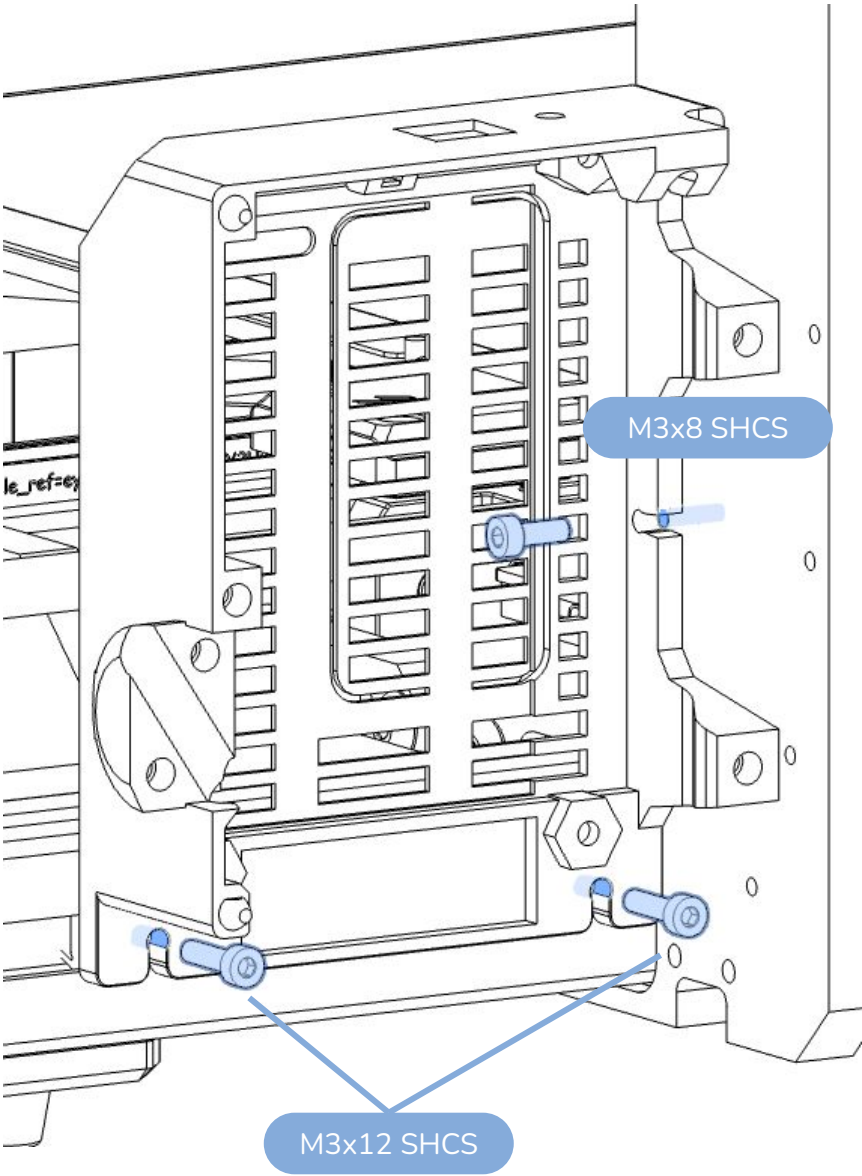
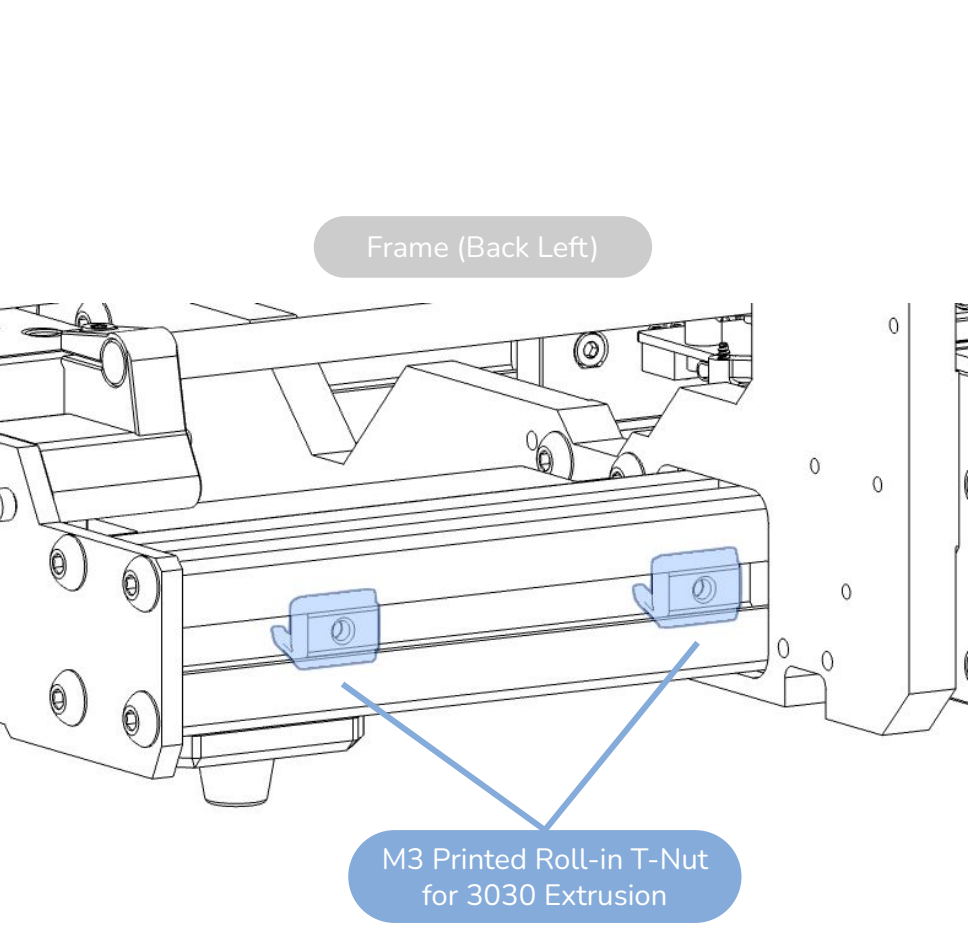
Einsy RAMbo: *mcu_box_einsy.stl* and *mcu_cover_einsy.stl*

BTT SKR Mini E3: *mcu_box_skr_mini_e3.stl* and *mcu_cover_skr_mini_e3.stl*

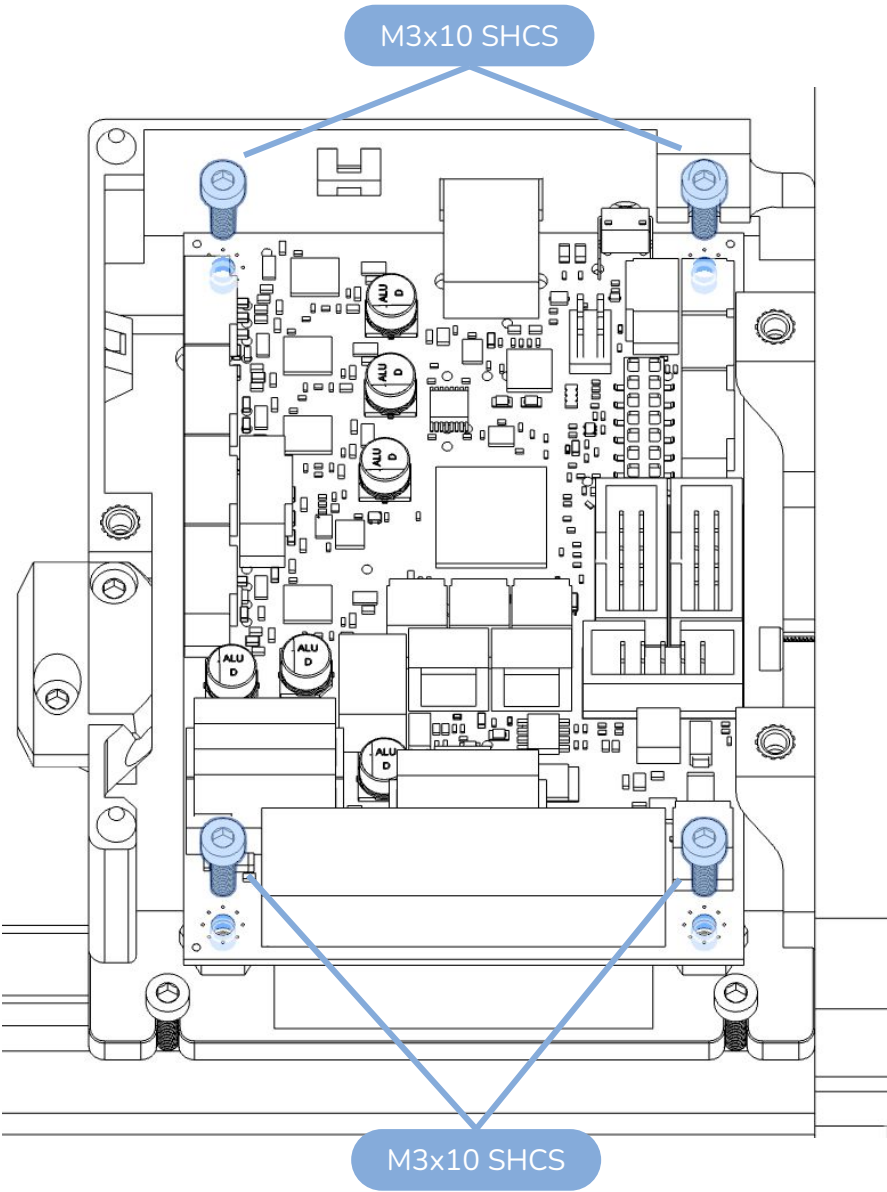
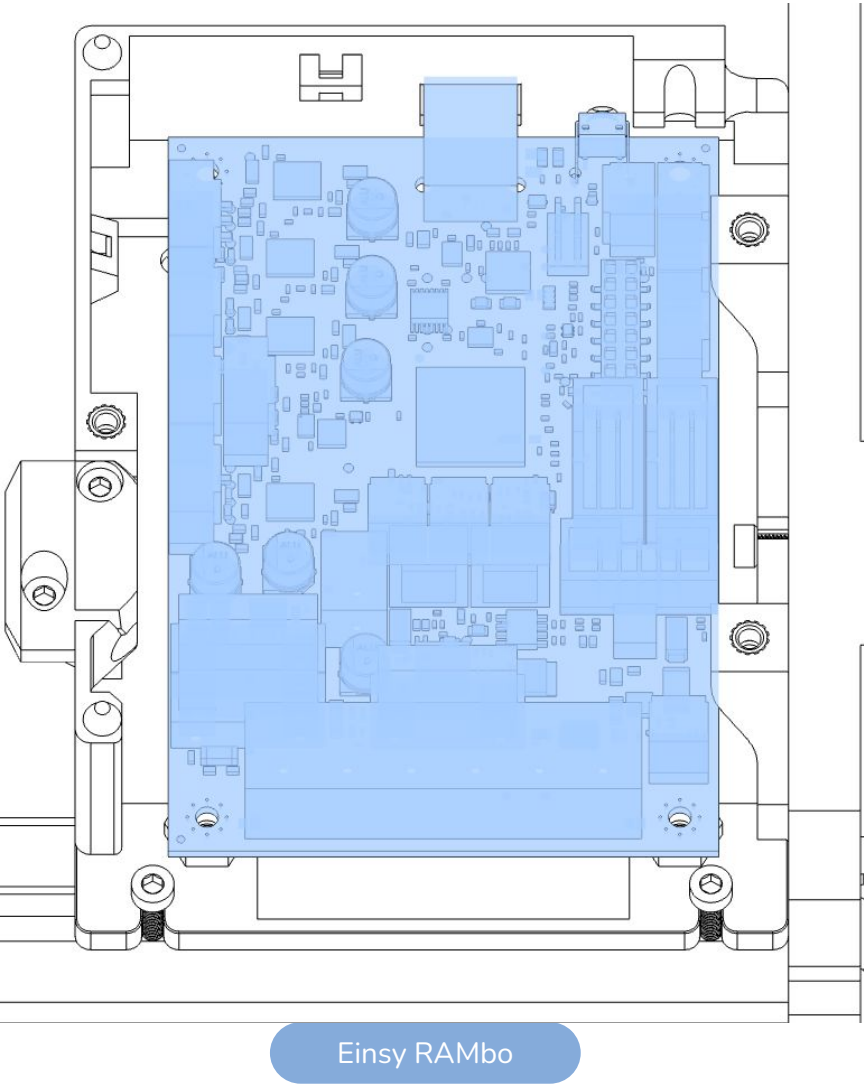
ELECTRONICS
Control Board



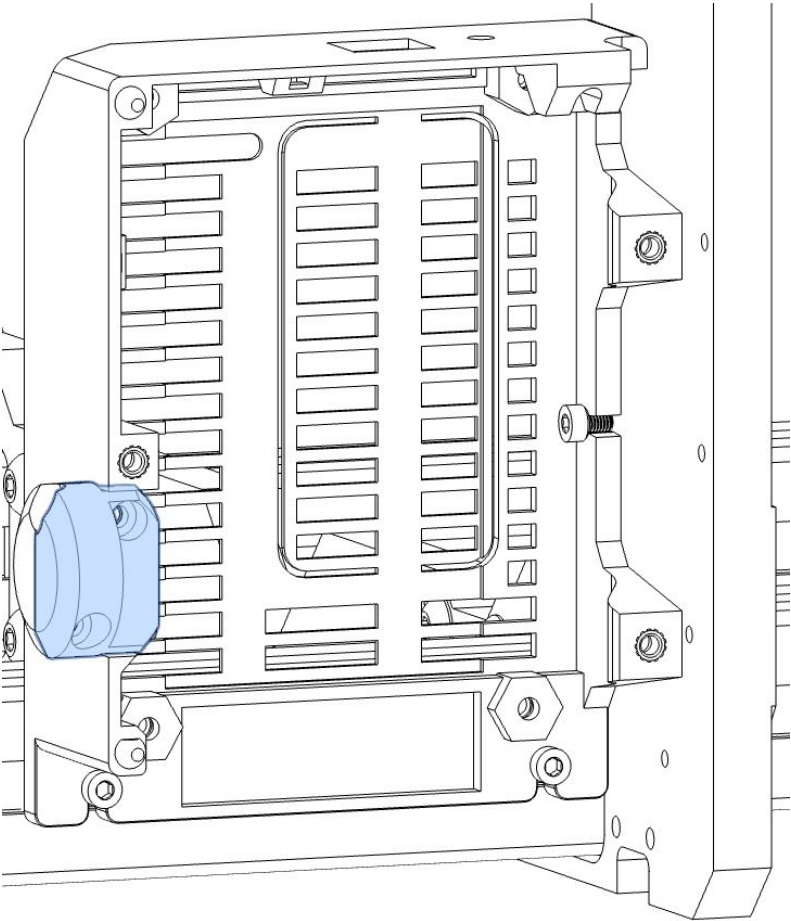
ELECTRONICS
Control Board



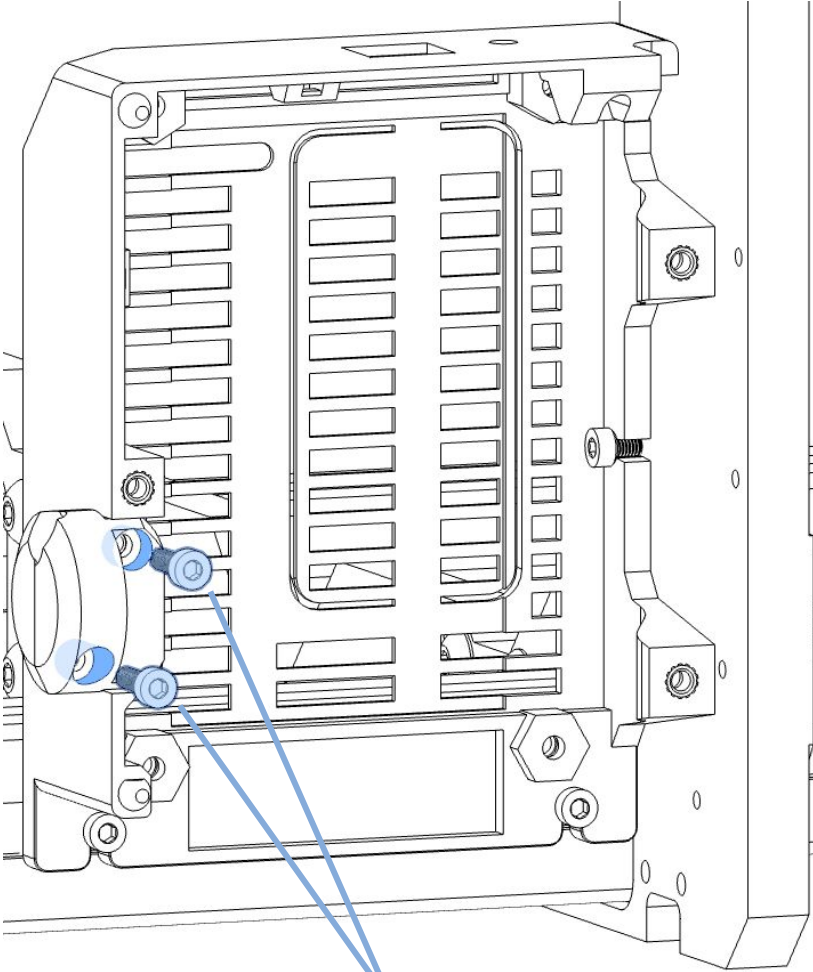
ELECTRONICS
Control Board



ELECTRONICS
Control Board

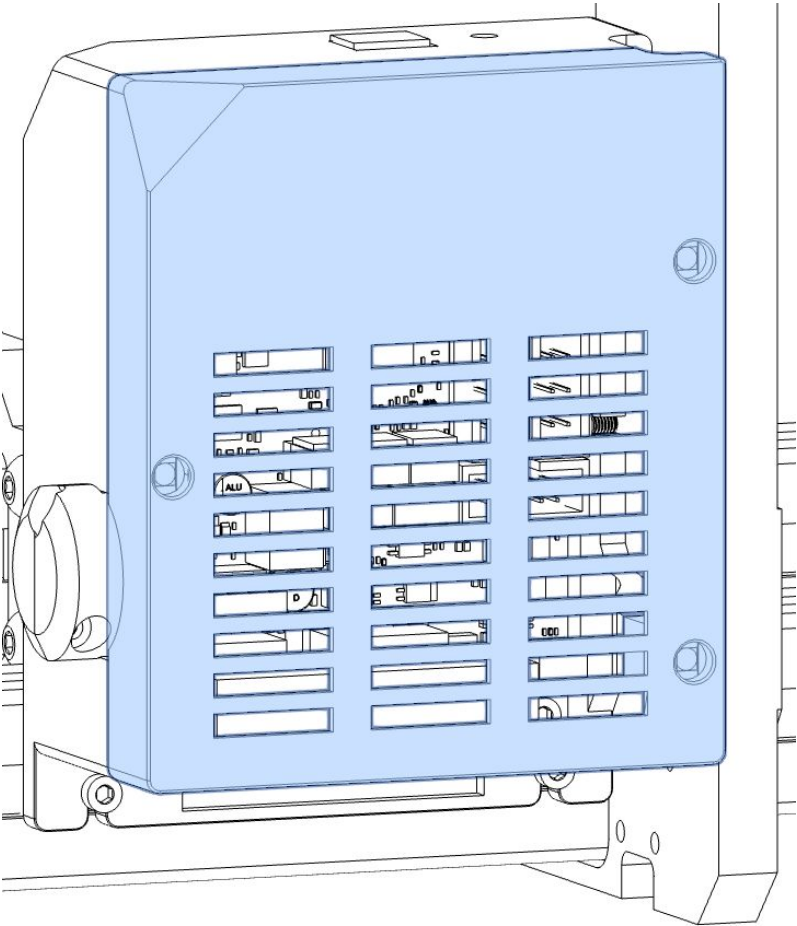


heatbed_cable_clip

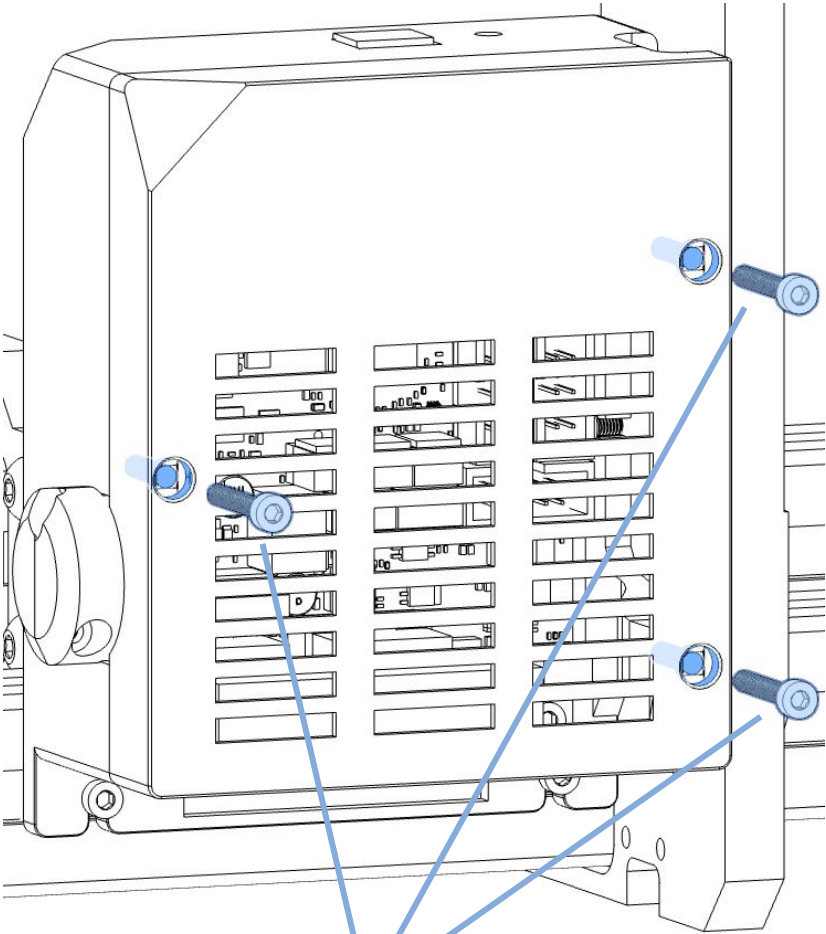


M3x10 SHCS

ELECTRONICS
Control Board



mcu_cover_einsy



M3x20 SHCS

ELECTRONICS

General Layout

POWER

LAYOUT AND WIRING

You have flexibility in the placement of certain components and how to wire them.

Prioritize clean runs and secure wires where possible. To receive a serial, wiring must be tidy and secured from any motion component.

A possible layout is shown here, but you may choose your own adventure.

KEY

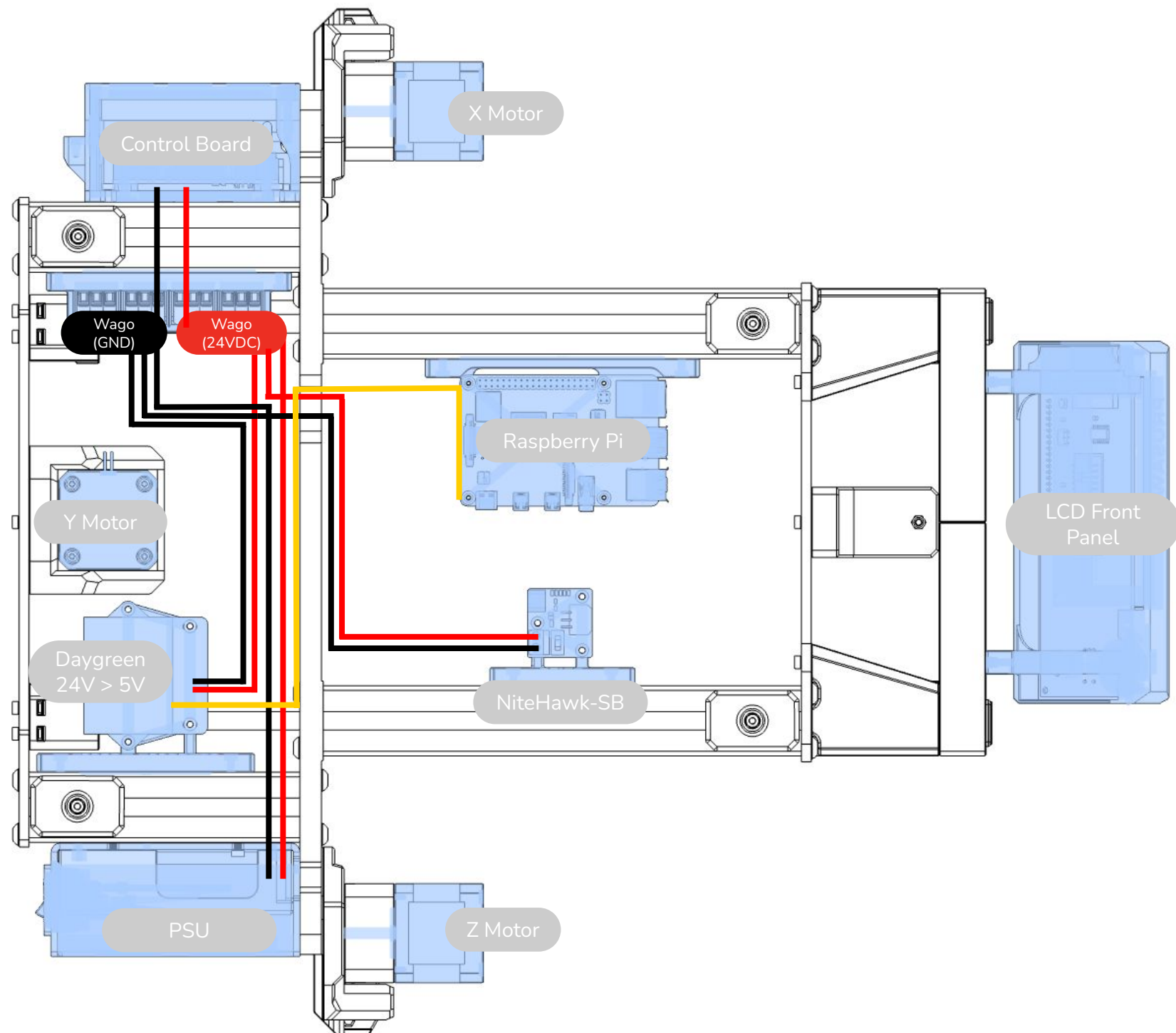
24VDC Positive



Ground



5VDC USB Power



ELECTRONICS
General Layout
DATA

LAYOUT AND WIRING

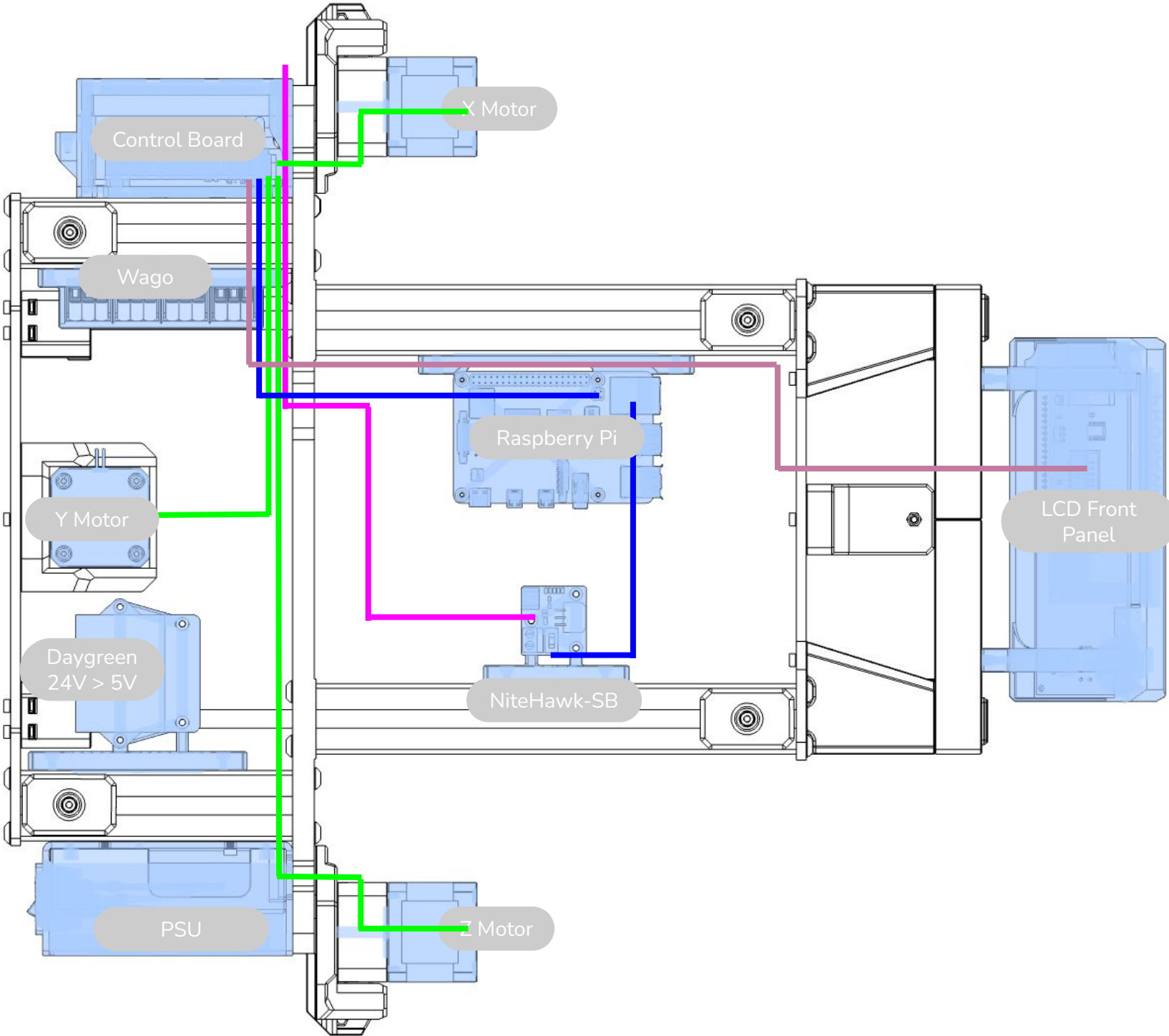
You have flexibility in the placement of certain components and how to wire them.

Prioritize clean runs and secure wires where possible. To receive a serial, wiring must be tidy and secured from any motion component.

A possible layout is shown here, but you may choose your own adventure.

KEY

- Stepper Motor Wire
- Toolhead Cable
- USB
- Display Cable



HEATBED WIRING

The Prusa Assembly Guides are a good reference for wiring and assembly of the heatbed cabling.

Prusa MK4S Heatbed Assembly:

https://help.prusa3d.com/guide/7-y-carriage-heatbed-assembly_707785#717694

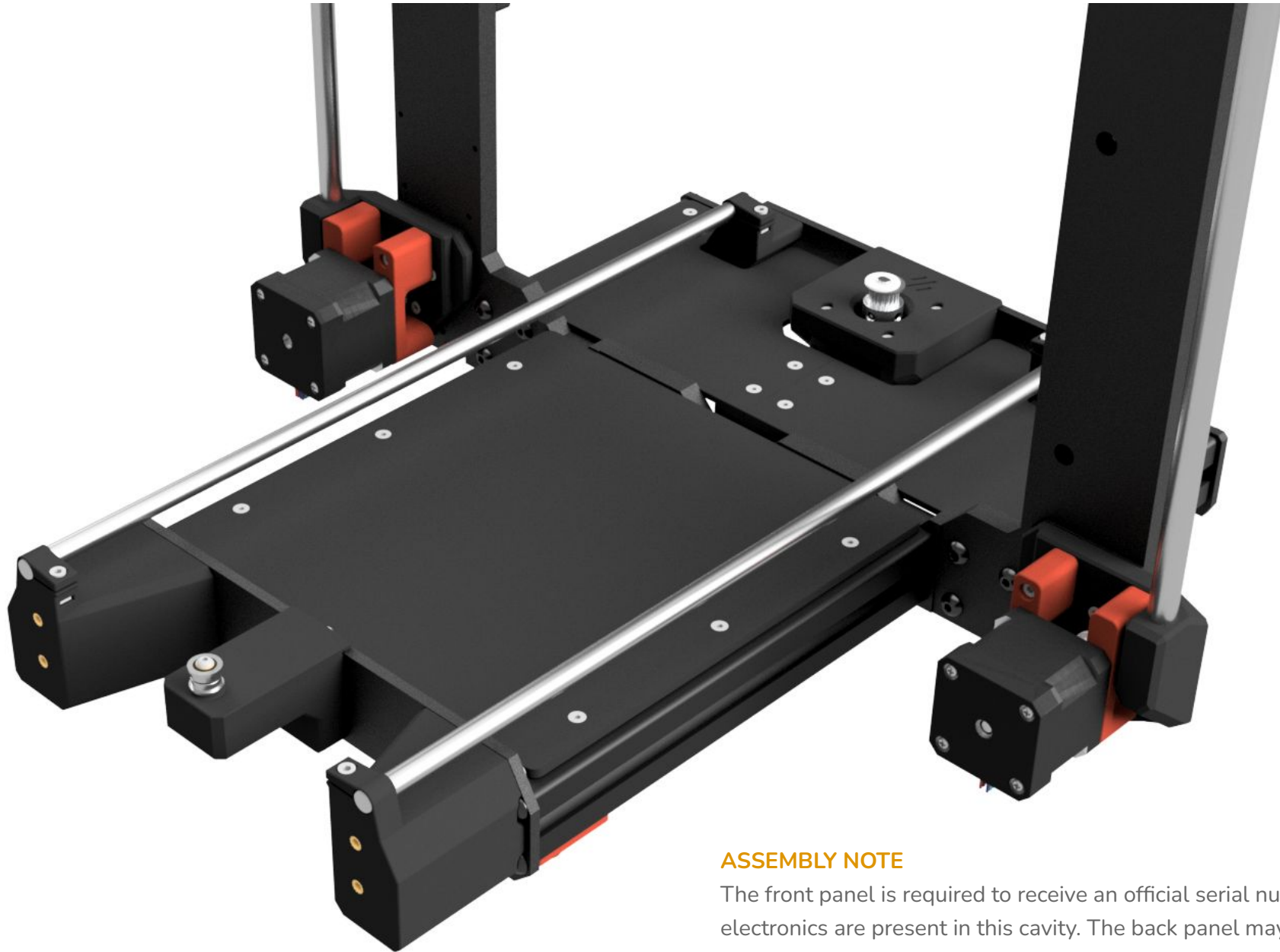
Reference Chapter 7, Steps 3 through 11.



TOOLHEAD WIRING

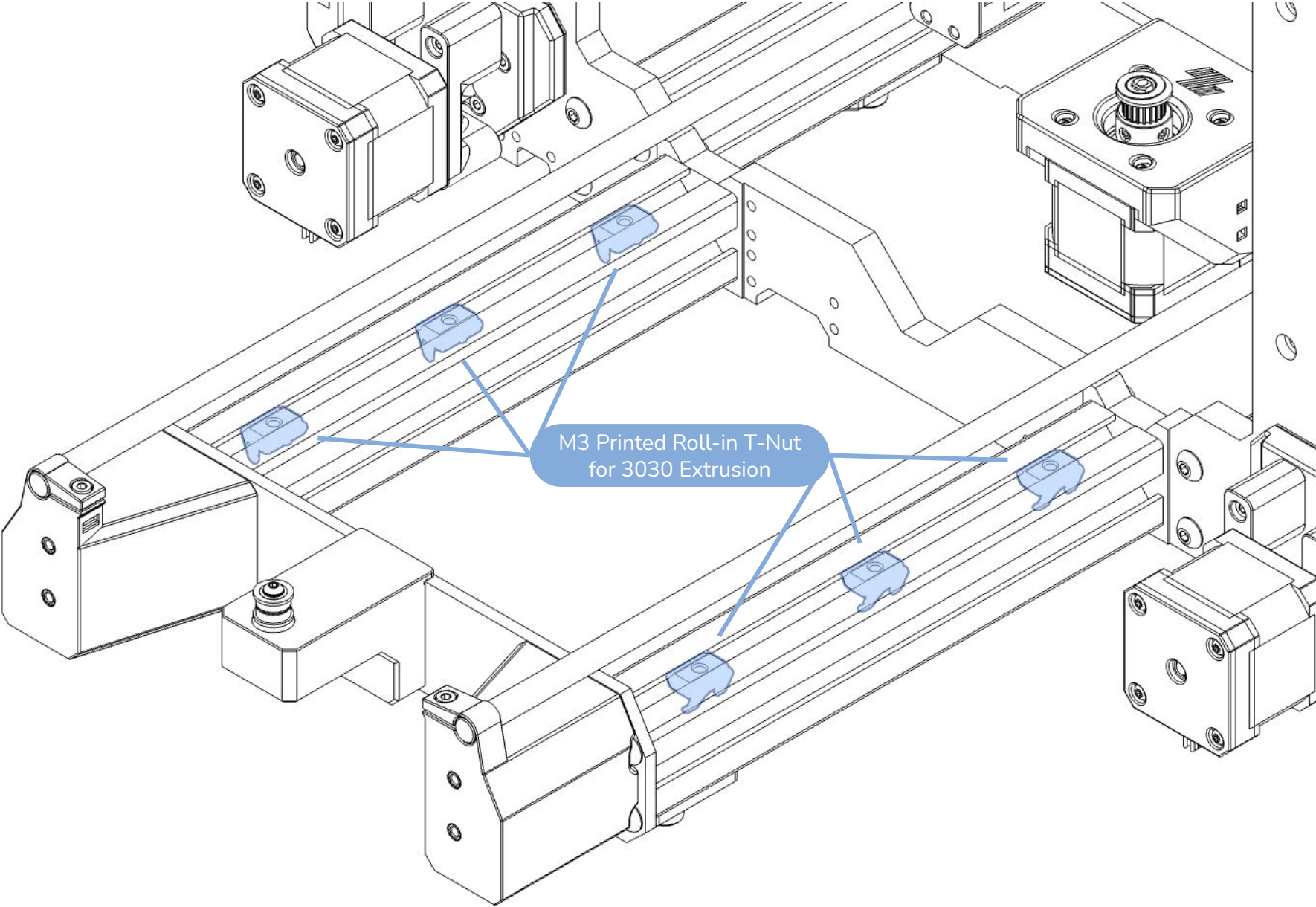
This will vary based on your chosen toolhead and toolhead board (if using).

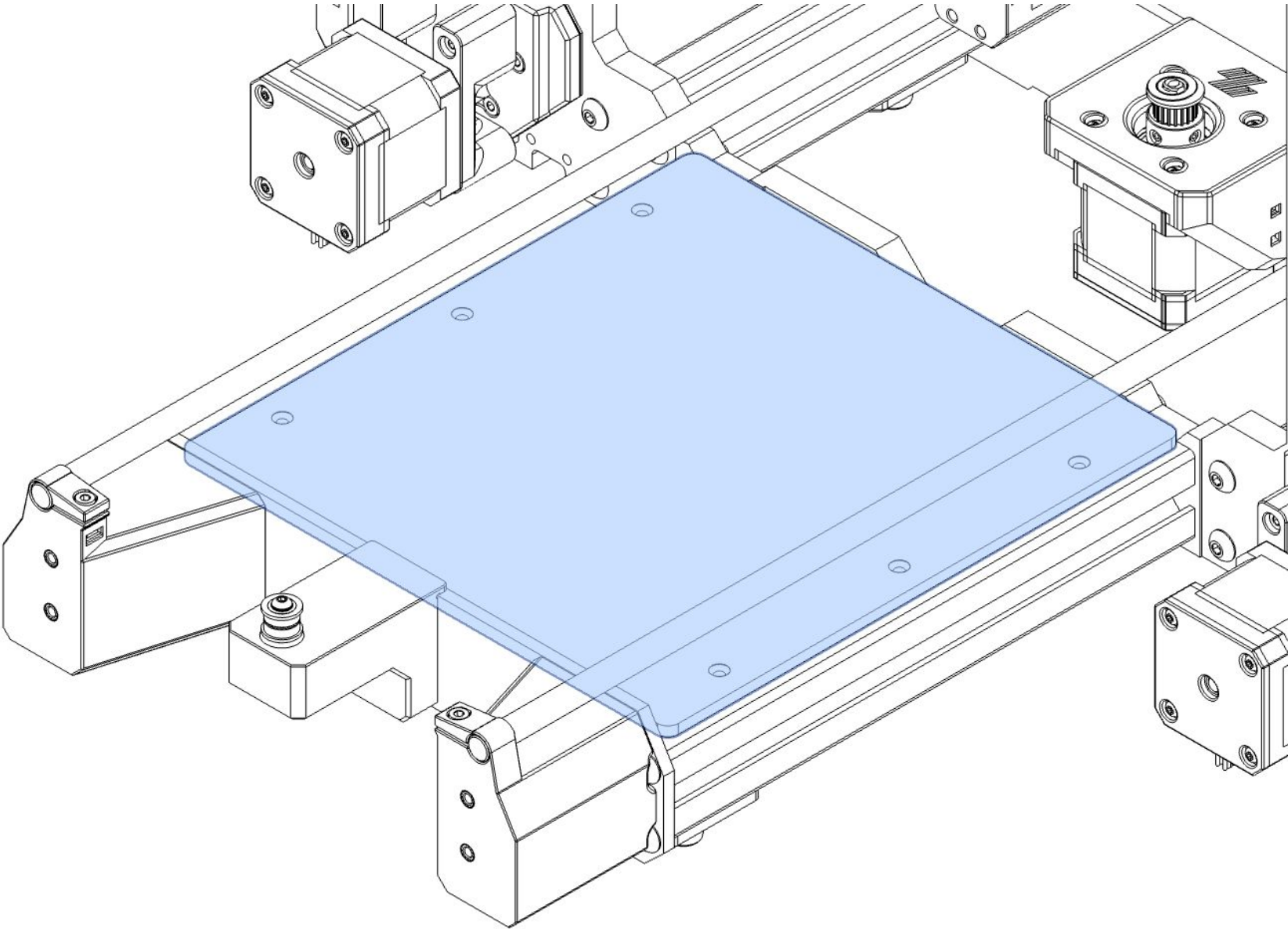
We suggest using the textile sleeve and nylon support to protect the wiring going to the toolhead. You can secure one end of the toolhead wiring to the recess on the door puller idler part. The other end can use the printed umbilical support on the toolhead.

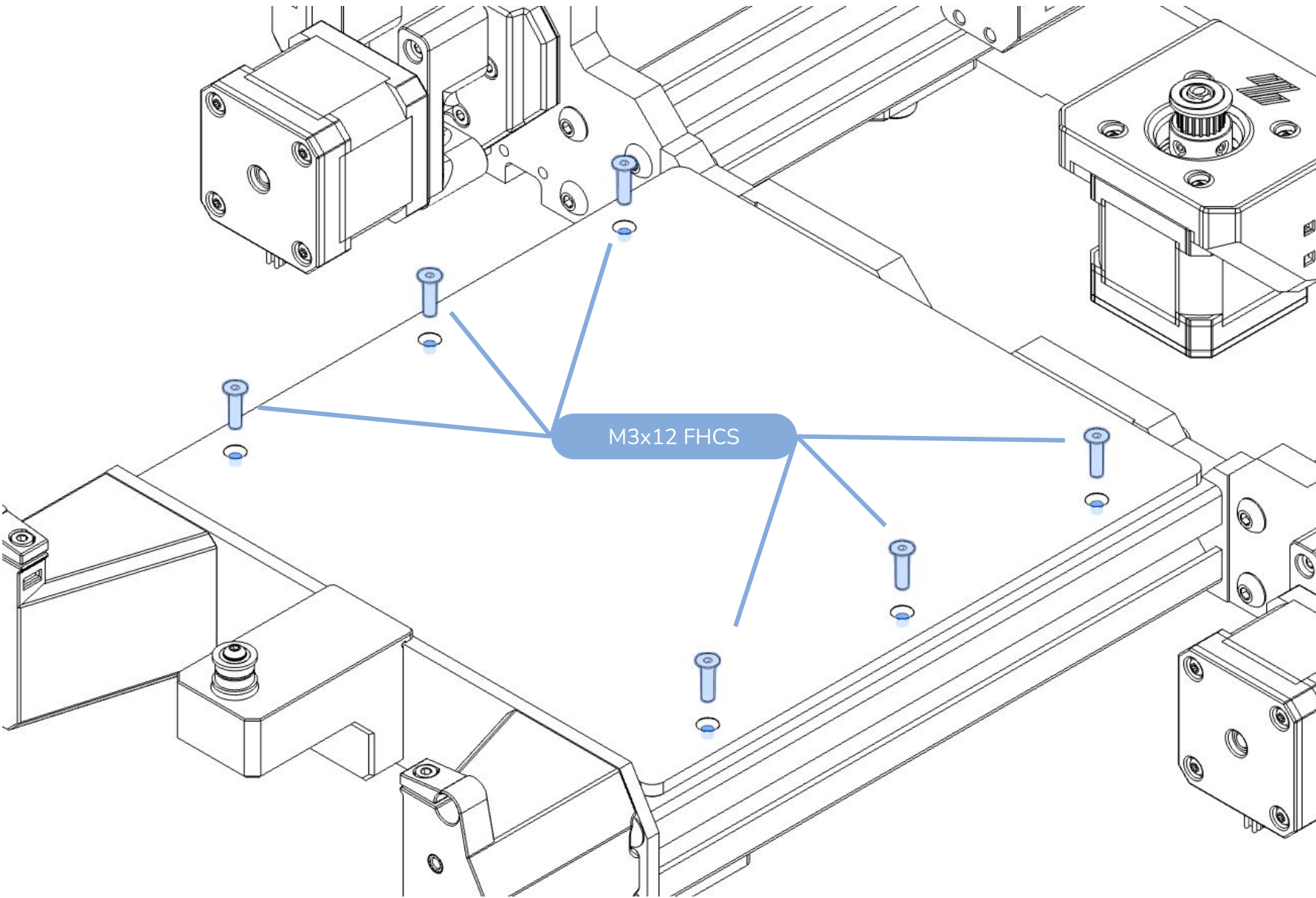


ASSEMBLY NOTE

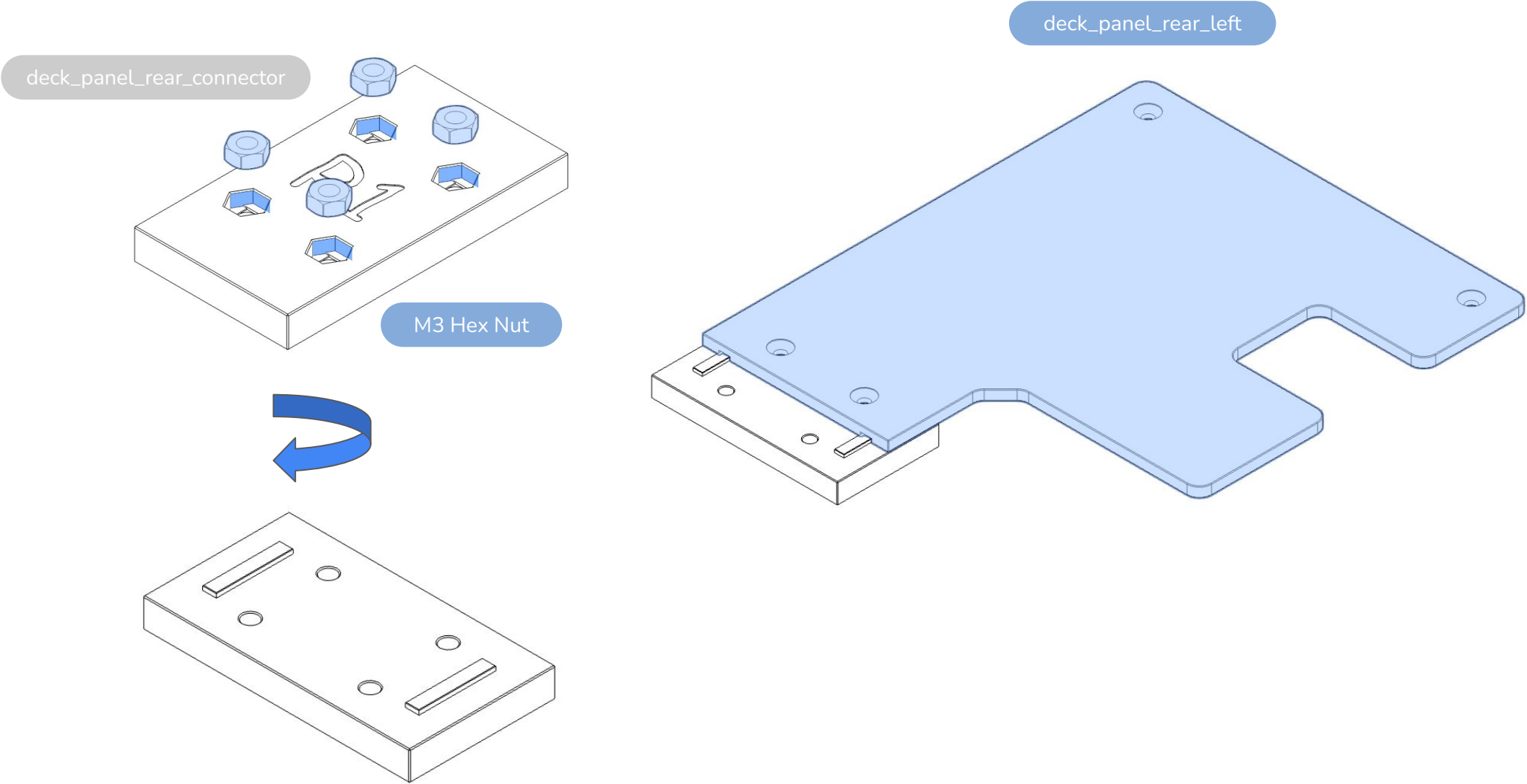
The front panel is required to receive an official serial number when electronics are present in this cavity. The back panel may be omitted if wiring is well secured and tidy, but is recommended.



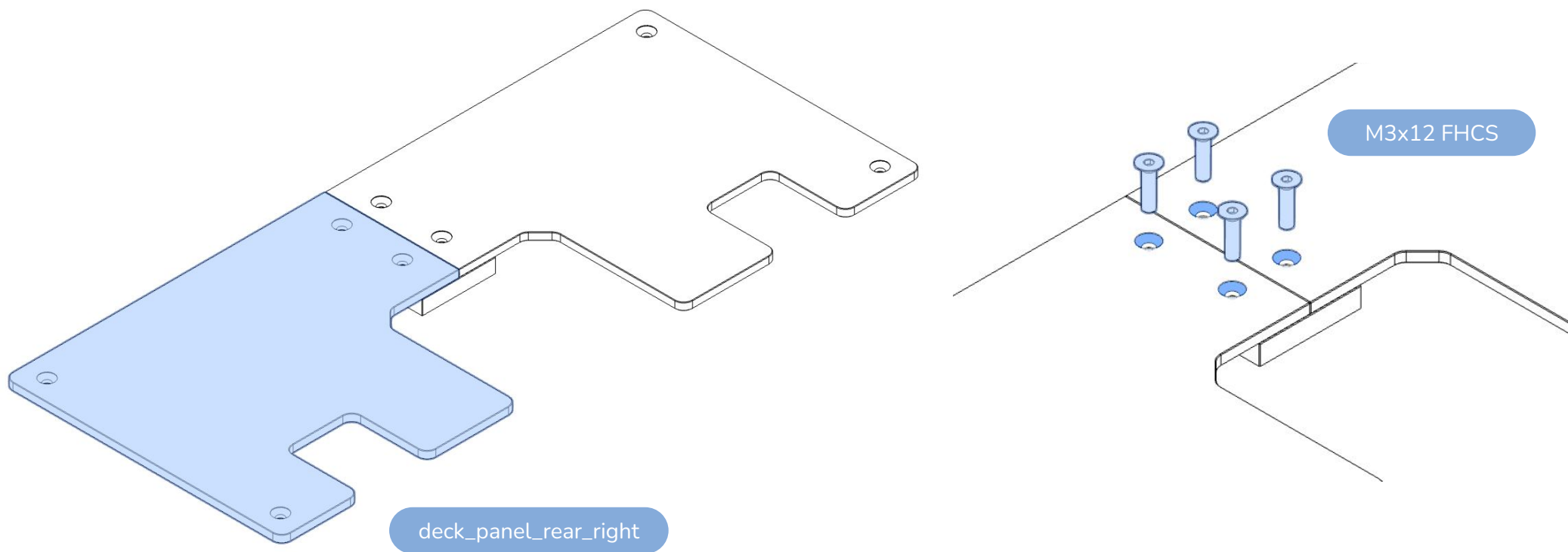




PANELS



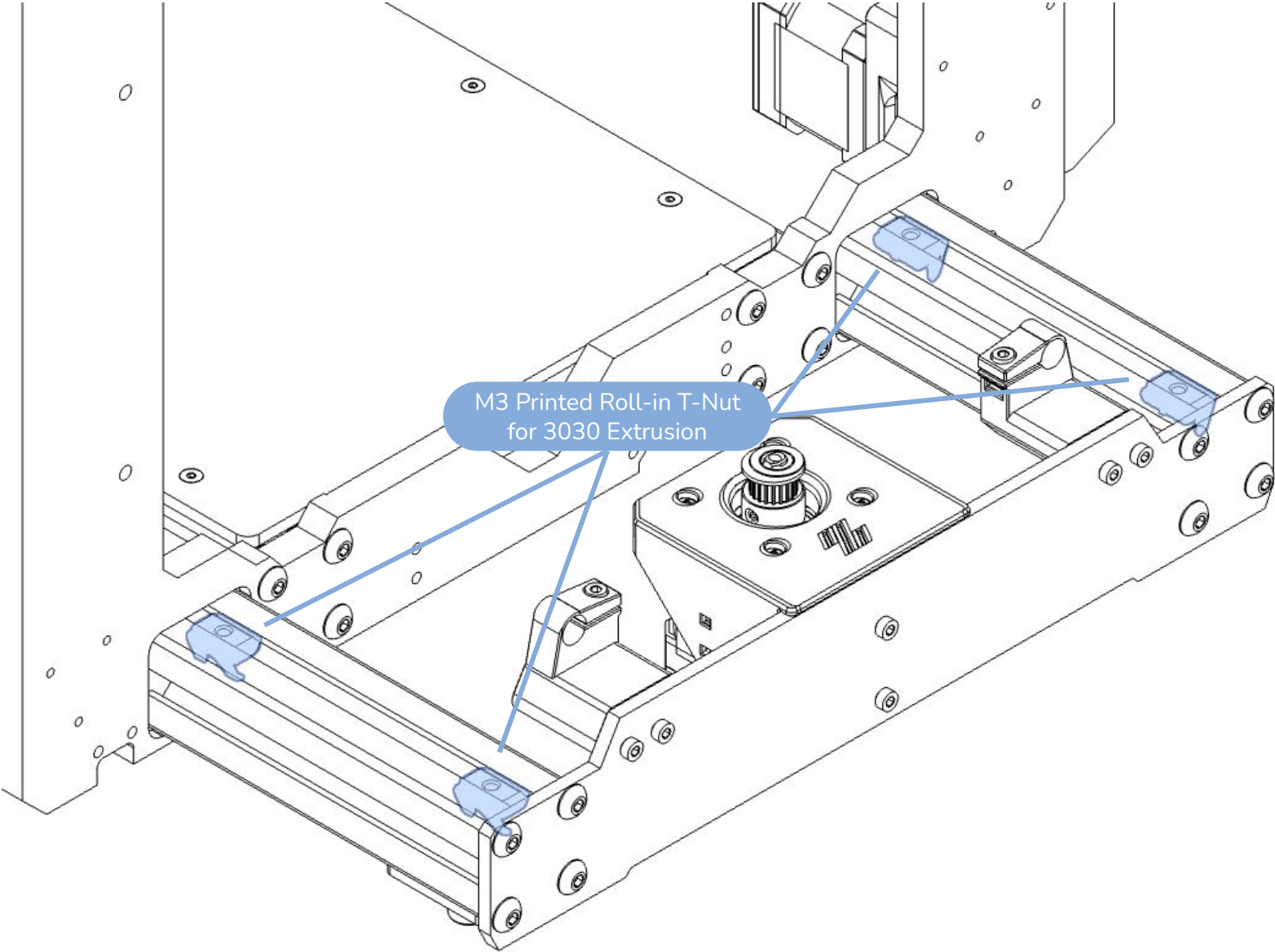
PANELS



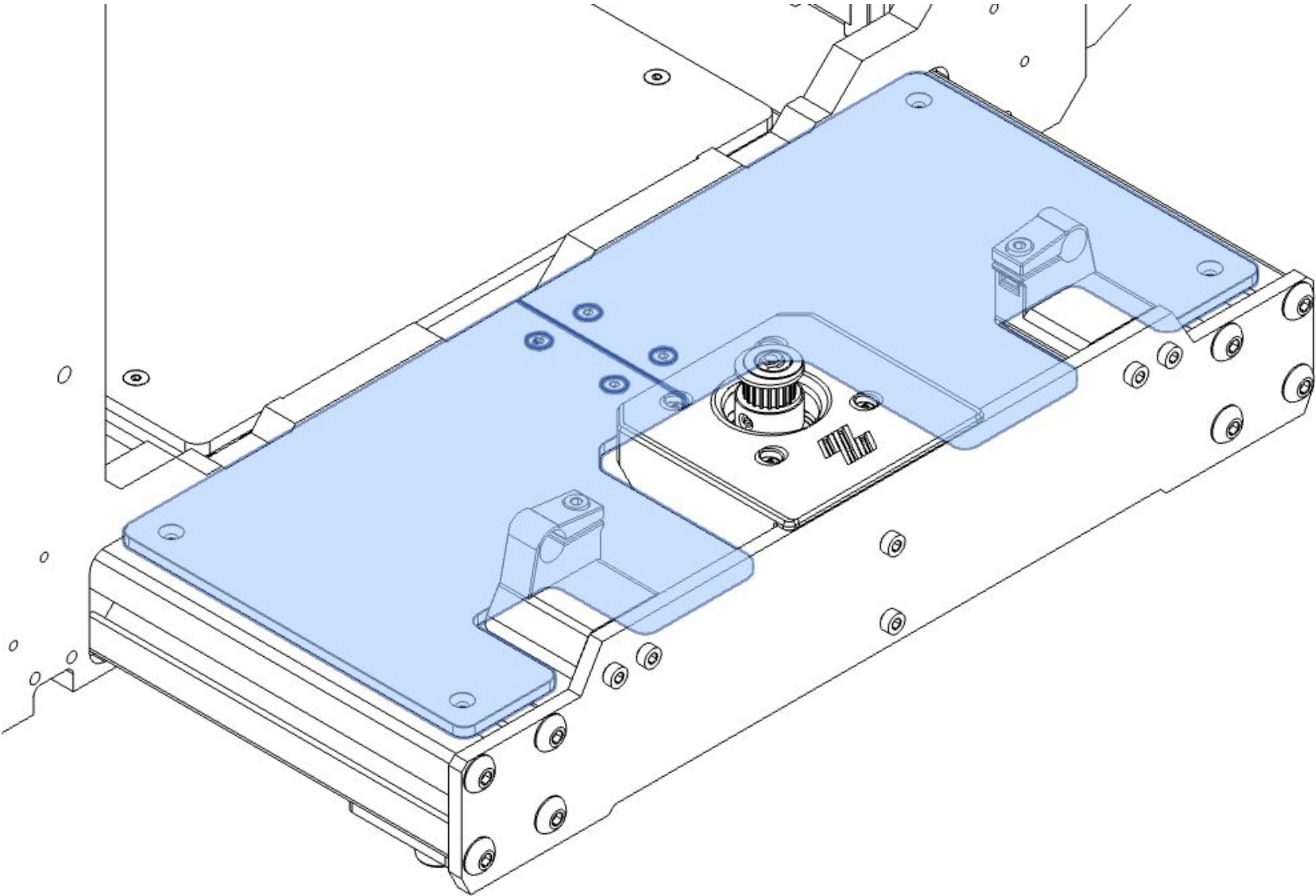
REAR PANEL

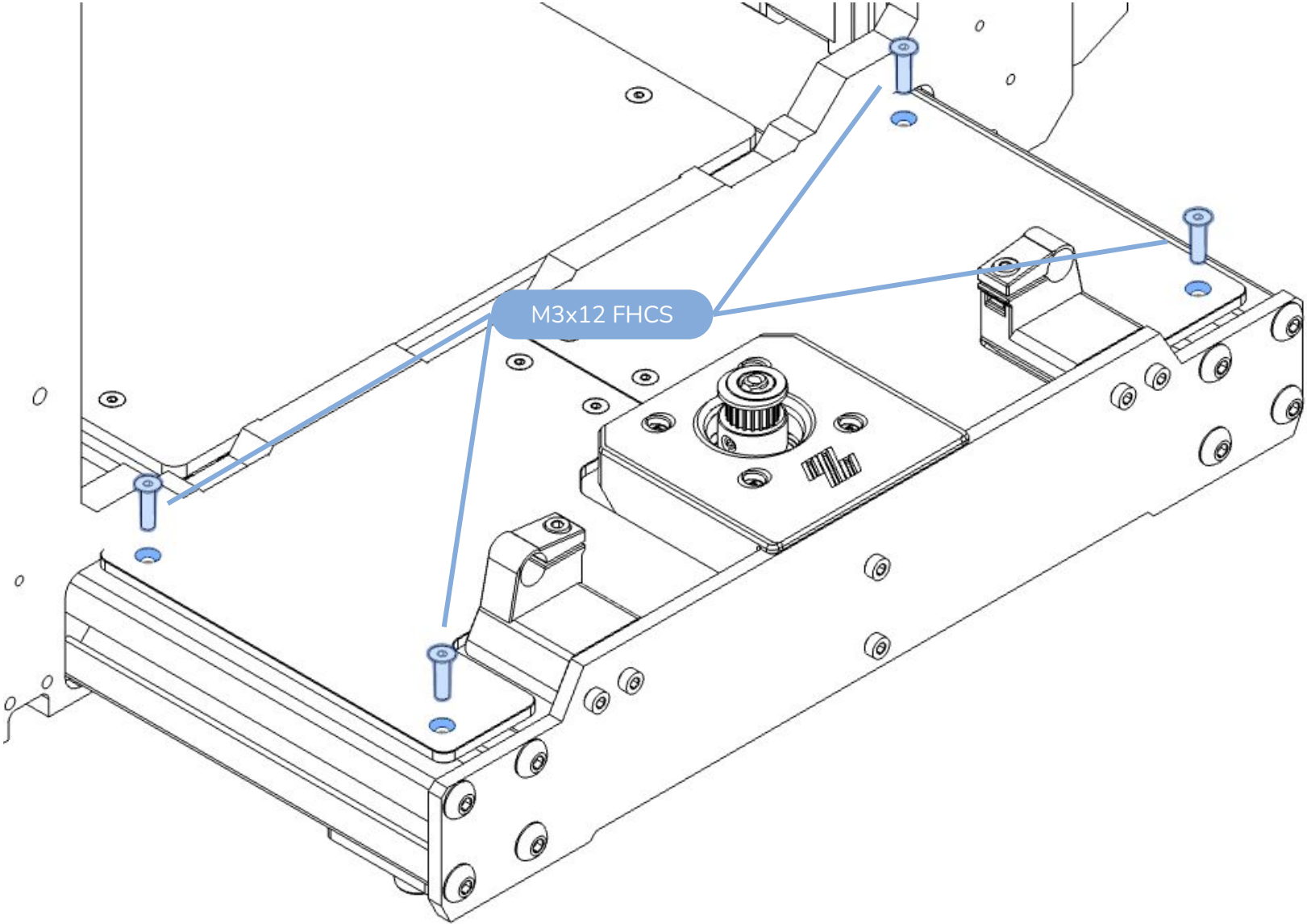
There are two variants of the right half of the rear panel:

- Prusa Black PSU
 - `deck_panel_rear_right_deltablackpsu`
- Prusa Silver PSU (or the Meanwell LRS-350-24 / RSP-320-24)
 - `deck_panel_rear_right_silverpsu_meanwellpsu`



M3 Printed Roll-in T-Nut
for 3030 Extrusion





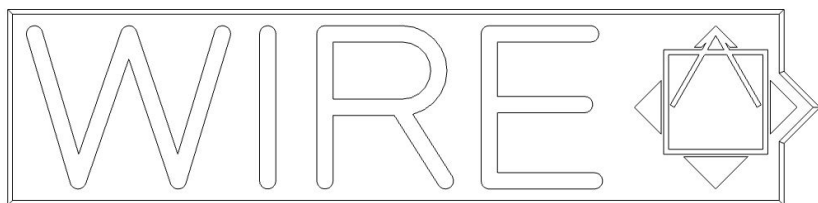
ACCESSORIES

Badge

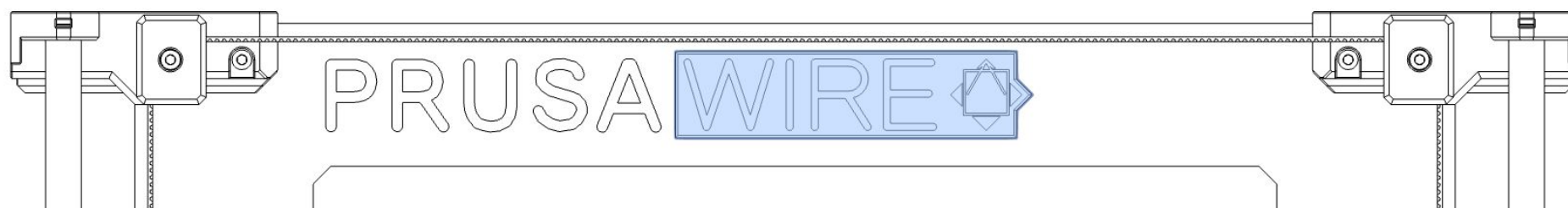
DID YOU KNOW?

VHB can be used in many structural applications. It is critical to cut the VHB to precisely 10mm x 80mm to maximize Core-XZ acceleration.

Prusawire Supplemental Badge



VHB



ACCESSORIES

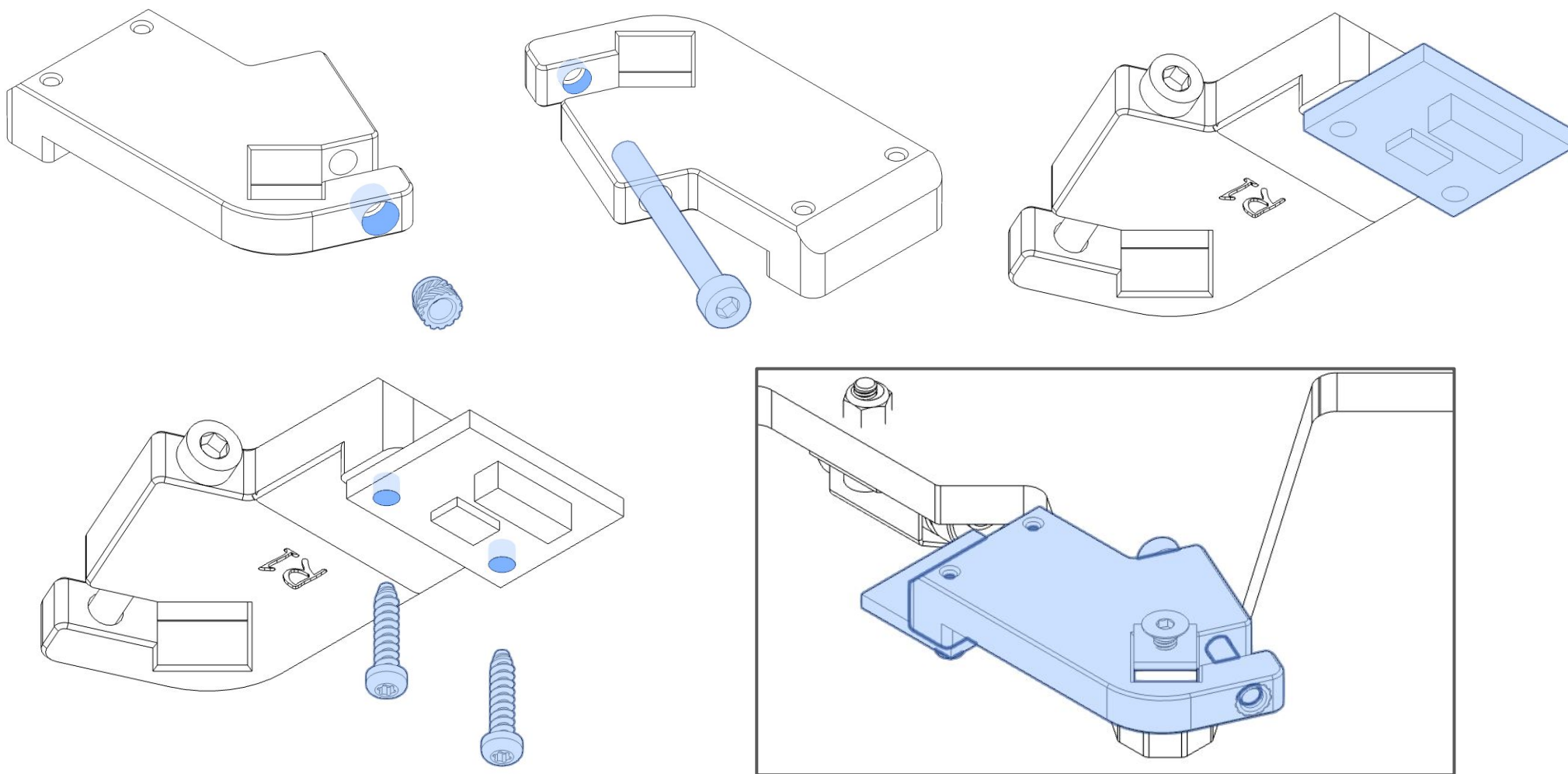
ADXL

I HAVE A NEED... A NEED FOR SPEED

While many toolhead boards feature an onboard ADXL/accelerometer, that doesn't help us with Y-Axis input shaping since Prusawire is a bed-slinger. We've included two ADXL mounts to attach to the front left Y-Carriage corner.

ADXL_Bed_Mount_by_Steve_Builds.stl when using MK3-style round bushings.

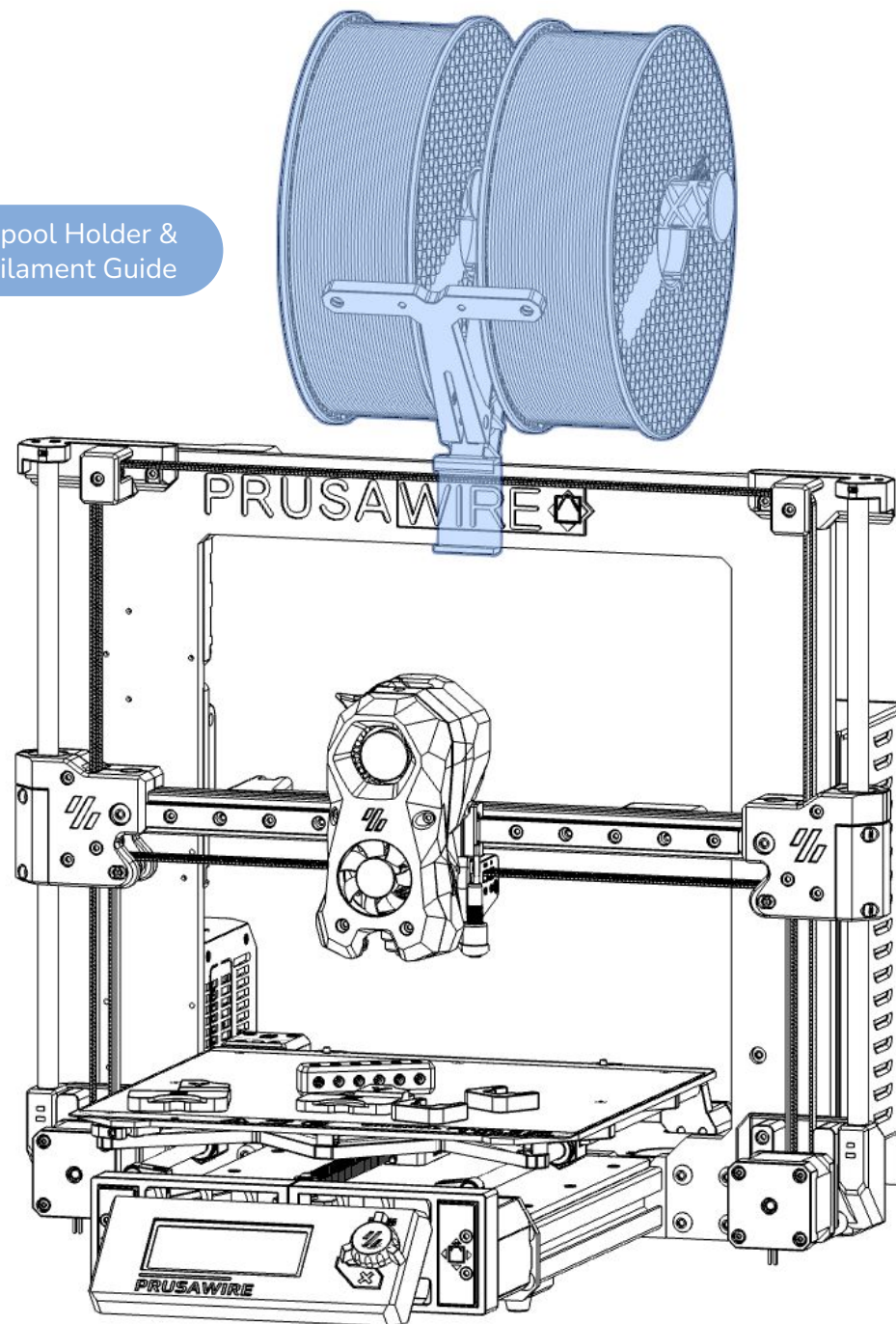
ADXL_Bed_Mount_Square_Expansion_Joints.stl when using MK4-style expansion joints. (shown below)



ACCESSORIES

Spool Holder

Spool Holder &
Filament Guide



NEXT STEPS

WHAT NEXT?

Congratulations on your assembled Prusawire!

Next you will want to:

- [Install Klipper and get the configurations installed](#)
- Test run your printer and start pushing plastic
- [Get a serial number assigned](#)
 - (You must post on the Positron3D Discord!)

You may also want to dig into [tuning and calibration](#).

Have you considered building a Micron?

[Prusawire Development: Behind The Scenes](#)

HOW TO GET HELP



<https://discord.com/invite/positron>





WEBSITE

<https://prusawire.positron3d.com>

GITHUB

<https://github.com/Positron3D/Prusawire>

DISCORD

<https://discord.com/invite/positron>
